Landau Geometry of Computational Resources

Effective semantics, formal geometry, and crystalline resource calculus

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Abstract

We construct foundations for a differential geometry of computational resources in effective domain and realizability settings. Exact rational cost profiles give a ring of bounded coefficients and an algebra of eventual germs. Embedded resource modules represent asymptotic budgets; a coherent locale classifies their families, and measured execution supplies the implementations witnessing feasibility. Compactness and supercompactness characterize finite dominating portfolios and optimal implementations.

Polynomial event valuations connect execution to differential geometry: their measured Taylor coefficients agree with geometric jets. Embedded budgets have formal deformation classifiers, and finite systems of budget and valuation constraints have intrinsic lifting obstructions. Overhead localization acts on these obstructions through their annihilators.

The primary infinitesimal geometry represents formal principal parts and completed diagonal cochains. An explicit contraction compares the latter with de Rham forms; integrable connections yield stratifications and crystalline jet coalgebras. The comparisons retain resource parameters, exact coefficients, and effective finite observations. A measured family exhibits polynomial bounds at every finite order without one polynomial budget for the complete coefficient family; its generating function also provides a nonzero logarithmic de Rham class and crystalline transport. Event instrumentations exhibit curvature, while logarithmic resource connections distinguish asymptotic pole orders.

Keywords: resource semantics; asymptotic complexity; effective domains; categorical semantics; synthetic differential geometry; deformation theory.
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Introduction

A differential geometry of computational resources must retain both the implementations that realize a bound and the coefficients through which that bound varies. We construct such a geometry from measured execution and asymptotic resource modules. Its formal neighbourhoods, differential forms, connections, and jets are related by explicit comparison maps; their finite coefficients remain tied to the underlying measurements.

The starting point is the relation $f = O(g)$. For rational profiles on a filtered input object, let K be the algebra of eventual germs and $B = O(1)$ its ring of bounded coefficients. Bounded division identifies $f = O(g)$ with membership in the embedded module Bg. The embedding records the costs admitted by a budget, while the exact coefficients support addition, multiplication, and differentiation. Growth ideals extend principal budgets to families with several or indefinitely many bounds.

Four connected contributions.

(1) *Resource spaces and computational witnesses.* The growth lattice L constructs a coherent locale $\mathfrak{C}_{\mathrm{Ev}}$ with

$$\mathrm{Loc}_{\mathcal{E}}(Z, \mathfrak{C}_{\mathrm{Ev}}) \cong \mathrm{Sub}_{\Delta_Z B}(\Delta_Z K).$$

A generalized point therefore specifies a family of embedded budgets. A measured span $V \xleftarrow{p} P \xrightarrow{\Phi} K$ supplies the implementations satisfying them. Its witness sheaf is a left Kan extension; its feasible open is supercompact exactly when an optimal implementation exists, and compact exactly when a finite dominating portfolio exists (Theorems 3.2, 4.14 and 5.2 and Proposition 4.19).

(2) *Measured variation and resource deformation.* Folding an execution history with truncated Taylor-valued event weights computes the geometric jet of its cost family (Theorem 6.1). For a principal marked budget M, finite logarithm identifies its deformation functor with the formal module determined by $T_M = \mathrm{Hom}_B(M, K/M)$. A feasible valuation $F(x)$ and a moving budget have linearized incidence

$$(v, \theta) \longmapsto qDF_x(v) - \theta(F(x)), \qquad q : K \twoheadrightarrow K/M.$$

Its kernel describes tangent directions and differences between lifts; its cokernel contains successive lifting obstructions. The construction includes finite budget diagrams and their overhead localization (Theorems 7.3, 7.14 and 7.21 and Proposition 7.18).

(3) *Formal and de Rham resource geometry.* The primary geometry is an internal étale site of nilpotent arrows. Representability identifies its infinitesimal function objects with principal parts and completed diagonal cochains. A finite equivariant contraction compares those cochains with algebraic de Rham forms, including localized coordinate cells. This constructs the differential theory over the exact resource coefficients and their classified parameters (Theorems 8.4, 9.6, 10.3 and 10.8).

(4) *Crystalline transport and effective coefficients.* Integrable connections are equivalent to formal stratifications. Principal-parts correspondences construct jets and differential operators; their completed system is the primary jet comonad, with algebraic stratifications giving its crystalline coalgebras. Effective finite observations and exact Taylor coefficients agree with these geometric constructions (Theorems 10.6 and 11.8 and Propositions 11.6 and 9.18).

Resource questions through the geometry. Finite-order feasibility asks which variations preserve a budget; completion asks whether compatible finite variations admit a common resource bound. De Rham forms express infinitesimal variation, and connections test its compatibility with transport. Crystalline jets collect the response coefficients and specify how they re-centre and compose. These questions concern the same measured families.

For example, a uniformly constructed family of finite programs has weighted costs $F_r(z) = \sum_{j=0}^{r}(n+1)^j z^j$. Every finite coefficient list has a polynomial budget, but the compatible formal family has no single polynomial growth degree. Its generating function gives a logarithmic de Rham cocycle, an integrable connection, and crystalline transport (Examples 11.9 and 9.12 and Proposition 10.10). Two event instrumentations detect order through a mixed jet and its curvature (Proposition 10.7). Coupled budgets show how an obstruction can persist when each budget separately moves (Example 7.19), and logarithmic residues distinguish asymptotic pole orders (Corollary 12.15).

Organization and mathematical settings. Sections 1–5 construct the effective coefficient setting, classified resources, and measured feasibility. Sections 6–7 connect history valuations to differential coefficients and constrained resource deformations. Sections 8–11 construct the geometric function objects, completion, de Rham theory, and crystalline jets, developing the measured examples through those comparisons. Section 12 gives numerical execution, size bounds, certificates, and logarithmic resource connections. Table 1 records the coefficient settings.

Table 1: Objects and their roles in the construction.

Notation	Setting	Role
$\mathcal{E},\ \Sigma \hookrightarrow \Omega$	Effective or relative realizability; finite-hit truth values	Computational names, partial returns and positive observations
$J_\Sigma,\ \mathcal{A}_\Sigma$	Selected ideal completion and its algebras	Admitted directed suprema and recursive execution
$S = \mathsf{Q}_\Sigma$	Effective nonnegative extended lower reals	Continuous quantitative interpretation
$K,\ B,\ \mathfrak{m}$	Exact rational germs, bounded germs, vanishing germs	Signed resource coefficients; $O(g) = Bg$, $o(g) = \mathfrak{m}g$
$L,\ \mathfrak{C}_{\mathrm{Ev}}$	Growth lattice and budget locale	Ordinary families of marked resources
$\mathscr{S} = \mathrm{Sh}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}})$	Budget sheaves, equivalently $[L, \mathcal{E}]$	Universal budget and implementation witnesses
$\mathscr{H}_{B,\mathrm{et}},\ \mathcal{O}_{\mathrm{inf}}$	Primary étale infinitesimal-arrow sheaves	Represented functions, formal nerve, de Rham comparison and jets
$\mathcal{G}_B,\ j_B M$	Based projective Weil diagrams	Finite resource coefficients and marked deformations

Conventions used throughout. Internal existence means a regular image. Formulas on local representatives define maps on quotients after their kernel-pair equations have been proved. Finite projectivity means a displayed finite-free retract, as specified in Section 1. Semantic incidence has values in Ω; positive observation requires its stated factorization through Σ. A formal coefficient family is an internal compatible family of finite stages; a uniform computational name for it is additional data. Computational environments and size maps are retained in implementation comparisons. The based geometry supplies finite module coefficients; primary representability supplies the function objects used for formal cochains.

Relation to published foundations and complexity theory. The computational construction draws on effective domain theory (Smyth 1977; Abramsky and Jung 1994) and realizability (Hyland 1982; Birkedal and van Oosten 2002; van Oosten 2008). Within that setting we define the positive-observation selection of ideal completion and prove the continuous presentation theorem needed for the resource algebra. The latter adapts the method of (Jung et al. 2008) to internal selected ideals. We also compare the selected-domain recursion with natural-number-indexed execution. The distinction between synthetic completeness notions and ordinary execution is discussed in (van Oosten and Simpson 2000).

The complexity interpretation connects three established strands. Blum's machine-independent measures (Blum 1967) provide the numerical comparison: our first-hit construction verifies the two measure properties, and speedup gives examples of failure of optimality. Schellekens's complexity spaces (Schellekens 1995) relate topological complexity to domain-theoretic completion. Resource monoids and resource-bounded realizability are developed in (Dal Lago and Hofmann 2011). Here the classifier records embedded asymptotic modules and their budget families, together with measured implementation witnesses. The optimality criteria and resource deformations are formulated for these objects.

The cost-sensitive adequacy method of Niu et al. (2024) relates abstract cost semantics to computational execution. Their partial cost monad and the cost–behaviour distinction in (Niu et al. 2022; Grodin et al. 2024) provide the relevant execution-semantic antecedents. Here retained histories give domain-and-value adequacy and valuation-independent erasure; Theorem 6.1 further identifies Taylor-valued instrumentation with geometric cost jets. These are comparisons for the constructed execution model and its polynomial valuations. Passing from runs to size bounds is related to cost-recurrence extraction (Danner et al. 2015). Representation-sensitive operator complexity (Kawamura and Cook 2012) motivates retaining input names and the measured overhead of changing them.

The resource algebra also connects with formal treatments of O-notation (Avigad and Donnelly 2004) and filter-based asymptotic verification (Guéneau et al. 2018). The latter provides a comparison for both the eventual filters and the finite resource certificates. Asymptotic gauges and moderate rings (Giordano and Luperi Baglini 2016), together with their functorial changes of scale (Luperi Baglini and Giordano 2017), are compared directly in Proposition 2.21.

Growth and derivations also occur in the asymptotic differential algebra of (Aschenbrenner et al. 2017). The present germ rings retain zero divisors and are not treated as ordered fields. Finally, algorithm portfolios (Gomes and Selman 2001) provide the computational comparison for our finite-portfolio property, which is characterized by compactness of the feasible open.

Geometric antecedents and the resource-specific comparisons. Kock's algebraic synthetic differential geometry (Kock 2006) provides the comprehensive and Weil identities used here. The symmetric-algebra and universal-derivation constructions belong to the differential and codifferential categorical framework (Blute et al. 2016, 2020). The contribution in Section 6 is their construction over the exact Landau coefficients and their comparison with the retained quantitative presentations. The primary infinitesimal-arrow site supplies the endpoint adjunctions and represented function objects needed for that comparison.

Principal parts, algebraic de Rham theory and crystalline stratifications have the classical foundations (Grothendieck 1966; Berthelot and Ogus 1978; Stacks Project Authors 2026a). Sections 9–11 identify their coefficient objects in the primary resource geometry and construct a finite, equivariant contraction compatible with completion and displayed projective retracts. The construction of jets from infinitesimal shape and their coalgebra interpretation has a direct synthetic precedent in Khavkine and Schreiber (2027). Our comparison specifies the algebraic endpoint model, the finite principal-parts coefficients, the effective-epimorphism domain of coalgebra descent, and the crystalline algebraic sector within the full jet-coalgebra category. The marked-resource deformation and moving-incidence calculations of Section 7 use these exact coefficient structures to describe variations of asymptotic budgets.

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1 Effective domains and exact quantitative presentations

The later resource constructions need three interfaces: partial and recursive execution, exact data with a continuous quantitative interpretation, and positive computational observations. This section builds them in the realizability setting. The selected ideal completion supplies the continuous operations used by execution. Exact positive presentations then connect that setting to signed resource algebra, while the observation and name constructions specify how its objects are presented computationally.

1.1 Published ambient models and internal reasoning

We work in the effective topos and its relative-realizability variants (Hyland 1982; Birkedal and van Oosten 2002; van Oosten 2008), using effective domain theory as the order-theoretic background (Smyth 1977; Abramsky and Jung 1994). The selected completion and continuous presentations required here are constructed below. The supporting presentation and coding arguments are given in Sections A and E.

Write $\mathcal{E}$ for the ambient elementary topos and $\mathbb{N}$ for its natural numbers object. Internal subsets, images, and quotients are those of $\mathcal{E}$. An internal family includes its free parameters: $x : Z \to X$ is a generalized element, and substitution is pullback in the corresponding slice. Effective quotients allow a map to descend exactly when it coequalizes the kernel pair of its presentation. A total functional relation determines a unique arrow, because its projection is both monic and regular epic.

The numerical realization is $\mathbf{Eff} = \mathrm{RT}(K_1)$, with the usual partial-recursive application. Its oracle versions use K_1^Z. For supplied function names we also use relative realizability $\mathrm{RT}(A_\#, A)$: names belong to the ambient partial combinatory algebra A, while uniform trackers belong to its elementary subalgebra $A_\#$. This includes the usual function-realizability and Kleene–Vesley presentations. The exact-completion presentation by partitioned assemblies (Birkedal 2000, Section 10.6, especially Theorem 10.6.7) provides the projective covers used in Section E. A partitioned assembly consists of a carrier and its naming map into A; a morphism has a single permitted tracker on all valid names.

Definition 1.1 (Finite-hit truth values and overtness)**.** The Rosolini object $i : \Sigma \hookrightarrow \Omega$ is the image of

$$h : 2^{\mathbb{N}} \longrightarrow \Omega, \qquad h(\alpha) \iff \exists n : \mathbb{N}\ \alpha(n) = 1.$$

A Σ-classified partial map is a span whose domain mono is classified by Σ. An object X is Σ-overt when existential quantification restricts uniformly to $\exists_X : \Sigma^X \to \Sigma$.

Proposition 1.2 (Dominance and uniform natural-number search)**.** *In the effective and relative-realizability models just described, Σ is closed under dependent conjunction, and $\mathbb{N}$ is Σ-overt. Thus Σ is a dominance and classified partial maps compose. Finite products, finite sums, regular-epimorphic images, and Σ-classified subobjects of overt objects are overt.*

Proof.

(1) **Construct finite truth operations.** Constant streams and finite interleaving give false, true, finite conjunction, and finite disjunction.

(2) **Prove closure under dependent conjunction.** To check dependent conjunction, pull two successive classified monos back to a partitioned projective cover P of their ambient object. Lift the first classifier through $2^{\mathbb{N}} \twoheadrightarrow \Sigma$, obtaining a total tracked stream α_p. On its hit assembly $H_\alpha = \{(p, n) : \alpha_p(n) = 1\}$, lift the second classifier to a total tracked stream $\beta_{p,n}$. This is possible on the displayed partitioned subassembly. The stream

$$\gamma_p(\langle n, k\rangle) = \begin{cases} \beta_{p,n}(k), & \alpha_p(n) = 1, \\ 0, & \alpha_p(n) = 0 \end{cases}$$

is total: its first branch is evaluated only on a valid hit name. Its hit image is the composite subobject. Its classifier therefore coequalizes the kernel pair of P and descends through $i : \Sigma \hookrightarrow \Omega$. This proves dominance in every parameter context.

(3) **Construct uniform natural-number search.** For $\exists_{\mathbb{N}}$, cover $\Sigma^{\mathbb{N}}$ projectively and lift evaluation on the product of that cover with $\mathbb{N}$ to streams $\alpha_{p,n}(k)$. The single total stream $\gamma_p(\langle n, k\rangle) = \alpha_{p,n}(k)$ represents the desired existential. Its classifier descends for the same reason. These constructions use the represented pairing, finite tests, and guarded evaluation, so their trackers remain in $A_\#$ on arbitrary supplied ambient names.

(4) **Preserve overtness under finite constructions.** Finite sums use disjunction; products and dependent sums use iterated restricted existential quantification.

(5) **Descend overtness along epimorphisms.** For an epic $q : X \twoheadrightarrow Y$, $\exists_y u(y) \leftrightarrow \exists_x u(qx)$ proves the image assertion.

(6) **Restrict overtness and verify coherence.** For a classified subobject, use dependent conjunction before the existential over its ambient overt object. The corresponding substitution, Frobenius, and Fubini equations hold in Ω and are reflected through the monomorphism i.

□

The subsequent internal constructions apply in the Rosolini-dominant recursion setting thereby realized; they retain arbitrary parameter contexts. In particular, a pointwise external list of semidecisions is not substituted for a map into $\Sigma^{\mathbb{N}}$. The dominance-based partial-element construction is standard (Escardó and Knapp 2017); explicitly,

$$L_\Sigma Y = \sum_{p:\Sigma} Y^p. \tag{1.1}$$

Its unit has true extent. Multiplication sends a nested partial element with extents p and $q(u)$ to extent $\exists u : p\, q(u)$ and the corresponding value. Strength retains an unused parameter. These formulas represent classified spans and prove the strong, commutative monad laws by reassociation of dependent extents.

1.2 The selected ideal completion

Ordinary ideal completion supplies directed suprema, but the computational construction must also track positive observations of those suprema. We therefore select ideals by their complete profile of positive upper tests. For an internal poset P, an ideal is an inhabited directed lower subset; $\mathrm{Idl}\, P$ denotes the object of all such ideals. Put

$$U_\Sigma P = \mathrm{Pos}(\mathcal{E})(P, \Sigma), \qquad \chi_I(u) = \exists p : P\, [I(p) \wedge u(p)].$$

Definition 1.3 (Admitted ideal)**.** Define $J_\Sigma P$ by the pullback

$$J_\Sigma P = \mathrm{Idl}\, P \times_{\Omega^{U_\Sigma P}} \Sigma^{U_\Sigma P}.$$

Thus admission is uniform factorization of the complete positive upper-observation profile. The underlying ideal membership can remain Ω-valued.

The defining factorization is the cartesian square

$$\begin{array}{ccc} J_\Sigma P & \longrightarrow & \Sigma^{U_\Sigma P} \\ \big\downarrow & & \big\downarrow \\ \mathrm{Idl}\, P & \xrightarrow[I \mapsto \chi_I]{} & \Omega^{U_\Sigma P}. \end{array}$$

The lower arrow is the complete upper-test profile; admission is its uniform factorization through Σ.

Theorem 1.4 (Selected completion and its algebra category)**.** *The formulas*

$$\eta_P(p) = {\downarrow} p, \qquad (J_\Sigma f)(I) = {\downarrow} f(I), \qquad \mu_P(\mathcal{I}) = \bigcup_{I \in \mathcal{I}} I$$

define a locally monotone monad on internal posets, with $J_\Sigma \eta_P \leq \eta_{J_\Sigma P}$ and a monad inclusion $J_\Sigma \hookrightarrow \mathrm{Idl}$. It preserves finite products and disjoint finite sums. Its algebra category $\mathcal{A}_\Sigma = \mathrm{Alg}(J_\Sigma)$ is cartesian closed; $[A, B]_\Sigma$ is the poset of maps preserving admitted suprema, with pointwise order and pointwise admitted suprema.

Proof.

(1) **Admit the monad operations.** For principal ideals $\chi_{\downarrow p}(u) = u(p)$. For lower images, $\chi_{\downarrow f(I)}(u) = \chi_I(uf)$. For multiplication define $\widehat{u}(I) = \chi_I(u)$, a monotone map $J_\Sigma P \to \Sigma$. Then

$$\chi_{\mu_P(\mathcal{I})}(u) = \chi_{\mathcal{I}}(\widehat{u}) \in \Sigma.$$

This proves closure of all three operations.

(2) **Verify the monad laws and KZ structure.** Their laws are the lower-image and union identities in ordinary ideal completion, reflected through the monic inclusion. If $K \in (J_\Sigma \eta_P)(I)$, then $K \subseteq\!\downarrow p \subseteq I$ for some $p \in I$; this proves the displayed KZ inequality. Its algebra equations identify an algebra map $a : J_\Sigma A \to A$ with the supremum of each admitted ideal, and its morphisms with maps preserving these suprema.

(3) **Preserve finite products.** An ideal of $P \times Q$ is the product of its two lower-image projections: directedness supplies a common bound for any two coordinate witnesses. Conversely

$$\chi_{I\times J}(u) = \chi_I\big(p \mapsto \chi_J(q \mapsto u(p,q))\big)$$

proves admission of that product. The terminal case follows from the unique ideal of 1.

(4) **Preserve disjoint finite sums.** An inhabited directed ideal of $P + Q$ lies in a unique summand; the local summand choice descends by uniqueness. Extending a test by false on the other summand proves admission in both directions. The empty poset has no ideals.

(5) **Construct the exponential algebra.** Let $H \subseteq B^A$ be the object of monotone supremum-preserving maps and $I \in J_\Sigma H$. Define $f_I(x) = \sup_{h\in I} h(x)$. For $J \in J_\Sigma A$, the two iterated suprema over I and J coincide: both are the supremum of the admitted lower image of $I \times J$ under evaluation. Thus f_I preserves the supremum of J, and the pointwise operation makes H an algebra.

(6) **Verify evaluation and currying.** The same calculation proves continuity of evaluation. For $k : C \times A \to B$ continuous, the ideal $\downarrow c \times J$ proves continuity of $k(c,-)$ in an arbitrary context $c : C$. The ideals $I\times \downarrow x$ prove continuity of its transpose $C \to H$. The ordinary evaluation and transposition identities restrict, giving the exponential universal property.

(7) **Retain parameter dependence.** Every construction used internal images, pullbacks, or uniquely specified suprema, so its equations retain their free parameters.

□

The Kock–Zöberlein structure. The inequality $J_\Sigma \eta \le \eta J_\Sigma$ identifies the selected completion as a Kock–Zöberlein monad on the locally ordered category of internal posets. Its algebra structure is characterized by the adjunction

$$a_A : J_\Sigma A \rightleftarrows A : \eta_A, \qquad a_A \dashv \eta_A, \qquad a_A \eta_A = \mathrm{id}_A\,.$$

Indeed $a_A(I) \le x$ holds exactly when $I \subseteq\!\downarrow x$. This characterizes each admitted supremum by its universal property. The algebra and algebra-morphism equations are therefore the following commutative diagrams:

$$\begin{array}{ccc} J_\Sigma J_\Sigma A & \xrightarrow{\mu_A} & J_\Sigma A \\ {\scriptstyle J_\Sigma a_A}\downarrow & & \downarrow{\scriptstyle a_A} \\ J_\Sigma A & \xrightarrow[a_A]{} & A \end{array} \qquad \begin{array}{ccc} J_\Sigma A & \xrightarrow{J_\Sigma f} & J_\Sigma B \\ {\scriptstyle a_A}\downarrow & & \downarrow{\scriptstyle a_B} \\ A & \xrightarrow[f]{} & B. \end{array} \tag{1.2}$$

The free-algebra adjunction gives $\mathcal{A}_\Sigma(J_\Sigma P, A) \cong \mathrm{Pos}(\mathcal{E})(P, U_{\mathcal{A}_\Sigma} A)$. It is the extension principle used below: an operation specified on generators extends uniquely when it respects the admitted suprema. The presentation theorem will apply this principle to linear and algebraic operations.

A discrete internal object X has only singleton inhabited directed ideals. Their unique elements define its algebra $\mathrm{Disc}\, X$. Every underlying map from it to an algebra preserves those ideals; hence $\mathrm{Disc} \dashv U_{\mathcal{A}_\Sigma}$. A preorder is treated by its effective antisymmetrization before applying J_Σ; monotone maps and the displayed ideal formulas descend.

Proposition 1.5 (Lifting and least solutions)**.** *The partiality monad L_Σ lifts to a strong commutative monad on $\mathcal{A}_\Sigma$. For every pointed algebra A,*

$$\mathrm{lfp}_A : [A, A]_\Sigma \longrightarrow A, \qquad f \longmapsto \bigvee_{n:\mathbb{N}} f^n(\bot)$$

is an algebra morphism selecting the least fixed point. The construction commutes with parameters and strict intertwining maps.

Proof.

(1) **Construct the distributive-law map.** Order partial elements by extension and increase of their values. For $I \in J_\Sigma(L_\Sigma P)$, its extent is $p_I = \chi_I(\mathrm{def}) \in \Sigma$. Under this extent form the lower ideal V_I generated by the defined values of members of I. It is inhabited and directed. For $u : P \to \Sigma$ monotone, the positive extension $\widetilde{u}(z) = \exists d : \mathrm{def}(z)\, u(z(d))$ is Σ-valued by dominance, and $\chi_{V_I}(u) = \chi_I(\widetilde{u})$ under p_I. Consequently (p_I, V_I) defines $\lambda_P : J_\Sigma L_\Sigma P \to L_\Sigma J_\Sigma P$.

(2) **Verify the distributive-law identities.** Principal ideals and total partial elements give the two unit identities. Flattening either nested ideals or nested extents gives the same existential extent and the same ideal of defined values; these are the two multiplication identities. Thus λ is a distributive law, and $L_\Sigma a \lambda_A$ is the lifted algebra structure.

(3) **Lift strength and commutativity.** Strength is continuous because, once one directed member is defined, the defined members are cofinal among both relevant coordinate families. Its equations, and commutativity, follow on extents and values.

(4) **Construct the least fixed point.** An increasing sequence (x_n) generates an admitted lower ideal: its pairing with u is $\exists_n u(x_n)$, using Proposition 1.2. This constructs the displayed least fixed point. Continuity gives the fixed-point equation; induction bounds all its iterates by any pre-fixed point.

(5) **Prove continuity of least fixed points.** For an admitted ideal I of endomorphisms put $g = \sup I$ and $y = \sup_{h \in I} \mathrm{lfp}(h)$. Given $f, h \in I$, a common upper member k gives $f(\mathrm{lfp}(h)) \le k(\mathrm{lfp}(k)) \le y$. Continuity and the pointwise supremum give $g(y) \le y$. Leastness and monotonicity yield $\mathrm{lfp}(g) = y$, proving the algebra-morphism assertion.

(6) **Verify naturality.** Strict intertwining identifies the zeroth iterates and then all successors; preservation of their supremum proves naturality.

□

Lemma 1.6 (Exact finite return)**.** *For $f : X \to L_\Sigma(Y + X)$, let D_Y, D_X be its exit and continuation domains, with source $c_X : D_X \hookrightarrow X$, continuation $n : D_X \to X$, and exit value $v : D_Y \to Y$. Define $R_0 = D_Y$ and $R_{k+1} = D_X \times_X R_k$, with source $\rho_{k+1} = c_X \pi_1$ and return value $v_{k+1} = v_k \pi_2$. The map $\sum_{k:\mathbb{N}} R_k \to X$ is a Σ-classified mono. Its returned partial map*

is the least solution of $a(x) = f(x)\,\mathrm{bind}\,[\eta, a]$ *(with the usual Kleisli ordering). For algebra morphisms* f*, it agrees with Proposition 1.5.*

Proof.

(1) **Construct the classified return domain.** Pullback and composition make each ρ_k a classified mono. Exit and continuation are disjoint, and $R_{k+1} \times_X R_{l+1} \cong D_X \times_X (R_k \times_X R_l)$. Induction proves disjointness of distinct lengths. Their single internally indexed dependent sum is therefore monic; its classifier is $\exists k\, \rho_k(x)$ and lies in Σ.

(2) **Identify the least solution.** The jth bottom approximant has domain $\sum_{k<j} R_k$ and value v_k on its kth component. The union of these domains gives the asserted partial map. Each solution agrees with v_k there by induction, so this map is least.

(3) **Compare continuous recursion and parameters.** For continuous f, the lifted supremum has the same extent and stabilized value. All recursions and images commute with parameters.

□

The completion now provides the domains on which the computational operations act. To relate their semantic elements to effective data, we next specify the available observations and the names that present the resulting effective points.

1.3 Observations, effective points, and names

An observation structure on a poset is an internal family of upper predicates containing the Σ-constants and closed under finite meets and overt-indexed unions. Its generated closure $\mathrm{cl}_\Sigma(G)$ is the intersection, inside the appropriate power object, of all such structures containing G. The definition is bounded: overt subfamilies are subobjects of the given predicate object, and overtness says that their existential map factors through Σ. A family in the intersection is mapped to an overt family in each member, proving closure of the intersection. Inverse image preserves generating observations and all these operations.

An observed dcpo $(D, \mathcal{O}_D)$ has $\mathcal{O}_D \hookrightarrow \mathrm{Sc}(D)$ an observation structure of Scott opens. Its maps are Scott-continuous maps whose inverse images preserve the specified observations; their category is $\mathbf{EDcpo}_\Sigma$. Define

$$\mathrm{Pt}_\Sigma\, D = D \times_{\Omega^{\mathcal{O}_D}} \Sigma^{\mathcal{O}_D}. \tag{1.3}$$

It has the uniform evaluation $\mathrm{Obs}_D : \mathrm{Pt}_\Sigma\, D \times \mathcal{O}_D \to \Sigma$.

The effective-point construction and its evaluation fit into

$$\begin{array}{ccc} \mathrm{Pt}_\Sigma\, D & \longrightarrow & \Sigma^{\mathcal{O}_D} \\ \big\downarrow & & \big\downarrow \\ D & \xrightarrow[\mathrm{ev}^\flat]{} & \Omega^{\mathcal{O}_D}. \end{array}$$

Transposing the upper map gives the positive evaluation.

Lemma 1.7 (Testing generators in context)**.** *When* $\mathcal{O}_D = \mathrm{cl}_\Sigma(G)$*, a parameterized* $x : Z \to D$ *factors through* $\mathrm{Pt}_\Sigma\, D$ *exactly when its evaluation on* $Z \times G$ *factors through* Σ*. The factorization is unique. Effective points are functorial and preserve the limits constructed with componentwise directed suprema and projection-generated observations.*

Proof.

(1) **Extend generator tests to all observations.** In context $z : Z$, the upper predicates whose value at $x(z)$ lies in Σ contain the Σ-constants and are closed under finite meets. An overt union of such predicates has value in Σ by restricted existential quantification. This is an observation structure in that context, hence contains $\mathrm{cl}_\Sigma(G)$ when it contains G.

(2) **Assemble the unique positive evaluation.** Monicity of $\Sigma \hookrightarrow \Omega$ and unique choice assemble the joint evaluation map $Z \times \mathcal{O}_D \to \Sigma$; transposition gives (1.3). The reverse implication is restriction.

(3) **Construct the functorial action.** For $F : D \to D'$ use $\mathrm{Obs}_{D'}(Fx, U) = \mathrm{Obs}_D(x, F^*U)$. Uniqueness gives the functor laws.

(4) **Identify effective points of limits.** At a limit, the generator condition says precisely that every projected point is effective, uniformly in the index; the pullback universal property supplies the limit comparison.

□

For a poset P the canonical observed free completion is

$$F_\Sigma P = (\mathrm{Idl}\, P, \{u^\dagger : u \in U_\Sigma P\}), \qquad u^\dagger(I) = \exists p\, [I(p) \wedge u(p)].$$

Restriction to principal ideals recovers u; dagger preserves finite meets and overt unions, and its predicates are Scott open. Thus the displayed family is an observation structure and $\mathrm{Pt}_\Sigma(F_\Sigma P) = J_\Sigma P$ by the two defining pullbacks. For a discrete C, $\mathrm{Idl}\, C \cong C$, whence $\mathrm{Pt}_\Sigma\, F_\Sigma(C_{\mathrm{disc}}) \cong C$; its observations are exactly the maps $C \to \Sigma$.

For completeness the semantic comparison on retracts is explicit. A J_Σ-Kleisli map $r : P \to J_\Sigma Q$ extends to $r^\# : \mathrm{Idl}\, P \to \mathrm{Idl}\, Q$, $I \mapsto \bigcup_{p \in I} r(p)$. The inverse image of $u^\dagger$ is $v^\dagger$, where $v(p) = \chi_{r(p)}(u) \in \Sigma$. Conversely an observed map between these free completions restricts on principal ideals to an admitted ideal profile and is recovered by directed extension. Hence the free observed category is the Kleisli category of J_Σ, equivalently the category of free J_Σ-algebras. Splitting an idempotent by its fixed subobject on both sides extends this comparison to their retracts. This is the precise comparison used when an algebra map is represented by a semantic observed map; a general observed domain retains its separate observation data.

In the realizability models, Proposition E.1 constructs regular-epimorphic name presentations $q_D : C_D \twoheadrightarrow \mathrm{Pt}_\Sigma\, D$ and tracked implementations. A partial implementation retains both the returned value equation and equality of its domain with the pullback of the semantic domain. Names, their sizes, and their costs are not identified merely because their denotations agree. For represented operators this distinction is central in the complexity theory of analysis (Kawamura and Cook 2012).

1.4 Lower-real scalars and continuous presentations

Resource measurements require continuous nonnegative quantities as well as exact rational profiles. Lower cuts provide the continuous scalars, with rational thresholds as their observations. Let Q be the poset of rounded lower cuts of $\mathbb{Q}$ containing all negative rationals. Zero is the negative cut and infinity is the whole rational object; suprema are unions with the negative cut. For a cut x put $R(x) = \{0\} \cup \{r \in \mathbb{Q}_{>0} : r < x\}$. This inhabited directed

family gives $x = \bigvee_{r \in R(x)} \underline{r}$. Define

$$x + y = \bigvee_{r \in R(x),\, s \in R(y)} \underline{r + s}, \qquad xy = \bigvee_{r \in R(x),\, s \in R(y)} \underline{rs}.$$

In particular $0 + y = y$ and $0 \cdot \infty = 0$. Rational interpolation identifies a positive strict threshold of these joins with a strict threshold below one of the displayed rational sums or products. Expanding nested operations and using common larger rational approximants therefore reduces associativity, distributivity, and commutativity to the rational identities. This proof uses the always available zero approximant and makes no decision whether a general cut is zero. The same finite-witness calculation proves Scott continuity and arbitrary-join preservation of multiplication in each variable (including the empty join).

Define $S = \mathsf{Q}_\Sigma$ by requiring that the rational strict-threshold profile of a cut factor uniformly through Σ. Rationals and finite rational arrays are overt: encode their numerator and positive denominator by natural numbers and use the epic-image part of Proposition 1.2. The threshold formulas therefore show that S is closed under the semiring operations. If $I \in J_\Sigma S$, the threshold of its supremum is

$$q < \sup I \iff \exists x\, [I(x) \wedge q < x] = \chi_I(x \mapsto (q < x)),$$

so S is a J_Σ-algebra and its operations are continuous. Positivity $\sigma(x) = (0 < x)$ preserves multiplicative tensor and arbitrary semantic joins. Its crisp section is $\tau(p) = \bigvee\{1 : p\}$, with $\sigma\tau = \mathrm{id}$; it preserves multiplication but is not claimed to preserve arithmetic addition. The effective version restricts to $p : \Sigma$. Thus the change from quantitative comparison to positive comparison used below follows from these explicit threshold maps.

Theorem 1.8 (Admitted presentations)**.** *For a preorder P and an admitted cover relation $\mathcal{U} \hookrightarrow P \times J_\Sigma P$, there is a generated algebra $\mathsf{Pres}_{J_\Sigma}(P, \mathcal{U})$ with a monotone generator map j and the parameter-uniform restriction isomorphism*

$$\mathcal{A}_\Sigma(\mathsf{Pres}_{J_\Sigma}(P, \mathcal{U}), A) \cong \{f : P \to U_{\mathcal{A}_\Sigma} A : f \text{ monotone}, \\ f(p) \leq a_A(J_\Sigma f(I)) \text{ for every } (p, I) \in \mathcal{U}\}. \tag{1.4}$$

Finite products of the generated algebras are generated under admitted suprema by tuples of generators. Finite-arity operations preserving the covers separately extend continuously, and their equations and inequalities extend from generators.

Proof.

(1) **Construct the generated algebra.** The full construction and extension argument are given in Section A. Briefly, form the complete lattice of cover-closed lower subsets of P, and take its least subobject containing the closed principal ideals and closed under intrinsic admitted suprema.

(2) **Extend maps and prove uniqueness.** Extension is constructed in the complete lattice of admitted-closed lower subsets of the target algebra and then restricted to the embedded target. Equality and inequality loci prove uniqueness and order reflection.

(3) **Establish product generation and parameter coherence.** Two successive section arguments prove finite-product generation. These are internal power-object constructions, so the universal property holds with parameters. This is the selected-ideal version of the dcpo presentation method of (Jung et al. 2008); the selection and parameter arguments are included here.

□

Write $\mathsf{Lin}_S^{J_\Sigma} = \mathsf{Lin}_S^{J_\Sigma}$ for continuous S-semimodule objects in $\mathcal{A}_\Sigma$ and continuous linear maps.

Proposition 1.9 (Continuous linear constructions)**.** *The category $\mathsf{Lin}_S^{J_\Sigma}$ has free continuous semimodules, a symmetric monoidal closed tensor representing continuous balanced bilinear maps, and presented quotient modules. Their constructions and coherence equations are uniform in parameters.*

Proof.

(1) **Construct free continuous semimodules.** For $A \in \mathcal{A}_\Sigma$, form finite formal S-linear combinations of elements of A, quotient by the semimodule equations, and impose the least compatible preorder containing the orders of A and S. Add covers

$$[\sup I] \triangleleft J_\Sigma[-](I) \quad (I \in J_\Sigma A), \qquad (\sup K)v \triangleleft J_\Sigma(s \mapsto sv)(K) \quad (K \in J_\Sigma S),$$

and close them under finite one-hole module contexts. The presentation theorem gives $\mathsf{L}_S A$. Maps out of it are exactly continuous maps $A \to UN$ extended linearly.

(2) **Construct the balanced tensor.** For $M \otimes_S^{J_\Sigma} N$ use generators $m \otimes n$, the balanced bilinear equations, and continuity covers in each input. The same theorem gives its bilinear universal property.

(3) **Construct the internal hom.** The subobject of $[UM, UN]_\Sigma$ cut out by linearity is closed under admitted pointwise suprema, by interchange of directed suprema; its pointwise operations give the continuous internal hom. Currying and evaluation give closedness.

(4) **Verify monoidal coherence.** The unit is S, with inverse to its action given by $m \mapsto 1 \otimes m$. The two associative tensor constructions represent the same trilinear maps. The symmetry and unit comparisons have the corresponding generator formulas, and all coherence diagrams agree on generators.

(5) **Construct quotient modules.** Quotients add the required equations as inequalities in both directions to the same presentation.

□

Throughout, an internally *finite-projective module* means a *displayed finite-free retract*: specified finite maps $M \xrightarrow{s} B^n \xrightarrow{r} M$ with $rs = \mathrm{id}_M$, or its idempotent matrix sr. Such a presentation constructs a finite dual basis, dualizability, and flatness. Indeed $\mathrm{Hom}_B(M, N) \cong M^\vee \otimes_B N$, where $M^\vee = \mathrm{Hom}_B(M, B)$; this functor is exact because $M^\vee$ is a finite-free retract. This is the finite-projective property used below. In an arbitrary topos it is not identified with unrestricted lifting against every regular epimorphism of internal modules. Standard finite-projective and flat-colimit calculations are cited in (Stacks Project Authors 2026a), with the internal finite-witness and displayed-retract arguments supplied where needed.

The continuous module construction gives the linear objects. Augmentations now record their quantitative grades, and the following adjunctions relate them to ordinary computational contexts. These are also the interfaces used by the resource-certificate interpretation.

1.5 The linear–nonlinear adjunctions

Put $\mathsf{Bud}_S^{J_\Sigma} = \mathsf{Lin}_S^{J_\Sigma}/S$, whose objects have a continuous linear augmentation $\mathrm{aug}_M : M \to S$ and whose arrows preserve it. Its tensor augmentation is $\mathrm{aug}_{M\otimes N}(m \otimes n) =$

$\mathrm{aug}_M(m)\,\mathrm{aug}_N(n)$. The unit (S, id) is terminal. The internal hom has the explicit presentation

$$[M,N]_{\mathsf{Bud}_S^{J_\Sigma}} = \{(r,f) \in S \times \underline{\mathrm{Hom}}_{\mathsf{Lin}_S^{J_\Sigma}}(M,N) : \mathrm{aug}_N f = r\ \mathrm{aug}_M\}, \qquad \mathrm{aug}(r,f) = r.$$

It is a continuous submodule. The balance equations and ordinary linear currying identify maps into it with augmentation-preserving maps out of the tensor.

Proposition 1.10 (Free augmented modules)**.** *There is a strong symmetric monoidal adjunction*

$$\mathsf{F}_S : \mathcal{A}_\Sigma \rightleftarrows \mathsf{Bud}_S^{J_\Sigma} : \mathsf{G}_S,$$
$$\mathsf{F}_S A = (\mathsf{L}_S A, \mathrm{aug}_A), \quad \mathrm{aug}_A([a]) = 1, \qquad \mathsf{G}_S M = \{m : \mathrm{aug}_M(m) = 1\}.$$

Composing with $\mathrm{Disc} \dashv U_{\mathcal{A}_\Sigma}$ *gives* $F_d = \mathsf{F}_S\,\mathrm{Disc} \dashv G_d = U_{\mathcal{A}_\Sigma}\mathsf{G}_S$*. Both constructions have their canonical parameter-substitution comparisons.*

Proof.

(1) **Construct the augmentation.** The constant map $A \to S$ with value 1 is continuous, so extends uniquely to aug_A.

(2) **Restrict the free-module adjunction.** The fibre $\mathrm{aug}_M^{-1}(1)$ is closed under admitted suprema, since augmentation preserves them. The free-module universal property restricts to
$$\mathsf{Bud}_S^{J_\Sigma}(\mathsf{F}_S A, M) \cong \mathcal{A}_\Sigma(A, \mathsf{G}_S M).$$

(3) **Establish strong monoidality.** The maps sending $[a] \otimes [b]$ to $[(a,b)]$ and conversely are obtained from the bilinear and free universal properties. They are inverse on generators; product generation proves they are inverse everywhere and proves all monoidal coherence equations. The unit comparison is $\mathsf{F}_S 1 \cong (S, \mathrm{id})$.

(4) **Pass to ordinary contexts.** Composition of adjunctions gives the second assertion, and Proposition A.5 proves substitution coherence.

□

Write $!_c = \mathsf{F}_S\mathsf{G}_S$ and $!_d = F_d G_d$. Ordinary dependent contexts are interpreted in the self-indexed category of $\mathcal{E}$: a family over Γ is an object of $\mathcal{E}/\Gamma$, substitution is pullback, Σ-types are composition, Π-types are the dependent right adjoints, and identity types are diagonals. Linear families use the displayed augmented-module construction in the same parameter slices. Choose the equivalent split presentation by families; the canonical Beck–Chevalley maps are then its substitution comparisons. The finite syntax and checking interpretation needed for resource certificates are given in Section F.

Proposition 1.11 (Comparison of the two budget exponentials)**.** *The counit* $\epsilon_A^{\mathrm{disc}} : \mathrm{Disc}\, U_{\mathcal{A}_\Sigma} A \to A$ *constructs a natural comonad morphism*

$$\lambda_M = \mathsf{F}_S(\epsilon^{\mathrm{disc}}_{\mathsf{G}_S M}) : !_d M \longrightarrow !_c M.$$

It commutes with parameter substitution. Thus the computational LNL structure and the ordinary-context dependent LNL structure have a specified comparison, while retaining their respective domains.

Proof.

(1) **Construct the comparison map.** The displayed counit is the identity on the underlying elements and is continuous from the discrete algebra. Applying the free functor gives λ.

(2) **Compare the counits.** The counit of the composite adjunction is the computational counit composed with λ, so $\epsilon^c \lambda = \epsilon^d$.

(3) **Compare the comultiplications.** The comultiplications satisfy

$$\delta^c_M \lambda_M = (!_c \lambda_M) \lambda_{!_d M} \delta^d_M.$$

On a free generator both sides insert that same generator twice; the triangle identities of Disc $\dashv U_{\mathcal{A}_\Sigma}$ and $\mathsf{F}_S \dashv \mathsf{G}_S$ identify the intervening units. Extension from generators proves the equation.

(4) **Verify substitution compatibility.** The same formulas hold in every slice; Proposition A.5 identifies their substitution comparisons.

□

For a budget morphism $f : M \to N$, the comparison of exponentials has the naturality square

$$\begin{array}{ccc} !_d M & \xrightarrow{\lambda_M} & !_c M \\ {\scriptstyle !_d f}\downarrow & & \downarrow{\scriptstyle !_c f} \\ !_d N & \xrightarrow[\lambda_N]{} & !_c N. \end{array}$$

The comonad-morphism equations proved above also identify its composites with the two counits and comultiplications.

The two exponentials just constructed organize the ordinary–linear interface. Later constructions have different roles: the symmetric-algebra monad T describes polynomial substitution and differentiation, and the shape operations $\mathfrak{I}_W$ and $\mathfrak{I}_{\mathrm{inf}}$ describe infinitesimal neighbourhoods in the based and primary geometries. The functors of Theorem 6.6 compare these structures with the quantitative layer. Resource certificates use F_d and its ordinary-context interpretation. Their grades, description lengths, and execution times are compared through the measured executions and proof transformations that carry those quantities.

1.6 Exact quantitative presentations

Differential resource algebra needs signed coefficients, while the continuous quantitative semantics includes an infinite scalar. The two obstructions below explain why the signed algebra must be built from exact data before continuous completion. Put $\mathscr{A}_+ = \mathrm{Mod}_{\mathbb{Q}_{\geq 0}}(\mathcal{E})$, $\mathsf{Lin}_S^{J_\Sigma} = \mathsf{Lin}_S^{J_\Sigma}$, and $\mathscr{A}_\pm = \mathrm{Mod}_{\mathbb{Q}}(\mathcal{E})$. Objects of $\mathscr{A}_+$ are algebraic semimodules, with no imposed order or cancellativity. This includes both exact cost profiles and the additive relations needed for universal differentials.

Proposition 1.12 (The augmented category cannot be the ordinary differential base)**.** *The nontrivial monoidal category* $\mathsf{Bud}_S^{J_\Sigma} = \mathsf{Lin}_S^{J_\Sigma}/S$*, with its existing tensor, cannot be made a commutative-monoid-enriched category with bilinear composition and tensor as required by an ordinary differential category.*

Proof.

(1) **Identify the terminal tensor unit.** Its tensor unit $I = (S, \mathrm{id})$ is terminal: the unique arrow $(M, a) \to I$ is a.

(2) **Derive collapse of an additive enrichment.** In any proposed additive enrichment, the zero endomorphism of I is therefore id_I. Bilinearity of tensor gives

$$\mathrm{id}_M = \mathrm{id}_M \otimes \mathrm{id}_I = \mathrm{id}_M \otimes 0 = 0$$

after the unit identifications. Every arrow is then zero, contradicting nontriviality.

(3) **Exhibit the augmentation mismatch.** For the existing pointwise addition the failure is already visible: augmentation-preserving f, g satisfy $a_N(f+g) = 2a_M$, and the zero linear map generally does not preserve augmentation.

□

Proposition 1.13 (The completed scalars have no nonzero signed envelope)**.** *Every unital semiring homomorphism $S \to R'$ to a ring has zero target. Moreover, the additive group completion of the underlying $\mathbb{Q}_{\geq 0}$-semimodule of any S-semimodule is zero.*

Proof.

(1) **Exclude a nonzero ring target.** The effective infinite scalar belongs to S, and $\infty + 1 = \infty$. Cancellation in R' forces $1 = 0$.

(2) **Collapse additive group completion.** For an S-semimodule, $\infty m + m = \infty m$, so the image of every m in its additive group completion is zero.

□

We therefore give the exact positive presentation two realizations: continuous scalar extension and signed completion. The common exact source supplies their comparison.

1.6.1 Continuous scalar extension and signed completion

Construction of the left-hand functor The rational scalar map $\iota : \mathbb{Q}_{\geq 0} \to S$ sends a rational to its principal lower cut. Write $U : \mathsf{Lin}_S^{J_\Sigma} \to \mathscr{A}_+$ for restriction of scalars followed by forgetting order and admitted suprema.

Construction 1.14. For $M \in \mathscr{A}_+$, form the algebraic S-semimodule generated by symbols $s \otimes m$, subject to additivity in each variable and

$$s \otimes (rm) = s\iota(r) \otimes m \qquad (r \in \mathbb{Q}_{\geq 0}).$$

Give it the least compatible preorder containing the scalar order of S. For every $K \in J_\Sigma S$ and every algebraic element v, impose the cover

$$(\sup K)v \;\triangleleft\; J_\Sigma(s \mapsto sv)(K).$$

Close these covers under finite one-hole module contexts. Apply Theorem 1.8 to $\mathsf{Pres}_{J_\Sigma}(P, C)$, including posetal reflection of the preorder. Denote the result by $\mathsf{E}_S M$.

Proposition 1.15. *There is a parameter-uniform adjunction $\mathsf{E}_S \dashv U$ with*

$$\mathsf{Lin}_S^{J_\Sigma}(\mathsf{E}_S M, N) \cong \mathscr{A}_+(M, UN).$$

The functor E_S is strong symmetric monoidal:

$$\mathsf{E}_S \mathbb{Q}_{\geq 0} \cong S, \qquad \mathsf{E}_S(M \otimes_{\mathbb{Q}_{\geq 0}} M') \cong \mathsf{E}_S M \otimes_S^{J_\Sigma} \mathsf{E}_S M'.$$

Proof.

(1) **Construct the scalar-extension adjunction.** An $\mathbb{Q}_{\geq 0}$-linear map $f : M \to UN$ extends algebraically by $s \otimes m \mapsto sf(m)$. It respects the generated preorder and the displayed covers by continuity of the scalar action on N. The admitted-presentation theorem gives a unique continuous linear extension. Restriction to $1 \otimes m$ is inverse to extension. Pointwise inequalities and free parameters are preserved by the same theorem.

(2) **Identify the balanced bilinear maps.** For the tensor assertion, a continuous balanced bilinear map out of $\mathsf{E}_S M \times \mathsf{E}_S M'$ is uniquely determined by an $\mathbb{Q}_{\geq 0}$-bilinear map out of $M \times M'$.

(3) **Extend separately and evaluate.** To extend such a map, first extend in M'; the resulting map into the continuous internal hom is $\mathbb{Q}_{\geq 0}$-linear in M. Extend that map in M, then evaluate. The internal hom and the parameter-uniform adjunction justify continuity in the other variable.

(4) **Identify tensors and verify coherence.** Consequently the two displayed tensor objects represent the same bilinear functor. Both unit objects represent selection of an element of N, giving $\mathsf{E}_S \mathbb{Q}_{\geq 0} \cong S$. All coherence diagrams agree on elementary generators, so uniqueness proves coherence.

□

The extension preserves addition and zero on morphisms: $\mathsf{E}_S(f+g) = \mathsf{E}_S f + \mathsf{E}_S g$ and $\mathsf{E}_S 0 = 0$. The equalities hold on exact generators and hence on the generated domain. Here M is algebraic. If a presentation also specifies order or continuity, its covers are imposed by a further presentation or retained in the map ρ of (1.5).

Construction of the right-hand functor

Proposition 1.16. *Additive group completion defines a strong symmetric monoidal left adjoint*

$$\mathrm{Gr} : \mathscr{A}_+ \longrightarrow \mathscr{A}_\pm.$$

It takes $\mathbb{Q}_{\geq 0}$ to $\mathbb{Q}$ and induces group completion of commutative semiring objects and their modules.

Proof.

(1) **Construct the group-completion quotient.** Construct $\mathrm{Gr}\, M$ from pairs (m, n) modulo

$$(m, n) \sim (m', n') \quad \Longleftrightarrow \quad \exists k \; m + n' + k = m' + n + k.$$

Addition is componentwise, and (m, n) represents $m - n$.

(2) **Extend scalars and establish the adjunction.** The $\mathbb{Q}_{\geq 0}$-action extends uniquely to a $\mathbb{Q}$-action, since the group completion of $\mathbb{Q}_{\geq 0}$ is $\mathbb{Q}$. Maps to a $\mathbb{Q}$-module correspond exactly to $\mathbb{Q}_{\geq 0}$-linear maps from M to its underlying monoid.

(3) **Prove additivity and tensor compatibility.** The functor is additive by its formula on differences. An $\mathbb{Q}_{\geq 0}$-bilinear map $M \times N$ to an additive group extends uniquely to a $\mathbb{Q}$-bilinear map $\mathrm{Gr}\, M \times \mathrm{Gr}\, N$. On differences the extension is

$$(m - n, p - q) \longmapsto b(m, p) - b(m, q) - b(n, p) + b(n, q).$$

This proves $\mathrm{Gr}(M \otimes_{\mathbb{Q}_{\geq 0}} N) \cong \mathrm{Gr}\, M \otimes_{\mathbb{Q}} \mathrm{Gr}\, N$ and its coherence.

(4) **Extend algebra and module structures.** Multiplication and module actions extend by the same bilinear formula. All quotients are internal effective quotients in $\mathcal{E}$.

□

For a commutative $\mathbb{Q}_{\geq 0}$-algebra A, the same proof gives functors on coefficient fibres

$$\mathrm{Mod}_A(\mathcal{E}) \longrightarrow \mathrm{Mod}_{\mathsf{E}_S A}(\mathsf{Lin}_S^{J_\Sigma}), \qquad \mathrm{Mod}_A(\mathcal{E}) \longrightarrow \mathrm{Mod}_{\mathrm{Gr}\, A}(\mathcal{E}).$$

They preserve relative tensor products. This follows either from their balanced-map universal properties or by applying the left adjoints to the defining balanced coequalizers. Continuous relative tensors and quotient modules are constructed by the same admitted presentations with the coefficient algebra in place of S.

1.6.2 The category of retained presentations

We now package an exact module together with its quantitative interpretation. Keeping both in one object lets a map carry the exact data needed for signed realization and the continuous data needed for quantitative semantics. Define

$$\mathscr{P} = (\mathsf{E}_S \downarrow \mathrm{id}_{\mathsf{Lin}_S^{J_\Sigma}}). \tag{1.5}$$

An object (M, N, ρ) consists of an exact positive module M, a quantitative module N, and a continuous linear map $\rho : \mathsf{E}_S M \to N$. A morphism (f, g) satisfies $g\rho = \rho' \mathsf{E}_S f$.

A morphism must preserve that interpretation. Its defining equation is the commuting square

$$\begin{array}{ccc} \mathsf{E}_S M & \xrightarrow{\rho} & N \\ {\scriptstyle \mathsf{E}_S f}\downarrow & & \downarrow{\scriptstyle g} \\ \mathsf{E}_S M' & \xrightarrow[\rho']{} & N'. \end{array} \tag{1.6}$$

The adjunction $\mathsf{E}_S \dashv U$ identifies this comma category with $(\mathrm{id}_{\mathscr{A}_+} \downarrow U)$: the transpose of ρ is the exact interpretation map $M \to UN$. Under this identification the same morphism is the commuting square

$$\begin{array}{ccc} M & \xrightarrow{\rho^\flat} & UN \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle Ug} \\ M' & \xrightarrow[(\rho')^\flat]{} & UN'. \end{array}$$

The exact component also has a signed realization. Together the two realization functors form the span

$$\mathsf{Lin}_S^{J_\Sigma} \xleftarrow{\pi_{\mathsf{Lin}_S^{J_\Sigma}}} \mathscr{P} \xrightarrow{\mathrm{Gr}\, \pi_{\mathscr{A}_+}} \mathscr{A}_\pm. \tag{1.7}$$

Its common source retains the exact leg needed for signed realization. The differential comparisons of Theorems 6.6 and 6.7 will equip both projections with their compatible algebra modalities and deriving transformations. Addition is componentwise. Strong monoidality of E_S defines the tensor componentwise, with the evident comparison map. Its unit is $(\mathbb{Q}_{\geq 0}, S, \mathrm{id})$ under $\mathsf{E}_S \mathbb{Q}_{\geq 0} \cong S$. These definitions make $\mathscr{P}$ a commutative-monoid-enriched symmetric monoidal category with finite biproducts (a semiadditive category). The codifferential axioms use this commutative-monoid enrichment.

The functor $M \mapsto (M, \mathsf{E}_S M, \mathrm{id})$ is fully faithful and strong monoidal. If U is the right adjoint of E_S, then $R_{\mathsf{Lin}_S^{J_\Sigma}}(N) = (UN, N, \epsilon_N)$ is a fully faithful right adjoint to quantitative projection. Indeed a map to $R_{\mathsf{Lin}_S^{J_\Sigma}}(N')$ is determined by its quantitative component, whose adjoint is the required exact component. This right adjoint is lax monoidal. Its signed component vanishes by the infinity calculation in Section 1.6; nonzero signed data are retained in the chosen exact presentation.

1.6.3 Raw cost profiles have a faithful quantitative interpretation

For an input object X, use its discrete admitted domain. The function object S^X is a continuous S-semimodule under pointwise operations. Exact profiles have a monomorphism

$$\mathbb{Q}_{\geq 0}{}^X \longrightarrow U(S^X), \qquad f \longmapsto \underline{f}.$$

Adjunction gives a canonical quantitative realization

$$\mathsf{E}_S(\mathbb{Q}_{\geq 0}{}^X) \longrightarrow S^X. \tag{1.8}$$

Because the composite with the exact-generator map is monic, the exact-generator map into $\mathsf{E}_S(\mathbb{Q}_{\geq 0}{}^X)$ is monic as well. Any exact submodule of raw profiles has the same argument, by restriction. Input substitution is defined on these raw profiles for every map of inputs.

1.6.4 Exact presentation of free augmented modules

For $A \in \mathcal{A}_\Sigma$, let $L_{\mathbb{Q}_{\geq 0}}(UA)$ be the algebraic free rational-positive module on its underlying object, and let $L_S^{J_\Sigma} A$ be the free continuous module of Proposition 1.9. Its generator map induces

$$\mathsf{E}_S L_{\mathbb{Q}_{\geq 0}}(UA) \longrightarrow L_S^{J_\Sigma} A.$$

This is a presentation object. Its exact augmentation sums rational coefficients, and its quantitative augmentation is the free augmentation of Proposition 1.10. These are compatible. The construction is strong monoidal on cartesian products of A, because both free-module comparisons send a pair of generators to their tensor. Projection gives F_S. The computational LNL adjunction and $!_c$ therefore have this specified exact lift of the free functor. Precomposition with Disc gives the corresponding exact lift of F_d; its comparison with the computational lift is induced by the same discrete counit as Proposition 1.11.

We now have a computational category with recursive execution, an observation and name interface, and exact presentations that admit both quantitative and signed realizations. The next section applies the exact side to cost profiles. Eventual comparison will turn those profiles into the modules used as resource budgets.

2 Landau algebra and observed resource modules

An asymptotic budget should record which cost profiles it contains and how those profiles combine. We obtain this algebraic description by identifying Landau domination with multiplication by bounded coefficients. The construction proceeds from exact profiles to eventual germs and then to growth classes. Keeping these stages distinct lets us recover both coarse bounds and finer information about coefficients and errors. The resulting embedded modules will be the objects classified in the next section.

2.1 Filters and eventual quotients

For $X \in \mathcal{E}$, form the object $\mathrm{Filt}(X)$ of predicates on subobjects closed upward and under finite intersection and containing X. The proper-filter subobject imposes $\neg\mathrm{Ev}(\varnothing)$. This is a construction inside the power objects of $\mathcal{E}$, so the following results are families over that subobject. Write $\mathrm{Ev}(P)$ for membership of a predicate P in the displayed filter.

A filtered map $u : (Y, \mathrm{Ev}_Y) \to (X, \mathrm{Ev}_X)$ is a map satisfying $\mathrm{Ev}_X(P) \Rightarrow \mathrm{Ev}_Y(u^{-1}P)$ for every subobject P. Substitution proves the identity and composition laws.

Proposition 2.1 (Germs and substitution)**.** *The relations* $\mathrm{Ev}(f = g)$ *on rational and nonnegative rational profiles are effective equivalence relations compatible with addition and multiplication. They construct internal algebras*

$$K^+ = (\mathbb{Q}_{\geq 0})^X/{=_{\mathrm{Ev}}}, \qquad K = \mathbb{Q}^X/{=_{\mathrm{Ev}}}.$$

Filtered maps induce algebra homomorphisms by precomposition.

Proof.

(1) **Verify the eventual equivalence relation.** Reflexivity uses $\mathrm{Ev}(X)$, symmetry uses symmetry of equality, and transitivity uses intersection and upward closure.

(2) **Descend the algebraic operations.** Intersect the extents of two representative equalities to verify each operation. The effective quotient therefore inherits the operations and their finite equations.

(3) **Construct filtered substitution.** The defining implication for a filtered map preserves representative equality and every operation. Identity and composition hold on representatives.

□

For $X = \mathbb{N}$, the cofinite filter is

$$\mathrm{Ev}(P) \equiv \exists N\, \forall n \geq N\, P(n).$$

For finite words, length gives $\mathrm{Ev}(P) \equiv \exists N\, \forall x\, (|x| \geq N \Rightarrow P(x))$. These filters are proper because a purported eventual empty extent, with cutoff N, is contradicted respectively by $n = N$ and by a word of length N. Thus the displayed formulas construct actual inhabitants of the proper-filter subobject. For a general size map the proper-filter fibre is retained as part of the context. Arbitrary input maps always act on raw profiles; a constant input map need not act on cofinite germs.

2.2 Exact, eventual, and growth information

The successive comparisons forget different information. The inclusion $(\mathbb{Q}_{\geq 0})^X \hookrightarrow S^X$ preserves exact pointwise profiles. Eventual equality then identifies profiles that agree on one eventual extent, and mutual domination makes the further growth quotient. For example, $n + 1$, $2(n + 1)$, and $n + 2$ have the same growth class but distinct exact germs. Leading coefficients and small relative errors must therefore be compared before passing to growth classes.

2.3 Landau comparison as quantitative enrichment

A growth comparison records whether a bounded multiplier exists. Before passing to that predicate, we can retain the size of the permitted multipliers in a quantitative hom. For nonnegative rational germs define

$$h_{\mathrm{Ev}}(f,g) = \bigvee \{\underline{r} : r \in \mathbb{Q}_{\geq 0},\ \mathrm{Ev}(rf \leq g)\} \in \mathsf{Q}. \tag{2.1}$$

The join is an internal lower cut: its strict rational thresholds are the existentially witnessed thresholds of the displayed principal cuts. It is a semantic scalar; an effective name is additional represented data.

Theorem 2.2 (Enriched Landau comparison)**.** *The homs* (2.1) *define a* Q*-enriched category. They satisfy*

$$0 < h(f,g) \iff \exists a \in \mathbb{N}_{\geq 1}\, \mathrm{Ev}(f \leq ag), \qquad h(f,g) = \infty \iff \forall k \geq 1\, \mathrm{Ev}(kf \leq g).$$

Proof.

(1) **Verify well-defined enriched composition.** Eventual equality of either argument preserves each witnessed inequality after finite intersection. The witness $r = 1$ gives $1 \leq h(f,f)$. If $\mathrm{Ev}(rf \leq g)$ and $\mathrm{Ev}(sg \leq k)$, their intersection witnesses $\mathrm{Ev}(rsf \leq k)$. Multiplication of lower reals preserves joins, hence $h(f,g)h(g,k) \leq h(f,k)$.

(2) **Identify positive comparison.** A positive threshold of the join supplies a rational witness $r > 0$ with $\mathrm{Ev}(rf \leq g)$. Choose an integer a with $ar \geq 1$. This gives $\mathrm{Ev}(f \leq ag)$. Conversely such an inequality supplies $r = 1/a$.

(3) **Identify infinite comparison.** Unboundedness of the cut supplies a witness $r > k$ for each positive integer k, yielding $\mathrm{Ev}(kf \leq g)$. The integer witnesses in the other direction make the cut unbounded.

□

Change of enrichment and separation. Positivity is a morphism of commutative quantales

$$\sigma : (\mathsf{Q}, \cdot, 1) \longrightarrow (\Omega, \wedge, \top), \qquad \sigma(q) = (0 < q) :$$

it preserves arbitrary joins and multiplication by the rational threshold formulas of Section 1. Change of enrichment along σ sends $h(f,g)$ to the comparison predicate $\sigma h(f,g) = [\![f = O(g)]\!]$. Transitivity is the image of enriched composition:

$$\begin{array}{ccc} h(f,g)\,h(g,k) & \leq & h(f,k) \\ \downarrow{\scriptstyle \sigma} & & \downarrow{\scriptstyle \sigma} \\ [\![f = O(g)]\!] \wedge [\![g = O(k)]\!] & \leq & [\![f = O(k)]\!]. \end{array}$$

Taking positivity retains the domination predicate and forgets the numerical value of the quantitative hom. Quotienting mutual domination then gives the growth poset. These predicates take values in Ω; the later admission construction selects their positive-observation profiles.

Write $f \preceq g$ for the positive comparison and L_{Ev} for its quotient by mutual comparison. Its join is $[f+g]$, its meet is $[\min(f,g)]$, and its bottom is $[0]$. Common upper bounds on f, g give an upper bound on $f+g$ after adding their constants; common lower bounds give

one for their minimum after taking the larger constant. The distributive identities follow from the pointwise identities for minimum and maximum, with $f + g$ growth-equivalent to $\max(f, g)$. Multiplication is monotone and distributes over joins.

Definition 2.3 (Admitted resources)**.** The resource object is $\mathsf{Res}_{\mathrm{Ev}} = J_{\Sigma}\mathsf{L}_{\mathrm{Ev}}$. Its members are the growth ideals selected by Definition 1.3. The full semantic ideal object is $\mathrm{Idl}(\mathsf{L}_{\mathrm{Ev}})$.

The positivity theorem above identifies $\sigma h(f, g)$ with the displayed growth comparison. Applying the pullback of Definition 1.3 to that order gives its admitted ideals. The full semantic ideal object retains ideals whose positive test profile need not factor uniformly through Σ.

For signed arguments, comparison uses $h(|f|, |g|)$. The hom $h(|f - g|, |g|)$ retains the exact difference used in asymptotic equivalence. On nonnegative constant ratios $f = cg$, with g eventually nonzero and $c > 0$ rational, one has $h(f, g) = 1/c$. Thus the enrichment retains numerical comparison information that positivity alone forgets.

2.4 Landau calculus

2.4.1 Definitions and their quantifiers

For signed germs put

$$f \preceq g \quad \Longleftrightarrow \quad \exists a \in \mathbb{N}_{\geq 1}\, \mathrm{Ev}(|f| \leq a|g|). \tag{2.2}$$

Positive rational constants give the same predicate by rational rounding. The notation $f = O(g)$ is shorthand for membership $f \in O(g)$, and similarly for the other class notations.

Definition 2.4. Define

$$\begin{aligned} f \in O(g) &\iff f \preceq g, & f \in \Omega(g) &\iff g \preceq f, \\ f \in \Theta(g) &\iff f \preceq g \ \wedge\ g \preceq f, & f \in o(g) &\iff \forall k \geq 1\, \mathrm{Ev}(k|f| \leq |g|), \\ f \in \omega(g) &\iff g \in o(f), & f \sim g &\iff f - g \in o(g). \end{aligned}$$

We use $f \asymp g$ as a synonym for $f \in \Theta(g)$. The symbol $\ll$ has its approximation-theoretic meaning; Landau domination uses $\preceq$. For nonnegative costs Ω is the eventual lower-bound convention of (Knuth 1976); $\sim$ is asymptotic equality in the sense of relative error (NIST 2026).

These division-free predicates include zero-support profiles. For instance $o(0) = O(0) = \{0\}$ as germs. The object $K_{\mathrm{nz}} = \{g : \mathrm{Ev}(g \neq 0)\}$ is constructed by its displayed predicate. On this object rational division, defined arbitrarily as zero off the support, gives the usual statements: $f \in o(g)$ exactly when $|f/g|$ tends to zero, and $f \sim g$ exactly when f/g tends to one. The support condition makes these formulas independent of the off-support extension.

The relations $O, \Omega, \Theta, o, \omega$ are recovered from the two directed enriched homs by positivity, mutual positivity, and the value ∞. Asymptotic equivalence is recovered from $h(|f-g|, |g|) = \infty$.

2.4.2 Arithmetic and replacement laws

Proposition 2.5 (Landau arithmetic)**.** *Domination is a preorder and direct smallness is transitive. Each of $O, \Omega, \Theta, o, \omega$ is unchanged by replacing either argument by a Θ-equivalent germ. For germs f, g, h, k one has*

$$\begin{aligned} O(g) + O(h) &\subseteq O(|g| + |h|), & O(g)O(h) &\subseteq O(gh), \\ o(g) + o(h) &\subseteq o(|g| + |h|), & o(g)O(h) &\subseteq o(gh), \\ O(g)o(h) &\subseteq o(gh). \end{aligned}$$

Furthermore $f \sim g$ is an equivalence relation, implies $f \in \Theta(g)$, and is preserved by multiplication of two asymptotically equivalent pairs. For nonnegative profiles, $f \sim g$ and $h \sim k$ imply $f + h \sim g + k$.

Proof.

(1) **Prove transitivity and replacement.** Multiply domination constants and intersect their extents. To compose smallness with domination by a, use the smallness inequality at ka. To compose two smallness relations, use smallness of one and the $k = 1$ bound of the other. These arguments also prove replacement under two-sided domination.

(2) **Establish the sum and product estimates.** For sums, use the two smallness inequalities at $2k$; for products, multiply the inequalities after intersecting their extents. Absolute values satisfy the needed triangle and product identities pointwise.

(3) **Deduce tightness from asymptotic equality.** If $f - g \in o(g)$, the inequality at $k = 2$ gives eventually $|g|/2 \le |f| \le 3|g|/2$. Hence $f \in \Theta(g)$.

(4) **Prove equivalence of asymptotic equality.** Replacement and $|f - g| = |g - f|$ give symmetry. For transitivity write $f - h = (f - g) + (g - h)$, replace the reference budget g by h, and use the sum law. Reflexivity is zero error.

(5) **Preserve asymptotic equality under products.** Write $f = g + r$ and $h = k + s$ with $r \in o(g)$ and $s \in o(k)$. Then $fh - gk = rk + gs + rs \in o(gk)$ by the product and sum laws.

(6) **Preserve asymptotic equality under nonnegative sums.** For nonnegative sums the error is $r + s$ and the reference budget is exactly $g + k$, giving the assertion. For signed sums the preceding estimate uses $|g| + |k|$; cancellation is retained in the actual reference profile.

□

The definition of $\sim$ uses the signed difference, not merely the two absolute values. For example $-g$ and g are Θ-equivalent, while their relative error is -2 on the eventually nonzero locus.

2.4.3 Logical refinements and uniformity

The alternative double-negation relation is retained explicitly:

$$f \in o_{\neg\neg}(g) \iff f \preceq g \ \wedge\ \forall k \ge 1\, \neg\neg \mathrm{Ev}(k|f| \le |g|).$$

The direct relation is also denoted $o_{\mathrm{ev}}(g)$ when comparison with an explicitly witnessed eventual presentation is useful; throughout this paper plain o means this direct relation. We have $o(g) \subseteq o_{\neg\neg}(g) \subseteq O(g)$. Finite conjunction is stable under double negation, so the arithmetic and replacement arguments above give the analogous closure statements for $o_{\neg\neg}$. Quantifiers are left in their displayed order.

For cofinite input, a chosen modulus is an element of

$$\mathrm{Mod}_o(f, g) = \{m : \mathbb{N}_{\ge 1} \to \mathbb{N} : \forall k\, \forall n \ge m(k),\ k|f(n)| \le |g(n)|\}.$$

It maps to the direct smallness predicate. A represented computable modulus carries its implementation as well. The modulus object and its represented points record these two levels of data.

For $f, g : A \times \mathbb{N} \to \mathbb{Q}$, uniform domination is the predicate

$$\exists c, N\, \forall a \in A\, \forall n \geq N, \quad |f(a,n)| \leq c|g(a,n)|.$$

Pointwise-in-A domination has $\forall a$ before $\exists c, N$. Uniform smallness has $\forall k\, \exists N\, \forall a\, \forall n \geq N$. All the arithmetic proofs preserve uniformity because only finitely many bounds are combined at each tolerance.

2.4.4 Oscillation and infinitely-often bounds

On cofinite numerical input define

$$f \in \Omega_{\mathrm{io}}(g) \iff \exists c \in \mathbb{Q}_{>0}\, \forall N\, \exists n \geq N, \quad |f(n)| \geq c|g(n)|.$$

For signed one-sided statements replace the last inequality by $f(n) \geq c|g(n)|$ or $f(n) \leq -c|g(n)|$, giving $\Omega_{\mathrm{io},+}$ and $\Omega_{\mathrm{io},-}$. These definitions use positive witnesses of arbitrarily late inputs. They are not defined by negating smallness.

Let $g(n) = n + 1$ and let $f(n) = n + 1$ on even inputs and 1 on odd inputs. Then $f \in O(g) \cap \Omega_{\mathrm{io}}(g)$, while $f \notin o(g)$ and $f \notin \Theta(g)$. Each failure has explicit arbitrarily large parity witnesses. If instead $f(n) = (n+1)^2$ on even inputs and 1 on odd inputs, then $f \notin O(g)$ and $f \notin \omega(g)$. Thus lower and strict lower bounds are actual predicates, not complements of upper bounds. The eventual/infinitely-often distinction is also the distinction emphasized in (Knuth 1976).

Filtered substitution preserves every direct eventual comparison, and the arithmetic proofs commute with it. An isomorphism of filtered inputs reflects the comparisons by its inverse. Infinitely-often comparisons use their own displayed witness transport and are not silently inferred from preservation of eventual filters.

The comparison laws can now be expressed inside the germ algebra. Bounded coefficients will describe O-bounds, vanishing coefficients will describe o-bounds, and their units will describe the corresponding equivalence relations. This converts the calculus of inequalities into module membership and group actions.

2.5 Bounded coefficients, vanishing errors, and units

2.5.1 Positive coefficients and bounded division

Within the exact positive germ algebra put

$$B^+ = \{b \in K^+ : b \in O(1)\}.$$

Bounds add and multiply, so B^+ is a unital $\mathbb{Q}_{\geq 0}$-algebra. It embeds into K^+. Both algebras are cancellative: equality $f + h = g + h$ on an eventual extent implies $f = g$ on the same extent. For a profile g let e_g denote the indicator germ of $g \neq 0$. Rational equality is decidable, so this is an actual bounded profile with $e_g^2 = e_g$. Write $z_g = 1 - e_g$ on the signed side.

Lemma 2.6 (Positive bounded division)**.** *For $f, g \in K^+$, $f = O(g)$ if and only if $f = bg$ for some $b \in B^+$. If e_g is the indicator germ of $g \neq 0$, then*

$$B^+ g \cong B^+ e_g, \qquad B^+ g + B^+ h = B^+(g + h), \qquad B^+ g \cap B^+ h = B^+ \min(g, h).$$

Proof.

(1) **Construct the bounded multiplier.** Define $b(x) = f(x)/g(x)$ where $g(x) \neq 0$ and $b(x) = 0$ otherwise. Rational equality is decidable. An eventual bound $f \leq ag$ makes b eventually bounded and gives $f = bg$ as germs. The converse is multiplication of the bound.

(2) **Identify the support module.** Multiplication by g gives the support isomorphism, with inverse given by the same division on its support.

(3) **Compute the module sum.** The two generators lie in $B^+(g + h)$ and their sum lies in their module sum.

(4) **Compute the intersection and descend representatives.** For intersections, combine the two eventual bounds with their maximum and apply bounded division to $\min(g, h)$. These arguments are local on representatives and descend through the effective quotient.

□

Proposition 2.7 (Signed coefficients)**.** *There are canonical algebra isomorphisms*

$$\operatorname{Gr} K^+ \cong K, \qquad \operatorname{Gr} B^+ \cong B = \{b \in K : b \in O(1)\}.$$

The algebra B is reduced and K is its total quotient algebra. For signed germs f, g,

$$f \in O(g) \iff f = bg \text{ for some } b \in B.$$

Proof.

(1) **Identify the signed coefficient rings.** Cancellativity embeds the group completions into K. Every signed rational profile has the canonical decomposition $f = f_+ - f_-$, where $f_+ = \max(f, 0)$ and $f_- = \max(-f, 0)$. Boundedness of $|f|$ bounds both parts, proving both isomorphisms.

(2) **Prove signed bounded division.** Division f/g on the support of g, extended by zero elsewhere, proves signed bounded division just as in Lemma 2.6.

(3) **Prove reducedness.** If $b^r = 0$ for a positive integer r, rational arithmetic gives $b = 0$ on the same eventual extent, proving reducedness.

(4) **Compute annihilators and regular elements.** The annihilator of b in B is Bz_b: an annihilating multiplier is zero on the support of b, and conversely. Consequently a regular b has $z_b = 0$ and is eventually nonzero. Its rational reciprocal is an inverse in K.

(5) **Construct the total quotient ring.** For arbitrary $f \in K$, the germ $s = (1 + |f|)^{-1}$ is regular and bounded, and sf is bounded. Thus every f is a fraction of bounded germs with regular denominator. Equality of these fractions is reflected in K by regularity, giving the localization universal property.

□

2.5.2 Units and the vanishing ideal

Lemma 2.8 (Bounded inverse)**.** *For $u \in B$, the following are equivalent:*

$$u \in B^\times; \qquad \exists a \in \mathbb{N}_{\geq 1} \operatorname{Ev}(1 \leq a|u|).$$

Thus the units are exactly the bounded germs bounded away from zero. In particular $B^\times = \Theta(1)$.

Proof.

(1) **Extract a lower bound from an inverse.** If $uv = 1$ and $|v| \le a$ eventually, then $1 \le a|u|$ on the intersection.

(2) **Construct and descend the reciprocal.** Conversely define $v = 1/u$ where $u \neq 0$ and $v = 0$ elsewhere. The displayed bound gives $|v| \le a$ and $uv = 1$ eventually. These equations descend to the germ ring. The last assertion includes the already specified boundedness of u.

□

Put $\mathfrak{m} = o(1)$ and $\mathfrak{m}_{\neg\neg} = o_{\neg\neg}(1)$. The sum inequalities at $2k$ and the product inequality at ka, for a bound a on a multiplier, prove that both are ideals of B. No infinite intersection of eventual extents is used.

Theorem 2.9 (Vanishing errors and the Jacobson radical)**.** *Define the unit-theoretic Jacobson radical by*

$$\mathrm{Jac}(B) = \{a \in B : \forall b \in B,\ 1 - ab \in B^{\times}\}.$$

Then

$$\mathrm{Jac}(B) = \mathfrak{m}.$$

In particular $1 + \mathfrak{m}$ *is a subgroup of* $B^{\times}$*.*

Proof.

(1) **Show that vanishing germs lie in the radical.** Let $a \in \mathfrak{m}$ and let $|b| \le c$ eventually. The smallness bound at $2c$ gives $|ab| \le 1/2$ eventually, so $|1 - ab| \ge 1/2$. The bounded-inverse lemma makes $1 - ab$ a unit.

(2) **Construct a cutoff multiplier.** Conversely, fix $a \in \mathrm{Jac}(B)$ and $k \ge 1$. On a representative define

$$b_k(x) = \begin{cases} a(x)^{-1}, & k|a(x)| > 1, \\ 0, & k|a(x)| \le 1. \end{cases}$$

This is bounded by k and descends to a bounded germ.

(3) **Deduce vanishing from radical membership.** The element $q_k = 1 - ab_k$ is the indicator of $k|a| \le 1$, hence idempotent. It is invertible by the definition of $\mathrm{Jac}(B)$. Multiplying $q_k^2 = q_k$ by its inverse gives $q_k = 1$, which is precisely $\mathrm{Ev}(k|a| \le 1)$. This holds for each k, proving $a \in \mathfrak{m}$.

(4) **Verify the subgroup of units.** For $a, b \in \mathfrak{m}$, one has $(1 + a)(1 + b) = 1 + (a + b + ab)$ with $a + b + ab \in \mathfrak{m}$. Also $(1 + a)^{-1} - 1 = -a(1 + a)^{-1} \in \mathfrak{m}$. These formulas prove the subgroup assertion.

□

The theorem uses a definition of the radical by units and finite rational tests. The quotient $B/\mathfrak{m}$ retains nontrivial idempotents. For example, on cofinite numerical input the parity indicator and its complement are neither zero nor vanishing: the tolerance $1/2$ and arbitrarily late inputs of each parity prove this directly. Its class is therefore an idempotent distinct from zero and one. The residue algebra is not treated as a field.

Theorem 2.10 (Multiplier and orbit formulas)**.** *For every signed germ* g*,*

$$O(g) = Bg, \qquad o(g) = \mathfrak{m}g, \qquad o_{\neg\neg}(g) = \mathfrak{m}_{\neg\neg}g,$$
$$\Theta(g) = B^{\times}g, \quad \{f : f \sim g\} = g + \mathfrak{m}g = (1 + \mathfrak{m})g.$$

Consequently

$$f \in \Omega(g) \iff Bg \subseteq Bf, \qquad f \in \omega(g) \iff g \in \mathfrak{m}f.$$

Proof.

(1) **Identify the bounded orbit.** Bounded division proves the first formula.

(2) **Identify the vanishing orbit.** For $f \in o(g)$ use the same ratio $a = f/g$ on the support of g, zero elsewhere. Every smallness inequality implies $k|a| \leq 1$ on its eventual extent, while the $k = 1$ inequality ensures $f = ag$ as germs. Thus $a \in \mathfrak{m}$. Conversely multiply the inequalities for a by $|g|$.

(3) **Identify the double-negated orbit.** For $o_{\neg\neg}$ the explicit domination supplies the germ equation and boundedness; applying double negation to each implication gives the remaining bounds.

(4) **Identify the supported unit orbit.** If $f \in \Theta(g)$, the two domination inequalities imply that f and g have the same support eventually. Define $u = f/g$ where both are nonzero and $u = 1$ elsewhere. On the intersection of these eventual extents it is bounded, its reciprocal is bounded, and $f = ug$. The bounded-inverse lemma gives $u \in B^\times$. Conversely a unit and its bounded inverse give both domination bounds. This argument includes $g = 0$.

(5) **Identify the asymptotic unit orbit.** Finally $f \sim g$ means $f - g \in \mathfrak{m}g$, proving the coset and orbit descriptions.

(6) **Recover the lower-bound relations.** The two reversed comparisons follow immediately from the definitions and bounded division.

□

For nonnegative profiles the unit orbit may be restricted to eventually positive units. The construction of u in the proof has this property. Signed absolute-value comparison also records sign-changing units, while the subgroup $1 + \mathfrak{m}$ retains the relative sign in asymptotic equivalence.

2.6 Resource modules and admitted growth ideals

Principal budgets have the form Bg. To include budgets with several or indefinitely many bounds, we collect growth classes into ideals. The following correspondence identifies these ideals with embedded B-submodules of K, so the marking records exactly which costs a budget contains. For $I \in \mathrm{Idl}(\mathsf{L}_{\mathrm{Ev}})$ define

$$M_I^+ = \{f \in K^+ : [f] \in I\}, \qquad M_I = \{f \in K : [|f|] \in I\}.$$

Here an ideal is inhabited, lower, and directed; equivalently it contains zero and is closed under finite joins.

Theorem 2.11 (Positive resource representation)**.** *There is an order isomorphism*

$$\mathrm{Idl}(\mathsf{L}_{\mathrm{Ev}}) \cong \mathrm{Sub}_{B^+}(K^+), \qquad I \longmapsto M_I^+ = \bigcup_{[g] \in I} B^+ g.$$

It preserves sums, intersections, and module products. Multiplication gives

$$M_I^+ \otimes_{B^+} M_J^+ \cong M_{IJ}^+.$$

Proof.

(1) **Identify ideals with positive resource modules.** A directed union of the displayed principal modules is a submodule. Conversely, for a submodule $M \subseteq K^+$, its growth classes form a lower set by bounded division and are directed by $f+g$. Both constructions are inverse.

(2) **Preserve sums, intersections, and products.** The formulas in Lemma 2.6 prove preservation of sums and intersections; finite products give the product statement.

(3) **Construct the principal tensor isomorphism.** Each B^+e is a retract of B^+ by multiplication with the idempotent e. For principal resources, the tensor identification is

$$B^+g \otimes_{B^+} B^+h \cong B^+e_g \otimes_{B^+} B^+e_h \cong B^+(e_g e_h) \cong B^+gh.$$

The middle map is multiplication; the inverse sends $be_g e_h$ to $be_g \otimes e_h$.

(4) **Verify naturality under inclusion.** If one principal module is contained in another, these maps agree after inclusion into K^+, so they form a natural isomorphism of the principal-stage diagrams.

(5) **Extend the tensor isomorphism to all resources.** General resources are their internal directed colimits. The balanced-map universal property makes tensor preserve these colimits in each variable. Passing to the colimits gives the general identification.

□

Theorem 2.12 (Signed resource lattice)**.** *The assignments*

$$I \longmapsto M_I, \qquad M \longmapsto \{[g] : g \geq 0,\ g \in M\}$$

are inverse order isomorphisms $\mathrm{Idl}(\mathsf{L}_{\mathrm{Ev}}) \cong \mathrm{Sub}_B(K)$*. Ideal joins correspond to module sums, meets to intersections, and ideal products to products inside* K*. For every* g*,*

$$\mathrm{Ann}_B(g) = Bz_g, \qquad Bg \cong B/Bz_g \cong Be_g,$$

and

$$Bg + Bh = B(|g| + |h|), \qquad Bg \cap Bh = B\min(|g|, |h|).$$

Proof.

(1) **Pass between signed modules and growth classes.** For every $f \in K$, the profile $\mathrm{sgn}(f)$ is bounded and hence defines an element of B. The identities

$$|f| = \mathrm{sgn}(f)\, f, \qquad f = \mathrm{sgn}(f)\, |f|$$

show that $f \in M$ if and only if $|f| \in M$ for every B-submodule $M \subseteq K$. Its nonnegative growth classes are lower by bounded division and directed by addition. Conversely the directed union of principal budgets is a submodule. These constructions are inverse on elements, giving the order isomorphism.

(2) **Identify principal supports.** The annihilator was calculated in Proposition 2.7; the idempotent splitting gives the principal isomorphism.

(3) **Establish the lattice formulas.** The sum and intersection arguments of Lemma 2.6 apply after absolute value. Finite sums and products describe the general lattice operations.

□

Theorem 2.13 (Conservative signed resource realization)**.** *There are canonical isomorphisms*

$$\operatorname{Gr} K^+ \cong K = \mathbb{Q}^X/{=_{\mathrm{Ev}}}, \qquad \operatorname{Gr} B^+ \cong B = \{b \in K : |b| = O(1)\}, \qquad \operatorname{Gr} M_I^+ \cong M_I \subseteq K.$$

On marked resource submodules these comparisons preserve and reflect inclusions and membership, and preserve sums, intersections, products, and tensor products.

Proof.

(1) **Identify the positive part after group completion.** Positive germs are cancellative: equality $f + h = g + h$ on an eventual set gives $f = g$ on that set. Their group completion therefore embeds into the signed germ algebra. Every signed rational profile splits canonically as

$$f = f_+ - f_-, \qquad f_+ = \max(f, 0), \quad f_- = \max(-f, 0).$$

This proves the first isomorphism.

(2) **Identify the signed coefficient ring.** If $|f|$ is bounded, so are f_+ and f_-, proving the second.

(3) **Identify each signed resource module.** If $f \in M_I$, then $f_\pm \le |f|$ and their growth belongs to I. Conversely the difference of two members of M_I^+ belongs to M_I because their sum dominates its absolute value. This proves the third, with the same ambient marking.

(4) **Reflect inclusion and membership.** A positive germ belongs to $\operatorname{Gr} M_I^+$ inside K exactly when its growth belongs to I. Thus inclusions and membership are reflected.

(5) **Transfer the lattice and tensor formulas.** Theorems 2.11 and 2.12 identify the same ideal lattice, so sums and intersections agree. Product and tensor comparisons follow from Theorem 2.11 and strong monoidality on coefficient fibres.

□

Corollary 2.14 (Monoidal resource representation)**.** *Let* $\mathsf{Emb}_B(K)$ *have objects marked submodules* $i_M : M \hookrightarrow K$ *and maps* $f : M \to N$ *satisfying* $i_N f = i_M$*. The ideal-to-module correspondence is a strong symmetric monoidal equivalence*

$$(\operatorname{Idl} L, \star, \downarrow [1]) \simeq (\mathsf{Emb}_B(K), \otimes_B, B).$$

The tensor marking is multiplication into K; it is monic by the resource tensor theorem. Joins correspond to embedded module sums, meets to intersections, and directed joins to directed unions.

Proof.

(1) **Establish the equivalence and tensor comparison.** The underlying equivalence is Theorem 2.12. Its tensor comparison is the already constructed multiplication $M_I \otimes_B M_J \cong M_{I \star J}$, and its unit comparison is $M_{\downarrow[1]} = B$.

(2) **Verify monoidal coherence.** Both paths in

$$\begin{array}{ccc} (M_I \otimes_B M_J) \otimes_B M_H & \xrightarrow{\cong} & M_{I\star J} \otimes_B M_H \\ \downarrow{\scriptstyle \cong} & & \downarrow{\scriptstyle \cong} \\ M_I \otimes_B (M_J \otimes_B M_H) & \xrightarrow[\cong]{} & M_{I\star J\star H} \end{array}$$

send $(f \otimes g) \otimes h$ to fgh in K; this proves associativity coherence. The unit and symmetry diagrams follow from $1f = f$ and $fg = gf$. Their markings determine the comparison maps uniquely.

(3) **Preserve lattice operations and directed colimits.** The lattice assertions are Theorem 2.12; the directed-colimit assertion follows from the membership formula defining M_I.

□

Restricting to $J_\Sigma \mathsf{L}_{\mathrm{Ev}}$ selects the resources whose positive test profiles are admitted. Natural-valued costs embed conservatively: their domination formula is unchanged by rational inclusion.

Theorem 2.15 (Constructive finite-generator splitting)**.** *Every finitely generated submodule $N \subseteq B^n$ has a displayed finite-free retract, hence a finite idempotent-matrix presentation. More precisely, if $w_1, \dots, w_m \in B^n$ generate N and $p : E \to N$ is a B-linear map with displayed lifts $v_i \in E$ satisfying $p(v_i) = w_i$, then these finite data construct a B-linear section $s : N \to E$ of p. All operations commute with substitution of coefficient and generator parameters.*

Proof.

(1) **Initialize the induction.** Induct on n. For $n = 0$ the zero module has its unique section.

(2) **Split a module with one coordinate.** For $n = 1$ put $h = \sum_i |w_i|$, so $N = Bh \cong Be_h$ by bounded division. Write $q(c, h) = c/h$ on the support of h, zero elsewhere, and define

$$s(c) = q(c, h) \sum_i \operatorname{sgn}(w_i) v_i \qquad (c \in Bh).$$

The quotient is a well-defined B-linear function of $c \in Bh$: if $c = bh$, then $q(c, h) = be_h$. Since $\sum_i \operatorname{sgn}(w_i) w_i = h$, we have $ps(c) = q(c, h)h = c$.

(3) **Split the final coordinate.** For $n > 1$ let $\pi : N \to B$ be the final-coordinate map, $a_i = \pi(w_i)$, and $h = \sum_i |a_i|$. The rank-one formula, using the same displayed lifts v_i, constructs $j : Bh \to E$ with $(\pi p)j = \mathrm{id}_{Bh}$. Put

$$q_0 = \mathrm{id}_E - j\pi p, \qquad q_0^2 = q_0, \qquad \pi p q_0 = 0.$$

(4) **Generate the kernel of the final coordinate.** The elements $w_i' = pq_0(v_i)$ have final coordinate zero and generate $N' = \ker(\pi : N \to Bh)$. Indeed, for $w \in N'$, generation of N gives locally $w = \sum_i b_i w_i$. Then

$$\sum_i b_i w_i' = \sum_i b_i (w_i - pj\pi w_i) = w - pj\pi w = w.$$

(5) **Apply the induction hypothesis.** The corestriction $pq_0 : E \to N'$ has displayed lifts v_i of these generators. Induction gives $s' : N' \to E$ with $pq_0 s' = \mathrm{incl}_{N'}$. Put $\bar{s}' = q_0 s'$, so $p\bar{s}' = \mathrm{incl}_{N'}$.

(6) **Reconstruct a section of the original surjection.** For $w \in N$ set

$$s(w) = j(\pi w) + \bar{s}'(w - pj(\pi w)).$$

The second argument lies in N', and direct substitution gives $ps(w) = pj(\pi w) + w - pj(\pi w) = w$.

(7) **Obtain finite projectivity and parameter stability.** Taking $E = B^m$ and p to be the canonical generating surjection gives $ps = \mathrm{id}_N$ and the idempotent sp on B^m, whose image is N. Every displayed operation is finite support division, finite arithmetic, or induction, and the equations hold uniformly in parameters.

□

Corollary 2.16 (The algebra needed by the later geometry)**.** *Every signed embedded resource is a directed union of finite projective principal modules. It is flat over B, and multiplication identifies $M \otimes_B N$ with $MN \subseteq K$. The algebra K is the total quotient algebra of B; B is reduced and semihereditary.*

Proof.

(1) **Identify principal projective modules.** For signed g, bounded division and its support idempotent give $Bg \cong Be_g$, an explicit finite-projective summand of B.

(2) **Express resources as directed unions.** For any finite family $f_1, \dots, f_n \in M$, the element $h = \sum_i |f_i| \in M$ satisfies $f_i \in Bh$, so the principal submodules form an internal directed system with union M.

(3) **Deduce flatness.** Every finite relation is witnessed at a common principal stage, and this stage is flat. The finite-witness filtered-colimit argument is the internal form of the standard exactness calculation (Stacks Project Authors 2026a, Tag 05UT).

(4) **Establish the tensor formulas.** The principal tensor multiplication isomorphisms commute with stage inclusions, and tensor preserves directed colimits. Their colimit therefore gives $M \otimes_B N \cong MN \subseteq K$.

(5) **Verify reducedness.** Reducedness follows pointwise from rational arithmetic on the same eventual set.

(6) **Identify the total quotient ring.** For any $f \in K$, the regular bounded germ $s = (1 + |f|)^{-1}$ satisfies $sf \in B$. Conversely the zero-locus indicator of a regular $b \in B$ is zero; its pointwise reciprocal, set to zero off its support, is an inverse in K. This proves the total-quotient description.

(7) **Deduce semihereditariness.** Every finitely generated ideal of B is one of the projective principal modules just described, proving semihereditariness.

□

In particular $f \in \Theta(g)$ says that f and g generate the same marked principal module. Asymptotic equivalence is equality with g in the quotient $Bg/\mathfrak{m}Bg$. The lower-bound incidences $\Omega(g)$ and $\omega(g)$ exclude zero when $g \neq 0$.

The lattice of all resources is a frame, since it is the ideal completion of the distributive growth lattice. The admitted subobject is selected by J_Σ. For numerical costs the passage to rational resources is the following explicitly typed comparison.

Proposition 2.17 (Conservative rational extension of growth ideals)**.** *Let $\mathsf{L}_{\mathrm{Ev}}^{\mathbb{N}}$ be the growth poset of nonnegative integer-valued profiles and let $\ell : \mathsf{L}_{\mathrm{Ev}}^{\mathbb{N}} \to \mathsf{L}_{\mathrm{Ev}}$ be induced by $\mathbb{N} \hookrightarrow \mathbb{Q}_{\geq 0}$. It is order-reflecting. For an ideal $I \in \mathrm{Idl}(\mathsf{L}_{\mathrm{Ev}}^{\mathbb{N}})$, its rational lower image is*

$$\ell_! I = \{q \in \mathsf{L}_{\mathrm{Ev}} : \exists n \in I,\ q \leq \ell(n)\}.$$

Then, for a natural-valued profile f,

$$[f]_{\mathbb{N}} \in I \quad \Longleftrightarrow \quad f \in M_{\ell_! I}.$$

The map $J_\Sigma \ell$ carries admitted ideals to admitted ideals and reflects their inclusion order; its underlying ideal map is $\ell_!$.

Proof.

(1) **Compare rational and integer bounds.** A rational eventual domination coefficient is bounded above by an integer coefficient; hence rational domination between integer profiles reflects integer domination. Therefore

$$\ell([f]_{\mathbb{N}}) \in \ell_! I \iff \exists n \in I\ ([f]_{\mathbb{N}} \le n) \iff [f]_{\mathbb{N}} \in I.$$

By definition of $M_{\ell_! I}$ this is the displayed comparison.

(2) **Reflect admitted resources and their operations.** Functoriality of J_Σ gives admission of the lower image. Moreover, for every f, $[f]_{\mathbb{N}} \in I \iff \ell([f]_{\mathbb{N}}) \in (J_\Sigma \ell)(I)$; these tests recover inclusion of ideals and prove order reflection. All statements are internal and uniform in the filter and input.

□

2.6.1 The observation domain of computational resources

The module classification describes resource membership semantically. We next specify which observations of a resource are positive computational tests. Put $L = \mathsf{L}_{\mathrm{Ev}}$. Its free observed domain from Section 1.3 has the type and observation family

$$F_\Sigma L = (\mathrm{Idl}\, L, \mathcal{O}_{F_\Sigma L}), \qquad \mathcal{O}_{F_\Sigma L} = \{u^\dagger : u \in \mathrm{Mon}(L, \Sigma)\},$$

where

$$u^\dagger(I) = \exists a : L\ [I(a) \wedge u(a)].$$

Its effective-point object is $\mathrm{Pt}_\Sigma(F_\Sigma L) = J_\Sigma L$. Thus the uniform positive pairing is a map

$$\chi : J_\Sigma L \times \mathrm{Mon}(L, \Sigma) \to \Sigma. \tag{2.3}$$

By contrast, the marked-resource membership and measured incidences have types

$$\mu : \mathrm{Idl}\, L \times K \to \Omega, \qquad \mu(I, f) \iff [|f|] \in I, \qquad (I, e) \mapsto \mu(I, \Phi(e)) : J_\Sigma L \times P \to \Omega.$$

Proposition 2.18 (Effective transport of resource observations)**.** *A monotone map $r : L \to L'$ induces an effective Scott map $F_\Sigma r : F_\Sigma L \to F_\Sigma L'$ and its admitted-point map $J_\Sigma r : J_\Sigma L \to J_\Sigma L'$. Explicitly,*

$$(F_\Sigma r)(I) = {\downarrow} r(I), \qquad \chi_{J_\Sigma r(I)}(u) = \chi_I(u \circ r).$$

Proof.

(1) **Construct the continuous ideal map.** The lower image is an inhabited directed ideal, and its formula commutes with directed unions.

(2) **Transport observations and profiles.** For $u : L' \to \Sigma$ monotone,

$$\exists b\, [b \in {\downarrow} r(I) \wedge u(b)] \iff \exists a\, [a \in I \wedge u(r(a))].$$

The forward implication uses $b \le r(a)$ and monotonicity; the reverse uses $b = r(a)$. The inverse image of $u^\dagger$ is therefore $(u \circ r)^\dagger$, proving observation preservation.

(3) **Transport resources and verify composition.** For admitted I the right side factors uniformly through Σ, proving the admitted-point assertion. Composition is literal substitution of tests.

□

Proposition 2.19 (Admitted membership is not generally positive)**.** *In the effective numerical fibre with the cofinite input filter, the predicate that a uniformly given exact profile lies in the principal budget* $\downarrow [1]$ *need not be* Σ*-classified.*

Proof.

(1) **Construct the clocked family of germs.** The bounded numerical interpreter of Propositions 12.2 and E.5 gives a Boolean test $H(e, e, n)$ for halting within n updates. Define the primitive-recursive family

$$r_e(n) = \begin{cases} 1, & \neg H(e, e, n), \\ n + 1, & H(e, e, n). \end{cases}$$

(2) **Characterize boundedness of the family.** If the diagonal computation never halts, r_e is constantly one. If it halts, r_e is eventually $n + 1$ and cannot be eventually bounded.

(3) **Deduce failure of semidecidability.** Hence $r_e = O(1)$ is exactly the complement of diagonal halting. A Σ factorization of this membership predicate would semidecide that complement, contradicting Lemma E.6.

(4) **Verify admission of the principal resource.** The principal ideal $\downarrow [1]$ is admitted by the unit of J_Σ.

□

The pairing (2.3) supplies the positive resource observations. Matrix equations over germ coefficients are interpreted in Ω; finite certificates have the checking and search maps constructed in Section F.

2.7 Gauge localization, soft notation, and coefficient descent

2.7.1 Comparison with asymptotic gauges

The asymptotic gauges of (Giordano and Luperi Baglini 2016) define moderate rings by domination by a member of a scale family. The following comparison uses rational-valued profiles and the same eventual filter as our germ algebra. It concerns moderate elements before a negligible-element quotient.

Definition 2.20 (Rational asymptotic gauge)**.** A rational asymptotic gauge is a family $\mathcal{G}$ of exact rational germs with a member tending to $+\infty$, such that for $g, h \in \mathcal{G}$ there are $k, l \in \mathcal{G}$ with $gh = O(k)$ and $|g| + |h| = O(l)$. Rational rescaling is absorbed by the O-relation. Its generated growth ideal is $I_{\mathcal{G}} = \downarrow \{[|g|] : g \in \mathcal{G}\}$.

Proposition 2.21 (Gauge-moderate germs)**.** *The family* $I_{\mathcal{G}}$ *is an inhabited directed growth ideal, and its marked resource module is*

$$M_{I_{\mathcal{G}}} = \{f \in K : \exists g \in \mathcal{G},\ f = O(g)\} = \bigcup_{g \in \mathcal{G}} B|g|.$$

It is a unital B*-subalgebra of* K*. Thus rational moderate profiles modulo eventual equality give exactly this marked algebra. If* $\mathcal{G}$ *and* $\mathcal{H}$ *dominate one another memberwise, their ideals and marked algebras coincide.*

Proof.

(1) **Verify directedness of the moderate resource.** The divergent member makes $\mathcal{G}$ inhabited and dominates 1. The sum condition provides a common upper growth bound for any two generators, so their lower closure is directed.

(2) **Identify its resource module.** Membership in M_I means precisely that the growth of the absolute value belongs to I; bounded division identifies this with membership in $B|g|$ for some g. This proves both equalities.

(3) **Construct the moderate subalgebra.** The sum and product conditions respectively close the displayed union under addition and multiplication; bounded multiplication and negation preserve domination. The divergent member puts 1, hence all of B, in this union.

(4) **Descend multiplication to eventual classes.** Eventual equality preserves every domination condition, proving the quotient statement. Memberwise domination gives inclusions of the generated lower ideals in both directions.

□

A general resource module need not have the gauge's multiplicative closure: on cofinite numerical input Bn contains n but not n^2. Also B itself contains no divergent germ. These facts distinguish the full resource classifier from its gauge-generated algebra sector. Admission of $I_{\mathcal{G}}$ is still the uniform positive-pairing property of Definition 1.3; it is not inferred solely from the algebraic gauge conditions. Colombeau constructions additionally quotient moderate elements by negligible ones, whereas K uses eventual equality. The comparison above therefore does not identify those quotient algebras with K or with its Weil thickenings. Functorial changes of asymptotic gauges are studied in (Luperi Baglini and Giordano 2017); the input and coefficient transports below retain their specified filter and module maps.

2.7.2 Overhead gauges

A chosen family of overheads changes which multipliers count as bounded. We encode that family before changing the coefficient ring. The gauge object is the subobject of $\mathcal{P}((\mathbb{Q}_{\geq 0})^X)$ consisting of multiplicative families $\mathcal{P}$ of profiles $p \geq 1$ containing 1. Its unit and finite products are part of this definition. For example, powers of the numerical bitlength profile below form such a family by NNO recursion. Its downward growth closure is closed under finite joins, since $p + q \leq 2pq$. Define

$$f \in O_{\mathcal{P}}(g) \iff \exists p \in \mathcal{P}, \quad f \in O(pg),$$

and define $\Omega_{\mathcal{P}}$ by reversal and $\Theta_{\mathcal{P}}$ by mutual comparison. The corresponding coefficient ring inside K is

$$B_{\mathcal{P}} = \bigcup_{p \in \mathcal{P}} Bp.$$

The displayed directed union is a ring: products multiply gauges, and sums use the gauge pq.

Lemma 2.22 (Admitted multiplication of growth ideals)**.** *Let $L = \mathsf{L}_{\mathrm{Ev}}$ with its monotone multiplication $m : L \times L \to L$. For $P, I \in J_\Sigma L$ put $P \star I = J_\Sigma m(P \times I)$. Then $P \star I \in J_\Sigma L$; these products are associative and commutative with principal unit $\downarrow [1]$. For every positive upper predicate $u : L \to \Sigma$,*

$$\chi_{P \star I}(u) = \chi_P(p \mapsto \chi_I(x \mapsto u(px))).$$

Consequently an admitted multiplicatively closed growth ideal P containing [1] *induces an inflationary idempotent operation $I \mapsto P \star I$ on admitted ideals.*

Proof.

(1) **Construct admitted multiplication.** Product preservation for J_Σ makes $P \times I$ an admitted ideal of $L \times L$; functoriality then admits its lower image under m.

(2) **Compute the multiplication formula.** Expanding the lower-image pairing gives the displayed iterated existential. Alternatively the inner pairing belongs uniformly to Σ by admission of I, is monotone in p, and is therefore a positive upper test admitted by P.

(3) **Verify algebraic laws and saturation.** Associativity, commutativity and the unit follow from the corresponding monoid identities on L by functoriality. If P contains [1] and is closed under products, then $P \star P = P$: one inclusion is multiplicative closure and the other uses the unit. Thus $I \leq P \star I$ and $P \star (P \star I) = P \star I$.

□

Theorem 2.23 (Gauge localization)**.** *Let $D = \{p^{-1} : p \in \mathcal{P}\} \subseteq B$. Then*

$$B_{\mathcal{P}} \cong D^{-1}B, \qquad O_{\mathcal{P}}(g) = B_{\mathcal{P}}g, \qquad B_{\mathcal{P}} \otimes_B M_I \cong M_{\mathcal{P}I}. \tag{2.4}$$

The same construction on exact positive coefficients commutes with Gr*. The induced resource saturation is an inflationary, idempotent, finite-meet-preserving operation.*

Proof.

(1) **Identify the gauge localization.** Each p^{-1} is bounded, regular, and invertible in K. A fraction b/p^{-1} is bp, and every element of the displayed union has this form. This proves the localization isomorphism and its universal property.

(2) **Compute localized resource modules and tensors.** Bounded division gives the second formula. The tensor-product theorem for marked resources, or its balanced-map proof on principal cells, gives the third, with marking $a \otimes m \mapsto am$.

(3) **Verify compatibility with group completion.** Positive fractions have the same presentation and their differences give precisely the signed fractions, proving compatibility with Gr.

(4) **Prove saturation and finite-meet preservation.** Containment of 1 makes saturation inflationary, and multiplicative closure makes it idempotent. Localization is flat: a finite relation of fractions is cleared by one product of denominators and is then a relation before localization. Thus it preserves the kernel representing an intersection of two submodules. This proves finite-meet preservation.

(5) **Restrict saturation to admitted resources.** For an admitted multiplicative growth ideal, the induced operation on admitted resources is precisely the J_Σ-product of Lemma 2.22. Its outputs are admitted by that lemma; finite-meet preservation on full resource modules is the localization calculation above.

□

Equivalently an admitted multiplicative ideal P containing [1] has coefficient algebra M_P. Its denominators can be taken as $(1+g)^{-1}$ for $[g] \in P$, $g \geq 0$:

$$f = \frac{f/(1+g)}{(1+g)^{-1}} \quad \text{whenever } f \in O(g).$$

The product of denominators for g, h is the denominator for $g + h + gh$. This proves the same localization directly from the ideal presentation.

2.7.3 Soft Landau notation

On numerical input put $\lambda(n) = 1 + \mathrm{bitlength}(n+1)$. This is an exact computable integer profile, comparable to $1 + \log_2(n+1)$ by fixed positive constants. Its finite powers form a gauge. Define

$$\widetilde{O}(g) = \bigcup_{k\geq 0} O(\lambda^k g), \quad f \in \widetilde{\Omega}(g) \iff g \in \widetilde{O}(f), \quad \widetilde{\Theta} = \widetilde{O} \cap \widetilde{\Omega}.$$

Then $\widetilde{O}(g) = B[\lambda]g$, where $B[\lambda] = B[(\lambda^{-1})^{-1}]$ inside K. In particular $f \in \widetilde{\Theta}(g)$ means equality of their principal modules after this localization. The profile uses finite bit-length computation, so no logarithm operation on arbitrary lower-real names is needed.

Proposition 2.24 (The admitted polylogarithmic gauge)**.** *In the realizability models of Section 1, the growth ideal*

$$P_\lambda = \downarrow \{[\lambda^k] : k \in \mathbb{N}\}$$

is J_Σ-admitted, contains $[1]$, and is closed under growth multiplication and finite joins. Consequently $I \mapsto P_\lambda \star I$ is a constructed inflationary, idempotent soft-resource operation on $J_\Sigma \mathsf{L}_{\mathrm{Ev}}$.

Proof.

(1) **Prove admission of the power ideal.** The powers are increasing because $\lambda \geq 1$, and their lower closure is an ideal. For a positive upper predicate $u : \mathsf{L}_{\mathrm{Ev}} \to \Sigma$,

$$\chi_{P_\lambda}(u) = \exists k \in \mathbb{N}\, u([\lambda^k]) \in \Sigma$$

by overtness of the NNO, uniformly in u.

(2) **Verify multiplicative closure and induced saturation.** The unit is the zeroth power; multiplication adds exponents. Also $[\lambda^j] \vee [\lambda^k] \leq [\lambda^{j+k}]$. The admitted-growth-product lemma now constructs the stated operation and proves its laws.

□

Proposition 2.25 (Vanishing under overhead inversion)**.** *If $s \in \mathfrak{m}$ is inverted by a gauge localization, then $\mathfrak{m}B_{\mathcal{P}} = B_{\mathcal{P}}$. In particular this occurs when $s = t$, $t(n) = 1/(n+1)$, is inverted, and when $s = \lambda^{-1}$ is inverted.*

Proof.

(1) **Show that inversion makes the ideal contain the unit.** The element $1 = s^{-1}s$ belongs to the extended ideal.

(2) **Verify the stated vanishing profiles.** The profiles t and λ^{-1} satisfy each vanishing bound by their explicit unbounded denominators.

□

Thus gauge-relative domination is calculated in the localized ring, while original small-error incidences remain recorded in the exact presentation over B. The extended ideal $\mathfrak{m}B_{\mathcal{P}}$ is not redefined to be the vanishing ideal of the original numerical filter. This formula specifies exactly how its marking changes.

2.8 Eventual quotients and continuous completion

Lemma 2.26 (Admitted rational approximation)**.** *In the realizability models of Section 1, put* $r_n = (n+1)/(n+2)$ *and* $I = {\downarrow}\{\underline{r_n} : n \in \mathbb{N}\} \subseteq S$*. Then* $I \in J_\Sigma S$ *and* $\sup I = 1$*.*

Proof.

(1) **Prove admission by a countable presentation.** The family is inhabited and increasing, hence its lower closure is an ideal. For every monotone $u : S \to \Sigma$,

$$\chi_I(u) = \exists s \in I\, u(s) \quad \Longleftrightarrow \quad \exists n \in \mathbb{N}\, u(\underline{r_n}).$$

The right-hand existential factors uniformly through Σ by Proposition 1.2. Hence I passes the defining J_Σ pairing test.

(2) **Compute the supremum.** Every $r_n \leq 1$, and for each positive rational $q < 1$ the Archimedean property of rational arithmetic gives an n with $q < r_n$. Equality of the resulting lower cuts gives $\sup I = 1$.

□

Proposition 2.27. *In the realizability models of Section 1, let* $X = \mathbb{N}$ *carry the cofinite filter, let* $t(n) = 1/(n+1)$*, and let* B^+ *be the semiring of eventually bounded nonnegative rational germs. In* B^+*,* $1 + t \neq 1$*, but their images in* $\mathsf{E}_S B^+$ *are equal.*

Proof.

(1) **Separate the germs before completion.** Pointwise $t > 0$, so the germ equality is false.

(2) **Construct the rational comparison.** Let x be the image of $1 + t$ and let e be the image of 1. For each rational $0 \leq r < 1$, the inequality $r(1 + t) \leq 1$ holds eventually. The nonnegative rational profile

$$c_r = \max\{1 - r(1 + t), 0\}$$

satisfies $r(1 + t) + c_r = 1$ as a germ. Therefore $rx \leq e$ in the positive order of $\mathsf{E}_S B^+$.

(3) **Use scalar continuity to identify the images.** The admitted ideal of Lemma 2.26 has supremum 1, so scalar continuity gives $x = \sup_n r_n x \leq e$. Conversely $e \leq x$ by positivity of the image of t. Antisymmetry gives $x = e$.

□

Continuous completion and eventual equality therefore have different effects on a cost profile. The construction keeps the exact profile and forms its eventual relations there; the presentation object in $\mathscr{P}$ records the quantitative interpretation alongside it. In particular, (1.8) remains available before the eventual quotient.

The resource algebra now supplies embedded modules, their tensor and lattice operations, and the coefficient localizations associated with overhead gauges. We next classify these resources in families, so that one construction can express a budget varying over a parameter space.

3 The universal structured resource space

The preceding section describes individual budgets as embedded modules. To vary a budget over parameters, we need a classifier carrying a universal family. The growth order already determines its topology: a basic open expresses availability of one growth class. For $L = \mathsf{L}_{\mathrm{Ev}}$, form the frame of upper predicates

$$\mathrm{Up}(L) = \{U : L \to \Omega : \forall a \le b,\ U(a) \Rightarrow U(b)\}.$$

Write $\mathfrak{C}_{\mathrm{Ev}}$ for the internal locale with $\mathcal{O}(\mathfrak{C}_{\mathrm{Ev}}) = \mathrm{Up}(L)$. We use the standard internal locale and localic-sheaf constructions (Vickers 1989; Mac Lane and Moerdijk 1992). The following results identify its generalized points with marked resource families and construct their universal module. Thus the geometry is determined by growth comparison itself.

3.1 Ideal classification and the Scott frame

Theorem 3.1 (Canonical classifying frame)**.** *For every internal frame H, frame maps $\mathrm{Up}(L) \to H$ are naturally equivalent to maps $I : L \to H$ satisfying*

$$I(0) = \top, \qquad a \le b \Rightarrow I(b) \le I(a), \qquad I(a) \wedge I(b) = I(a \vee b). \tag{3.1}$$

Consequently

$$\mathrm{Pt}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}}) \cong \mathrm{Idl}(L), \qquad \mathcal{O}(\mathfrak{C}_{\mathrm{Ev}}) \cong \mathrm{Sc}(\mathrm{Idl}(L)).$$

The principal opens $U_a = \{b : a \le b\}$ are supercompact, are closed under finite intersections, and generate the frame. Thus $\mathfrak{C}_{\mathrm{Ev}}$ is a coherent internal locale.

Proof.

(1) **Extract an ideal from a frame map.** Every upper predicate has the canonical expansion $U = \bigvee_{a \in U} U_a$. Because L has bottom and finite joins,

$$U_0 = \top, \qquad U_a \wedge U_b = U_{a \vee b}.$$

Given a frame map $\theta : \mathrm{Up}(L) \to H$, put $I_\theta(a) = \theta(U_a)$. The displayed identities and $U_b \subseteq U_a$ for $a \le b$ prove (3.1).

(2) **Construct a frame map from an ideal.** Conversely, given I with those equations, define

$$\theta_I(U) = \bigvee_{a \in U} I(a).$$

Arbitrary-join preservation is immediate. Frame distributivity gives

$$\theta_I(U) \wedge \theta_I(V) = \bigvee_{a \in U, b \in V} I(a \vee b) = \theta_I(U \cap V).$$

For the last equality, $a \vee b \in U \cap V$ gives one inequality, while each $c \in U \cap V$ is obtained with $a = b = c$ for the other. Preservation of top follows from $I(0) = \top$.

(3) **Verify the classification and its points.** The inverse equations are

$$\theta_{I_\theta}(U) = \bigvee_{a \in U} \theta(U_a) = \theta(U), \qquad I_{\theta_I}(a) = \bigvee_{b \ge a} I(b) = I(a).$$

For $H = \Omega$, conditions (3.1) say exactly that I is an inhabited directed lower ideal: bottom belongs to it, and the join of two members is a member. This gives the asserted internal point comparison.

(4) **Identify the Scott frame.** For the Scott-frame comparison, map an upper predicate U to

$$\widehat{U}(I) \iff \exists a \in I\, U(a).$$

It is upper and preserves inhabited directed joins of ideals, which are unions. Conversely send a Scott-open W to $u_W(a) = W(\downarrow a)$. Each ideal is the directed union $I = \bigvee_{a\in I} \downarrow a$, so Scott openness gives $W(I) \iff \exists a \in I\, W(\downarrow a)$. Also $\widehat{U}(\downarrow a) \iff U(a)$ by upperness. These are the two inverse equations.

(5) **Prove supercompactness and coherence.** Finally, if $U_a \le \bigvee_j V_j$, evaluating at a gives locally a witness j with $a \in V_j$; upperness gives $U_a \le V_j$. This proves supercompactness internally. The principal opens form a finite-meet-stable generating family, including top, so the locale is coherent.

□

3.2 Embedded modules and the universal resource family

The ideal classification now extends to families of embedded modules. For an internal locale Z, write $p_Z : \mathrm{Sh}_{\mathcal{E}}(Z) \to \mathcal{E}$ for its localic geometric morphism and $\Delta_Z = p_Z^*$ for inverse image. A budget over Z is a module subobject in $\mathrm{Sh}_{\mathcal{E}}(Z)$; this allows its membership conditions to vary locally with the parameter.

The family-valued moduli problem is the contravariant indexed poset

$$\mathcal{B} : \mathrm{Loc}_{\mathcal{E}}^{\mathrm{op}} \longrightarrow \mathbf{Pos}, \qquad \mathcal{B}(Z) = \mathrm{Sub}_{\Delta_Z B}(\Delta_Z K), \qquad \mathcal{B}(f) = f^*.$$

Each object retains its embedding into the ambient module. The next theorem represents this indexed poset by $\mathfrak{C}_{\mathrm{Ev}}$. Its naturality says that reindexing a budget family is composition with its classifying map, and hence determines the universal family.

Theorem 3.2 (Localic resource-module classification)**.** *There is a natural order isomorphism*

$$\mathrm{Loc}_{\mathcal{E}}(Z, \mathfrak{C}_{\mathrm{Ev}}) \;\cong\; \mathrm{Sub}_{\Delta_Z B}(\Delta_Z K), \tag{3.2}$$

where the right-hand side is the poset of $\Delta_Z B$-submodules of the fixed ambient module $\Delta_Z K$. Under it, the identity of $\mathfrak{C}_{\mathrm{Ev}}$ classifies a canonical marked module

$$\mathscr{M} \hookrightarrow \Delta_{\mathfrak{C}_{\mathrm{Ev}}} K \qquad \textit{in } \mathrm{Sh}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}}). \tag{3.3}$$

For a generalized ideal point I, its pullback is $M_I \hookrightarrow K$.

Proof.

(1) **Express subobjects by open-valued membership.** The localic subobject-classifier formula $p_{Z*}\Omega_{\mathrm{Sh}_{\mathcal{E}}(Z)} \cong \mathcal{O}(Z)$ and the adjunction $\Delta_Z \dashv p_{Z*}$ give

$$\mathrm{Sub}_{\mathrm{Sh}_{\mathcal{E}}(Z)}(\Delta_Z K) \cong \mathcal{E}(K, \mathcal{O}(Z)).$$

Thus a subobject has an open-valued membership function $\lambda : K \to \mathcal{O}(Z)$. It is a module precisely when the equalities and implications expressing membership of zero, addition, negatives, and multiplication by B hold in this frame-valued logic.

(2) **Descend membership to growth classes.** The sign multiplier $\mathrm{sgn}(f) \in B$ satisfies $|f| = \mathrm{sgn}(f) f$ and $f = \mathrm{sgn}(f)|f|$; hence $\lambda(f) = \lambda(|f|)$. If $f, g \ge 0$ have equal growth classes, the bounded-division theorem supplies multipliers in both directions, giving $\lambda(f) = \lambda(g)$. Therefore positive membership descends across the effective growth quotient $K^+ \twoheadrightarrow L$ to a map $I_\lambda : L \to \mathcal{O}(Z)$.

(3) **Verify the ideal equations.** It satisfies $I_\lambda(0) = \top$. Bounded division gives lowerness. On positive elements, module closure gives $\lambda(f) \wedge \lambda(g) \leq \lambda(f+g)$, and both f, g are bounded multiples of $f+g$, giving

$$\lambda(f) \wedge \lambda(g) = \lambda(f+g).$$

Since $[f+g] = [f] \vee [g]$, these are precisely (3.1).

(4) **Recover the module from its ideal.** Conversely an $\mathcal{O}(Z)$-valued ideal I defines $\lambda_I(f) = I([|f|])$. The triangle inequality, $|bf| \preceq |f|$ for $b \in B$, and (3.1) prove closure under all module operations. The two constructions are inverse on their membership predicates.

(5) **Classify modules by locale maps.** By Theorem 3.1, the resulting $\mathcal{O}(Z)$-valued ideals are precisely frame maps $\mathrm{Up}(L) \to \mathcal{O}(Z)$, equivalently locale maps $Z \to \mathfrak{C}_{\mathrm{Ev}}$. Every step is natural in Z by pullback of membership predicates.

(6) **Identify the universal module and its fibres.** At $Z = \mathfrak{C}_{\mathrm{Ev}}$, the identity frame map defines (3.3), with membership of $f \in K$ given by the open

$$[\![f \in \mathscr{M}]\!] = U_{[|f|]}. \tag{3.4}$$

Evaluating this at an ideal point I gives $[|f|] \in I$, exactly the membership predicate of M_I from Section 2.6. This proves the fibre assertion.

□

For $b : Z \to \mathfrak{C}_{\mathrm{Ev}}$, the classified family is

$$\mathscr{M}_b = b^*\mathscr{M} \hookrightarrow \Delta_Z K, \qquad \mathscr{M}_{b\circ f} = f^*\mathscr{M}_b. \tag{3.5}$$

Thus the category of marked budget families and cartesian maps is $\mathrm{Loc}_{\mathcal{E}}/\mathfrak{C}_{\mathrm{Ev}}$, with universal family corresponding to $\mathrm{id}_{\mathfrak{C}_{\mathrm{Ev}}}$.

3.3 Generalized points as families of budgets

The classification can be used directly to evaluate a resource condition on a varying family. Retain the indexed budget functor $\mathcal{B}$ and the coordinate identities

$$U_a \wedge U_b = U_{a\vee b}, \qquad U_0 = \top, \qquad U_a(I) \iff a \in I, \qquad M_{\downarrow[g]} = Bg = O(g).$$

Write $I \models U$ for evaluation of an open U at an ideal point I. This notation expresses an internal truth value.

The budget functor of points is the classification

$$h_{\mathfrak{C}_{\mathrm{Ev}}}(Z) = \mathrm{Loc}_{\mathcal{E}}(Z, \mathfrak{C}_{\mathrm{Ev}}) \cong \mathrm{Sub}_{\Delta_Z B}(\Delta_Z K) \tag{3.6}$$

of Theorem 3.2. The following example evaluates a resource condition on a nonconstant family.

A nonconstant family of budgets. To exhibit a nonconstant budget family, take the Sierpiński locale $Z = \mathbb{S}$ itself, and write $u \in \mathcal{O}(Z)$ for its distinguished proper open. Let $a_0 = [1]$ and $a_1 = [n+1]$; thus $a_0 < a_1$ in the numerical cofinite growth order. Construct the following $\mathcal{O}(Z)$-valued ideal on L:

$$I_Z(c) = [\![c \leq a_0]\!]_Z \ \vee \ (u \wedge [\![c \leq a_1]\!]_Z), \qquad c \in L. \tag{3.7}$$

Here $[\![P]\!]_Z$ denotes the constant open of Z classified by the internal truth value P; it is not assumed that arbitrary order relations on L are decidable. Since $a_0 \le a_1$, the expression in (3.7) sends 0 to top, is lower in c, and satisfies $I_Z(c) \wedge I_Z(d) = I_Z(c \vee d)$. Indeed the component with no factor u requires both $c, d \le a_0$, whereas all remaining terms require u and both $c, d \le a_1$. By Theorems 3.1 and 3.2 it defines a generalized point $b : Z \to \mathfrak{C}_{\mathrm{Ev}}$. Its budget is constant-bounded at the closed specialization and linear-bounded on the distinguished open u.

$$b^* U_{a_1} = [\![a_1 \le a_0]\!]_Z \vee (u \wedge [\![a_1 \le a_1]\!]_Z) = u. \tag{3.8}$$

Thus a resource condition can hold on a proper open of the parameter locale. The family-valued classification records this varying locus, whereas an individual ideal point records only its specialization. In Section 12.4 the same family will be probed by a measured numerical implementation.

Theorem 3.3 (Flat universal resource module)**.** *The universal module $\mathscr{M}$ is an internal directed colimit of finite projective principal modules and is flat over $\Delta_{\mathfrak{C}_{\mathrm{Ev}}} B$.*

Proof.

(1) **Construct projective principal stages.** Work in the internal logic of $\mathrm{Sh}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}})$. The bounded-division maps and support idempotents are obtained by pullback of their parameter-uniform constructions in Lemma 2.6 and Proposition 2.7. Every local element $f \in \mathscr{M}$ generates $Bf \cong Be_f$, a split summand of the rank-one free module. Its inclusion into $\mathscr{M}$ is the multiplication map with this marking.

(2) **Express the universal module as a filtered colimit.** Principal submodules are directed: $f, g \in \mathscr{M}$ have common upper principal stage $B(|f| + |g|) \subseteq \mathscr{M}$. They cover $\mathscr{M}$, because $f \in Bf$. Their internal filtered colimit is therefore $\mathscr{M}$.

(3) **Deduce flatness from finite relations.** Each principal stage is projective and flat. A finite module relation in this colimit is, locally, represented at finitely many stages; directedness supplies one common principal stage, where projectivity resolves it. This finite-equation proof of filtered-colimit flatness applies in the internal module category (cf. (Stacks Project Authors 2026a, Tag 05UT)), proving flatness.

□

3.4 The ringed resource topos and its diagram presentation

The universal family now equips the budget classifier with coefficients. Put $\mathscr{S} = \mathrm{Sh}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}})$, $\bar{B} = \Delta_{\mathfrak{C}_{\mathrm{Ev}}} B$ and $\bar{K} = \Delta_{\mathfrak{C}_{\mathrm{Ev}}} K$. The structured resource space is $(\mathscr{S}, \bar{B})$ with its marked module $\mathscr{M} \hookrightarrow \bar{K}$. A cartesian map $(Z, N) \to (Z', N')$ is a locale map $f : Z \to Z'$ with $N = f^* N'$ as marked submodules. By Theorem 3.2, these families form a category equivalent to $\mathrm{Loc}_{\mathcal{E}} /\mathfrak{C}_{\mathrm{Ev}}$. Allowing $N \subseteq f^* N'$ instead gives the indexed poset of resources.

The marking fixes the image in the ambient module. In particular, $i\alpha = i$ forces a marked automorphism α to be the identity. Choices of a presentation can still have change-of-frame maps. The following diagram description makes both the coefficients and their budget transitions explicit.

Theorem 3.4 (Resource sheaves as budget diagrams)**.** *There is an equivalence of topoi over $\mathcal{E}$*

$$\mathrm{Sh}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}}) \simeq [L, \mathcal{E}], \qquad F \longmapsto (a \longmapsto F(U_a)). \tag{3.9}$$

The right side is the category of internal covariant diagrams on the internal poset L. Under this equivalence $\bar{B}, \bar{K}$ are constant diagrams and

$$\mathscr{M}(a) = M_a := M_{\downarrow a}, \qquad (a \leq b) \longmapsto (M_a \hookrightarrow M_b).$$

At a generalized ideal point I, inverse image is $I^*F \cong \varinjlim_{a \in I} F(a)$.

Proof.

(1) **Construct a sheaf from a budget diagram.** The principal opens form a basis, and $a \leq b$ gives the restriction $F(U_a) \to F(U_b)$. Supercompactness says that a covering sieve on this basis contains the identity internally; thus no additional matching condition is imposed on a diagram. Explicitly, extend a diagram D by

$$\widetilde{D}(U) = \{(x_a)_{a \in U} : x_a \in D(a),\ D(a \leq b)x_a = x_b \text{ for } a \leq b \text{ in } U\}.$$

(2) **Verify gluing and the inverse equivalence.** These compatible families glue uniquely across unions of opens. On U_a, the element a is initial in its indexing poset, so $\widetilde{D}(U_a) \cong D(a)$. Principal decomposition of an open proves the other inverse equation. All constructions take place in the internal logic, including in parameter slices.

(3) **Identify constant and universal module diagrams.** Constant diagrams represent $\Delta_{\mathfrak{C}_{\mathrm{Ev}}}$, and (3.4) gives the stated module diagram.

(4) **Compute the fibre at an ideal.** The inverse image at I is the colimit over its principal neighbourhoods; I is inhabited and directed. Equivalently this colimit preserves finite limits by taking a common stage for a finite tuple and its equations, and sends U_b to the truth value $b \in I$. It is therefore the inverse image classified by I.

□

Evaluation at a generalized ideal point is compatible with the diagram presentation:

$$\begin{array}{ccc} \mathrm{Sh}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}}) & \xrightarrow{\simeq} & [L, \mathcal{E}] \\ {\scriptstyle I^*}\downarrow & & \downarrow{\scriptstyle \varinjlim_{a \in I}} \\ \mathcal{E} & = & \mathcal{E}. \end{array}$$

The commutativity is the canonical natural isomorphism of inverse-image functors, including their action on marked modules and witness sheaves.

This equivalence preserves the observation data specified in Section 2.6.1: the effective points are $J_\Sigma L$ with their positive-test pairing. Membership in a principal open U_a remains the corresponding Ω-valued resource condition.

The individual resource operations now lift to universal families. Tensor multiplication, sums, and intersections must preserve the markings as well as the underlying modules. The next results construct these operations on the classifier and compare them with admission.

3.5 Multiplication, lattice structure, and admission

Theorem 3.5 (Localic resource multiplication)**.** *There is a commutative monoid structure*

$$\mu : \mathfrak{C}_{\mathrm{Ev}} \times \mathfrak{C}_{\mathrm{Ev}} \to \mathfrak{C}_{\mathrm{Ev}}, \qquad e : 1 \to \mathfrak{C}_{\mathrm{Ev}},$$

whose generalized-point formulas are $\mu(I,J) = I \star J = \downarrow \{ab : a \in I, b \in J\}$ *and* $e = \downarrow [1]$. *On principal opens,*

$$\mu^* U_g = \bigvee_{g \le ab} (\pi_1^* U_a \wedge \pi_2^* U_b). \tag{3.10}$$

For the universal module there is a canonical coherent isomorphism of marked modules over $\mathfrak{C}_{\mathrm{Ev}} \times \mathfrak{C}_{\mathrm{Ev}}$*:*

$$\mu^* \mathscr{M} \;\cong\; \pi_1^* \mathscr{M} \otimes_{\Delta B} \pi_2^* \mathscr{M}. \tag{3.11}$$

Finite lattice joins and meets of resources likewise define locale maps, represented by sums and intersections of their universal modules.

Proof.

(1) **Construct the product ideal and locale map.** On $\mathfrak{C}_{\mathrm{Ev}} \times \mathfrak{C}_{\mathrm{Ev}}$, let I_1, I_2 denote the two universal frame-valued ideals. Define

$$(I_1 \star I_2)(g) = \bigvee_{g \le ab} \big(I_1(a) \wedge I_2(b)\big).$$

This is inhabited because both ideals contain zero, and is lower by construction. For two witnesses $a_1 b_1, a_2 b_2$, the join witnesses $a = a_1 \vee a_2$ and $b = b_1 \vee b_2$ lie in the respective ideals and dominate both products. The ideal-classification theorem therefore supplies a locale map μ with exactly the inverse-image formula (3.10).

(2) **Verify the monoid laws.** Multiplication of growth classes proves associativity and commutativity; multiplication by [1] proves the unit laws. Uniqueness in the classifying property turns these into equations of locale maps.

(3) **Construct the principal tensor comparison.** The pullback $\mu^* \mathscr{M}$ is the resource module of the product ideal. Multiplication in ΔK supplies a balanced map from $\pi_1^* \mathscr{M} \times \pi_2^* \mathscr{M}$ into it. At principal stages it induces the isomorphisms

$$Bf \otimes_B Bg \cong Be_f \otimes_B Be_g \cong Be_f e_g \cong Bfg.$$

Their inverses are the support-idempotent inverse maps of Theorem 2.12; their naturality under inclusion into K was verified in Section 2.6.

(4) **Extend the comparison and verify coherence.** Tensor preserves the internally directed colimits of these stages. Passing to the colimits proves (3.11). Both associativity composites take $f \otimes g \otimes h$ to fgh, and the unit and symmetry composites have the analogous formulas. Equality on principal stages proves coherence.

(5) **Construct the lattice operations.** For meets, the inverse image of a principal open is $U_g \mapsto \pi_1^* U_g \wedge \pi_2^* U_g$. For joins, it is

$$U_g \longmapsto \bigvee_{g \le a \vee b} (\pi_1^* U_a \wedge \pi_2^* U_b).$$

These formulas classify intersection and sum of resource modules by Theorem 2.12; preservation of frame operations follows from the classifying ideal property, proving the final assertion.

□

The universal tensor isomorphism preserves the markings:

$$\begin{array}{ccc} \pi_1^*\mathscr{M} \otimes_{\Delta B} \pi_2^*\mathscr{M} & \xrightarrow{\cong} & \mu^*\mathscr{M} \\ \downarrow & & \downarrow \\ \Delta K \otimes_{\Delta B} \Delta K & \xrightarrow[\text{mult}]{} & \Delta K. \end{array}$$

All objects in this square live over $\mathfrak{C}_{\mathrm{Ev}} \times \mathfrak{C}_{\mathrm{Ev}}$. Associativity of this square is the family-valued version of the monoidal coherence in Corollary 2.14.

The effective points of this locale are selected by the *existing* admission doctrine, not by declaring every principal open effective.

Proposition 3.6 (Admitted points and positive observation)**.** *The comparison* $\mathrm{Pt}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}}) \cong \mathrm{Idl}(L)$ *identifies* $J_\Sigma L$ *with the subobject of points* I *for which*

$$\chi_I(u) = \exists a : L\,[I(a) \wedge u(a)] \in \Sigma \quad \textit{uniformly in monotone } u : L \to \Sigma.$$

Localic multiplication restricts on this subobject to the admitted product $J_\Sigma m$ *of Lemma 2.22. The canonical map from the discrete locale on* $J_\Sigma L$ *to* $\mathfrak{C}_{\mathrm{Ev}}$ *pulls the universal module back to* $I \mapsto M_I$*.*

Proof.

(1) **Identify the admitted points.** The first assertion is exactly the defining pullback of the selected monad of Definition 1.3; the uniform factorization is unchanged under the internal point isomorphism.

(2) **Prove closure under multiplication.** For an upper positive predicate u, the admitted-product calculation reads

$$\chi_{I \star J}(u) = \chi_I(a \mapsto \chi_J(b \mapsto u(ab))).$$

The inner pairing is uniformly Σ-valued and monotone; the outer pairing is Σ-valued by admission. Thus the product is admitted and agrees with $J_\Sigma m$ by its principal-weight calculation.

(3) **Identify the pulled-back universal family.** The canonical map of discrete locales is the transpose of the inclusion $J_\Sigma L \hookrightarrow \mathrm{Pt}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}})$. Pulling (3.4) back along it gives $I(a)$ on the principal growth class a, proving the family statement. The exact localic relation between admitted and semantic resources is calculated in Proposition 3.7.

□

Proposition 3.7 (Admitted points are locale-dense)**.** *Let* $a : \mathrm{Disc}(J_\Sigma L) \to \mathfrak{C}_{\mathrm{Ev}}$ *be the canonical locale map from admitted points. Its inverse-image map is*

$$a^* : \mathrm{Up}(L) \longrightarrow \Omega^{J_\Sigma L}, \qquad (a^*U)(I) = \exists x \in I\, U(x).$$

It preserves arbitrary joins and finite meets, and is order-reflecting. Consequently the localic image nucleus $a_* a^*$ *is the identity on* $\mathrm{Up}(L)$*: the localic image is all of* $\mathfrak{C}_{\mathrm{Ev}}$*.*

Proof.

(1) **Verify preservation of frame operations.** Joins commute with existential quantification. For finite meets, two witnesses $x, y \in I$ have their join $x \vee y \in I$, and upperness of U, V gives the common witness; the reverse inclusion follows by projecting a witness. This proves that a^* is a frame homomorphism.

(2) **Prove order reflection using principal ideals.** Every principal ideal $\downarrow x$ is J_Σ-admitted: its pairing with a positive upper test u is $u(x)$, as in Section 1. Evaluation on it gives

$$(a^*U)(\downarrow x) \iff U(x).$$

Hence $a^*U \leq a^*V$ implies $U \leq V$, so a^* is order-reflecting.

(3) **Establish density by the frame adjunction.** Since a^* preserves arbitrary joins, it has a right adjoint a_*. The adjunction unit gives $U \leq a_*a^*U$, while the counit gives $a^*a_*a^*U \leq a^*U$; order reflection yields $a_*a^*U \leq U$. Thus $a_*a^* = \mathrm{id}$. This proves density of the admitted points.

□

Finite diagrams of the classifier parameterize compatible budgets for compiler maps and resource deformations.

3.6 Classifying finite diagrams of budgets

Theorem 3.8 (Finite resource-diagram classifiers)**.** *The equalizer*

$$\mathrm{Flag}_2 = \mathrm{Eq}(\vee, \pi_2 : \mathfrak{C}_{\mathrm{Ev}}^2 \rightrightarrows \mathfrak{C}_{\mathrm{Ev}})$$

classifies pairs of marked modules $M_0 \subseteq M_1$ over every localic parameter. More generally, a finite diagram of inclusions between resource modules has a classifier obtained by finite limits from products of $\mathfrak{C}_{\mathrm{Ev}}$. Finite equations and inclusions between resource expressions formed using sums, intersections, and products have classifiers by the same construction.

Proof.

(1) **Classify a single inclusion.** A map to $\mathfrak{C}_{\mathrm{Ev}}^2$ classifies two marked modules. Factoring through the equalizer says $M_0 + M_1 = M_1$, equivalently $M_0 \subseteq M_1$, by Theorem 3.5.

(2) **Classify finite diagrams and resource equations.** For a finite graph of inclusions, impose this equation for each edge and take their common equalizer. Commutativity is automatic for inclusions into the fixed ambient module. Resource expressions are interpreted by the already constructed locale operations; their equations are equalizers, and $R \subseteq S$ is the equation $R + S = S$. These constructions commute with parameter pullback because their stated universal properties do.

□

For a monotone growth transformation $r : L \to L'$, the inverse-image frame map r^{-1} constructs $s_r : \mathfrak{C}_{\mathrm{Ev}} \to \mathfrak{C}'_{\mathrm{Ev}}$. The same construction gives a locale

$$\mathfrak{R}_r = \mathrm{Eq}(\vee \circ (s_r\pi_1, \pi_2), \pi_2 : \mathfrak{C}_{\mathrm{Ev}} \times \mathfrak{C}'_{\mathrm{Ev}} \rightrightarrows \mathfrak{C}'_{\mathrm{Ev}}), \tag{3.12}$$

classifying $\downarrow r(I) \subseteq J$. An implemented compiler will act over this budget-compatibility locale in Section 4.10, with its code supplied separately.

3.7 Gauge saturation as a classifying sublocale

Coefficient localization closes a resource under chosen overheads. We now identify the families fixed by this operation and their classifying sublocale. Let $P \in \mathrm{Idl}(L)$ be a multiplicatively closed ideal containing [1], regarded as a point of the object of such ideals. Its coefficient ring is the localization $B_P = M_P \subseteq K$ of Section 2.7. Define

$$S_P : \mathrm{Idl}(L) \longrightarrow \mathrm{Idl}(L), \qquad I \longmapsto P \star I.$$

Theorem 3.9 (Gauge saturation and the localized moduli problem)**.** *The map S_P is Scott-continuous, inflationary and idempotent. Consequently its inverse-image operation*

$$j_P = S_P^{-1} : \mathcal{O}(\mathfrak{C}_{\mathrm{Ev}}) \longrightarrow \mathcal{O}(\mathfrak{C}_{\mathrm{Ev}})$$

is a nucleus. It is determined on the principal opens by

$$j_P(U_g) = \bigvee_{\substack{p\in P,\ a\in L \\ g\le pa}} U_a. \tag{3.13}$$

The nucleus constructs a sublocale $\iota_P : \mathfrak{C}_{\mathrm{Ev}}^{(P)} \hookrightarrow \mathfrak{C}_{\mathrm{Ev}}$ satisfying, naturally in every internal locale Z,

$$\mathrm{Loc}_{\mathcal{E}}(Z, \mathfrak{C}_{\mathrm{Ev}}^{(P)}) \cong \mathrm{Sub}_{\Delta_Z B_P}(\Delta_Z K). \tag{3.14}$$

Its points are precisely the P-saturated ideals $I = P \star I$, and its universal marked module is $\iota_P^* \mathscr{M}$ with its canonical $\Delta_{\mathfrak{C}_{\mathrm{Ev}}^{(P)}} B_P$-action. Resource multiplication restricts to a commutative monoid on $\mathfrak{C}_{\mathrm{Ev}}^{(P)}$ with unit P, and its universal marked tensor comparison is balanced over B_P.

Proof.

(1) **Construct the gauge nucleus.** Products of ideals distribute over directed unions, so S_P preserves directed joins. The inequalities $I \le P\star I$ and $P\star(P\star I) = I\star(P\star P) = P\star I$ follow from the unit and product closure of P. Under the Scott-frame identification in Theorem 3.1, inverse image along S_P preserves arbitrary joins and finite meets. Since S_P is inflationary and idempotent, its inverse image is also inflationary and idempotent; hence j_P is a nucleus.

(2) **Compute the nucleus formula.** Expanding $g \in P \star I$ gives

$$g \in P \star I \iff \exists p \in P\, \exists a \in I\ (g \le pa),$$

which proves (3.13).

(3) **Characterize factorization through the sublocale.** A classifying map $f : Z \to \mathfrak{C}_{\mathrm{Ev}}$ corresponds to an open-valued ideal I_f and its marked module $M_{I_f} \subseteq \Delta_Z K$. It factors through the nucleus sublocale precisely when $f^* j_P(U) = f^* U$ for every open U. Since the U_g generate, this is equivalent to

$$(P \star I_f)(g) = I_f(g) \quad \text{for all } g \in L.$$

(4) **Identify localized modules and their universal family.** The module representation and the localization theorem identify this equation with $B_P M_{I_f} = M_{I_f}$. Since $1 \in B_P$, this is exactly closure under the B_P-action in $\Delta_Z K$. The two classifications therefore restrict to the bijection (3.14); pullback proves its naturality and supplies the stated universal module.

(5) **Restrict multiplication and its unit.** If I, J are saturated, associativity gives $P \star (I \star J) = (P \star I) \star J = I \star J$, so multiplication restricts. Its unit is $S_P(\downarrow [1]) = P$.

(6) **Restrict the coherent tensor comparison.** As $B_P \otimes_B B_P \cong B_P$ by localization, for two B_P-modules the balanced tensors over B and B_P agree. The universal tensor comparison of Theorem 3.5 therefore restricts to one balanced over B_P. Its associativity, unit and symmetry follow from multiplication of marked elements.

□

Proposition 3.10 (Reflection into overhead-stable resources)**.** *Extension of marked modules and restriction of their coefficient action give an adjunction with fully faithful right adjoint*

$$B_P(-) : \mathsf{Emb}_B(K) \rightleftarrows \mathsf{Emb}_{B_P}(K) : \mathrm{incl}.$$

Its unit is the marked inclusion $M \hookrightarrow B_P M$, and its counit is the identity on a B_P-submodule. Under Corollary 2.14, the induced idempotent monad is $S_P(I) = P \star I$. This reflection preserves resource products with their localized unit, and finite intersections.

Proof.

(1) **Prove the reflection adjunction.** For an embedded B_P-module N, closure under B_P gives

$$B_P M \subseteq N \quad \Longleftrightarrow \quad M \subseteq N.$$

This is the adjunction, including uniqueness of its marked maps.

(2) **Verify idempotence and the comparison formulas.** The multiplication isomorphism $B_P \otimes_B B_P \cong B_P$ gives idempotence. The tensor and intersection comparisons are the resource product and flat localization formulas of Theorem 2.23.

(3) **Identify the commuting classification square.** In particular the square of functors

$$\begin{array}{ccc} \operatorname{Idl} L & \xrightarrow{\simeq} & \mathsf{Emb}_B(K) \\ {\scriptstyle S_P}\downarrow & & \downarrow{\scriptstyle B_P(-)} \\ (\operatorname{Idl} L)^{S_P} & \xrightarrow[\simeq]{} & \mathsf{Emb}_{B_P}(K) \end{array}$$

commutes up to the canonical marked isomorphism. Here the lower left object denotes the fixed ideals, with their inherited maps.

□

The adjunction identifies the same operation in three forms: extension to B_P-coefficients, reflection into overhead-stable budgets, and the classifying nucleus $j_P = S_P^{-1}$. Its unit embeds each budget into its saturation, compatibly with the marking:

$$\begin{array}{ccc} M & \hookrightarrow & B_P M \\ & \searrow & \downarrow \\ & & K \end{array} \qquad \begin{array}{ccc} B_P \otimes_B M & \xrightarrow{\cong} & B_P M \\ \downarrow & & \downarrow \\ B_P \otimes_B K & \xrightarrow[\cong]{} & K. \end{array}$$

Corollary 3.11 (Functorial overhead geometries)**.** *For multiplicative ideals $P \subseteq Q$ containing* [1]*, there is a canonical inclusion $\mathfrak{C}_{\mathrm{Ev}}^{(Q)} \hookrightarrow \mathfrak{C}_{\mathrm{Ev}}^{(P)} \hookrightarrow \mathfrak{C}_{\mathrm{Ev}}$, compatible with multiplication and the localized universal marked modules. For numerical cofinite input, the explicitly admitted bitlength ideal P_λ of Proposition 2.24 constructs $\mathfrak{C}_{\mathrm{Ev}}^{(P_\lambda)}$, which classifies budgets closed under polylogarithmic overhead.*

Proof.

(1) **Derive the inclusion of fixed-resource locales.** The equations $Q \star P = Q$ and $Q \star Q = Q$ yield $S_Q S_P = S_Q$. Every Q-fixed ideal is therefore P-fixed, which gives the sublocale inclusion by the defining nuclei.

(2) **Verify coefficient and tensor compatibility.** Compatibility of coefficient actions and tensors follows from $B_P \subseteq B_Q \subseteq K$.

(3) **Specialize to the polylogarithmic gauge.** For P_λ use the explicitly constructed admitted multiplicative ideal and $B_{P_\lambda} = B[\lambda]$ of Section 2.7.

□

Theorem 3.12 (The localized growth order)**.** *For the multiplicative resource ideal P, define a preorder on L by*

$$a \preceq_P b \quad \Longleftrightarrow \quad \exists p \in P\ (a \le pb). \tag{3.15}$$

Let L/P be its separated quotient and $q_P : L \to L/P$ the quotient map. Finite joins descend to L/P, and there are natural isomorphisms

$$\mathrm{Idl}(L/P) \cong \{I \in \mathrm{Idl}(L) : P \star I = I\}, \qquad \mathcal{O}(\mathfrak{C}_{\mathrm{Ev}}^{(P)}) \cong \mathrm{Up}(L/P). \tag{3.16}$$

In particular the localized principal opens $U^{(P)}_{q_P(a)}$ are supercompact and form a finite-meet-stable basis of $\mathfrak{C}_{\mathrm{Ev}}^{(P)}$.

Proof.

(1) **Verify the localized preorder.** Reflexivity uses $[1] \in P$. If $a \le pb$ and $b \le qc$, then $a \le pqc$ with $pq \in P$, proving transitivity.

(2) **Descend finite joins to the quotient.** To descend joins, replace p, q by $p \vee [1], q \vee [1] \in P$: their product dominates both, so $a \preceq_P b$ and $c \preceq_P d$ imply $a \vee c \preceq_P b \vee d$. The quotient is therefore a finite-join semilattice with bottom $q_P(0)$.

(3) **Identify saturated ideals.** A lower ideal I of L is saturated precisely when it is lower also for $\preceq_P$: if $b \in I$ and $a \le pb$, saturation puts pb and hence a in I; the converse follows by setting $a = pb$. Since q_P preserves finite joins, inverse image along q_P identifies ideals of L/P with exactly these saturated ideals.

(4) **Compute the nucleus on upper predicates.** For an upper predicate U on L, evaluate its nucleus image at a using the principal ideal $\downarrow a$:

$$(j_P U)(a) \iff \exists p \in P\ U(pa).$$

(5) **Identify fixed opens and prove coherence.** The equation $j_P U = U$ is equivalent to U being upward closed also under $\preceq_P$. Indeed, $pa \preceq_P a$ and hence upward closure for $\preceq_P$ makes $U(pa) \Rightarrow U(a)$; conversely, if $a \le pb$ and $U(a)$, original upperness gives $U(pb)$, and the fixed-point equation gives $U(b)$. Such predicates are precisely the pullbacks of upper predicates on the separated quotient. This proves the second isomorphism in (3.16). Applying Theorem 3.1 to the finite-join semilattice L/P proves the principal-open and coherence assertions.

□

3.8 The multiplicative quantale and its support spectrum

The resource product also defines a spectrum. Calculating it will distinguish the support information carried by multiplication from the growth information carried by $\mathfrak{C}_{\mathrm{Ev}}$. Write $\mathfrak{Q} = \mathrm{Idl}(L)$, ordered by inclusion, with resource product $\star$ and unit $\downarrow [1]$. This product is distinct from the finite meet of the Scott frame $\mathcal{O}(\mathfrak{C}_{\mathrm{Ev}})$. Let $\mathrm{Idem}(B) = \{e \in B : e^2 = e\}$, with Boolean operations $e \wedge f = ef$, $e \vee f = e + f - ef$, and $\neg e = 1 - e$.

Theorem 3.13 (Quantale and constructive multiplicative spectrum)**.** *The complete lattice* $\mathfrak{Q}$ *is a commutative unital quantale: multiplication preserves arbitrary joins in each variable. Its unital multiplicative frame-valued characters, that is, maps* $\chi : \mathfrak{Q} \to H$ *to an internal frame* H *satisfying*

$$\chi(\bigvee_i I_i) = \bigvee_i \chi(I_i), \quad \chi(I \star J) = \chi(I) \wedge \chi(J), \quad \chi(\downarrow [1]) = \top,$$

are classified by the Stone locale of $\mathrm{Idem}(B)$*:*

$$\mathcal{O}(\mathrm{Sp}_{\otimes}(\mathfrak{Q})) \cong \mathrm{Idl}(\mathrm{Idem}(B)). \tag{3.17}$$

More precisely every such character satisfies

$$\chi(\downarrow [g]) = \chi(\downarrow [e_g]), \qquad \chi(I) = \bigvee_{[g] \in I} \chi(\downarrow [e_g]). \tag{3.18}$$

Thus the unital multiplicative spectrum sees support idempotents, whereas the original classifying locale retains growth comparisons.

Proof.

(1) **Verify the quantale operations.** The join of resource ideals is generated by finite joins of their members. Multiplying such finite joins distributes over their products in L, giving

$$I \star \bigvee_j J_j = \bigvee_j (I \star J_j).$$

The zero ideal is absorbing and $\downarrow [1]$ is a unit; associativity and commutativity are those of profile multiplication.

(2) **Extract the principal character equations.** A join-preserving χ is determined by its values $d_g = \chi(\downarrow [g])$, because every ideal is the join of its principal ideals. Finite joins and resource products imply

$$d_0 = \bot, \qquad d_1 = \top, \qquad d_{a \vee b} = d_a \vee d_b, \qquad d_{ab} = d_a \wedge d_b.$$

(3) **Reduce characters to support idempotents.** For a nonnegative germ g let $g^\dagger$ be its rational reciprocal on its nonzero support and zero elsewhere. It is a germ in K^+ with $gg^\dagger = e_g$ and $ge_g = g$. The multiplicative equations give

$$d_g = d_g \wedge d_{e_g} \leq d_{e_g}, \qquad d_{e_g} = d_g \wedge d_{g^\dagger} \leq d_g.$$

Hence $d_g = d_{e_g}$. These idempotents are invariant under growth equivalence, and $e_{g+h} = e_g \vee e_h$ and $e_{gh} = e_g e_h$. The d_e therefore form a Boolean-algebra homomorphism $\mathrm{Idem}(B) \to H$ (the images of e and $1 - e$ are complements).

(4) **Recover a character from Boolean support data.** Conversely any such map φ defines a quantale character by $\chi(I) = \bigvee_{[g] \in I} \varphi(e_g)$: finite joins handle ideal generation, and frame distributivity handles products. These constructions are inverse.

(5) **Classify the support spectrum.** Finally, the ideal completion of a Boolean algebra is the free frame on its bounded-distributive-lattice operations: a map $\varphi : E \to H$ preserving finite meets, joins and bounds extends uniquely to $\widehat{\varphi}(J) = \bigvee_{e \in J} \varphi(e)$. Its inverse is restriction to principal ideals. Taking $E = \mathrm{Idem}(B)$ proves the frame classification (3.17) by its universal property. This is the point-free support/Pierce spectrum; the general quantalic spectrum perspective is discussed in (Manuell 2022).

□

For cofinite numerical input, the parity indicator yields a nontrivial complemented open in the support spectrum. The support spectrum records these idempotent decompositions; $\mathfrak{C}_{\mathrm{Ev}}$ records growth budgets.

3.9 Filtered-input substitution and coherence

For a filtered map $u : (Y, \mathrm{Ev}_Y) \to (X, \mathrm{Ev}_X)$, substitution gives $B_X \to B_Y$ and $K_X \to K_Y$. On a principal resource,

$$B_Y \otimes_{B_X} B_X g \cong B_Y e_{g\circ u} \cong B_Y(g \circ u).$$

Taking directed colimits proves

$$B_Y \otimes_{B_X} M_I \cong M_{\downarrow u^* I}. \tag{3.19}$$

The map sends $b \otimes m$ to $b(m \circ u)$.

Both composite comparisons for successive substitutions or localizations have this same formula on elementary tensors. Their equality proves unit and associativity coherence. Raw profiles still reindex along every input map, as in (1.8).

Proposition 3.14 (Natural base change of classifying resources)**.** *For a filtered map $u : (Y, \mathrm{Ev}_Y) \to (X, \mathrm{Ev}_X)$, write $\varphi_u : L_X \to L_Y$ for precomposition of growth profiles. The frame map $\varphi_u^{-1} : \mathrm{Up}(L_Y) \to \mathrm{Up}(L_X)$ defines $s_u : \mathfrak{C}_X \to \mathfrak{C}_Y$, acting on ideal points by $I \mapsto\downarrow \varphi_u(I)$. There is a natural isomorphism of marked modules over $\mathfrak{C}_X$,*

$$\Delta_{\mathfrak{C}_X} B_Y \otimes_{\Delta_{\mathfrak{C}_X} B_X} \mathscr{M}_X \;\cong\; s_u^* \mathscr{M}_Y, \tag{3.20}$$

where the marking on the left is multiplication after substitution into $\Delta_{\mathfrak{C}_X} K_Y$. These isomorphisms satisfy the identity and composition laws.

Proof.

(1) **Compute the locale inverse image.** The inverse-image formula preserves arbitrary unions and finite intersections. On the principal opens of $\mathfrak{C}_Y$ it is

$$s_u^* U_h = \bigvee_{\substack{g \in L_X \\ h \le \varphi_u(g)}} U_g.$$

(2) **Construct the comparison on principal resources.** Work internally in $\mathrm{Sh}_{\mathcal{E}}(\mathfrak{C}_X)$. Principal resources generate $\mathscr{M}_X$ locally and form a directed family, by Theorem 3.3. For such a local generator g, support idempotents give the explicit marked comparison

$$B_Y \otimes_{B_X} B_X g \cong B_Y e_{g\circ u} \cong B_Y(g \circ u) \hookrightarrow K_Y.$$

The map takes $b \otimes m$ to $b(m \circ u)$.

(3) **Prove injectivity of the total marking.** To verify injectivity after passage to the directed union, let $z = \sum_i b_i \otimes m_i$ be a finite tensor expression that maps to zero in K_Y. Locally the finitely many m_i are contained in a common principal stage: the principal budget generated by $\sum_i |m_i|$. The displayed principal-stage comparison is an isomorphism onto a submodule of K_Y, so $z = 0$ at that stage. Kernel membership is local, making the total marking monic.

(4) **Identify its image with the pulled-back module.** For surjectivity onto $s_u^*\mathscr{M}_Y$, evaluate the membership predicate of $h \in K_Y$:

$$\begin{aligned}[\![h \in s_u^*\mathscr{M}_Y]\!] &= s_u^* U_{[|h|]} \\ &= \bigvee_{\substack{g \in L_X \\ [|h|] \le \varphi_u(g)}} U_g.\end{aligned}$$

Thus membership locally supplies $g \in \mathscr{M}_X$ for which $h \in B_Y(g \circ u)$. Bounded division produces the multiplier $b \in B_Y$ with $h = b(g \circ u)$, represented by $b \otimes g$. The image of the monic total marking is therefore precisely $s_u^*\mathscr{M}_Y$, proving the marked-module isomorphism.

(5) **Verify composition coherence.** For successive filtered maps, both comparison composites send an elementary tensor to the same iterated substituted product. Equality on the directed principal stages proves composition coherence, and the identity case is immediate.

□

Filtered substitution intertwines resource representation and scalar extension by the natural comparison

$$\begin{array}{ccc} \mathrm{Idl}(L_X) & \xrightarrow{\simeq} & \mathsf{Emb}_{B_X}(K_X) \\ {\scriptstyle \downarrow\varphi_u(-)}\big\downarrow & & \big\downarrow{\scriptstyle B_Y \otimes_{B_X} -} \\ \mathrm{Idl}(L_Y) & \xrightarrow[\simeq]{} & \mathsf{Emb}_{B_Y}(K_Y). \end{array}$$

The right-hand marking is induced by $K_X \to K_Y$. The theorem proves its required monicity on these resources.

Proposition 3.15 (Extension and restriction of input budgets)**.** *For the filtered substitution $u : Y \to X$, the growth map $\varphi_u : L_X \to L_Y$ preserves 0 and finite joins. There is a locale map $t_u : \mathfrak{C}_Y \to \mathfrak{C}_X$ with*

$$t_u^* U_a^X = U^Y_{\varphi_u(a)}, \qquad t_u(J) = \varphi_u^{-1}(J), \qquad s_u(I) \subseteq J \iff I \subseteq t_u(J).$$

In the diagram presentations,

$$s_u^* G = G \circ \varphi_u, \qquad (t_u^* F)(b) = \varinjlim_{\varphi_u(a) \le b} F(a). \tag{3.21}$$

The coefficient map $B_X \to B_Y$ equips t_u with a ringed-topos map. Its module pullback has a canonical monomorphism into $\mathscr{M}_Y$; at J its marked image is $M_{s_u t_u(J)} \subseteq M_J$.

Proof.

(1) **Construct the reverse locale map and ideal adjunction.** Precomposition preserves zero and sums of profiles, hence finite joins of growth classes. The assignment on principal opens preserves their finite-meet relations, so Theorem 3.1 constructs t_u. Its point formula and the displayed ideal adjunction follow by membership.

(2) **Identify the diagram inverse images.** Precomposition of diagrams sends the subterminal U_b to $\varphi_u^{-1}(U_b)$, proving the first formula in (3.21). For the second, the comma poset $\{a : \varphi_u(a) \le b\}$ contains 0 and is directed by joins. Its colimit preserves finite limits by the finite-witness argument in Theorem 3.4. The resulting left Kan extension preserves colimits and sends U_a to $U_{\varphi_u(a)}$, so it is exactly t_u^*.

(3) **Construct the ringed pullback.** It sends the constant coefficient ring B_X to the constant ring B_X; the substitution map to B_Y defines the ringed structure. Module pullback is $\bar{B}_Y \otimes_{t_u^* \bar{B}_X} t_u^*(-)$.

(4) **Compute the resource comparison and its fibres.** On a principal stage a, the marked scalar-extension comparison of Proposition 3.14 identifies its value with $M_{\varphi_u(a)}$. At budget b, these form a directed union inside M_b, proving monicity. At an ideal J this union has growth ideal $\downarrow \varphi_u(\varphi_u^{-1} J)$.

(5) **Verify coherence and observation transport.** Identity and composition follow on elementary tensors and principal opens. The forward map s_u retains the effective observation transport of Proposition 2.18; no assertion that t_u preserves admission is needed or made.

□

3.10 Finite principal-open descent

To use the resource coefficients on affine charts, we need to glue modules and their maps across localizations. For a commutative coefficient algebra A, finite lists of generators and equations construct the category of finitely presented A-algebras. A finite principal cover is displayed by $s_1, \dots, s_r$ and a unit-ideal witness $1 = \sum_i a_i s_i$. The next result supplies the gluing calculation used later for differential coefficients.

Proposition 3.16 (Module gluing)**.** *For every A-module M such a cover gives an equalizer*

$$M \longrightarrow \prod_i M_{s_i} \rightrightarrows \prod_{i,j} M_{s_i s_j}.$$

These equalizers commute with restriction and refinement.

Proof.

(1) **Construct unit combinations of denominator powers.** This is the finite standard-open Čech equalizer (see (Stacks Project Authors 2026a, Tag 01X9)). Its denominator calculation is finite, and hence applies internally. First, for every $d \geq 1$, $1 = \sum_i c_i s_i^d$ follows by expanding $(\sum_i a_i s_i)^{r(d-1)+1}$ and grouping each monomial by an index whose exponent is at least d.

(2) **Prove injectivity of the equalizer map.** If m maps to zero in every M_{s_i}, choose a common d with $s_i^d m = 0$. The unit combination yields $m = 0$, proving injectivity.

(3) **Choose compatible denominator representatives.** For compatible local elements choose a common denominator exponent n and representatives $x_i = m_i / s_i^n$ with $m_i \in M$. Compatibility on every overlap gives a common q such that

$$(s_i s_j)^q (s_j^n m_i - s_i^n m_j) = 0 \qquad (i, j).$$

(4) **Construct the global glued element.** Put $d = n + q$ and $\widetilde{m}_i = s_i^q m_i$. Then $s_j^d \widetilde{m}_i = s_i^d \widetilde{m}_j$. Choose $1 = \sum_i c_i s_i^d$ by the unit-combination calculation above and set $m = \sum_i c_i \widetilde{m}_i$. For each j,

$$s_j^d m = \sum_i c_i s_i^d \widetilde{m}_j = \widetilde{m}_j,$$

so $m/1 = x_j$ in M_{s_j}.

(5) **Verify uniqueness and refinement coherence.** This constructs the gluing map; injectivity proves uniqueness, restriction, and refinement coherence. All common exponents arise from finite maxima.

□

The gluing condition is the equalizer diagram

$$M \longrightarrow \prod_i M_{s_i} \rightrightarrows \prod_{i,j} M_{s_i s_j}.$$

The parallel arrows restrict to the two factors of each overlap; refinement compares the equalizers through localization.

We now have a classifier of budget families, its marked coefficient module, and compatible resource operations and transports. The remaining computational question is which of these budgets contain the cost of an implementation of a given problem. Answering it requires the measured execution construction that follows.

4 Effective implementations and universal incidence

The budget classifier tells us what a resource constraint is. To ask whether a problem satisfies it, we must supply implementations and measure their execution. We first construct returning histories and their resource valuations, then use finite descriptions and input names to fix the measured implementation family. Its denotation and cost maps form the span from which universal feasibility is obtained.

The partiality and cost construction is related to cost-sensitive synthetic domain semantics (Niu et al. 2024). Here the exact-return argument establishes the required comparison in the selected effective-domain setting. The subsequent sheaf construction carries the resulting descriptions and observation maps over the budget classifier.

4.1 Finite words in the computational category

Histories must record finitely many events in their execution order. We construct their word objects in $\mathcal{A}_\Sigma$, the computational category of Section 1. The classified domains of partial execution remain subobjects in $\mathcal{E}$.

Lemma 4.1 (Internally indexed disjoint sums)**.** *Every object of $\mathcal{E}$ with its discrete order has a canonical J_Σ-algebra. For an internally given family $(P_n)_{n:\mathbb{N}}$ of posets, with the disjoint order on its dependent sum, there is an isomorphism*

$$J_\Sigma\Big(\sum_{n:\mathbb{N}} P_n\Big) \cong \sum_{n:\mathbb{N}} J_\Sigma P_n$$

compatible with principal ideals and ideal unions. Thus an internal family of computational algebras has a componentwise disjoint-sum algebra.

Proof.

(1) **Identify admitted ideals in a discrete poset.** An inhabited directed ideal in a discrete poset has exactly one element: two members with a common upper bound are equal. Unique existence gives its element and identifies $J_\Sigma A$ with A.

(2) **Identify the unique summand and its admitted ideal.** In a disjoint sum, members with a common upper bound have the same index. Inhabitedness and directedness therefore specify a unique n. Restrict the ideal to P_n. To prove its admission, extend

each monotone test $P_n \to \Sigma$ by false on the other summands, using decidable equality on $\mathbb{N}$. Conversely, every monotone positive test on the sum restricts to a test on the chosen fibre. These constructions identify the pairing predicates and are inverse.

(3) **Verify monad and algebra compatibility.** Principal ideals and unions retain the unique index, proving the monad compatibility and hence the algebra assertion. The sum is one internally constructed family, not an external countable coproduct.

□

Proposition 4.2 (Computational finite words)**.** *For $A \in \mathcal{A}_\Sigma$, put*

$$A^* = \sum_{n:\mathbb{N}} A^{[n]}, \qquad [n] = \{j : \mathbb{N} : j < n\},$$

ordered componentwise at each fixed length. This is a computational algebra, and empty word, singleton, and concatenation are algebra morphisms

$$[] : 1 \to A^*, \qquad [-] : A \to A^*, \qquad +\!+ : A^* \times A^* \to A^*.$$

It is the free monoid on A in $\mathcal{A}_\Sigma$. In particular, for a monoid (C, e, μ) in $\mathcal{A}_\Sigma$ and an algebra morphism $w : A \to C$, its unique monoid extension $\overline{w} : A^* \to C$ is constructed by

$$\overline{w}([]) = e, \qquad \overline{w}(a :: s) = \mu(w(a), \overline{w}(s)).$$

Proof.

(1) **Construct the word algebra.** Finite powers carry the product algebra of Theorem 1.4, uniformly by recursion on the length. Apply Lemma 4.1.

(2) **Prove continuity of concatenation and the unit.** An admitted ideal of a pair of words has fixed lengths in both components, so preservation of its supremum by concatenation reduces to the finite-product algebra formula. The empty word and singleton satisfy the same equation.

(3) **Construct the continuous fold and prove uniqueness.** At each length the displayed fold is a finite composite of algebra morphisms. The disjoint-sum algebra therefore makes it an algebra morphism on all words. Induction proves preservation of concatenation, and proves uniqueness from the empty and singleton clauses. These constructions retain all free parameters.

□

4.2 Resource-valued partiality

A returning computation now carries both a value and a resource. Sequential execution combines resources using a monoid; commutativity is unnecessary because the execution order is retained. Let (C, e, μ) be an internal monoid in $\mathcal{A}_\Sigma$, with algebra morphisms $e : 1 \to C$ and $\mu : C \times C \to C$, and write $cd = \mu(c, d)$. Define

$$L_{\Sigma,C}A = L_\Sigma(C \times A), \qquad L_{\Sigma,C}u = L_\Sigma(\mathrm{id}_C \times u), \qquad \eta^C_A(a) = \eta_{C\times A}(e, a).$$

For $u \in L_{\Sigma,C}A$ and $g : A \to L_{\Sigma,C}B$, define resource-valued bind by

$$u \operatorname{bind}_C g = u \operatorname{bind}_\Sigma ((c, a) \mapsto g(a) \operatorname{bind}_\Sigma ((d, b) \mapsto \eta_{C\times B}(cd, b))). \tag{4.1}$$

Here $\operatorname{bind}_\Sigma$ is the partiality bind of (1.1). The strength has type

$$X \times L_{\Sigma,C}A \longrightarrow L_{\Sigma,C}(X \times A), \qquad (x, u) \longmapsto u \operatorname{bind}_\Sigma ((c, a) \mapsto \eta(c, (x, a))).$$

Theorem 4.3 (Constructed resource partiality)**.** *These maps make $L_{\Sigma,C}$ a strong monad on $\mathcal{A}_\Sigma$. Cost erasure*

$$\epsilon_A^C = L_\Sigma \pi_A : L_{\Sigma,C} A \to L_\Sigma A$$

is a strong monad morphism. A monoid morphism $h : C \to C'$ induces the strong monad morphism $L_\Sigma(h \times \mathrm{id}_A) : L_{\Sigma,C} A \to L_{\Sigma,C'} A$. For commutative C, the constructed monad is commutative.

Proof.

(1) **Verify continuity of the resource operations.** All operations are composites of the lifted partiality operations, strength, products, and e, μ, so they are algebra morphisms.

(2) **Compute composite domains and costs.** Write a partial value as $(p, (c, a))$ in context $p : \Sigma$, and the next partial value as $(q, (d, b))$ in that context. Its composite has extent $\exists t : p\ q(t)$ and value (cd, b). Dominance classifies this extent.

(3) **Verify associativity, units, and strength.** Three successive computations have the same composite domain under both associations; their values differ only by $(cd)r = c(dr)$. The unit equations are $ec = c = ce$. Strength retains the same unused parameter throughout these domains and values, so the strength equations of Proposition 1.5 prove its laws.

(4) **Verify erasure and change-of-cost morphisms.** Erasure forgets precisely the accumulated monoid value. Applying h to a product gives $h(cd) = h(c)h(d)$ and $h(e) = e$, proving the two monad-morphism claims.

(5) **Prove commutativity for independent computations.** For independent calls the two orders have the same domain by commutativity of L_Σ and costs cd and dc; commutativity of C proves the final assertion.

□

The notation $L_{\Sigma,C}$ distinguishes resource-valued partiality from the symmetric-algebra monad T and the infinitesimal-shape modality $\Im_W$. The monoids $(\mathbb{N}, 0, +)$, $(\mathbb{N}, 0, \max)$, and their product are constructed using the discrete algebra of Lemma 4.1. They accumulate declared event costs, recorded peak occupancy, and joint time–space values, respectively.

The resource-valued bind describes finite sequential composition. Recursive execution requires the same account of costs on every returning branch. We obtain it by attaching resource values to the finite return objects of the selected-domain recursion construction.

4.3 Weighted exact returns and iteration

For an equation morphism $f : X \to L_{\Sigma,C}(Y + X)$ in $\mathcal{A}_\Sigma$, the map

$$\Psi : [X, L_{\Sigma,C}(Y + X)]_\Sigma \times [X, L_{\Sigma,C} Y]_\Sigma \longrightarrow [X, L_{\Sigma,C} Y]_\Sigma, \quad \Psi(f, a)(x) = f(x)\ \mathrm{bind}_C\ [\eta_Y^C, a]$$

is an algebra morphism. The second function object is pointed by the everywhere undefined map. The parameterized least-fixed-point operation of Proposition 1.5 therefore constructs

$$f^{\ddagger_C} = \mathrm{lfp}(\Psi(f, -)) : X \to L_{\Sigma,C} Y.$$

The operation $f \mapsto f^{\ddagger_C}$ is itself an algebra morphism between the displayed function objects.

To calculate it, represent f by the partial-map span

$$X \xleftarrow{m} D \xrightarrow{o} C \times (Y + X), \qquad D = D_Y + D_X.$$

Its branches have maps in $\mathcal{E}$

$$\begin{aligned} c_Y : D_Y \hookrightarrow X, \quad w_Y : D_Y \to C, \quad v : D_Y \to Y, \\ c_X : D_X \hookrightarrow X, \quad w_X : D_X \to C, \quad n : D_X \to X. \end{aligned}$$

Both source monos are Σ-classified. Retain exactly the return objects of Lemma 1.6, and attach the indicated valuations:

$$\begin{aligned} R_0 &= D_Y, & R_{k+1} &= D_X \times_{n,X,\rho_k} R_k, \\ \rho_0 &= c_Y, & \rho_{k+1} &= c_X \pi_1, \\ v_0 &= v, & v_{k+1} &= v_k \pi_2, \\ a_0 &= w_Y, & a_{k+1} &= \mu(w_X \pi_1, a_k \pi_2). \end{aligned}$$

Theorem 4.4 (Resource-valued finite return)**.** *The domain of $f^{\ddagger_C}$ is the Σ-classified mono*

$$D_f = \sum_{k:\mathbb{N}} R_k \hookrightarrow X,$$

and its value on R_k is (a_k, v_k). Consequently

$$\epsilon_Y^C f^{\ddagger_C} = (\epsilon_{Y+X}^C f)^{\ddagger}. \tag{4.2}$$

These constructions and equations commute with substitution of parameters.

Proof.

(1) **Construct the classified finite-return domain.** Lemma 1.6 applies to the erased equation. Pullback and composition prove that each ρ_k is Σ-classified. Disjointness of exit and continuation, followed by induction, proves disjointness of distinct return lengths. Hence the dependent sum is monic. Its classifier $\exists k : \mathbb{N}\ \rho_k(x)$ lies in Σ by Proposition 1.2.

(2) **Identify the finite approximants and their costs.** Starting $\Psi(f, -)$ at bottom gives $b_0 = \bot$ and $b_{j+1} = \Psi(f, b_j)$. Induction shows that b_j has domain $\sum_{k<j} R_k$ and value (a_k, v_k) on R_k. The return branch contributes $w_Y e = w_Y$; the continuation branch contributes the current cost followed by the recursive cost. Once defined, the value remains unchanged in later approximants.

(3) **Recover the partial map and parameter coherence.** Their admitted supremum is therefore the displayed partial map, proving the first claim in $\mathcal{A}_\Sigma$. Erasure leaves precisely the original return span, including its domain. All pullbacks, finite recursions, and classified existential images retain their parameters, as does the cost recursion.

□

Equivalently, define

$$\widetilde{f} : C \times X \to L_\Sigma((C \times Y) + (C \times X))$$

by executing $f(x)$ and sending $(d, \mathsf{inl}\, y)$ to $\mathsf{inl}(cd, y)$ and $(d, \mathsf{inr}\, x')$ to $\mathsf{inr}(cd, x')$. The constructed iteration satisfies

$$f^{\ddagger_C} = \widetilde{f}^{\ddagger} \langle e!, \mathrm{id}_X \rangle.$$

Indeed a return in R_k started with accumulator c ends with ca_k, by induction on k. This identifies the two constructions. The return length is the projection $D_f \to \mathbb{N}$; the predicates R_k need not be Boolean. Unit cost on every call gives $k+1$ on R_k. Unit cost on continuation and zero on return gives k; we specify the convention in each concrete measurement.

Theorem 4.5 (Resource-valued Elgot structure)**.** *Write $g \diamond_C f$ for Kleisli composition and $g \oplus_C h = [L_{\Sigma,C}(\mathsf{inl})g, L_{\Sigma,C}(\mathsf{inr})h]$. For $f : X \to L_{\Sigma,C}(Y + X)$, $g : Y \to L_{\Sigma,C}Z$, and $h : X \to L_{\Sigma,C}((Y + X) + X)$, the constructed operation satisfies*

$$f^{\ddagger_C} = [\eta^C_Y, f^{\ddagger_C}] \diamond_C f, \tag{4.3}$$

$$g \diamond_C f^{\ddagger_C} = ((g \oplus_C \eta^C_X) \diamond_C f)^{\ddagger_C}, \tag{4.4}$$

$$(h^{\ddagger_C})^{\ddagger_C} = (L_{\Sigma,C}[\mathrm{id}_{Y+X}, \mathsf{inr}]h)^{\ddagger_C}. \tag{4.5}$$

For $u : Z \to X$ and $k : Z \to L_{\Sigma,C}(Y + Z)$,

$$fu = L_{\Sigma,C}(\mathrm{id}_Y + u)k \quad \Longrightarrow \quad f^{\ddagger_C} u = k^{\ddagger_C}.$$

Thus $L_{\Sigma,C}$ is a complete Elgot monad. Monoid morphisms commute with this iteration. Coproduct feedback is the algebra morphism

$$\mathrm{Tr}^U_{+,C}(f) = [\eta^C_Y, (f\iota_U)^{\ddagger_C}] \diamond_C f\iota_X \quad (f : X + U \to L_{\Sigma,C}(Y + U)). \tag{4.6}$$

It satisfies the symmetric coproduct-trace equations.

Proof.

(1) **Verify the fixed-point equation.** Exposing the first transition proves the fixed-point equation.

(2) **Verify naturality by matching histories.** For naturality, both sides execute the same returning f-history and then g. Their partial domains agree, including failure of g, and their ordered products append the same g-cost.

(3) **Verify codiagonal by cutting and flattening histories.** For codiagonal, tag each continuation by its inner or outer occurrence of X, and cut a finite history at the outer tags. This gives exactly the successive inner returns of the left side. Flattening the blocks is inverse. Each original h-call occurs once in its original order, so associativity and the unit law identify the costs.

(4) **Transport return domains and costs.** For uniformity, the commuting equation transports transitions along u with identical tags, costs, and outputs. Induction identifies each exact return domain with the corresponding pullback and identifies its cost. The classified union proves the equation.

(5) **Verify change-of-cost compatibility and continuity.** A monoid morphism preserves the displayed cost recurrence, which proves its iteration compatibility. The operation is an algebra morphism by the construction using Ψ.

(6) **Verify the trace equations.** Formula (4.6) first enters the equation and then iterates the interface branch. For each trace equation, matching the finite-return components retains every primitive call in order: naturality adjoins the same entry and exit calls, dinaturality moves the interface cut, and vanishing cuts or flattens the same blocks. The cost products are consequently equal. Superposing leaves the unused component unchanged. The symmetry in yanking is a pure map followed by η^C, so its cost is the monoid unit. These arguments prove all trace equations, including equality of classified domains, in every parameter context.

□

Functorial change of resource monoid. Changing a resource valuation should commute with recursive execution. The monoid-morphism clause of the theorem gives exactly this compatibility: the construction is a functor to strong complete Elgot monads and iteration-preserving strong monad morphisms,

$$C \longmapsto L_{\Sigma,C}, \qquad h : C \to C' \longmapsto \alpha_h = L_\Sigma(h \times -).$$

Functoriality follows from functoriality of L_Σ; preservation of iteration is the monoid-morphism assertion just proved. In particular the following square commutes:

$$\begin{array}{ccc} [X, L_{\Sigma,C}(Y+X)]_\Sigma & \xrightarrow{\ddagger_C} & [X, L_{\Sigma,C}Y]_\Sigma \\ \downarrow{\scriptstyle (\alpha_h)_*} & & \downarrow{\scriptstyle (\alpha_h)_*} \\ [X, L_{\Sigma,C'}(Y+X)]_\Sigma & \xrightarrow[\ddagger_{C'}]{} & [X, L_{\Sigma,C'}Y]_\Sigma. \end{array} \tag{4.7}$$

Here $(-)_*$ means postcomposition, and all four objects are the constructed computational function objects. Erasure is the case of the unique monoid morphism $C \to 1$.

4.4 Histories and their resource valuations

We can measure several resources from the same execution by recording its events before applying a valuation. For $A \in \mathcal{A}_\Sigma$ and $f : X \to L_\Sigma(A \times (Y + X))$, replace each event by its singleton word and execute in the free monoid:

$$f_* = L_\Sigma([-] \times \mathrm{id}_{Y+X})f, \qquad H_f = f_*^{\ddagger_{A^*}} : X \to L_\Sigma(A^* \times Y).$$

This is a constructed returning-history map with the original domain.

The event weight $w : A \to C$ extends through the free-monoid unit:

$$\begin{array}{ccc} A & \xrightarrow{[-]} & A^* \\ & \searrow^{w} & \downarrow{\scriptstyle \overline{w}} \\ & & C. \end{array} \tag{4.8}$$

The dashed arrow is the unique monoid morphism extending the event weight. It induces $\alpha_{\overline{w}} : L_{\Sigma,A^*} \to L_{\Sigma,C}$. Thus Equation (4.7) says that folding the returning history gives the same result as accumulating its weights during execution. The following theorem records this comparison.

Theorem 4.6 (Finite-history valuation)**.** *For an algebra morphism $w : A \to C$ and its monoid extension $\overline{w} : A^* \to C$,*

$$L_\Sigma(\overline{w} \times \mathrm{id}_Y)H_f = \big(L_\Sigma(w \times \mathrm{id}_{Y+X})f\big)^{\ddagger_C}. \tag{4.9}$$

Proof.

(1) **Identify the word and cost recurrences.** The word on R_0 is the singleton return label. On R_{k+1} it is the continuation label followed by the word on R_k. The fold recurrence is precisely the recurrence for a_k in Theorem 4.4.

(2) **Compare values on every return summand.** Induction identifies the values on every summand of the common domain.

□

State events are recorded by a typed history map. Given $b : X \to L_\Sigma(Y + X)$, strength constructs

$$X \to L_\Sigma((X \times (Y + X)) \times (Y + X)), \qquad x \mapsto b(x) \operatorname{bind}_\Sigma (z \mapsto \eta((x, z), z)).$$

Its words record actual state events. A finite word may contain domain values or supplied names, so validity of an arbitrary proposed history is not thereby decidable. For a total algebra morphism $\nu : A^* \to C$, $L_\Sigma(\nu \times \mathrm{id})H_f$ is a measured execution even if ν is not a monoid morphism. Only a monoid fold inherits the sequential product law of (4.9). In particular, peak occupancy uses the maximum of recorded occupancy values, including the initial and final states when relevant; allocation increments alone do not give that measurement.

Proposition 4.7 (Universal classification of event weights)**.** *For an event algebra $A \in \mathcal{A}_\Sigma$ and an internal monoid (C, e, μ) in $\mathcal{A}_\Sigma$, the computational exponential $\mathcal{W}_C = [A, C]_\Sigma$ represents all continuous event weights $w : A \to C$. There is a canonical, parameter-uniform valuation map*

$$\mathrm{fold}_C : \mathcal{W}_C \times A^* \to C$$

whose value on $(w, [a_1, \ldots, a_n])$ is the ordered finite product $w(a_1) \cdots w(a_n)$, with empty product e. This represents monoid morphisms from A^* to C in every parameter context; its pullback at w is the history valuation of Theorem 4.6.

Proof.

(1) **Construct the parametrized fold.** Cartesian closure of $\mathcal{A}_\Sigma$ constructs $\mathcal{W}_C$ and its algebra-morphism evaluation $\mathrm{ev} : \mathcal{W}_C \times A \to C$. The finite-word object is an NNO-indexed disjoint sum of finite powers. Iterate μ and ev on each finite-power summand, retaining the parameter w. The formulas give the displayed map from the disjoint sum and preserve admitted suprema componentwise by Lemma 4.1 and Proposition 4.2.

(2) **Verify the monoid equations.** Induction on word length proves

$$\mathrm{fold}_C(w, []) = e, \qquad \mathrm{fold}_C(w, s{+}{+}t) = \mathrm{fold}_C(w, s)\, \mathrm{fold}_C(w, t).$$

(3) **Prove the universal classification of weights.** A monoid map from A^* is determined by its value on singleton words and hence by an arrow $A \to C$. Restriction and the displayed fold are inverse by the same induction, uniformly in free parameters. Evaluating at a fixed w recovers its unique monoid extension. Thus a measured history valuation is a point of this classification object. A chosen resource interpretation still selects which point is measured.

□

The execution operations are now available. To form a complexity judgment, we must specify which descriptions and primitive operations are being measured. Fixing that presentation also fixes what is charged by an event valuation and what is supplied by the environment.

4.5 Implementation descriptions

Construction 4.8 (Fixed computational environments)**.** The finite description construction of Construction E.2 has typed contexts of supplied name and operation parameters. For such a context, form its environment object Θ by the indicated finite products and dependent sums of parameter carriers. Varying a finite context uses the corresponding internally constructed sum of these objects. A description in an environment refers to its supplied parameters by their typed slots. Closed constructor data and implemented programs are still constructed by their finite syntax. Every unrestricted opaque literal is recorded in the environment; it is not an extra choice made while taking an implementation fibre.

Pairing a well-typed description with its environment constructs $E_{X,Y} \to \Theta$. Interpretation is a morphism over Θ, obtained by the finite recursion of Lemma E.3. The primitive signature and its selected execution descriptions are retained in this context; by Proposition 4.7, selecting a history valuation is precisely selecting an event-weight point of a constructed computational exponential. We work internally at a generic $\theta : \Theta$, suppressing θ in the formulas. Thus $E_{X,Y}$ below means $E_{X,Y,\theta}$, and an implementation quantifier fixes θ. Products, function objects, images, and quantifiers in this construction are those of $\mathcal{E}/\Theta$.

An implemented substitution of environments precomposes these interpretations. A substitution which itself performs charged work is represented by its execution description and contributes its history; ordinary reindexing alone supplies no assertion that this work is free. The numerical instance has the fixed register signature and no oracle parameters. The relative instance retains its declared oracle in Θ. Program-generated function values are obtained by evaluating their generating descriptions, with that evaluation recorded.

Use the finite typed descriptions with supplied literal parameters constructed in Construction E.2 and Lemma E.3. Fix a typed fragment of its implemented signature. Its primitive operations have types $b_j : U_j \to L_\Sigma V_j$, and the fragment includes the products, sums, composition, and execution operations it uses. The well-typed subobject of $\mathsf{Tree}_{\mathcal{S}}$ carries these descriptions, their interpretation, and their name presentation. Literal carriers remain supplied parameters; a numerical enumeration of them is not part of the construction.

For the finitely many primitive operation tags form the computational event object $A = \sum_j (U_j \times V_j)$. Define the logged primitive by

$$b_j^H(u) = b_j(u) \,\mathrm{bind}_\Sigma\, (v \mapsto \eta([\iota_j(u, v)], v)) : U_j \to L_\Sigma(A^* \times V_j).$$

Pure structural maps are followed by η^{A^*}, so have empty word. Interpret composition by $\diamond_{A^*}$, coproduct branching by copairing, and recursive execution by $\ddagger_{A^*}$ or (4.6). A specified sequential pair of computations $f : U \to L_{\Sigma,A^*}V$, $g : U \to L_{\Sigma,A^*}W$ has interpretation

$$u \mapsto f(u) \,\mathrm{bind}_{A^*}\, (v \mapsto g(u) \,\mathrm{bind}_{A^*}\, (w \mapsto \eta^{A^*}(v, w))).$$

Thus its evaluation order is retained in the word. These clauses are operations on the same finite descriptions; no quotient by unweighted semantic equality is performed.

The substitution here is substitution in the description's typed operation parameters. For a context of arrow parameters

$$G = \prod_i [U_i, L_{\Sigma,A^*}V_i]_\Sigma,$$

interpretation of a description $t : X \to Y$ has type $[\![t]\!]_H : G \to [X, L_{\Sigma,A^*}Y]_\Sigma$. For substitutions σ_i from a parameter context G', put $\sigma_H = \langle [\![\sigma_i]\!]_H \rangle_i : G' \to G$. The substitution equation is

$$[\![t[\sigma]]\!]_H = [\![t]\!]_H \sigma_H.$$

Ordinary data parameters are retained in the same constructor maps; their evaluation uses the stated sequencing convention.

Proposition 4.9 (Interpretation of retained descriptions)**.** *For each input and output pair in the fragment, structural recursion constructs maps in* $\mathcal{E}$

$$[\![-]\!]_H : E_{X,Y} \to [X, L_\Sigma(A^* \times Y)]_\Sigma, \qquad \mathrm{ev}_H : E_{X,Y} \times X \to L_\Sigma(A^* \times Y).$$

They commute with the supplied finite-tree substitutions. Erasing words recovers the original interpretation of every description, including its classified partial domain. A fixed algebra morphism $\nu : A^* \to \mathbb{N}$ *constructs its measured evaluator*

$$\mathrm{ev}_\nu = L_\Sigma(\nu \times \mathrm{id}_Y)\,\mathrm{ev}_H : E_{X,Y} \times X \to L_\Sigma(\mathbb{N} \times Y). \tag{4.10}$$

Proof.

(1) **Interpret the typed constructors.** Each constructor clause above is a morphism on the appropriate computational function objects: composition and pairing use cartesian closure and strength; iteration and feedback use Theorem 4.5. Interpret a well-typed finite tree by its constructors, retaining its literal parameters. Induction on its typing derivation gives the displayed function-object type. This is the finite structural recursion on the tree carrier of Construction E.2.

(2) **Verify substitution by structural induction.** Substitution at a variable evaluates the substituted tree, substitution at a literal retains its interpretation, and a compound constructor applies the same operation to the substituted children. Induction proves the substitution equation.

(3) **Verify erasure and the evaluation maps.** Erasure commutes with the primitive clauses, unit, bind, strength, and iteration by Theorems 4.3 and 4.4; induction therefore gives the original domain and value. Evaluation and postcomposition give the last two maps.

□

The signature, valuation, and environment θ remain fixed when we quantify over $E_{X,Y,\theta}$. This specifies what each event measures. An opaque function literal is an oracle for an invocation-cost measurement; measuring its internal work instead requires its execution description and the events of that description. The numerical instance constructs the configuration update in Construction 12.1. Projective lifting supplies a map on names and its domain comparison; the history of an execution supplies its cost.

Fixing the environment is essential to the complexity interpretation. If arbitrary functions could be varied as literals while application had unit cost, the description $x \mapsto f(x)$ would give every such denotation a one-call bound. Here an opaque oracle f is fixed by θ. To measure its internal execution we substitute its supplied description and charge the resulting events. The language and the environment together determine the implementations under comparison.

The word fold is stable under the cut and concatenation operations in the weighted trace proof. Counting outer loop iterations alone is not: regrouping two primitive calls into one body changes that count. Nor does equality after the unweighted denotation quotient imply equal cost. Resource-preserving transformations preserve the recorded word, or carry a proved comparison of its selected valuation.

4.6 Measured name presentations

A represented input may have several names, and execution on those names may have different costs. We therefore carry the name maps through the measurement. In the generic fixed environment, retain the input and output presentations

$$q_X : C_X \twoheadrightarrow X_0 = \mathrm{Pt}_\Sigma\, D_X, \qquad q_Y : C_Y \twoheadrightarrow Y_0 = \mathrm{Pt}_\Sigma\, D_Y.$$

Use the projective presentations and their implemented structural maps constructed in Proposition E.1, with partitioned refinements after pullback. Each primitive retains its actual partial name map and the equality of its domain with the pullback of the semantic domain. Finite products and sums of these named sorts provide the typed execution fragment. The discrete admitted algebra on a name carrier allows the finite-word and weighted-execution constructions to be applied to these actual maps. Let $\widehat{A}$ be their event object: an invocation event records the primitive tag and its input and output names; an unfolded primitive instead records its displayed internal execution events.

Theorem 4.10 (Measured name-presentation comparison)**.** *Finite recursion on the retained implementation descriptions constructs*

$$\widehat{\mathrm{ev}}_H : E \times C_X \longrightarrow L_\Sigma(\widehat{A}^* \times C_Y). \tag{4.11}$$

Write $\mathrm{ev}^{\mathrm{sem}} : E \times X_0 \rightharpoonup Y_0$ *for the interpretation of Lemma E.3 on effective points. Then, as partial maps,*

$$q_Y \ \mathrm{erase}\, \widehat{\mathrm{ev}}_H = \mathrm{ev}^{\mathrm{sem}}(\mathrm{id}_E \times q_X), \qquad \widehat{D} = (\mathrm{id}_E \times q_X)^* D_{\mathrm{sem}}. \tag{4.12}$$

For the fixed implemented total valuation $\widehat{\nu} : \widehat{A}^* \to \mathbb{N}$*, let* $P \hookrightarrow E$ *classify total named executions. The returned values and measured histories construct*

$$p_0 : P \to Y_0^{X_0}, \qquad \widehat{\Phi}_0 : P \to \mathbb{N}^{C_X}, \qquad P \xrightarrow{p \twoheadrightarrow} V \hookrightarrow Y_0^{X_0}. \tag{4.13}$$

The cost descends to $\mathbb{N}^{X_0}$ *exactly on the subobject of* P *where*

$$q_X c = q_X c' \quad \Longrightarrow \quad \widehat{\Phi}_0(e)(c) = \widehat{\Phi}_0(e)(c') \qquad (c, c' : C_X). \tag{4.14}$$

All constructions commute with substitution of the retained environment and finite-description parameters.

Proof.

(1) **Interpret the measured name primitives.** Log the actual name primitives and apply the clauses of Proposition 4.9. Their selected events determine the measurement; lifting a semantic arrow alone has assigned no runtime.

(2) **Verify adequacy for finite constructors.** Erasure of a primitive is its retained name implementation, whose domain-and-value square is that of Proposition E.1. Composition pulls back the second domain along the first returned name; the two primitive adequacy squares identify this pullback with the semantic composite domain. Its returned value satisfies the composite square. Coproduct branching and the stated sequential pairing give the same conclusion by their finite case and bind clauses.

(3) **Verify adequacy on exact return domains.** For iteration, induction on the number of transitions identifies each exact finite-return domain with the pullback of the corresponding semantic return domain, and identifies the returned values.

(4) **Pass to total return domains and substitution.** The union of these domains is the classified return domain on both sides by Theorem 4.4. Pullback and the natural-number existential of Proposition 1.2 identify those unions. This proves (4.12), including domains, for every finite description. The same constructor induction proves substitution.

(5) **Construct the totality locus and descend values.** Write $\widehat{d} : E \times C_X \to \Sigma$ for the return classifier. Define $P(e) \iff \forall c : C_X\ \widehat{d}(e, c)$. Since q_X is a regular epimorphism, the domain equation identifies this with $\forall x : X_0\ d_{\mathrm{sem}}(e, x)$. On $P \times C_X$ the evaluator is total. Its returned name, followed by q_Y, is invariant on the kernel pair of $\mathrm{id}_P \times q_X$, by value adequacy. Effective quotient descent therefore gives the unique map $P \times X_0 \to Y_0$.

(6) **Construct the implementation image and measured cost.** Transposition and regular image factorization give p_0, p, V. Folding its retained history gives the total map $P \times C_X \to \mathbb{N}$ and hence $\widehat{\Phi}_0$.

(7) **Characterize cost descent and parameter coherence.** Finally form the subobject of P defined by (4.14). On that subobject the measured map coequalizes the same kernel pair and thus descends uniquely to X_0. Conversely every descended cost coequalizes that pair. Pullback stability of these quotients, dependent-product Beck–Chevalley, and the finite interpretation equations prove parameter coherence.

□

The return denotation descends by its extensionality equation. Cost descends precisely when it is constant on names of the same input. For a fixed total implementation e, the coequalizer criterion for q_X expresses that condition:

$$\begin{array}{ccccc} C_X \times_{X_0} C_X & \underset{\pi_2}{\overset{\pi_1}{\rightrightarrows}} & C_X & \xrightarrow{\ q_X\ } & X_0 \\ & & \Big\downarrow{\scriptstyle \widehat{\Phi}_0(e)} & \swarrow{\scriptstyle \Phi_0^{\mathrm{ext}}(e)} & \\ & & \mathbb{N} & & \end{array} \tag{4.15}$$

The dashed map exists uniquely when the two cost composites from the kernel pair agree. The value leg has its own descent equation, Equation (4.12). Until the cost equation also holds, the name carrier remains the domain of the measured profile.

Example 4.11 (Value descent without cost descent)**.** Take $q_X : \mathbb{N} \times \mathbb{N} \twoheadrightarrow \mathbb{N}$, $q_X(n, k) = n$. In the retained signature a description can perform k implemented idle invocations and return 0. For their counting valuation its cost is k. Its erased name map $(n, k) \mapsto 0$ has true domain and commutes with q_X and the identity presentation of the output. It therefore implements the constant semantic primitive; unfolding this particular implementation records its name-dependent waiting history. The returned value descends to $\mathbb{N}$, but the cost fails (4.14). This is a name-level measurement, not an extensional function of n.

4.7 Total implementations and the measured image

We now collect the total implementations of the fixed measured evaluator and separate their denotations from their costs. First take actual execution input and return carriers X, Y. Write $E = E_{X,Y}$, or use the numerical family of Construction 12.1, with evaluator (4.10). If $d : E \times X \to \Sigma$ classifies its domain, totality defines the subobject

$$P \hookrightarrow E, \qquad P(e) \iff \forall x : X\ d(e, x).$$

Totality is an Ω-valued predicate. On a returning execution write

$$H_e(x) \in A^*, \qquad T_{\nu,e}(x) = \nu(H_e(x)). \tag{4.16}$$

Both expressions have exactly the return domain of the represented execution.

Proposition 4.12 (Measured implementation image)**.** *The construction gives canonical arrows*

$$p_0 : P \to Y^X, \qquad \Phi_0 : P \to \mathbb{N}^X, \qquad P \xrightarrow{p \twoheadrightarrow} V \hookrightarrow Y^X, \quad V = \mathrm{im}(p_0).$$

Their uncurried values are the returned value and the selected finite resource measurement. The cost map has domain P.

Proof.

(1) **Restrict the evaluator to its totality locus.** Restriction of the evaluator to $P \times X$ has true extent, so its partial-map span is a unique total map $P \times X \to \mathbb{N} \times Y$.

(2) **Construct the implementation image and cost map.** Transposing its two projections gives Φ_0, p_0. Regular image factorization in $\mathcal{E}$ gives p and V.

□

For represented inputs, use Theorem 4.10 instead: the denotation lies in $Y_0^{X_0}$ and the raw measurement in $\mathbb{N}^{C_X}$. From now on X in a cost profile denotes the actual execution input carrier, so $X = C_X$ in this case. The notation $p : P \twoheadrightarrow V$, $\Phi_0 : P \to \mathbb{N}^X$ covers both constructions. All these objects remain over the fixed environment θ of Construction 4.8.

The order of this construction matters: the language, environment, and event valuation are fixed before the denotation image is formed. Otherwise arbitrary clocks or unit-cost semantic primitives would trivialize the measurement. For example, the clock constantly equal to true with return value $g(x)$ has first hit zero for every total $g : X \to Y$. The concrete numerical clocks of Proposition 12.2 instead come from the specified transition system. The general return theorem Theorem 4.4 does not require those clocks.

For Landau comparison, rational inclusion and the eventual quotient of Proposition 2.1 give the exact cost map

$$\Phi : P \xrightarrow{\Phi_0} \mathbb{N}^X \longrightarrow {\mathbb{Q}_{\geq 0}}^X \xrightarrow{q_+} K^+.$$

Each measurement is fixed before taking the existential over its implementation fibre. Joint resources use a product valuation on the same histories and retain the same implementation witness.

The proper-filter context used in K is a context on this actual input carrier X. Bounds specified by an extensional input or by an input-size map are pulled back to X. See Proposition 12.5.

The measured implementation span. For a fixed computational environment, a total implementation determines both a denotation and a measured cost. These assignments have the same source and form the span in $\mathcal{E}/\Theta$

$$V \xleftarrow{\ p\ } P \xrightarrow{\ \Phi\ } K. \tag{4.17}$$

Its apex retains the total implementations. To impose a budget, we will pull its marked module back along Φ, then compose with p to obtain the family of implementations meeting that budget. Taking the image will give the feasible denotations.

At the raw-profile level the cost endpoint is $\mathbb{N}^X$. Rational inclusion and the germ quotient change it to K while preserving the apex. The pair of legs also factors through its regular image:

$$P \xrightarrow{\quad} \mathscr{R}_{p,\Phi} \hookrightarrow V \times K, \qquad P \xrightarrow{(p,\Phi)} V \times K.$$

The relation records which denotation–cost pairs occur. The span carries the witnesses of those pairs, including distinct implementations with the same denotation and cost. The following slice construction preserves this information until the feasibility image is taken.

Proposition 4.13 (Witness transport and its feasibility image)**.** *The measured span induces the slice functor*

$$\mathsf{Wit}_{p,\Phi} = p_! \Phi^* : \mathcal{E}/K \longrightarrow \mathcal{E}/V, \qquad \mathsf{Wit}_{p,\Phi}(M \to K) = (P \times_K M \to V).$$

Here $p_!$ is dependent sum in the slice. On marked subobjects, taking the image gives the monotone map

$$\mathsf{Feas}_{p,\Phi} = \exists_p \Phi^* : \mathrm{Sub}(K) \longrightarrow \mathrm{Sub}(V).$$

Both constructions commute with substitution of environment parameters. A commuting regular-epimorphic refinement of the apex induces a regular-epimorphic map of witness families and preserves their feasibility images.

Proof.

(1) **Construct witness transport and its feasibility image.** The pullback and subsequent projection are

$$\begin{array}{ccc} P \times_K M & \longrightarrow & M \\ \downarrow & & \downarrow \\ P & \xrightarrow[\Phi]{} & K \\ {\scriptstyle p}\downarrow & & \\ V & & \end{array} \tag{4.18}$$

and define $p_! \Phi^*$. Their regular image is $\exists_p \Phi^* M$ by the definition of existential image.

(2) **Verify parameter substitution.** Beck–Chevalley for pullbacks and stability of regular images give parameter substitution.

(3) **Prove invariance under a covering presentation.** If $r : P' \twoheadrightarrow P$ satisfies $p' = pr$ and $\Phi' = \Phi r$, the square

$$\begin{array}{ccc} P' \times_K M & \longrightarrow & P \times_K M \\ \downarrow & & \downarrow \\ P' & \xrightarrow[r]{} & P \end{array}$$

is a pullback. Its upper arrow is therefore regular epic. The composites to V have the same image.

□

For simultaneous measurements $\Phi_T : P \to K_T$ and $\Phi_S : P \to K_S$, the right leg is their pairing and the joint witness family is the pullback

$$\begin{array}{ccc} Z_{M_T,M_S} & \longrightarrow & M_T \times M_S \\ \downarrow & & \downarrow \\ P & \xrightarrow[(\Phi_T,\Phi_S)]{} & K_T \times K_S. \end{array}$$

The pullback imposes both constraints on the same implementation. Its image in V is therefore simultaneous feasibility. This fixes the order of the operations: combine the resource conditions first, then quantify over their shared witness.

4.8 Universal implementation incidence and its fibres

The measured span can now be applied to the universal budget family. Write its fixed-environment maps as $p : P \twoheadrightarrow V$, $\Phi : P \to K^+ \hookrightarrow K$, and $\kappa : P \to L$; with the parameter displayed they are $p_\theta : P_\theta \twoheadrightarrow V_\theta$ and Φ_θ. For represented inputs, these are the maps of Theorem 4.10, with germs formed on the name carrier. In the ringed resource topos define

$$\mathscr{Z} = \Delta_{\mathfrak{C}_{\mathrm{Ev}}} P \times_{\bar{K}} \mathscr{M}, \qquad \mathscr{F} = \mathrm{im}(\mathscr{Z} \to \Delta_{\mathfrak{C}_{\mathrm{Ev}}} V). \tag{4.19}$$

This applies Proposition 4.13 in $\mathscr{S}$. The cartesian square selects the implementations meeting the universal budget, and the regular image records their feasible denotations:

$$\begin{array}{ccc} \mathscr{Z} & \longrightarrow & \mathscr{M} \\ \downarrow & & \downarrow \\ \Delta P & \xrightarrow[\Delta\Phi]{} & \bar{K} \\ {\scriptstyle \Delta p}\downarrow & & \\ \Delta V & & \end{array} \qquad \mathscr{Z} \twoheadrightarrow \mathscr{F} \hookrightarrow \Delta V. \tag{4.20}$$

At an ideal point I, write

$$Z_I = I^*\mathscr{Z} = P \times_K M_I, \qquad \mathcal{C}_{\mathrm{cost}}(I) = I^*\mathscr{F} = \mathrm{im}(Z_I \to V). \tag{4.21}$$

This defines semantic budget fibres; restricting I to $J_\Sigma L$ gives the admitted computational instances.

Theorem 4.14 (Classified computational upper bounds)**.** *The incidence and its image are the budget diagrams*

$$\mathscr{Z}(a) = \{e : P : \kappa(e) \le a\}, \qquad \mathscr{F}(a) = \{v : V : \exists e \in P_v,\ \kappa(e) \le a\}.$$

The open-valued classifier of $\mathscr{F} \hookrightarrow \Delta V$ is

$$\mathcal{U}_v = \bigvee_{e \in P_v} U_{\kappa(e)}. \tag{4.22}$$

At every ideal point,

$$I^*\mathscr{F} = \mathcal{C}_{\mathrm{cost}}(I) = \{v : \exists e \in P_v,\ \kappa(e) \in I\}. \tag{4.23}$$

Regular-epimorphic refinements of implementation covers preserving denotation and measured cost leave this image unchanged.

Proof.

(1) **Compute incidence and feasibility at each budget.** Apply Theorem 3.4 to the pullback in (4.19). Images are computed at each budget, giving the two diagrams. Membership of e has extent $U_{\kappa(e)}$; existential quantification along p gives (4.22).

(2) **Compute the fibres.** Geometric inverse image preserves pullbacks and regular images, proving the fibre formula.

(3) **Prove invariance under presentation refinement.** For a refinement $r : P' \twoheadrightarrow P$, $\exists_r \top = \top$; composition of images proves the invariance.

□

The budget fibres satisfy $\mathcal{C}_{\mathrm{cost}}(I) = \bigvee_{a \in I} \mathcal{C}_{\mathrm{cost}}(\downarrow a)$ by interchanging their two existential quantifiers. Their incidence has type $J_\Sigma L \times V \to \Omega$, with the separately specified positive observations of Section 2.6.1.

4.9 Complexity judgments as subobject adjunctions

An upper bound asks for an implementation whose cost meets the bound. A lower bound constrains every implementation of the same denotation. The adjunctions $\exists_p \dashv p^* \dashv \forall_p$ express these different quantifiers over the fixed implementation family. Their classifiers are Ω-valued; positive observations retain the separately specified meaning of Section 2.6.1. For a nonnegative bound g, form

$$A_g = \{e : P : \Phi(e) \in Bg\}, \qquad D_g = \{e : P : g \in B\Phi(e)\},$$

and define

$$\mathrm{Upper}_g = \exists_p A_g, \qquad \mathrm{Lower}_g = \forall_p D_g, \qquad \mathrm{Tight}_g = \mathrm{Upper}_g \wedge \mathrm{Lower}_g\,. \tag{4.24}$$

Their internal formulas are respectively $\exists e \in P_v\, \Phi(e) \in O(g)$ and $\forall e \in P_v\, \Phi(e) \in \Omega(g)$. With the environment displayed, the first is

$$\mathrm{Upper}_g^\theta(v) \iff \exists e \in P_{\theta,v}\ \Phi_\theta(e) \in Bg,$$

and the lower and tight judgments use that same $P_{\theta,v}$. The lower-bound constants and eventual extents occur inside $\forall e$; the variant with uniform constants is a different dependent-quantifier expression.

Proposition 4.15 (Complexity bounds as inclusions of feasibility opens)**.** *For $g \in K^+$ and $a = [g]$, the upper, lower and tight judgments have the geometric descriptions*

$$\begin{aligned} \mathrm{Upper}_g(v) &\iff U_a \subseteq \mathcal{U}_v, \\ \mathrm{Lower}_g(v) &\iff \mathcal{U}_v \subseteq U_a, \\ \mathrm{Tight}_g(v) &\iff \mathcal{U}_v = U_a. \end{aligned} \tag{4.25}$$

These equivalences are internal and uniform in the original denotation, bound and computational-environment parameters.

Proof.

(1) **Reflect growth order through principal opens.** The principal opens satisfy $U_b \subseteq U_a \iff a \leq b$: evaluate an inclusion at b for one direction, and use upperness for the other.

(2) **Characterize the upper-bound judgment.** If $\kappa(e) \leq a$, then $U_a \subseteq U_{\kappa(e)} \subseteq \mathcal{U}_v$. Conversely, evaluation of $U_a \subseteq \mathcal{U}_v$ at a and (4.22) give $\exists e \in P_v\ \kappa(e) \leq a$, precisely the upper judgment. Equivalently, supercompactness of U_a extracts a member of the canonical implementation cover.

(3) **Characterize the lower and tight judgments.** For the lower judgment,

$$\begin{aligned} \mathcal{U}_v \subseteq U_a &\iff \forall e \in P_v\ U_{\kappa(e)} \subseteq U_a \\ &\iff \forall e \in P_v\ a \leq \kappa(e) \iff \mathrm{Lower}_g(v). \end{aligned}$$

Combining the two comparisons proves the tight statement.

□

The feasibility open therefore determines every principal upper, lower and tight judgment. A tight bound is the assertion that this open is the corresponding principal open. The implementation cover will identify principality with an attained optimal growth class in Theorems 5.2 and 5.3. The lower statement in (4.25) is an inclusion between constructed opens; its proof retains the universal implementation quantifier and its individual domination witnesses.

Proposition 4.16 (Matching bounds and shared witnesses)**.** *A tight bound supplies an implementation with cost in $\Theta(g)$. On the eventually nonzero comparison locus, a lower bound excludes an implementation in $o(g)$. An existential implementation in $\Theta(g)$ does not imply the universal lower bound. For two incidence subobjects $A, D \hookrightarrow P$, the shared-witness classification is $\exists_p(A \wedge D)$, which lies below $(\exists_p A) \wedge (\exists_p D)$; equality is not automatic.*

Proof.

(1) **Combine bounds using a shared implementation.** Frobenius reciprocity and the counit of $p^* \dashv \forall_p$ give

$$(\exists_p A_g) \wedge (\forall_p D_g) = \exists_p(A_g \wedge p^* \forall_p D_g) \leq \exists_p(A_g \wedge D_g).$$

Mutual domination gives $\Theta(g)$.

(2) **Exclude simultaneous smallness and a nonzero lower bound.** If a small implementation also satisfies $g \preceq \Phi(e)$ with constant a, smallness at $2a$ gives eventually $2|g| \leq |g|$, contradicting the eventually nonzero extent and properness of the filter.

(3) **Distinguish an implementation cost from a universal bound.** A constant-output function has both an immediate-return implementation of bounded cost and one scanning every input letter, of cost $\Theta(|x| + 1)$. The latter does not make that bound a universal lower bound.

(4) **Establish the shared-witness inclusion.** Finally monotonicity of images gives the shared-witness inclusion; taking the two images separately permits different implementations.

□

All these predicates are defined in $\mathcal{E}$ with their displayed parameters. Constant-diagram transport preserves their particular quantifications: a map of constant objects has the same fibre at every budget and Weil stage, and a constant predicate has the same universal clause on that fibre. Parameter substitution inside a topos obeys the dependent-product Beck–Chevalley law, by its right-adjoint universal property. For a cover refinement $r : P' \twoheadrightarrow P$,

both $\exists_r r^*$ and $\forall_r r^*$ are the identity on subobjects of P, so pulled-back upper and lower judgments are invariant. This follows from the inhabited fibres of the regular epi.

The open $\mathcal{U}_v$ is Scott open by its principal cover. Universal resource assertions need not be: in the cofinite numerical fibre put $g_k(n) = (n+1)^k$ and $I_m = {\downarrow}\,[g_m]$. The assertion $\forall k\,[g_k] \in I$ holds at $\bigcup_m I_m$ but at no I_m, since $[g_{m+1}] \not\leq [g_m]$. Universal judgments therefore retain their dependent-product type instead of being identified with feasibility opens.

4.10 The implementation sheaf and compiler morphisms

Proposition 4.17 (Substitution of fixed environments)**.** *For a parameter substitution $s : \Theta' \to \Theta$, reindexing the retained execution construction gives the same measured evaluator as interpreting its substituted descriptions. It carries P, V, p, Φ, their implementation incidence and feasibility image, and their upper and lower judgments to the corresponding constructions over Θ'. The implementation-sheaf, compiler, and optimality constructions below are likewise performed in this fixed-environment family.*

Proof.

(1) **Transport interpretation through environment substitution.** The interpretation statement is the finite substitution equation of Proposition 4.9 and Theorem 4.10.

(2) **Identify totality, denotations, and cost profiles.** Pullback between parameter slices preserves finite limits and regular images. The dependent-product Beck–Chevalley equation for the input projection identifies the totality subobjects; exponentiation in the slice identifies their transposed denotations and cost profiles. The quotient and image comparisons then identify p, V, Φ.

(3) **Transport feasibility and lower judgments.** The same pullback and image equations identify incidence and upper feasibility. Beck–Chevalley for p identifies the universal lower judgment.

(4) **Verify parameter coherence of the remaining constructions.** All budget and coefficient constructions retain these parameters; the sheaf constructions below have the same pointwise formulas in each slice. In particular an implementation quantifier is never replaced by an existential over environments. The displayed comparisons satisfy identity and composition by their universal properties.

□

In particular the denotation objects and their implementation covers form a cartesian square under $s : \Theta' \to \Theta$:

$$\begin{array}{ccc} P_{\Theta'} & \longrightarrow & P_{\Theta} \\ {\scriptstyle p_{\Theta'}}\downarrow & & \downarrow{\scriptstyle p_{\Theta}} \\ V_{\Theta'} & \longrightarrow & V_{\Theta}. \end{array}$$

Pullback of the measured cost leg gives the compatible incidence square; Beck–Chevalley identifies its existential and universal judgments.

Fixing a denotation in the universal incidence leaves a family of implementations indexed by budgets. We now describe this witness sheaf explicitly. Its image is the feasible open, while its sections retain the choices of implementation needed for the later optimality criterion.

Theorem 4.18 (The universal sheaf of feasible implementations)**.** *For $v : V$ the pullback $\mathscr{W}_v$ of $\mathscr{Z}$ along $\Delta\{v\} \to \Delta V$ is the dependent coproduct*

$$\mathscr{W}_v \cong \coprod_{e \in P_v} U_{\kappa(e)} \qquad \text{in } \mathrm{Sh}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}}). \tag{4.26}$$

Its image over $\mathfrak{C}_{\mathrm{Ev}}$ is $\mathcal{U}_v$, and its restriction to that open is a regular epimorphism $\mathscr{W}_v \twoheadrightarrow \mathcal{U}_v$. By the sheaf–local-homeomorphism correspondence, this constructs an internal étale locale $E_v \to \mathfrak{C}_{\mathrm{Ev}}$ whose fibre over a resource ideal I is $\{e \in P_v : \kappa(e) \in I\}$.

Proof.

(1) **Identify the implementation sheaf by membership.** Over an implementation $e : P_v$, the defining pullback $\mathscr{Z} = \Delta P \times_{\Delta K} \mathscr{M}$ restricts to the membership subterminal $[\![\Phi(e) \in \mathscr{M}]\!] = U_{\kappa(e)}$. Dependent disjoint union over P_v gives (4.26).

(2) **Compute feasibility and the regular epimorphism.** The image of its projection is the join of these membership opens, exactly $\mathcal{U}_v$ by Theorem 4.14. Factoring through the image gives the regular epi.

(3) **Construct the associated locale and its fibres.** The standard internal sheaf–étale-locale equivalence for a localic sheaf topos (Vickers 1989; Mac Lane and Moerdijk 1992) constructs E_v; pullback along I computes the displayed fibre.

□

Under Theorem 3.4, this is the internal diagram

$$\mathsf{Impl}_v : L \to \mathcal{E}, \qquad \mathsf{Impl}_v(a) = \{e \in P_v : \kappa(e) \leq a\},$$

whose transition maps retain the same implementation as the budget increases. Its ideal fibre is the directed colimit over $a \in I$.

Each implementation is available at its own cost and at every larger budget. Extending this generating data along the cost map produces the entire witness diagram. The following universal property says that a transformation out of the diagram is determined by what it does to those generators.

Proposition 4.19 (The free budget extension of an implementation family)**.** *Write $\kappa_v : \mathrm{Disc}(P_v) \to L$ for the cost functor from the discrete internal category of implementations. The witness diagram is the left Kan extension of its terminal diagram:*

$$\mathsf{Impl}_v \cong \mathrm{Lan}_{\kappa_v} \mathbf{1}, \qquad \mathsf{Impl}_v(a) \cong \coprod_{e \in P_v} L(\kappa_v(e), a).$$

Here $L(b, a)$ is the subterminal object expressing $b \leq a$. For every internal diagram $D : L \to \mathcal{E}$ there is a natural isomorphism

$$\mathrm{Nat}(\mathsf{Impl}_v, D) \cong \prod_{e \in P_v} D(\kappa_v(e)). \tag{4.27}$$

Proof.

(1) **Identify the budget-indexed implementation family.** At a, the displayed dependent coproduct has exactly the elements $e \in P_v$ with $\kappa_v(e) \leq a$, and its transition maps retain e.

(2) **Extract data at each implementation's cost.** For a transformation α, evaluate α at the element e of $\mathsf{Impl}_v(\kappa_v(e))$ to obtain $d_e \in D(\kappa_v(e))$.

(3) **Construct the inverse and prove the universal property.** Conversely transport d_e along $D(\kappa_v(e) \le a)$ whenever e is feasible at a. These operations are inverse by naturality. They are internal dependent operations, so require no selection from unrelated fibres. This proves the Kan-extension universal property.

□

The Kan-extension formula retains each implementation across all budgets that admit it; its regular image is the feasible open.

4.11 A problem probes the locale by an open-valued observable

The witness sheaf describes implementations at each budget. Its support answers the corresponding question about the problem: on which budget families does an implementation exist? For $v : V$, use the fibre P_v of the measured span (4.17), with witness diagram $\mathrm{Lan}_{\kappa_v} \mathbf{1}$. Its support is

$$\mathcal{U}_v = \bigvee_{e \in P_v} U_{\kappa(e)} \quad \in \mathcal{O}(\mathfrak{C}_{\mathrm{Ev}}). \tag{4.28}$$

Thus $I \models \mathcal{U}_v$ if and only if $\exists e \in P_v\ \Phi(e) \in M_I$. This is the same cost incidence as in Theorem 4.14, with neither names nor implementation costs quotiented by denotation.

Proposition 4.20 (The functor-of-points feasibility probe)**.** *For each represented denotation $v : V$, there is a locale map $\chi_v : \mathfrak{C}_{\mathrm{Ev}} \to \mathbb{S}$ to the Sierpiński locale, uniquely classified by $\mathcal{U}_v$. On every test locale Z it induces a natural transformation*

$$h_{\mathfrak{C}_{\mathrm{Ev}}}(Z) \longrightarrow \mathcal{O}(Z), \qquad b \longmapsto b^*\mathcal{U}_v. \tag{4.29}$$

For the budget family $\mathscr{M}_b \subseteq \Delta_Z K$, the open $b^*\mathcal{U}_v$ is the regular-image locus of the pullback

$$\Delta_Z P_v \times_{\Delta_Z K} \mathscr{M}_b \longrightarrow 1_Z, \tag{4.30}$$

where the first map to $\Delta_Z K$ is $\Delta_Z \Phi$. Consequently the functor of points of the open sublocale $\mathcal{U}_v \hookrightarrow \mathfrak{C}_{\mathrm{Ev}}$ is the subfunctor

$$h_{\mathcal{U}_v}(Z) \cong \{b : Z \to \mathfrak{C}_{\mathrm{Ev}} : b^*\mathcal{U}_v = \top_Z\} \ \hookrightarrow h_{\mathfrak{C}_{\mathrm{Ev}}}(Z). \tag{4.31}$$

For an ideal point I, the specialization is

$$\chi_v(I) = \top \quad \Longleftrightarrow \quad v \in \mathcal{C}_{\mathrm{cost}}(I) \quad \Longleftrightarrow \quad \exists e \in P_v\ \kappa(e) \in I.$$

Proof.

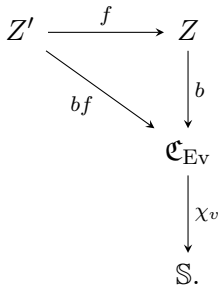
(1) **Classify the feasibility open and verify naturality.** The Sierpiński classification of opens gives the unique χ_v. Under the natural identification $\mathrm{Loc}_{\mathcal{E}}(Z, \mathbb{S}) \cong \mathcal{O}(Z)$, composition with χ_v is exactly (4.29). For a map $f : Z' \to Z$, $(bf)^*\mathcal{U}_v = f^*(b^*\mathcal{U}_v)$, proving naturality.

(2) **Pull back the witness family and its image.** The universal incidence is the pullback $\mathscr{Z} = \Delta P \times_{\bar{K}} \mathscr{M}$. Pulling its v-fibre back along b gives (4.30). Geometric inverse image preserves pullbacks and regular images, so the image open is $b^*\mathcal{U}_v$.

(3) **Characterize feasible factorization and ideal fibres.** An open immersion $\mathcal{U}_v \hookrightarrow \mathfrak{C}_{\mathrm{Ev}}$ has the asserted factorization property for locale maps, proving (4.31). The last equivalences are the fibre formula in Theorem 4.14.

□

For $f : Z' \to Z$, the problem observable respects the parameter triangle



Its open is $(bf)^*\mathcal{U}_v = f^*(b^*\mathcal{U}_v)$. The pullback witness family supplies its local implementation witnesses.

Equivalently, the Yoneda correspondence identifies the probe with

$$\mathrm{Nat}(h_{\mathfrak{C}_{\mathrm{Ev}}}, h_{\mathbb{S}}) \;\cong\; \mathrm{Loc}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}}, \mathbb{S}) \;\cong\; \mathcal{O}(\mathfrak{C}_{\mathrm{Ev}}),$$

where $h_{\mathbb{S}}(Z) = \mathcal{O}(Z)$. In this precise sense *problems probe the complexity locale*: the generalized points are budget families, while a problem gives an open-valued observable on those families. By Theorem 5.1, the same observable is the point $c(v)$ of the constructed exponential locale $\mathfrak{F} = \mathbb{S}^{\mathfrak{C}_{\mathrm{Ev}}}$, with $c : \mathrm{Disc}(V) \to \mathfrak{F}$ natural in the denotation parameter.

The witness sheaf remains distinct from its feasible open. By Theorem 4.18, it has the presentation

$$\mathscr{W}_v \cong \coprod_{e \in P_v} U_{\kappa(e)}, \qquad \mathscr{W}_v \twoheadrightarrow \mathcal{U}_v.$$

After pullback along $b : Z \to \mathfrak{C}_{\mathrm{Ev}}$, the equation $b^*\mathcal{U}_v = \top_Z$ says that the witness projection is locally inhabited over Z. A section supplies the stronger datum of compatible implementation witnesses throughout the family. The distinction between these two statements will govern the optimality result.

4.12 A constant problem and redundant programs

The distinction between a program's cost and a problem's complexity already appears for a constant function. Work on $X = \mathbb{N}$ with its cofinite filter. Select an implemented unit-valued idle primitive a that leaves the result unchanged, and count its recorded invocations with $w(\mathsf{a}) = 1$ and empty-word cost 0. This is the history-fold measurement of Propositions 4.2 and 4.7.

Let $v : \mathbb{N} \to \mathbb{N}$ be the constant-zero function. The pure constant-return description e_0 has an empty event word on every input, and therefore

$$p(e_0) = v, \qquad H_{e_0}(n) = [], \qquad T_{e_0}(n) = 0, \qquad \kappa(e_0) = [0].$$

A second finite description e_1 performs n consecutive calls to a and then returns zero. The iteration is NNO recursion applied to the implemented primitive; the log equation and finite induction give

$$p(e_1) = v, \qquad H_{e_1}(n) = \mathsf{a}^n, \qquad T_{e_1}(n) = n, \qquad \kappa(e_1) = [n].$$

Here a^n is the finite word of charged idle events.

The first program contributes $U_{[0]} = \top$ to the feasibility open. Since every growth ideal contains its bottom element, it follows that

$$\mathcal{U}_v = \top_{\mathfrak{C}_{\mathrm{Ev}}}. \tag{4.32}$$

The additional contribution $U_{[n]} \subseteq U_{[0]}$ leaves this open unchanged. Thus the problem is feasible at the zero-cost budget although e_1 has linear event cost. For the corresponding bounded and linear machine-update example, see Proposition 4.16.

We next compare witness families by translating their implementations. The Kan-extension property reduces the semantic comparison to the cost of each translated generator. A finite compiler description then supplies the implementation of that translation.

4.13 Implemented natural transformations

Proposition 4.21 (Lax cost squares classify witness transformations)**.** *For a monotone budget transformation $r : L_v \to L_w$ there is a natural isomorphism*

$$\mathrm{Nat}(\mathsf{Impl}_v, \mathsf{Impl}_w \circ r) \cong \{\gamma : P_v \to P_w : \kappa_w \gamma \leq r\kappa_v\}. \tag{4.33}$$

These correspondences preserve identities and composition.

Proof.

(1) **Apply the Kan-extension classification.** Apply Equation (4.27) with $D = \mathsf{Impl}_w \circ r$. Its right side is the dependent product of the objects $\{d \in P_w : \kappa_w(d) \leq r(\kappa_v(e))\}$, which is precisely the object of displayed maps γ.

(2) **Identify the action and its composition laws.** The transformation sends e at every feasible budget to $\gamma(e)$. The two descriptions therefore have the same identity and composition.

□

The compiler's cost bound is the inequality in the lax square

$$\begin{array}{ccc} P_v & \xrightarrow{\gamma} & P_w \\ {\scriptstyle \kappa_v}\downarrow & \leq & \downarrow{\scriptstyle \kappa_w} \\ L_v & \xrightarrow[r]{} & L_w. \end{array} \qquad \kappa_w \gamma \leq r\kappa_v. \tag{4.34}$$

Monotonicity transports the bound to every larger budget. The resulting components send the same implementation to the same compiled code, which proves naturality. We now retain finite-code realizations of these maps.

Construction 4.22 (The category of measured compiler comparisons)**.** Work at the fixed environment of Construction 4.8. Source and target signatures may share that environment; a comparison using an implemented change of environment retains the specified map and is formed after the corresponding pullback. Its execution cost is obtained from that change's retained description. Use the data extension of the implementation signature in Section 4.5, retaining its code carriers as sorts as in Construction E.2. For two code carriers E_v, E_w, let $T_{v,w}$ be their constructed carrier of well-typed finite transformation descriptions, with interpreted partial map

$$\mathrm{ev}^{\mathrm{tr}}_{v,w} : T_{v,w} \times E_v \to L_\Sigma E_w.$$

This interpretation is the finite-tree construction of Lemma E.3; its identity and sequential-composition constructors give actual descriptions of the identity and composite code maps. In the numerical instance $T_{v,w} = \mathbb{N}$ with its partial numerical evaluator on program codes; the identity and composition wrappers are those of Lemma E.3 and Proposition E.5 and

Construction 12.1. For the measured fibre inclusions $i_v : P_v \hookrightarrow E_v$, $i_w : P_w \hookrightarrow E_w$, form the subobject

$$\mathrm{CompCode}(P_v, P_w) = \{t : T_{v,w} : \forall e : P_v \ \exists d : P_w,\ \mathrm{ev}^{\mathrm{tr}}_{v,w}(t, i_v e) = \eta_{E_w}(i_w d)\}.$$

The condition means that the partial domain is true and its returned value belongs to P_w; it is an Ω-valued condition. Restriction to this true domain and transposition construct

$$\gamma : \mathrm{CompCode}(P_v, P_w) \to (P_w)^{P_v}.$$

Define the raw bound subobject

$$R(v,w) = \left\{ (t,r) : \begin{array}{ll} t \in \mathrm{CompCode}(P_v, P_w), & r \in \mathrm{Mon}(L_v, L_w), \\ \forall e : P_v, & \kappa_w(\gamma_t e) \le r(\kappa_v(e)) \end{array} \right\}.$$

The hom object of measured comparisons is its regular image

$$\mathrm{MComp}(v,w) = \mathrm{im}\left(R(v,w) \xrightarrow{(\gamma,\pi_2)} (P_w)^{P_v} \times \mathrm{Mon}(L_v, L_w) \right). \tag{4.35}$$

Thus a comparison retains an actual code-output map and its resource transformation, with local evidence of a finite implementation. Its identity and composition are

$$1_v = (\mathrm{id}_{P_v}, \mathrm{id}_{L_v}), \qquad (\delta, s) \circ (\gamma, r) = (\delta\gamma, sr).$$

They land in the image because the finite constructors implement identity and composition. Their bounds follow from

$$\kappa_z(\delta\gamma e) \le s(\kappa_w(\gamma e)) \le sr(\kappa_v(e)).$$

Associativity and units are equations of interpreted functions and monotone maps, proving the internal category laws for every retained internal family of objects. The codes themselves remain a presentation of this category; raw composition wrappers need not be equal as syntax. Nor are two maps identified merely because their outputs have equal denotations: distinct codes in P_w can have different costs. The quotient in (4.35) uses equality in $(P_w)^{P_v}$, not equality after $p_w : P_w \to V_w$.

Theorem 4.23 (Compiler action on the implementation moduli)**.** *For a measured comparison $(\gamma, r) : v \to w$ in* MComp*, the implementation map $\gamma : P_v \to P_w$ and monotone resource transformation $r : L_v \to L_w$ satisfy*

$$\kappa_w(\gamma e) \le r(\kappa_v(e)) \qquad (e \in P_v). \tag{4.36}$$

The growth transformation induces a locale map $s_r : \mathfrak{C}_v \to \mathfrak{C}_w$, acting on ideals by $I \mapsto\downarrow r(I)$. The compiler induces a morphism of witness sheaves over s_r,

$$\mathscr{W}_v \longrightarrow s_r^* \mathscr{W}_w,$$

and its regular image gives

$$\mathcal{U}_v \le s_r^* \mathcal{U}_w. \tag{4.37}$$

Identity and composed comparisons in MComp *act by identity and composition of these morphisms. This action is independent of the choice of finite code representing the same interpreted comparison in Construction 4.22.*

Proof.

(1) **Construct the compiler transformation.** Under Theorem 3.4, s_r^* is precomposition with r. The lax comparison therefore gives the required witness transformation by Proposition 4.21. Its explicit component at budget a sends e to γe, with

$$\kappa_w(\gamma e) \le r(\kappa_v(e)) \le r(a),$$

so this is a natural transformation $\mathsf{Impl}_v \to \mathsf{Impl}_w \circ r$.

(2) **Transport feasibility and verify coherence.** Taking images proves (4.37). Both composites send an implementation to the same composite code and use the same composed monotone bound, proving identity, composition, and independence of the finite description of the interpreted map.

□

For a generalized budget $b : Z \to \mathfrak{C}_v$, the same compiler therefore gives

$$b^*\mathscr{W}_v \longrightarrow (s_r b)^*\mathscr{W}_w, \qquad b^*\mathcal{U}_v \le (s_r b)^*\mathcal{U}_w.$$

These comparisons commute with further substitution of parameter locales because geometric inverse image preserves the witness map and its image. The family statement retains the actual code map γ; an inclusion of feasibility opens alone supplies no compiler.

Over the locale $\mathfrak{R}_r$ of (3.12), this comparison further gives $\pi_1^*\mathscr{W}_v \to \pi_2^*\mathscr{W}_w$. Indeed $\kappa_v(e) \in I$ implies $\kappa_w(\gamma e) \in\!\downarrow r(I) \subseteq J$. The map still retains γe; it is not obtained by choosing a witness from the target feasibility image. Observation transport is the equation $\chi_{J_\Sigma r(I)}(u) = \chi_I(u \circ r)$ of Proposition 2.18, and implemented name maps for γ are supplied by the finite compiler descriptions above. The budget-name operation is a separate constructed implementation, as follows.

Proposition 4.24 (Named transport of admitted budgets)**.** *The observed map $F_\Sigma r$ has the admitted-point map $J_\Sigma r$ and, on the projective name presentations of Proposition E.1, an implementation*

$$\widehat{J_\Sigma r} : C_L \to C_{L'}, \qquad q_{L'}\widehat{J_\Sigma r} = J_\Sigma r\, q_L. \tag{4.38}$$

For a family of monotone r, this construction is uniform after retaining a projective name for that family parameter. The compiler code for γ and this budget tracker jointly present the action on named implementations and admitted budgets.

Proof.

(1) **Construct the observation-preserving budget map.** Put $R = \mathrm{Mon}(L, L')$. The underlying uniform map

$$R \times J_\Sigma L \longrightarrow J_\Sigma L', \qquad (r, I) \longmapsto J_\Sigma r(I)$$

is constructed by the lower-image formula. Its observation equation is

$$\chi_{J_\Sigma r(I)}(u) = \chi_I(u \circ r),$$

uniformly in r, I, u. This is the observation-preserving comparison required by Proposition E.1.

(2) **Lift budget transport to retained names.** Take its constructed projective covers $C_R \twoheadrightarrow R$, $q_L : C_L \twoheadrightarrow J_\Sigma L$, and $q_{L'} : C_{L'} \twoheadrightarrow J_\Sigma L'$. The product of the displayed partitioned-assembly presentations is projective. Pull back $q_{L'}$ along the preceding map on $C_R \times C_L$ and use Proposition E.1. It gives a tracked lift $C_R \times C_L \to C_{L'}$, whose commutative square is (4.38) with the parameter name retained.

(3) **Verify parameter substitution on refinements.** For a dependent parameter pullback, take a partitioned refinement of the actual input presentation before extracting the tracker. The same equation holds on that refinement; comparisons of choices use their common projective refinement from Proposition E.1.

(4) **Verify coherence after name realization.** Identity and sequential composition of these retained name maps give implementations of $J_\Sigma\,\mathrm{id}$ and $J_\Sigma(sr)$; their squares follow by composition and functoriality of J_Σ. Their positive observations compose by substitution of tests. Different lifts can return different names for the same budget, so coherence is asserted after the displayed name-realization maps, not as equality of unrelated trackers. Together with the finite code-output tracker for γ, these are the two components of the named compiler action.

□

Neither projective lifting nor the observation equation assigns the budget tracker's runtime. A runtime comparison for that operation is obtained by retaining and measuring its execution description. Likewise, the semantic bound parameter r is supplied data of a comparison; it is not computed merely by executing the program transformation γ.

4.14 Measured changes of input representation

Keep X_0, Y_0 and the environment fixed. Use two input presentations $q_X : C_X \twoheadrightarrow X_0$ and $q'_X : C'_X \twoheadrightarrow X_0$. Form the object of retained conversion descriptions whose total returned-name map $r : C'_X \to C_X$ satisfies $q_X r = q'_X$. Its defining predicates are formed as subobjects, just as for CompCode; its generic element carries its actual history H_r and selected measurement. For resource transport, take the subobject on which the two supplied proper-filter contexts make r a filtered-input map of Section 3.9. This retains the eventual-substitution witness needed to form $r^* : K_X \to K_{C'_X}$.

Proposition 4.25 (Conversion overhead from retained histories)**.** *Sequential composition constructs an implementation e^r on C'_X with the same extensional denotation as e. Its domain is the pullback of the original domain along r. In the retained writer interpretation its history and additive-time profile are*

$$H_{e^r}(c') = H_r(c') +\!\!+ H_e(r(c')), \qquad T_{e^r} = T_r + T_e \circ r. \tag{4.39}$$

On the total-implementation subobject, its exact germ is $\Phi(e^r) = h + r^*\Phi(e)$, where $h = [T_r]$. If $\varphi_r : L_X \to L_{C'_X}$ is the induced growth map, the monotone budget transformation

$$a \longmapsto [h] \vee \varphi_r(a)$$

constructs the corresponding measured compiler comparison.

Proof.

(1) **Identify the composite domain and output.** The code-composition constructor first runs the converter and then e. Totality of the converter and the bind domain equation give the stated domain. The two value-adequacy squares and $q_X r = q'_X$ identify the extensional output.

(2) **Compute the exact history and cost.** Writer bind concatenates the two histories, proving the first equation; the additive fold proves the second. Passing to germs gives the exact equality.

(3) **Derive the growth bound and its retained witness.** For nonnegative profiles, the growth class of a sum is the join of their classes. Thus the cost of e^r has class $[h] \vee \varphi_r(\kappa(e))$, and monotonicity gives the required compiler bound at every larger budget. The conversion and composition descriptions are retained witnesses of its implementation.

□

For a different history valuation the formula is $\nu(H_r +\!\!+ H_e \circ r)$; no additive identity is inferred. In a numerical compilation every actual dispatch and simulation instruction belongs to the compiled history. Its bound is computed from that full history, including any overhead of realizing the writer-level composition. Opposite representation conversions give opposite measured comparisons with their own overheads. This supplies the quantitative comparison between presentations; equality of their unmeasured denotations alone does not identify their complexity profiles.

The measured span now supplies a witness sheaf and a feasible open for each represented problem, together with the action of implemented compilers. The next section studies the feasible open itself. Its covering properties will say whether one implementation, or a finite portfolio, accounts for all feasible budgets.

5 Feasibility moduli and optimality

A problem determines an open of feasible budgets. We first classify these open-valued specifications and then compare their covers with the implementation witnesses that generate them. One witness accounting for the whole open is an optimal implementation; finitely many such witnesses form a dominating portfolio. The same construction applies to joint resources, provided every constraint is imposed on one shared implementation.

5.1 The classifying locale of feasibility specifications

The environment parameter remains fixed throughout this section: V, P_v mean $V_\theta, P_{\theta,v}$. Its classifying maps and optimality statements are families over that parameter, with the substitution comparisons of Proposition 4.17. Let $\mathbb{S}$ denote the Sierpiński locale, which classifies opens, and let $K_c(\mathfrak{C}_{\mathrm{Ev}})$ be the internal lattice of compact opens. Its elements are precisely finite unions of principal opens: compactness extracts a finite subcover from the principal decomposition, and finite unions of compact opens are compact. Finite unions and intersections make $K_c(\mathfrak{C}_{\mathrm{Ev}})$ a bounded distributive lattice.

Theorem 5.1 (Feasibility specifications are classified)**.** *The locale $\mathfrak{F}$ with frame $\mathcal{O}(\mathfrak{F}) = \mathrm{Up}(K_c(\mathfrak{C}_{\mathrm{Ev}}))$ has the natural universal property*

$$\mathrm{Loc}_{\mathcal{E}}(Z, \mathfrak{F}) \cong \mathcal{O}(Z \times \mathfrak{C}_{\mathrm{Ev}}). \tag{5.1}$$

Thus it constructs the exponential $\mathbb{S}^{\mathfrak{C}_{\mathrm{Ev}}}$. Its universal open is

$$\mathscr{U} = \bigvee_{a \in L} V_{U_a} \times U_a \ \in \mathcal{O}(\mathfrak{F} \times \mathfrak{C}_{\mathrm{Ev}}), \qquad V_k = \{\ell \in K_c(\mathfrak{C}_{\mathrm{Ev}}) : k \subseteq \ell\}.$$

The measured incidence determines a canonical map $c : \mathrm{Disc}(V) \to \mathfrak{F}$ classifying $v \mapsto \mathcal{U}_v$.

Proof.

(1) **Describe opens of the product by monotone functions.** For $H = \mathcal{O}(Z)$, the frame of the product $Z \times \mathfrak{C}_{\mathrm{Ev}}$ is naturally the frame of monotone maps $w : L \to H$. Indeed a rectangle $z \times U_a$ corresponds to the function $b \mapsto \bigvee_{a \le b} z$, and every monotone function is $\bigvee_a w(a) \times U_a$.

(2) **Verify the frame-coproduct universal property.** For frame maps $f : H \to H'$ and $h : \mathrm{Up}(L) \to H'$, the unique extension is $w \mapsto \bigvee_a f(w(a)) \wedge h(U_a)$. Finite meets are preserved using $U_a \cap U_b = U_{a \vee b}$, and joins and top are preserved by principal decomposition. This proves the frame-coproduct universal property, including the inverse formulas.

(3) **Pass from classifying ideals to monotone functions.** By Theorem 3.1, a map $Z \to \mathfrak{F}$ is an H-valued ideal j on $K_c(\mathfrak{C}_{\mathrm{Ev}})$. It determines $w(a) = j(U_a)$.

(4) **Construct the inverse ideal and verify independence.** Conversely, for monotone w, put

$$j\left(\bigcup_{i=1}^{n} U_{a_i}\right) = \bigwedge_{i=1}^{n} w(a_i).$$

This is independent of the finite presentation: if $U_a \subseteq \bigcup_i U_{b_i}$, supercompactness gives internally an i with $b_i \le a$, hence $\bigwedge_i w(b_i) \le w(a)$. Applying this to each generator proves order reversal and independence; the empty union has value $\top$, and unions become meets.

(5) **Identify the universal open and the classification map.** The constructions are inverse and natural in H. Their composition proves (5.1), and sends the universal ideal to the displayed open $\mathscr{U}$. Apply it to the open relation (4.22) over $\mathrm{Disc}(V)$ to obtain c.

□

Let $\chi_{\mathscr{F}}$ classify the universal measured feasibility family on $\mathrm{Disc}(V) \times \mathfrak{C}_{\mathrm{Ev}}$. The exponential transpose c is characterized by

$$\begin{array}{ccc} \mathrm{Disc}(V) \times \mathfrak{C}_{\mathrm{Ev}} & \xrightarrow{c \times \mathrm{id}} & \mathfrak{F} \times \mathfrak{C}_{\mathrm{Ev}} \\ & \searrow^{\chi_{\mathscr{F}}} & \downarrow{\scriptstyle \mathrm{ev}} \\ & & \mathbb{S}. \end{array}$$

Here ev classifies the universal open $\mathscr{U}$. This triangle relates the problem-indexed observable to the representing locale of feasibility specifications.

The point $c(v)$ records the feasible specification. Its measured instance also carries the epimorphism $\mathscr{W}_v \twoheadrightarrow \mathcal{U}_v$, which records the implementations. Selecting a section of that epimorphism is a further question. Effective names and positive observations retain the structures already constructed; the classification alone assigns neither to an arbitrary point of $\mathfrak{F}$. The principal opens contributed by implementations cover the feasible open. If one member covers the whole, its implementation is optimal. If a finite subfamily covers it, those implementations suffice as a portfolio. Supercompactness and compactness express these two possibilities intrinsically.

5.2 Optimization and topological compactness

The implementation comparison is with abstract complexity measures (Blum 1967); finite portfolios have also been studied as algorithm-selection objects (Gomes and Selman 2001). Here the relevant properties will be expressed by compactness of a feasibility open. Fix

a denotation $v : V$ and write $P_v = \{e : P : p(e) = v\}$. The universal cost open of Theorem 4.14 is $\mathcal{U}_v = \bigvee_{e \in P_v} U_{\kappa(e)}$. We use the internal frame definitions: an open W is *supercompact* if $W \leq \bigvee_{i:J} W_i$ implies $\exists i : J \ (W \leq W_i)$, and *compact* if it implies $\exists n \in \mathbb{N}\, \exists i_1, \ldots, i_n : J \ W \leq \bigvee_{k=1}^{n} W_{i_k}$. The empty family is included in these definitions.

Theorem 5.2 (Optimal algorithms and finite portfolios)**.** *The classified feasibility open satisfies*

$$\mathcal{U}_v \text{ is supercompact} \iff \exists e_0 \in P_v \ \forall e \in P_v \ \kappa(e_0) \leq \kappa(e). \tag{5.2}$$

It is compact precisely when a finite asymptotically dominating portfolio exists:

$$\begin{aligned} \mathcal{U}_v \text{ is compact} \iff \exists n \in \mathbb{N}\, \exists e_1, \ldots, e_n \in P_v \ \forall e \in P_v \\ \exists k \in \{1, \ldots, n\} \ \kappa(e_k) \leq \kappa(e). \end{aligned} \tag{5.3}$$

In particular the empty portfolio accounts for the empty feasibility open. Both equivalences are internal and stable under substitution of denotation parameters.

Proof.

(1) **Reflect growth order through principal opens.** The principal opens satisfy

$$U_b \leq U_a \iff a \leq b. \tag{5.4}$$

For the forward implication evaluate inclusion at $\downarrow b$; the reverse implication is upperness.

(2) **Deduce supercompactness from an optimal implementation.** If an e_0 as in (5.2) exists, (5.4) makes every $U_{\kappa(e)}$ lie below $U_{\kappa(e_0)}$, so $\mathcal{U}_v = U_{\kappa(e_0)}$ is supercompact.

(3) **Extract an optimal implementation.** Conversely, apply supercompactness to the canonical cover $\mathcal{U}_v = \bigvee_{e \in P_v} U_{\kappa(e)}$. It supplies e_0 with $\mathcal{U}_v \leq U_{\kappa(e_0)}$. For every e, the principal-open reflection then yields $\kappa(e_0) \leq \kappa(e)$.

(4) **Extract a finite dominating portfolio.** Compactness of $\mathcal{U}_v$ applied to the same cover produces a finite list $e_1, \ldots, e_n$ with $\mathcal{U}_v \leq \bigvee_k U_{\kappa(e_k)}$. For any e, supercompactness of $U_{\kappa(e)}$ extracts one k with $U_{\kappa(e)} \leq U_{\kappa(e_k)}$; the reflection equation gives the displayed domination.

(5) **Deduce compactness from a portfolio.** Conversely, a finite dominating portfolio makes $\mathcal{U}_v = \bigvee_k U_{\kappa(e_k)}$, a finite union of compact principal opens. All arguments use finite lists and internal existential elimination.

□

Theorem 5.3 (Optimal implementations are sections)**.** *For $v : V$, there is a natural isomorphism of internal $\mathcal{E}$-objects*

$$\mathrm{Sect}(\mathscr{W}_v \to \mathcal{U}_v) \cong \{e_0 \in P_v : \forall e \in P_v, \ \kappa(e_0) \leq \kappa(e)\}. \tag{5.5}$$

Here Sect *represents sections in all parameter contexts, equivalently $p_{\mathfrak{C}_{\mathrm{Ev}^*}}(\mathscr{W}_v^{\mathcal{U}_v})$. Consequently the witness epimorphism splits internally exactly when $\mathcal{U}_v$ is supercompact.*

Proof.

(1) **Show that a section selects one common implementation.** In Theorem 3.4, a section assigns a feasible implementation e_a to every feasible budget a, with $e_a = e_b$ when $a \leq b$. Any two feasible budgets have the feasible common upper bound $a \vee b$; hence all selected implementations agree. Since $v \in V = \mathrm{im}(p_0)$, the fibre P_v is internally inhabited. Thus there is a unique common implementation e_0; unique existence descends through the inhabited parameter cover.

(2) **Prove optimality of the selected implementation.** At $a = \kappa(e)$ feasibility gives $\kappa(e_0) \le \kappa(e)$.

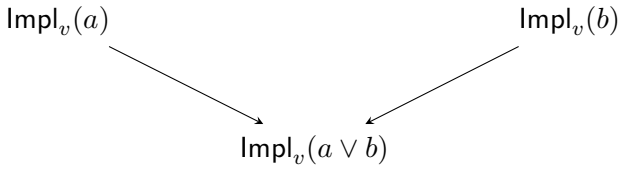
(3) **Construct the inverse section and verify naturality.** Conversely such an e_0 is available at every feasible budget and gives a constant compatible section. The two operations are inverse and commute with parameters. The final assertion is Theorem 5.2.

□

An optimal implementation is therefore exactly a splitting

$$\mathscr{W}_v \underset{s}{\overset{\pi}{\rightleftarrows}} \mathcal{U}_v \qquad \pi s = \mathrm{id}_{\mathcal{U}_v}\,.$$

A section makes compatible choices at all feasible budgets. In the diagram presentation, the common upper budget compares the choices made at a and b:



Both transition maps retain the implementation code, so compatibility forces those choices to agree. This is why a section supplies one optimal witness throughout the feasible open.

The optimality criterion can also be evaluated after allowing chosen overheads. Pulling the witness family to the gauge sublocale gives the same questions relative to the localized notion of resource.

5.3 Gauge-relative feasibility

Corollary 5.4 (Soft optimality and portfolios)**.** *Let $\mathcal{U}_v^{(P)}$ be the restriction of the measured feasibility open to $\mathfrak{C}_{\mathrm{Ev}}^{(P)}$. Then*

$$\mathcal{U}_v^{(P)} = \bigvee_{e \in P_v} U^{(P)}_{q_P(\kappa(e))}.$$

It is supercompact precisely when an implementation e_0 satisfies $\kappa(e_0) \preceq_P \kappa(e)$ for all $e \in P_v$. It is compact precisely when finitely many implementations dominate every implementation in this gauge-relative preorder. For $P = P_\lambda$, this is optimality and finite-portfolio completeness for soft-O comparison.

Proof.

(1) **Compute feasibility in the localized growth order.** Under (3.16), membership of a cost in a saturated budget is membership of its q_P-class in the corresponding ideal of L/P. This gives the displayed open formula.

(2) **Characterize soft optimality and finite portfolios.** Principal opens of L/P are supercompact; applying the proof of Theorem 5.2 to that principal cover proves both equivalences. The bitlength gauge identifies $\preceq_{P_\lambda}$ with soft domination by Section 2.7.

□

5.4 Joint-resource moduli and Pareto portfolios

Separate resource bounds may be witnessed by different programs. Joint feasibility instead requires one program meeting every bound. Let L_T, L_S be growth lattices for time and space measurements of the same implementation object, with classifying locales $\mathfrak{C}_T, \mathfrak{C}_S$. The product construction below retains that common implementation while varying both budgets.

Theorem 5.5 (Product moduli and joint feasibility)**.** *There is a canonical frame isomorphism*

$$\mathcal{O}(\mathfrak{C}_T \times \mathfrak{C}_S) \cong \mathrm{Up}(L_T \times L_S). \tag{5.6}$$

Each principal rectangle $U_a^T \times U_b^S$ is supercompact. For joint measured costs $\kappa_T : P_v \to L_T$, $\kappa_S : P_v \to L_S$, the simultaneous feasibility open is

$$\mathcal{U}_v^{T,S} = \bigvee_{e \in P_v} (U^T_{\kappa_T(e)} \times U^S_{\kappa_S(e)}). \tag{5.7}$$

It is supercompact exactly when one implementation is simultaneously asymptotically optimal in both costs. It is compact exactly when finitely many implementations jointly dominate every implementation: for each e there is a member e_i of the finite list satisfying

$$\kappa_T(e_i) \le \kappa_T(e), \qquad \kappa_S(e_i) \le \kappa_S(e).$$

The witness sheaf is the coproduct of the principal rectangles in (5.7); hence both budget tests refer to the same implementation.

Proof.

(1) **Construct the frame map on principal rectangles.** Every upper predicate on $L_T \times L_S$ is the union of principal rectangles. Given frame maps $h_T : \mathrm{Up}(L_T) \to H$ and $h_S : \mathrm{Up}(L_S) \to H$, define

$$h(W) = \bigvee_{(a,b) \in W} (h_T(U_a^T) \wedge h_S(U_b^S)).$$

(2) **Verify the product universal property.** The identity $(U_a^T \times U_b^S) \wedge (U_c^T \times U_d^S) = U^T_{a \vee c} \times U^S_{b \vee d}$ and frame distributivity prove that h preserves finite meets; principal decomposition proves join preservation. Restriction along the two projections gives the inverse bijection. This is the coproduct universal property of the frame of the localic product, proving (5.6).

(3) **Prove supercompactness of principal rectangles.** Evaluating a covering family at (a, b) proves supercompactness of each principal rectangle as in Theorem 3.1.

(4) **Construct joint feasibility and its witness sheaf.** The simultaneous incidence is the pullback of the two membership predicates along the diagonal implementation map, followed by the regular image to the budget product. The image is exactly (5.7); taking its implementation fibre gives the stated witness sheaf.

(5) **Characterize joint optimality and portfolios.** The supercompactness and compactness proofs now repeat Theorem 5.2 with the product order and principal rectangles. Both comparisons are coordinatewise, so they yield exactly the two displayed cost inequalities.

□

In particular, intersecting the two separately existential feasibility predicates may use different implementations. Equation (5.7) is the geometric construction of simultaneous time–space feasibility.

The section characterization of Theorem 5.3 also applies to the product budget order: a section of the joint witness sheaf is exactly one implementation optimal in every displayed cost. Its proof uses the componentwise join as the common upper budget.

5.5 A compact two-resource portfolio

The next example calculates a compact feasible open for which neither implementation is optimal in both resources. We use a selected two-description presentation and two event-count valuations, so the entire implementation fibre can be inspected. Work on $X = \mathbb{N}$ with its cofinite filter, and fix two implemented unit-valued idle primitives with distinct tags a, b and identical extensional effect. Define the product valuation by

$$w(\mathsf{a}) = (1,0), \qquad w(\mathsf{b}) = (0,1), \qquad w([]) = (0,0),$$

and extend it to words by componentwise addition. These are two points of the history-weight object of Proposition 4.7. They count the selected events; total machine time and peak-space occupancy are separate measurements.

Finite NNO iteration and multiplication construct two fixed total descriptions e_A, e_B with the same output, namely the identity function $v(n) = n$, and respective logged histories

$$H_{e_A}(n) = \mathsf{a}^n\,\mathsf{b}^{n^2}, \qquad H_{e_B}(n) = \mathsf{a}^{n^2}\,\mathsf{b}^n. \tag{5.8}$$

The exponents here are finite repetition counts implemented by the retained arithmetic and iteration constructors. In each case the program returns its unchanged input after the logged calls. The history fold therefore gives *exact* raw cost pairs

$$\big(T_{1,A}(n), T_{2,A}(n)\big) = (n, n^2), \qquad \big(T_{1,B}(n), T_{2,B}(n)\big) = (n^2, n).$$

Put $a = [n+1]$ and $b = [(n+1)^2]$ in the numerical growth lattice. Then $a < b$: the inequality $a \le b$ holds, while $b \not\le a$ is witnessed by arbitrarily large numerical inputs. The growth pairs of the two implementations are (a, b) and (b, a), which are incomparable in the coordinatewise order.

Take the two-description subpresentation $E_{\mathrm{port}} = \{e_A, e_B\}$. Its evaluator is the copairing of the two total evaluations, its denotation image contains just v, and its implementation object consists of the two displayed members. The measured-image and incidence constructions therefore give its complete joint feasible open:

$$\mathcal{U}_v^{\mathrm{port}} = (U_a^{(1)} \times U_b^{(2)}) \;\vee\; (U_b^{(1)} \times U_a^{(2)}) \quad \in \mathcal{O}(\mathfrak{C}_{\mathrm{Ev}}^{(1)} \times \mathfrak{C}_{\mathrm{Ev}}^{(2)}). \tag{5.9}$$

The budget points $(\downarrow a, \downarrow b)$ and $(\downarrow b, \downarrow a)$ are feasible, whereas $(\downarrow a, \downarrow a)$ is not feasible in this presentation. Each program meets one of the smaller bounds, but neither meets both. Choosing different witnesses for the two resources would therefore give the wrong joint judgment.

For clarity, put $I_a = {\downarrow}\, a$ and $I_b = {\downarrow}\, b$. Direct evaluation of the two principal rectangles gives the following complete table for the displayed portfolio:

joint budget	e_A feasible	e_B feasible	some implementation feasible
(I_a, I_a)	no	no	no
(I_a, I_b)	yes	no	yes
(I_b, I_a)	no	yes	yes
(I_b, I_b)	yes	yes	yes

Each entry tests both resource components of one of the two measured cost pairs. The table evaluates the complete open of this finite presentation.

The open in (5.9) is compact, being a union of two supercompact principal rectangles. It is not supercompact. Indeed, if one of the two rectangles contained the union, it would contain the other rectangle. The budget point (I_b, I_a) belongs to the second rectangle but not the first, while (I_a, I_b) belongs to the first but not the second. Thus the displayed two-rectangle cover has no single covering member. By Theorems 5.2 and 5.5, the two codes form a finite dominating portfolio, whereas neither code is simultaneously optimal in both measurements *within this selected subpresentation*. This computes both sides of the compactness criterion on an implemented instance.

In the full description language, this calculated open is a subopen of $\mathcal{U}_v^{(1,2)}$. Other implementations can enlarge it: an uncharged pure identity description makes the full open top for these idle-event counters. The example thus establishes the portfolio criterion for its specified two-description fibre. The product theorem also applies to the time and occupancy measurements of Construction 12.1; identifying a particular frontier there additionally requires the corresponding universal lower-bound analysis.

The classifier now describes budget families, their feasible opens, and their implementation witnesses. Keeping a measured history fixed and varying its valuation gives the resource families studied next.

6 Differential calculus of resource constraints

The measured span separates a program's denotation from its resource valuation. Keeping its execution history fixed while varying event weights produces a family of resource values. We begin with this family and its budget equation, then construct the finite coefficient calculus that differentiates them.

6.1 Measured valuation families and budget variations

For a fixed total implementation e with finitely many measured event types, let $N_{e,i}$ be its exact event-count profiles. Additive rational weights give the measured family

$$F_e(w) = \sum_i N_{e,i} w_i.$$

This follows by folding the returning history (Theorem 4.6). Passing to germs and extending coefficients gives the B-linear map $F_e : B^k \to K$ with the same formula. A polynomial parametrization $w = w(z)$ therefore gives a polynomial resource law. Its finitely many coefficients lie in the principal resource generated by the sum of their absolute values, so it is calculated on a finite-projective resource cell.

For an embedded budget $M \subseteq K$, put $q_M : K \to K/M$. Feasibility is the equation $q_M F_e(z) = 0$. Over the dual numbers, polynomial expansion gives

$$q_M F_e(z + \varepsilon v) = q_M F_e(z) + \varepsilon q_M DF_{e,z}(v).$$

At a feasible point, the fixed-budget tangent condition is therefore $q_M DF_{e,z}(v) = 0$. If a finite-projective budget also varies, its tangent class is $\theta \in \mathrm{Hom}_B(M, K/M)$ and the condition becomes

$$q_M DF_{e,z}(v) = \theta(F_e(z)).$$

Theorems 7.2 and 7.6 construct the budget tangent and prove this equation by comparing coefficients.

Theorem 6.1 (Measured Taylor coefficients and geometric jets)**.** *Let H_e be the returning history of a retained implementation, with finitely many measured event types and polynomial weights $w_i(z) \in \mathbb{Q}[z_1, \dots, z_m]$. Write*

$$\tau_r : \mathbb{Q}[z] \longrightarrow \mathbb{Q}[z, \varepsilon_1, \dots, \varepsilon_m]/(\varepsilon_1, \dots, \varepsilon_m)^{r+1}, \qquad p(z) \longmapsto p(z + \varepsilon).$$

Folding H_e with the weights $\tau_r w_i$ gives the Taylor coefficients of its measured cost through order r:

$$\operatorname{fold}(\tau_r w, H_e) = \tau_r \operatorname{fold}(w, H_e). \tag{6.1}$$

The two executions have the same return domain and returned value. For a total implementation on the retained input carrier, taking exact germs identifies these coefficients with the order-r geometric jet of $F_e(z) = \sum_i N_{e,i} w_i(z)$. The comparisons commute with truncation, polynomial parameter substitution, and the retained environment maps. Their finite coefficient operations have uniform implementations in the order and the displayed polynomial data.

Proof.

(1) **Compare Taylor substitution with finite history folds.** The map τ_r is the algebra homomorphism specified by its displayed values on generators. In particular it preserves the additive monoid of weights. On a returning word $[a_1, \dots, a_k]$, both sides of (6.1) are $\sum_{j=1}^{k} w_{a_j}(z + \varepsilon)$ modulo total degree $r + 1$. Induction on word length proves the equality.

(2) **Extend the comparison to iteration and erasure.** The finite-return components of Theorem 4.4 have this same word fold; their common classified union proves the equality for iteration. Valuation changes only the resource component, so erasure and its domain agree by Theorems 4.3 and 4.10.

(3) **Descend the coefficients to germs.** At every actual input the equality is coefficientwise in exact rational polynomials. The germ quotient preserves the finite sums and products, hence carries it to K.

(4) **Identify the geometric jet on a principal cell.** The finitely many coefficients of F_e lie in the principal cell generated by the sum of their absolute values. The polynomial realization and principal-parts comparisons of Theorems 6.20 and 11.3 and Proposition 11.5 identify substitution by $z + \varepsilon$ on that cell with its geometric jet. Thus the geometric coefficients are the measured coefficients.

(5) **Verify substitution and truncation coherence.** Truncation deletes higher monomials, and polynomial substitution commutes with the displayed algebra homomorphisms. The history substitution theorem gives the environment comparison.

(6) **Construct uniform coefficient operations.** Monomial enumeration, rational arithmetic, and finite list folds construct the coefficient operations by NNO recursion; the retained arithmetic name maps therefore implement them uniformly. The Taylor coefficients of positive cost valuations may be signed, and are interpreted by the exact signed coefficient construction.

□

The theorem supplies the execution interpretation of the differential calculus: finite response coefficients are obtained by instrumenting the same returning history. Its compatible truncations will give the measured formal families used in the completion and crystalline comparisons.

Example 6.2 (A measured polynomial resource family)**.** Work on $\mathbb{N}$ with the cofinite filter and the unit-valued idle events a, b of Section 5.5. Finite iteration constructs a fixed total program that emits one a and then $(n+1)^2$ copies of b, and returns its input. Assign weights 1 and λ^2, respectively. The history fold gives the exact nonnegative rational cost

$$T_\lambda(n) = 1 + (n+1)^2\lambda^2 \qquad (\lambda \in \mathbb{Q}).$$

Writing $t(n) = 1/(n+1)$, its polynomial coefficient extension is

$$F : B \longrightarrow K, \qquad F(\lambda) = 1 + t^{-2}\lambda^2. \tag{6.2}$$

The base value $F(0) = 1$ belongs to B. At first order, $F(d) = 1$ in $K[d]/d^2$. At second order the additional coefficient is t^{-2}, whose class in K/B is nonzero. We will use this same family to calculate fixed, moving, and coupled-budget obstructions in Examples 7.10, 7.16 and 7.19.

Free commutative algebras represent the polynomial expressions above. Their universal derivatives compare the exact, quantitative, and signed interpretations. Based Weil stages then represent finite coefficient variations, and module realization carries the budget equation into the differential geometry.

6.2 The differential extension and its comparison maps

6.2.1 Free continuous commutative algebra

Theorem 6.3 (Free continuous commutative algebra)**.** *The forgetful functor* $\mathrm{CAlg}(\mathsf{Lin}_S^{J_\Sigma}) \to \mathsf{Lin}_S^{J_\Sigma}$ *has a constructed left adjoint. Its underlying monad* $T = \mathrm{Sym}_S^{J_\Sigma}$ *is an algebra modality on* $\mathsf{Lin}_S^{J_\Sigma}$.

Proof.

(1) **Present the finite polynomial algebra.** For $M \in \mathsf{Lin}_S^{J_\Sigma}$, begin with finite commutative polynomial expressions over S in symbols $[m]$, imposing

$$[m + m'] = [m] + [m'], \qquad [sm] = s[m], \qquad [0] = 0,$$

and the commutative unital algebra equations. Give this algebra the least compatible preorder incorporating the orders of S and M.

(2) **Construct the continuous algebra.** Impose covers

$$[\sup I] \triangleleft J_\Sigma[-](I) \quad (I \in J_\Sigma M), \qquad \sup K \triangleleft J_\Sigma \iota_S(K) \quad (K \in J_\Sigma S),$$

where ι_S inserts scalar constants. Close the covers under finite one-hole algebra contexts. Theorem 1.8 and Lemma A.4 construct a continuous algebra TM. Scalar contexts give joint continuity of scalar action, and multiplication contexts give continuous multiplication. Equations extend from generators by product generation.

(3) **Prove the free-algebra universal property.** A continuous linear map $M \to A$ into a continuous commutative algebra extends algebraically to the finite expressions. It respects every cover by continuity, including contextual covers. It therefore extends uniquely to a continuous algebra homomorphism $TM \to A$. This proves the free-algebra universal property.

(4) **Construct the algebra modality and verify its laws.** The unit η_M inserts a linear generator; $\mu_M : TTM \to TM$ substitutes polynomials. The monad laws follow because both sides have the same action on free generators. The multiplication $m_M : TM \otimes TM \to TM$ and unit $u_M : S \to TM$ are natural, and μ_M is an algebra map. Thus this is an algebra modality.

□

The same finite-expression construction, without admitted covers, gives $T_+ = \mathrm{Sym}_{\mathbb{Q}_{\geq 0}}$ on $\mathscr{A}_+$ and $T_{\mathbb{Q}} = \mathrm{Sym}_{\mathbb{Q}}$ on $\mathscr{A}_{\pm}$.

6.2.2 Universal derivative and its identities

Finite products in $\mathsf{Lin}_S^{J_\Sigma}$ are also biproducts: $(m, n) = (m, 0) + (0, n)$, and all structure maps are continuous linear. For a continuous algebra A and a continuous A-module N, form the square-zero extension

$$A \ltimes N = A \oplus N, \qquad (a, n)(b, n') = (ab, an' + bn), \qquad 1 = (1, 0). \tag{6.3}$$

The finite equations prove associativity and commutativity; its operations are continuous because the module action and addition are continuous.

Theorem 6.4 (Constructed deriving transformation)**.** *There is a natural continuous linear map*

$$d_M : TM \longrightarrow TM \otimes_S M$$

which is the universal continuous S-derivation of the free algebra TM. It satisfies the constant, Leibniz, linear, monadic chain, and interchange identities.

Proof.

(1) **Construct the universal derivation by a square-zero extension.** Regard $TM \otimes M$ as the free TM-module on M. The continuous linear map

$$M \longrightarrow TM \ltimes (TM \otimes M), \qquad m \longmapsto (\eta_M m, 1 \otimes m)$$

extends uniquely to an algebra map. Its first component is id_{TM} by freeness. Call the second component d_M. Equation (6.3) gives

$$\begin{aligned} d_M(1) &= 0, \qquad & d_M(ab) &= a\, d_M b + b\, d_M a, \\ d_M(\eta_M m) &= 1 \otimes m. \end{aligned} \tag{6.4}$$

(2) **Prove the representing property.** More generally, a continuous linear $v : M \to N$ into any TM-module defines an algebra map with generator $(\eta m, v(m))$. This proves

$$\mathrm{Der}_S^{J_\Sigma}(TM, N) \cong \mathsf{Lin}_S^{J_\Sigma}(M, N) \cong \mathrm{Mod}_{TM}(TM \otimes M, N),$$

so d_M is universal.

(3) **Verify naturality.** For $f : M \to N$, both $(Tf \otimes f)d_M$ and $d_N Tf$ are derivations along Tf and send a generator m to $1 \otimes f(m)$. Universality proves naturality.

(4) **Prove the monadic chain identity.** Use composition in the usual right-to-left order. The chain identity is

$$d_M \mu_M = (m_M \otimes 1_M)(\mu_M \otimes d_M) d_{TM}. \tag{6.5}$$

Both sides are derivations from the free algebra TTM, with target module structure through μ_M. At its generator $p \in TM$, the left side is $d_M p$; the right side is also $d_M p$. The universal property therefore proves (6.5).

(5) **Prove interchange on monomials.** On a finite monomial,

$$d_M(m_1 \cdots m_n) = \sum_i \left(\prod_{j \neq i} m_j \right) \otimes m_i. \tag{6.6}$$

Applying d_M again to its first factor sums over ordered pairs of distinct positions. Interchanging the two selected positions permutes this finite sum. Hence $(d_M \otimes 1_M)d_M$ is invariant under switching its final two factors.

(6) **Extend the identities to the presented domain.** Both sides are continuous linear maps, so the identity extends from finite expressions to the entire presented domain. This is the interchange identity. The other displayed identities were already proved by square-zero multiplication and hold on completed elements as well.

□

These universal properties and identities establish a codifferential category in the terminology of (Blute et al. 2016, 2020). In the admitted setting, its free continuous algebra and derivative have been constructed explicitly, so the later realization comparisons can be checked on their generators.

6.2.3 Universal differentials of a general algebra

Proposition 6.5 (Universal continuous differentials)**.** *Every continuous commutative algebra A has a constructed module $\Omega^{1,J_\Sigma}_{A/S}$ representing continuous S-derivations. For a continuous A-module N,*

$$\mathrm{Mod}_A(\Omega^{1,J_\Sigma}_{A/S}, N) \cong \mathrm{Der}^{J_\Sigma}_S(A, N) \cong \mathrm{CAlg}(\mathsf{Lin}^{J_\Sigma}_S)_{/A}(A, A \ltimes N).$$

Proof.

(1) **Present the module of continuous differentials.** Start with the free continuous A-module $A \otimes_S A$ on the underlying S-module of A, writing $da = 1 \otimes a$. Impose the module congruence

$$d1 = 0, \qquad d(ab) = a\,db + b\,da.$$

Construct its continuous quotient by the admitted-presentation theorem, using equations as inequalities in both directions and imposing the original module's continuity covers.

(2) **Prove the universal bijections.** Maps from this quotient are exactly continuous S-linear maps satisfying the displayed relations. This is the first bijection. Equation (6.3) proves the second.

(3) **Construct relative differentials.** Relative differentials Ω_{A/A_0} are constructed by additionally imposing $d(a_0) = 0$ for the specified coefficient map $A_0 \to A$.

□

The algebraic versions over $\mathbb{Q}_{\geq 0}$ and $\mathbb{Q}$ use the same generators and relations. One must use congruences, not subtract generators, on the semiring side.

6.2.4 The differential comparisons commute

Theorem 6.6 (Comparison of modalities and derivations)**.** *There are natural coherent isomorphisms*

$$\begin{aligned} \mathsf{E}_S(T_+M) &\cong T(\mathsf{E}_S M), \\ \mathrm{Gr}(T_+M) &\cong T_{\mathbb{Q}}(\mathrm{Gr}\, M). \end{aligned} \tag{6.7}$$

They commute with algebra multiplication, monad unit and multiplication, and the deriving transformations. For an exact commutative $\mathbb{Q}_{\geq 0}$-algebra A there are also natural isomorphisms

$$\begin{aligned} \mathsf{E}_S\Omega^1_{A/\mathbb{Q}_{\geq 0}} &\cong \Omega^{1,J_\Sigma}_{\mathsf{E}_S A/S}, \\ \mathrm{Gr}\,\Omega^1_{A/\mathbb{Q}_{\geq 0}} &\cong \Omega^1_{\mathrm{Gr}\,A/\mathbb{Q}}, \end{aligned} \tag{6.8}$$

and corresponding relative-coefficient comparisons.

Proof.

(1) **Extend the module adjunction to algebras.** Strong monoidality equips $\mathsf{E}_S A$ with a continuous algebra structure. Write $U_+ : \mathrm{CAlg}(\mathsf{Lin}_S^{J_\Sigma}) \to \mathrm{CAlg}(\mathscr{A}_+)$ for restriction of scalars on algebras, $U_{\mathsf{Lin}_S^{J_\Sigma}} : \mathrm{CAlg}(\mathsf{Lin}_S^{J_\Sigma}) \to \mathsf{Lin}_S^{J_\Sigma}$ for forgetting the algebra operations, and $U : \mathsf{Lin}_S^{J_\Sigma} \to \mathscr{A}_+$ as above. The module adjunction restricts to an algebra adjunction: an exact algebra map $A \to U_+ N$ extends to $\mathsf{E}_S A \to N$, and its multiplication and unit equations hold on exact generators and hence by admitted generation. Conversely scalar restriction preserves them.

(2) **Compare the free-algebra modalities.** Consequently, for $N \in \mathrm{CAlg}(\mathsf{Lin}_S^{J_\Sigma})$,

$$\begin{aligned} \mathrm{CAlg}(\mathsf{Lin}_S^{J_\Sigma})(\mathsf{E}_S T_+ M, N) &\cong \mathrm{CAlg}(\mathscr{A}_+)(T_+M, U_+N) \\ &\cong \mathscr{A}_+(M, U U_{\mathsf{Lin}_S^{J_\Sigma}} N) \\ &\cong \mathsf{Lin}_S^{J_\Sigma}(\mathsf{E}_S M, U_{\mathsf{Lin}_S^{J_\Sigma}} N) \\ &\cong \mathrm{CAlg}(\mathsf{Lin}_S^{J_\Sigma})(T(\mathsf{E}_S M), N). \end{aligned}$$

This is the universal property of $T(\mathsf{E}_S M)$. The group-completion comparison has the same proof.

(3) **Verify compatibility of the modality structures.** The isomorphisms send each exact generator to the corresponding linear generator. The unit, substitution, algebra multiplication, and all their coherence diagrams therefore agree by their universal properties.

(4) **Compare the deriving transformations.** Both functors preserve finite biproducts and tensor products, and hence square-zero extensions. Under the free-algebra comparison, the image of the algebra map defining d is the algebra map sending a generator to $(\eta m, 1 \otimes m)$. Uniqueness proves compatibility with d, including on the continuous side's admitted closure.

(5) **Compare the representing modules of derivations.** For general differentials, let N be an $\mathsf{E}_S A$-module. The adjunction on module fibres gives

$$\mathrm{Mod}_{\mathsf{E}_S A}(\mathsf{E}_S \Omega^1_{A/\mathbb{Q}_{\geq 0}}, N) \cong \mathrm{Mod}_A(\Omega^1_{A/\mathbb{Q}_{\geq 0}}, UN) \cong \mathrm{Der}_{\mathbb{Q}_{\geq 0}}(A, UN).$$

In the last expression the A-action is through $A \to U\mathsf{E}_S A$. A derivation on A is an algebra map $A \to U(\mathsf{E}_S A \ltimes N)$ with prescribed first component. Its adjoint is an algebra section $\mathsf{E}_S A \to \mathsf{E}_S A \ltimes N$, and thus a continuous derivation. This proves the first comparison by representability. Group completion gives the second in the same way.

(6) **Deduce the relative comparisons.** Requiring vanishing on a coefficient algebra commutes with these correspondences, proving the relative assertions.

□

To differentiate a retained presentation, its exact and quantitative derivatives must have the same interpretation. The free-algebra comparison supplies this compatibility. With $\kappa_M : \mathsf{E}_S T_+ M \cong T\mathsf{E}_S M$, it is the square

$$\begin{array}{ccc} \mathsf{E}_S(T_+M) & \xrightarrow{\mathsf{E}_S d^+_M} & \mathsf{E}_S(T_+M \otimes_{\mathbb{Q}_{\geq 0}} M) \\ \downarrow{\scriptstyle \kappa_M} & & \downarrow{\scriptstyle \cong} \\ T(\mathsf{E}_S M) & \xrightarrow[d_{\mathsf{E}_S M}]{} & T(\mathsf{E}_S M) \otimes_S \mathsf{E}_S M. \end{array} \tag{6.9}$$

The corresponding square for Gr has the same construction:

$$\begin{array}{ccc} \mathrm{Gr}(T_+M) & \xrightarrow{\mathrm{Gr}\, d^+_M} & \mathrm{Gr}(T_+M \otimes_{\mathbb{Q}_{\geq 0}} M) \\ \downarrow{\scriptstyle \cong} & & \downarrow{\scriptstyle \cong} \\ T_{\mathbb{Q}}(\mathrm{Gr}\, M) & \xrightarrow[d_{\mathrm{Gr}\, M}]{} & T_{\mathbb{Q}}(\mathrm{Gr}\, M) \otimes_{\mathbb{Q}} \mathrm{Gr}\, M. \end{array}$$

The squares identify the derivatives under both realizations. They therefore supply the compatibility needed to define the derivative on a retained presentation.

Together these universal properties give a span preserving the differential-algebra structure. The signed leg still acts on the exact presentation, as required by Section 1.6.

6.2.5 The differential category of presentations

Theorem 6.7 (Codifferential structure on retained presentations)**.** *The category $\mathscr{P}$ in (1.5) is a commutative-monoid-enriched codifferential category with finite biproducts. Its tensor and modality are constructed componentwise using the comparisons of Theorem 6.6. Projection $\pi_{\mathsf{Lin}_S^{J_\Sigma}} : \mathscr{P} \to \mathsf{Lin}_S^{J_\Sigma}$ and signed realization $\mathrm{Gr}\, \pi_{\mathscr{A}_+} : \mathscr{P} \to \mathscr{A}_\pm$ are strong monoidal and preserve the algebra modality and deriving transformation.*

Proof.

(1) **Construct the additive monoidal structure.** An arrow $(M, N, \rho) \to (M', N', \rho')$ is a pair (f, g) with $g\rho = \rho' \mathsf{E}_S f$. Addition and zero are componentwise. Define

$$(M, N, \rho) \otimes (M', N', \rho') = (M \otimes M', N \otimes N', (\rho \otimes \rho')\phi^{-1}),$$

where $\phi : \mathsf{E}_S M \otimes \mathsf{E}_S M' \cong \mathsf{E}_S(M \otimes M')$. The unit is $(\mathbb{Q}_{\geq 0}, S, \mathrm{id})$ under the unit comparison. Coherence and bilinearity follow componentwise.

(2) **Define the modality componentwise.** Write $\kappa_M : \mathsf{E}_S T_+ M \cong T\mathsf{E}_S M$. Set

$$T_{\mathscr{P}}(M, N, \rho) = (T_+M, TN, T\rho\, \kappa_M).$$

The modality unit, substitution, multiplication, and deriving map are respectively the pairs of those maps on $\mathscr{A}_+$ and $\mathsf{Lin}_S^{J_\Sigma}$.

(3) **Verify compatibility and the differential laws.** Finite biproducts, their injections and projections, and the hom-monoid zeros are componentwise because E_S preserves them; the defining compatibility equations are stable under addition. Naturality and the differential comparison show that the structural maps satisfy the commuting-square condition for arrows of $\mathscr{P}$. Their laws hold componentwise. For example, the square for (d_M, d_N) follows by first commuting $\mathsf{E}_S d_M$ with κ_M, then using naturality of d along ρ. This proves the theorem.

(4) **Prove freeness on retained presentations.** In fact the modality is the free commutative algebra modality: the two free extensions of a compatible pair of maps remain compatible by uniqueness on generators.

□

For a presentation (M, N, ρ), the structural map of its lifted free algebra is the composite in

$$\mathsf{E}_S T_+ M \xrightarrow[\cong]{\kappa_M} T\mathsf{E}_S M \xrightarrow{T\rho} TN. \qquad \text{(}T\rho\,\kappa_M\text{)}$$

The comma-square condition for its derivative is

$$\begin{array}{ccc} \mathsf{E}_S T_+ M & \xrightarrow{T\rho\,\kappa_M} & TN \\ {\scriptstyle \mathsf{E}_S d^+_M}\downarrow & & \downarrow{\scriptstyle d_N} \\ \mathsf{E}_S(T_+ M \otimes_{\mathbb{Q}_{\geq 0}} M) & \xrightarrow[(T\rho\,\kappa_M)\otimes\rho]{} & TN \otimes_S N. \end{array}$$

The lower label uses the strong monoidal comparison to identify its source with $\mathsf{E}_S T_+ M \otimes_S \mathsf{E}_S M$. Thus the derivative is a morphism of retained presentations, and both realization legs of Equation (1.7) preserve it.

6.3 Finite-projective stages for resource geometry

6.3.1 Rational stages and finite-projective closure

A rational Weil presentation is a finite-dimensional augmented commutative $\mathbb{Q}$-algebra $W = \mathbb{Q} \oplus \mathfrak{n}$, specified by a rational basis, a multiplication table, and r with $\mathfrak{n}^{r+1} = 0$. Let $C_W = B \otimes_{\mathbb{Q}} W$. The *rational-stage category* $\mathsf{W}_B^{\mathrm{rat}}$ has these objects and all B-algebra maps between them, encoded by finite lists of images satisfying their multiplication equations. Tensor product gives coproducts of these stages, but the category with all B-algebra maps is not, in general, closed under equalizers.

Example 6.8 (A coefficient-dependent rational equalizer)**.** In the cofinite numerical fibre, let e be the parity idempotent, which is neither zero nor one. On $C = B[\varepsilon]/\varepsilon^2$ the maps $u(\varepsilon) = \varepsilon$ and $v(\varepsilon) = e\varepsilon$ have algebraic equalizer

$$\mathrm{Eq}(u, v) = B \oplus Be\varepsilon.$$

Its nilpotent summand is rank one on the support of e and rank zero on the complementary support. It is finite projective over B, but is not $B \otimes_{\mathbb{Q}} W$ for a fixed rational Weil algebra W: on the two idempotent components the augmentation ranks of such a rational extension would agree. Thus the full rational-stage category cannot serve as the stage category for all coefficient-dependent finite Weil limits.

Theorem 6.9 (Finite-projective Weil-stage closure)**.** *There is a small internal category* $\mathsf{W}_B^{\mathrm{proj}}$ *whose objects are augmented commutative* B*-algebras* $C = B \oplus N$ *with* $N^{r+1} = 0$ *and a displayed finite-free retract (idempotent-matrix presentation) of* N*. It contains* $\mathsf{W}_B^{\mathrm{rat}}$ *as a full subcategory and is closed under finite limits and Weil tensor coproducts. Each stage is finite projective in this explicit sense and flat over* B*. These presentations and closure operations are uniform in parameters.*

Proof.

(1) **Construct the internal category of Weil stages.** A stage code consists of an integer n, an idempotent $p \in M_n(B)$, a finite bilinear multiplication table on $N = \mathrm{im}(p)$, and a positive integer r witnessing $N^{r+1} = 0$. Commutativity and associativity are finite matrix equations. A stage morphism is similarly a finite matrix representing a multiplication-preserving map between these augmentation modules. Matrix identity, substitution, and composition construct the internal category. The validity conditions are matrix equations over B, interpreted as in Section 2.6.1. The ring is $B \oplus N$, with multiplication $(b, x)(c, y) = (bc, by + cx + xy)$.

(2) **Construct coproducts and finite products.** The tensor coproduct of $B \oplus N$ and $B \oplus N'$ has underlying module

$$B \oplus N \oplus N' \oplus (N \otimes_B N'),$$

a finite-free retract. The augmentation ideal is nilpotent: a sufficiently long product contains either more than r factors from the first augmentation ideal or more than r' from the second. The terminal stage is B. Products in the augmented-algebra category are fibre products over B, with augmentation ideal $N \oplus N'$.

(3) **Split the image of the difference map.** For equalizers take two stage maps $u, v : C \rightrightarrows C'$. Every such map preserves augmentation: its composite with $C' \to B$ kills the nilpotent ideal of C, because B is reduced. Thus $d = u - v$ restricts to a B-linear map $d : N \to N'$. Its image $J \subseteq N'$ is generated by images of the displayed finite generators of N. Embed N' as a displayed summand of a finite free module. Apply Theorem 2.15 to the corestriction $d : N \twoheadrightarrow J$, with those images and their specified preimages in N. It constructs a section $s : J \to N$ satisfying $ds = \mathrm{id}_J$.

(4) **Construct the equalizer idempotent.** The endomorphism

$$p_N = \mathrm{id}_N - sd : N \longrightarrow N$$

satisfies

$$p_N^2 = \mathrm{id}_N - 2sd + s(ds)d = p_N, \qquad dp_N = d - dsd = 0.$$

For $x \in \ker d$ one has $p_N x = x$, so $\mathrm{im}(p_N) = \ker d$. Therefore the algebraic equalizer is

$$\mathrm{Eq}(u, v) = B \oplus \mathrm{im}(p_N),$$

a displayed finite-free retract of C as a module.

(5) **Verify algebra closure and finite-limit stability.** It is closed under multiplication because u and v are algebra maps; its augmentation ideal inherits nilpotence from N. Finite limits follow from products and equalizers, with the idempotent presentations retained. All calculations are finite, compatible with parameter substitution, and use the displayed lifts.

□

For the remainder of the paper put $\mathsf{W}_B = \mathsf{W}_B^{\mathrm{proj}}$ and

$$\mathcal{G}_B = [\mathsf{W}_B, \mathcal{E}].$$

The functor topos is internal and covariant, equivalently the presheaf topos on the opposite stage category; finite limits, colimits, and images are computed stagewise. Write

$$\mathcal{R}_B(C) = C, \quad j_B M(C) = C \otimes_B M, \quad D_A(C) = \mathrm{Hom}_{B\text{-alg}}(A, C) \qquad (A, C \in \mathsf{W}_B).$$

The resource line is a ring object and $j_B M$ an $\mathcal{R}_B$-module. Each D_A is a representable infinitesimal object.

A budget family and an infinitesimal coefficient stage can vary together. The next construction combines these two parameters, so each budget carries its realized resource module at every Weil stage.

6.4 Relative geometry of budgets and Weil coefficients

Theorem 6.10 (Budget–Weil presentation)**.** *For the finite-matrix stage presentations constructed above,*

$$\mathsf{W}_{\bar{B}}^{\mathrm{proj}} \cong \Delta_{\mathfrak{C}_{\mathrm{Ev}}} \mathsf{W}_B^{\mathrm{proj}}, \qquad \mathcal{G}_{\mathfrak{C}_{\mathrm{Ev}}} := [\mathsf{W}_{\bar{B}}^{\mathrm{proj}}, \mathscr{S}] \simeq [L \times \mathsf{W}_B^{\mathrm{proj}}, \mathcal{E}].$$

At a probe (a, A) the structure ring and universal resource are

$$\mathcal{R}_{\bar{B}}(a, A) = A, \qquad \mathfrak{M}(a, A) = A \otimes_B M_a. \tag{6.10}$$

For $a \leq b$ and $u : A \to A'$, the transition sends $c \otimes m$ to $u(c) \otimes m$ using $M_a \hookrightarrow M_b$.

Proof.

(1) **Transport the finite stage presentations.** A stage code comprises natural numbers, finite coefficient lists, idempotent and multiplication matrices, and finite equations. The nilpotence bound is a displayed natural number and its equations are checked on the finite generators. The object of codes and the object of arrows are therefore built from finite limits and natural-number-indexed finite lists of coefficients. The constant-diagram functor preserves these constructions; in particular it preserves their equality subobjects. This identifies both stage categories, including identities and composition.

(2) **Identify budget diagrams of Weil diagrams.** An internal diagram over the resulting constant category is equivalently a budget diagram of Weil diagrams: the two actions commute. Currying gives the asserted equivalence.

(3) **Compute the probe formulas.** Tensor and module operations are computed at each probe, giving (6.10).

□

Write $i_W : \mathscr{S} \to \mathcal{G}_{\mathfrak{C}_{\mathrm{Ev}}}$ for constancy in the Weil variable and $\Gamma_W Z(a) = Z(a, B)$. The restricted based-Weil adjunctions are $i_W \dashv \Gamma_W \dashv i_W \dashv \Gamma_W$. They calculate reduction on fixed-base probes at each budget. The finite-hit dominance of $\mathscr{S}$ is $\Delta_{\mathfrak{C}_{\mathrm{Ev}}}\Sigma$: constant diagrams preserve $2^{\mathbb{N}}$ and its finite-hit image, and composition of the classified monos is checked at each budget. Applying the same construction in the Weil variable gives $\Sigma_{\mathcal{G}_{\mathfrak{C}_{\mathrm{Ev}}}} = i_W \Delta_{\mathfrak{C}_{\mathrm{Ev}}}\Sigma$. For every positive test $u : Z \to \Sigma_{\mathcal{G}_{\mathfrak{C}_{\mathrm{Ev}}}}$,

$$u = i_W(\Gamma_W u)\eta_Z. \tag{6.11}$$

This follows from $\Gamma_W \dashv i_W$ exactly as in Proposition 6.12. Budget locality is retained; infinitesimal coefficients are invisible to these positive tests. For a name cover $q_D : C_D \twoheadrightarrow \mathrm{Pt}_\Sigma D$ and an implemented map $\widehat{F} : C_D \to C_{D'}$, let $k = i_W \Delta_{\mathfrak{C}_{\mathrm{Ev}}}$. The transported cover remains a regular epimorphism and

$$(kq_{D'})(k\widehat{F}) = k(\mathrm{Pt}_\Sigma F)(kq_D), \qquad k\,\mathrm{Obs}_D : k\,\mathrm{Pt}_\Sigma D \times k\mathcal{O}_D \to k\Sigma.$$

These equations are evaluated at each probe as the original name-realization and observation equations; pointwise finite limits and images preserve their types. For a partial implementation the same argument preserves its Σ-classified domain span and its value map. Thus the relative geometry retains both domains and returned values.

6.5 Reduction, observations, and Weil coefficient calculus

6.5.1 Reduction and infinitesimal shape

Every B-algebra map $C = B \oplus N \to B$ kills the nilpotent augmentation ideal because B is reduced. It therefore equals the augmentation. Thus B is terminal as well as initial in the full projective Weil-stage category W_B. Write $i : \mathcal{E} \to \mathcal{G}_B$ for constant objects and $\Gamma(F) = F(B)$.

Theorem 6.11 (The ambidextrous based-Weil restriction)**.** *On the based-Weil functor topos, the functor i is fully faithful and there is an ambidextrous adjoint quadruple*

$$i \dashv \Gamma \dashv i \dashv \Gamma.$$

The endofunctor $\mathfrak{I}_W = i\Gamma$ is left exact and idempotent, with natural maps

$$\epsilon_X : \mathfrak{I}_W X \to X, \qquad \eta_X : X \to \mathfrak{I}_W X, \qquad \eta_X \epsilon_X = \mathrm{id}_{\mathfrak{I}_W X}.$$

It carries the induced reduction comonad and infinitesimal-shape monad structures. The associated triple of adjoint idempotent endofunctors has coincident underlying functor $\mathfrak{I}_W$.

Proof.

(1) **Construct the adjunction from initiality.** Initiality of B gives

$$\mathrm{Nat}(iA, F) \cong \mathrm{Hom}_{\mathcal{E}}(A, F(B)) :$$

a map at B extends at C through $F(B \to C)$.

(2) **Construct the adjunction from terminality.** Terminality gives

$$\mathrm{Nat}(F, iA) \cong \mathrm{Hom}_{\mathcal{E}}(F(B), A) :$$

a map at B extends at C through $F(C \to B)$.

(3) **Verify the adjunction quadruple and full faithfulness.** Both inverse equations are naturality along those stage maps. These two adjunctions repeat to give the quadruple. Since $\Gamma i = \mathrm{id}$, i is fully faithful.

(4) **Identify the units, counits, and modal structures.** At C the maps ϵ_X, η_X are respectively $X(B \to C)$ and $X(C \to B)$. Their composite on $X(B)$ is the identity. Evaluation and constant formation preserve finite limits, giving left exactness; $\Gamma i = \mathrm{id}$ gives idempotence. The units, counits, and their triangle identities give the stated monad, comonad, and adjoint structures.

□

The ambidexterity here has a specific source: both endpoints of a based probe reduce to the fixed B-point. It describes the based-Weil restriction of the primary étale endpoint geometry of Theorem 8.4. The comparison in Theorem 9.4 places the following Weil coefficient and microlinearity calculations within that larger geometry. For the line,

$$\Im_W \mathcal{R}_B = iB, \qquad \eta_{\mathcal{R}_B}(C) : C \to B$$

is augmentation; for every representable probe, $\Im_W D_A = iD_A(B) = 1$. Thus reduction extracts the reduced point of each infinitesimal probe.

The reduction $B \to B/\mathfrak{m}$ has a different type: it removes numerical errors vanishing along the eventual filter. The map η_X removes formal nilpotent directions. Both are used with their displayed coefficient objects.

6.5.2 Computational observations inside the cohesive geometry

The functor $i : \mathcal{E} \to \mathcal{G}_B$ preserves finite limits, images, and the NNO stagewise. Between constant objects it preserves exponentials:

$$(iY)^{iX} \cong i(Y^X).$$

Indeed a natural map $h_C \times iX \to iY$ is determined by its value on (id_C, x); naturality along $u : C \to C'$ forces the value on (u, x) to be the same. This proves the isomorphism in every parameter context.

Proposition 6.12 (Positive observations factor through reduction)**.** *The canonical finite-hit Rosolini truth-value object of $\mathcal{G}_B$ is*

$$\Sigma_{\mathcal{G}_B} \cong i\Sigma.$$

It is a dominance. Every positive computational test has the unique factorization

$$u : Z \to \Sigma_{\mathcal{G}_B}, \qquad u = i(\Gamma u)\eta_Z : Z \to i\Gamma Z \to i\Sigma.$$

Consequently its pullbacks along the two projections of $Z \times_{\Im_W Z} Z$ are equal.

Proof.

(1) **Construct the geometric positive truth-value object.** Constant-subobject classification gives a monomorphism $i\Omega_{\mathcal{E}} \to \Omega_{\mathcal{G}_B}$: at each stage, evaluation of the classified natural family on its identity arrow recovers the original truth value. The preceding exponential calculation gives $(i2)^{i\mathbb{N}} \cong i(2^{\mathbb{N}})$. The finite-hit predicate is the same natural-number existential at every stage; its classifier is the constant finite-hit classifier followed by this monomorphism. Images are stagewise, so its image is $i\Sigma$.

(2) **Factor positive observations through reduction.** The adjunction $\Gamma \dashv i$ identifies a map $Z \to i\Sigma$ with its component $\Gamma Z \to \Sigma$ and gives the displayed factorization. A subobject so classified is the pullback along η_Z of its reduced classified subobject.

(3) **Verify dominance and formal-disk invariance.** For two composable such subobjects, left exactness of Γ identifies the reduced composite; dominance in $\mathcal{E}$ classifies it, and pullback along η_Z recovers the original composite. Truth and pullback stability follow by the same argument. Thus $i\Sigma$ is a dominance. The two formal-disk projections have the same composite with η_Z, proving the last assertion.

□

For $j_B M$, reduction is augmentation $C \otimes_B M \to M$, since $\Gamma j_B M = M$. Canonical positive tests therefore observe the reduced value and are constant along its formal disk. Infinitesimal coefficients are instead read by the algebraic coefficient calculations below.

Computational names, natural numbers, and denotations are constant objects of this geometry. The coefficient calculation in Theorem 6.13 also expresses their infinitesimal discreteness as $(iX)^{D_A} \cong iX$. Resource values lie in the nonconstant modules $j_B M$. These two roles allow an implementation witness to stay fixed while its resource constraint varies infinitesimally.

6.5.3 Weil coefficient identity and comparison of stages

Theorem 6.13 (Exact realization and Weil coefficients)**.** *The functor $j_B : \mathrm{Mod}_B(\mathcal{E}) \to \mathrm{Mod}_{\mathcal{R}_B}(\mathcal{G}_B)$ is exact, fully faithful, strong symmetric monoidal, and reflects exactness and isomorphisms. For every $F \in \mathcal{G}_B$ and projective stage A,*

$$F^{D_A}(C) \cong F(C \otimes_B A),$$

and therefore

$$(j_B M)^{D_A}(C) \cong (C \otimes_B A) \otimes_B M \cong (C \otimes_B M) \otimes_B A. \tag{6.12}$$

All comparisons are natural in the stages, their maps, and the module, and commute with augmentation and Weil substitution.

Proof.

(1) **Compute exponentials by Weil disks.** Writing $h_C = \mathrm{Hom}_{B\text{-alg}}(C, -)$, tensor coproduct and Yoneda give

$$h_C \times D_A \cong h_{C \otimes_B A}, \qquad F^{D_A}(C) = \mathrm{Nat}(h_C \times D_A, F) \cong F(C \otimes_B A).$$

This also proves the displayed coefficient formula.

(2) **Prove exactness of realization.** Every stage is finite projective and hence flat, so each $C \otimes_B -$ is exact, making j_B exact stagewise.

(3) **Prove full faithfulness and reflection.** A natural $\mathcal{R}_B$-linear map $j_B M \to j_B N$ is determined by its component at B: naturality along $B \to C$ determines its values on $1 \otimes m$, and C-linearity determines all $c \otimes m$. Conversely any B-linear map gives such a natural transformation by scalar extension, proving full faithfulness. Evaluation at B reflects exactness and isomorphisms.

(4) **Verify strong monoidality and coherence.** The associativity of balanced tensor gives $j_B(M \otimes_B N) \cong j_B M \otimes_{\mathcal{R}_B} j_B N$ and $j_B B \cong \mathcal{R}_B$; uniqueness on elementary tensors proves unit, symmetry, and associativity coherence.

□

Proposition 6.14 (Restriction to rational Weil probes)**.** *The full inclusion $k : \mathsf{W}_B^{\mathrm{rat}} \hookrightarrow \mathsf{W}_B^{\mathrm{proj}}$ induces restriction $k^* : \mathcal{G}_B \to [\mathsf{W}_B^{\mathrm{rat}}, \mathcal{E}]$ with coherent isomorphisms*

$$k^* j_B^{\mathrm{proj}} M \cong j_B^{\mathrm{rat}} M, \qquad k^* D_{C_W}^{\mathrm{proj}} \cong D_{C_W}^{\mathrm{rat}}, \qquad k^* i^{\mathrm{proj}} X \cong i^{\mathrm{rat}} X.$$

It preserves the rational Weil coefficient and exponential identities and commutes with evaluation at B and the resulting infinitesimal-shape operation.

Proof.

(1) **Compare restriction and Weil exponentiation.** All three functors have the displayed stagewise formulas. For rational stages C_W, C_V, their tensor coproduct $C_{W\otimes V}$ is again rational; applying the exponential calculation of Theorem 6.13 before or after restriction therefore gives the same expression $F(C_W \otimes_B C_V)$.

(2) **Compare reduction and verify coherence.** Evaluation at B commutes with restriction, and constant functors do so stagewise. These identities give the stated shape comparison, while equality of the elementary-tensor formulas gives all coherence diagrams.

□

6.5.4 Finite Kock–Lawvere identities and microlinearity

For $A = B[\varepsilon]/\varepsilon^2$, put $D = D_A$. Yoneda identifies $D \subseteq \mathcal{R}_B$ with the square-zero elements. Formula (6.12) gives

$$\mathcal{R}_B^D \cong \mathcal{R}_B \times \mathcal{R}_B, \qquad (j_B M)^D \cong j_B M \times j_B M,$$

where (a, b) corresponds to the map $d \mapsto a + db$. The same formula for every finite-projective probe is the finite Weil–Kock–Lawvere coefficient identity (Kock 2006).

Theorem 6.15 (Rational and projective microlinearity)**.** *For every B-module M, the augmented coefficient functor*

$$W \longmapsto ((j_B M)^{D_{C_W}} \to j_B M)$$

preserves finite limits of rational Weil diagrams with rational structure maps. For every embedded resource module $M \subseteq K$, the functor

$$A \longmapsto ((j_B M)^{D_A} \to j_B M)$$

preserves finite limits of the full finite-projective Weil-stage category. In both statements the codomain is $\mathcal{G}_B/j_B M$ and the displayed maps are induced by augmentation.

Proof.

(1) **Preserve rational Weil limits by split exactness.** Rational Weil algebras with rational algebra homomorphisms have finite limits: the product over $\mathbb{Q}$ is an augmented fibre product, and the equalizer is a rational linear kernel. The latter has a basis constructed by Gaussian elimination. Its finite exact sequences split over $\mathbb{Q}$, so tensoring with any $\mathbb{Q}$-module preserves these limits. At a stage C the coefficient functor is $W \mapsto (C \otimes_B M) \otimes_{\mathbb{Q}} W$ over $C \otimes_B M$; this proves the first assertion stagewise.

(2) **Establish flatness of resource coefficients.** For the second, Theorem 2.12 writes each embedded resource as an internally directed colimit of finite-projective principal modules. Its internal filtered-colimit flatness proof in Section 2.6 makes M flat. Each stage C is finite projective and flat, so $T = C \otimes_B M$ is flat over B.

(3) **Preserve projective Weil limits stagewise.** The formula at C is now $A \mapsto T \otimes_B A$ with augmentation to T. Finite limits of projective augmented Weil algebras are computed on the underlying B-modules by fibre products and kernels of differences. Flatness preserves these finite module limits, including the augmentation maps. The resulting comparisons are natural at stages and therefore are isomorphisms in $\mathcal{G}_B/j_B M$.

□

Proposition 6.16 (Failure for arbitrary coefficient modules)**.** *For the numerical cofinite ring B and regular nonunit $t(n) = 1/(n+1)$, the augmented coefficient functor of Theorem 6.15 need not preserve projective Weil equalizers for $M = B/tB$.*

Proof.

(1) **Compute the original equalizer.** On $A = B[\varepsilon]/\varepsilon^2$, take $u(\varepsilon) = t\varepsilon$ and $v(\varepsilon) = 0$. Regularity of t gives $\mathrm{Eq}(u, v) = B$.

(2) **Exhibit failure of the equalizer comparison.** After tensoring with $M = B/tB$, the two maps coincide and their equalizer becomes $M \oplus M\varepsilon$, whereas tensoring the original equalizer gives only M. The two are not isomorphic as the indicated augmented equalizer comparison: its image is the constant-coefficient summand and misses ε. The embedded resources of Theorem 6.15 do not have this obstruction because they are flat.

□

At $C = B[\varepsilon]/\varepsilon^2$ the generic infinitesimal ε is nonzero: coefficient extraction would otherwise force $1 = 0$ in B, contradicting properness of the filter. Thus the constructed model has nontrivial infinitesimal directions.

6.5.5 Coefficient change and reduced-base cohesion

Finite Weil coefficient realization commutes with a ring map $\beta : B \to B'$ on rational probes: for any rational Weil W,

$$(B' \otimes_{\mathbb{Q}} W) \otimes_{B'} (B' \otimes_B M) \cong B' \otimes_B ((B \otimes_{\mathbb{Q}} W) \otimes_B M).$$

Both sides send an elementary coefficient–Weil–module tensor to the same expression; the unit and substitution diagrams follow from tensor coherence. The adjoint quadruple of Theorem 6.11 has the further property that the coefficient stage is terminal, which is a consequence of reducedness rather than of tensor coherence alone.

Proposition 6.17 (A coefficient reduction producing nilpotents)**.** *In the cofinite numerical fibre, let $t(n) = 1/(n+1)$ and $z(n) = 1/(1 + \lfloor\sqrt{n+1}\rfloor)$. Then $z \notin tB$ but $z^2 \in tB$. Thus $B' = B/tB$ has a nonzero square-zero element, and its coefficient stage is not terminal in the category of B'-Weil stages.*

Proof.

(1) **Construct a nonzero square-zero coefficient.** Write $m = \lfloor\sqrt{n+1}\rfloor$. The inequalities

$$\frac{z(n)^2}{t(n)} = \frac{n+1}{(1+m)^2} \leq 1, \qquad \frac{z(n)}{t(n)} = \frac{n+1}{1+m} \geq \frac{m^2}{1+m}$$

show that z^2/t is bounded but z/t is unbounded along square inputs. Hence the class $\bar{z} = z + tB$ is nonzero and satisfies $\bar{z}^{\,2} = 0$.

(2) **Show failure of terminality.** The two B'-algebra maps

$$B'[\varepsilon]/\varepsilon^2 \rightrightarrows B', \qquad \varepsilon \longmapsto 0, \quad \varepsilon \longmapsto \bar{z}$$

are distinct, so B' is not terminal among those stages. This explicitly separates coefficient-tensor comparisons from the reduced-base cohesion calculation.

□

6.5.6 Explicit linear adjoints

Let $\mathscr{M}_B = \mathrm{Mod}_{\mathcal{R}_B}(\mathcal{G}_B)$ and $\Gamma_{\mathrm{lin}}(F) = F(B)$ with its B-module structure.

Theorem 6.18 (Linear realization adjoints)**.** *There is a constructed adjoint triple*

$$\Lambda_{\mathrm{lin}} \dashv j_B \dashv \Gamma_{\mathrm{lin}}, \qquad \Lambda_{\mathrm{lin}} j_B \cong \mathrm{id} \cong \Gamma_{\mathrm{lin}} j_B.$$

Proof.

(1) **Construct the right-adjoint correspondence.** A B-linear $v : M \to F(B)$ extends at C by $c \otimes m \mapsto cF(B \to C)(v(m))$. Naturality and linearity prove it is inverse to evaluation, giving $j_B \dashv \Gamma_{\mathrm{lin}}$.

(2) **Construct the coend candidate.** For the left adjoint put $C^\vee = \mathrm{Hom}_B(C, B)$ with right C-action $(\phi c)(x) = \phi(cx)$, and construct

$$\Lambda_{\mathrm{lin}} F = \mathrm{coeq}\left(\bigoplus_{u:C\to C'} (C')^\vee \otimes_C F(C) \rightrightarrows \bigoplus_{C} C^\vee \otimes_C F(C) \right). \tag{6.13}$$

(3) **Define the balanced coequalizer maps.** One map sends $\phi \otimes x$ to $(\phi u) \otimes x$ in the C-summand; the other to $\phi \otimes F(u)x$ in the C'-summand. The right C-action on $(C')^\vee$ uses u, so both maps are balanced. The indexing objects are the object and arrow objects of the internal stage category; internal free modules and quotients construct these sums and this coequalizer.

(4) **Establish the stagewise duality correspondence.** Finite projectivity gives

$$\mathrm{Hom}_C(F(C), C \otimes_B M) \cong \mathrm{Hom}_B(C^\vee \otimes_C F(C), M).$$

It sends v to $\phi \otimes x \mapsto (\phi \otimes 1)v(x)$. For a dual basis (c_i, ϕ_i) the inverse sends h to $x \mapsto \sum_i c_i \otimes h(\phi_i \otimes x)$. The dual-basis identities and balancing prove the inverse equations and C-linearity.

(5) **Prove the left adjunction and composite identities.** Naturality of the family v is exactly equalization of the two maps in (6.13). The coequalizer universal property proves $\Lambda_{\mathrm{lin}} \dashv j_B$. Full faithfulness of j_B gives the two composite identities.

□

The coefficient model now lets us express resource operations as geometric maps. We first identify the coordinate algebra of a finite-projective module. Its dual appears because coordinates are linear functions on that module. Universal differentials then describe the first-order change of those functions.

6.6 Affine resource coordinates and universal differentials

6.6.1 Dual coordinate modules

Proposition 6.19 (Finite-cell differential comparison)**.** *Let M be a finite projective B-module and set $M^\vee = \mathrm{Hom}_B(M, B)$. Its affine coordinate algebra is $\mathrm{Sym}_B(M^\vee)$, and for every B-algebra C,*

$$\mathrm{Hom}_{B\text{-}alg}(\mathrm{Sym}_B(M^\vee), C) \cong \mathrm{Hom}_B(M^\vee, C) \cong C \otimes_B M.$$

Under this identification, the universal algebraic derivative gives the coefficient of ε in evaluation on $C[\varepsilon]/\varepsilon^2$.

Proof.

(1) **Establish the finite-duality bijections.** The first bijection is the free-algebra universal property. For a finite dual basis (m_i, λ_i), the inverse of the second sends a functional $f : M^\vee \to C$ to $\sum_i f(\lambda_i) \otimes m_i$; the dual-basis identity proves both inverse equations and independence of the presentation.

(2) **Extract the dual-number derivative.** A polynomial coordinate p satisfies

$$p(x + \varepsilon v) = p(x) + \varepsilon\, dp_x(v)$$

because multiplying factors chooses either no ε or exactly one ε. This is precisely formula (6.6). Addition and relations preserve the equality. Thus the induced map of Kähler modules, transposed to tangent vectors, agrees with the dual-number coefficient.

□

Positive principal resources $B^+g \cong B^+e_g$ are dualizable, with dual B^+e_g. Strong monoidal functors preserve their evaluation and coevaluation maps and hence their duals. Theorem 6.6 therefore compares these coordinate algebras and their derivatives before and after signed realization.

For a general resource $M = \bigcup_g Bg$, a finite tuple of inputs lies in a common finite projective cell. A finite polynomial resource operation lands in the finite principal cell generated by the sum of the absolute values of its finitely many coefficient-budget products. The comparison above applies on each such cell. Cell inclusions commute with evaluation and derivation by naturality, so these comparisons assemble over the directed resource presentation. This proves the derivative comparison for resource arithmetic and polynomial overhead cells on this directed presentation.

Theorem 6.20 (Canonical polynomial lifting to Weil geometry)**.** *Let E, N be finite-projective B-modules. A polynomial resource map $p : E \to N$, specified by its coordinate-algebra map*

$$p^* : \mathrm{Sym}_B(N^\vee) \longrightarrow \mathrm{Sym}_B(E^\vee),$$

induces canonically a geometric map $\widehat{p} : j_B E \to j_B N$. It preserves identities and polynomial composition, and

$$D\widehat{p}(x, v) = dp_x(v).$$

These maps and derivatives commute with finite-projective resource cell inclusions whenever the polynomial expressions and their coefficient budgets are transported by those inclusions.

Proof.

(1) **Construct the geometric polynomial map.** For each Weil stage C, finite duality identifies $C \otimes_B E$ with the C-points of $\mathrm{Sym}_B(E^\vee)$, and similarly for N. Precomposition with p^* therefore gives $C \otimes_B E \to C \otimes_B N$, naturally in every stage map. This is the required geometric transformation.

(2) **Verify functoriality.** Algebra-map composition establishes functoriality and identity preservation.

(3) **Identify the derivative by finite duality.** For the dual-number coefficient $x + \varepsilon v$, a coordinate $\lambda \in E^\vee$ evaluates to $\lambda(x) + \varepsilon\lambda(v)$. Expanding the finite polynomial $p^*(\mu)$ for each $\mu \in N^\vee$ extracts the term containing exactly one ε, which is $dp_x(v)$ by the universal Leibniz formula. Finite duality detects equality of the resulting N-valued coefficients, proving the derivative identity.

(4) **Extend compatibility over principal cells.** Every finite polynomial expression in an embedded resource uses finitely many arguments and coefficient budgets. Their absolute sum supplies a common finite-projective principal cell by Theorem 2.12; naturality of polynomial evaluation and the differential comparison proves compatibility over directed cell presentations.

□

Proposition 6.21 (Finite polynomial laws on arbitrary modules)**.** *For B-modules H, N, a finite family of B-multilinear maps $m_r : H^{\otimes_B r} \to N$ for $0 \le r \le d$ (with $m_0 \in N$) constructs the polynomial law $p(x) = \sum_{r=0}^{d} m_r(x, \dots, x)$ and a canonical natural geometric lift $\widehat{p} : j_B H \to j_B N$, satisfying*

$$D\widehat{p}(x, v) = \sum_{r=1}^{d} \sum_{j=1}^{r} \widetilde{m}_r(x, \dots, x, \underset{j}{v}, x, \dots, x),$$

where $\widetilde{m}_r$ is the scalar-extended multilinear map at the generalized Weil stage. On finite-projective modules this agrees with the coordinate-algebra construction of Theorem 6.20.

Proof.

(1) **Extend the multilinear coefficients to Weil stages.** For each Weil stage C, balanced tensor extension of m_r produces a C-multilinear map $(C \otimes_B H)^{\otimes_C r} \to C \otimes_B N$. Evaluating the finite sum diagonally defines $\widehat{p}_C : C \otimes_B H \to C \otimes_B N$.

(2) **Verify naturality and extract the derivative.** Tensor functoriality makes these maps natural in every B-algebra stage morphism. Evaluating at $x + \varepsilon v$, the square-zero multiplication removes all terms with two or more occurrences of ε. The terms with exactly one occurrence give the displayed formula, by coefficient uniqueness.

(3) **Compare with the finite-duality construction.** For finite projective H, N, both stagewise constructions are polynomial evaluations with the same coordinate values, so finite duality identifies them.

□

An arbitrary numerical value family also has a canonical, but different, realization. The constant-object/evaluation adjunction of Theorem 6.11 gives

$$\mathcal{G}_B(iX, j_B K) \cong \mathcal{E}(X, K), \qquad f \longmapsto (x \mapsto 1 \otimes f(x)).$$

The source iX is infinitesimally discrete: $(iX)^D \cong iX$. This realization therefore preserves the values of an arbitrary numerical family without differentiating its discrete input. Polynomial resource variables have a different role: the lifting theorem realizes their polynomial maps geometrically and identifies their derivatives.

6.6.2 Relative differential modules

For a commutative B-algebra A, form the free A-module on symbols da and impose additivity, $d(ba) = b\,da$ for $b \in B$, and $d(aa') = a\,da' + a'\,da$. The quotient $\Omega^1_{A/B}$ represents B-derivations by direct evaluation of these generators and relations. Equivalently a derivation is a section of $A \ltimes N \to A$ as a B-algebra map. For a finite projective coordinate cell this gives

$$\Omega^1_{\mathrm{Sym}_B(M^\vee)/B} \cong \mathrm{Sym}_B(M^\vee) \otimes_B M^\vee.$$

Dual bases identify its evaluation pairing with the tangent coefficient of Proposition 6.19.

The absolute module $\Omega^1_{B/\mathbb{Q}}$ is obtained by fixing only the rational constants. The comparison (6.8) identifies it with signed realization of $\Omega^1_{B^+/\mathbb{Q}_{\geq 0}}$. These constructions retain their coefficient maps. For instance every relative derivation over B kills a scale $t \in B$, whereas its absolute differential dt is the specified generator in $\Omega^1_{B/\mathbb{Q}}$.

Localization is compatible with relative differentials:

$$\Omega^1_{A_s/B} \cong A_s \otimes_A \Omega^1_{A/B}.$$

To prove this, a derivation extends uniquely by $d(s^{-1}) = -s^{-2}ds$ and the quotient rule $d(a/s^r) = s^{-r}da - ras^{-r-1}ds$. Clearing denominators verifies independence of the fraction, and the Leibniz law verifies the extension. Representability proves the isomorphism. The gluing equalizer of Proposition 3.16 therefore also glues these differential modules and their maps.

6.7 Finite tangent calculus

Proposition 6.22 (Differentiation in the realized geometry)**.** *Every map $F : j_B M \to j_B N$ in $\mathcal{G}_B$ has a unique first-order expansion*

$$F(x + \varepsilon v) = F(x) + \varepsilon\, DF(x, v).$$

Its derivative is linear in v, obeys the chain rule, and agrees with Proposition 6.19 on polynomial resource cells.

Proof.

(1) **Obtain existence and uniqueness of the derivative.** The dual-number coefficient identity gives existence and uniqueness.

(2) **Prove additivity in the tangent direction.** For addition in v, use the Weil algebra with two generators $\varepsilon_1, \varepsilon_2$ and all their pairwise products zero. Apply F to $x+\varepsilon_1 v+\varepsilon_2 w$. Projections to each dual-number axis identify its two linear coefficients as $DF(x, v)$ and $DF(x, w)$. The algebra map sending both ε_1 and ε_2 to ε then proves additivity.

(3) **Prove scalar linearity.** The stage map $\varepsilon \mapsto a\varepsilon$ proves scalar linearity, including generalized coefficients $a \in C$.

(4) **Prove the chain rule.** Composing two expansions proves the chain rule by uniqueness of coefficients.

(5) **Compare algebraic and geometric derivatives.** The finite polynomial calculation in Proposition 6.19 gives the comparison.

(6) **Obtain higher symmetry and Taylor expansions.** Iterated probes and permutation of their generators give the higher symmetry identities; truncated probes give finite Taylor expansions with the rational factorials available.

□

For a linear map L, its coefficient formula is $DL(x, v) = L(v)$. For addition and multiplication in the resource line it gives

$$D(+)((x, y), (v, w)) = v + w, \qquad D(\cdot)((x, y), (v, w)) = yv + xw.$$

For a composite the chain rule is $D(GF)(x, v) = DG(Fx, DF(x, v))$. The tangent functor is $T_{\tan}X = X^D$ with projection induced by $0 : 1 \to D$. On realized modules it is the usual

pair of base point and direction. Its zero section, addition of directions, and scalar action are the coefficient maps just constructed.

Iterated tangent uses $\mathbb{Q}[\varepsilon_1, \varepsilon_2]/(\varepsilon_1^2, \varepsilon_2^2)$. Its coefficient of $\varepsilon_1\varepsilon_2$ is invariant under interchanging the two generators. This proves symmetry of the second base-point derivative. Iteration and adjacent transpositions prove symmetry of every higher base-point derivative, with directions held fixed. These are equations of natural maps, so include generalized coefficients at every stage.

We can now apply the differential calculus to resource relations. The algebraic descriptions of O, o, Θ, and $\sim$ use multiplier witnesses. Realizing those multiplication maps first retains the witnesses needed to differentiate the constraints.

6.8 Realization of the Landau incidences

Multiplication makes $j_B K$ an $\mathcal{R}_B$-algebra. Write $U = \mathcal{R}_B^\times$ and $\mathcal{M} = j_B\mathfrak{m}$. The subobject $1 + \mathcal{M}$ is a subgroup of U. To see this at $C = B \oplus N$, augment $a \in C \otimes_B \mathfrak{m}$ to $a_0 \in \mathfrak{m}$. Then $1 + a_0$ is a unit of B, while $a - a_0$ lies in the nilpotent augmentation ideal. A finite geometric inverse makes $1 + a$ a unit of C, and $(1 + a)^{-1} - 1 = -a(1 + a)^{-1}$ lies in the same extended ideal.

Define geometric binary relations as the images of

$$\begin{aligned}
\mathcal{R}_B \times j_B K &\longrightarrow j_B K \times j_B K, \quad (b, g) \longmapsto (bg, g),\\
\mathcal{M} \times j_B K &\longrightarrow j_B K \times j_B K, \quad (a, g) \longmapsto (ag, g),\\
U \times j_B K &\longrightarrow j_B K \times j_B K, \quad (u, g) \longmapsto (ug, g),\\
\mathcal{M} \times j_B K &\longrightarrow j_B K \times j_B K, \quad (a, g) \longmapsto ((1 + a)g, g).
\end{aligned}$$

They realize respectively $O, o, \Theta, \sim$. Reversing the first two gives the realizations of Ω, ω. These formulas specify the geometric relations also on nonconstant points.

Theorem 6.23 (Exact Landau incidence)**.** *Pullback of each of these six geometric relations along $i(K \times K) \to j_B K \times j_B K$ is the corresponding constant exact Landau relation. This statement is uniform in exact computational parameters.*

Proof.

(1) **Extend exact witnesses to all Weil stages.** An exact multiplier, unit, or small multiplier extends to every stage and witnesses the forward inclusion by Theorem 2.10.

(2) **Reflect membership through augmentation.** Conversely, augment a stagewise witness to B. Augmentation takes an arbitrary scalar to a bounded coefficient, a unit to a unit, and an element of $C \otimes_B \mathfrak{m}$ to an element of $\mathfrak{m}$. Its equation becomes the exact equation on f, g.

(3) **Identify incidence and the reversed comparisons.** Thus membership is reflected. Images and the resulting pullbacks are computed stagewise. Reversal gives the two lower-bound cases.

□

Differentiating a witnessed incidence varies both its value and its multiplier. The resulting equations describe how the resource relation is preserved to first order; taking its image recovers the corresponding relation on values.

6.9 Differentials of the Landau incidences

Proposition 6.24 (Tangents of multiplier orbits)**.** *For an exact reference germ g, let $\widehat{\Theta}_g$ be the image of $U = \mathcal{R}_B^\times \to j_B K$, $u \mapsto ug$, and put $\widehat{\sim}_g = g + j_B(\mathfrak{m}g)$. At the common point g,*

$$T_g\widehat{\Theta}_g \cong j_B(Bg), \qquad T_g\widehat{\sim}_g \cong j_B(\mathfrak{m}g). \tag{6.14}$$

Proof.

(1) **Extract a tangent direction from a unit witness.** A tangent in the first image at stage C is a point over $C[\varepsilon]/(\varepsilon^2)$ reducing to g. By the regular-image presentation, after a regular-epimorphic refinement it has a unit witness $u = u_0 + \varepsilon u_1$. Its value is $ug = u_0 g + \varepsilon u_1 g$. Reduction gives $u_0 g = g$, and coefficient extraction places the direction $u_1 g$ in $C \otimes_B Bg$.

(2) **Descend the direction through the witness cover.** This membership descends because $C \otimes_B Bg \hookrightarrow C \otimes_B K$ is monic and subobject inclusions are reflected by regular-epimorphic covers.

(3) **Realize every supported tangent direction.** Conversely, for every $b \in C$, the element $1 + \varepsilon b$ is a unit with inverse $1 - \varepsilon b$ and has orbit value $g + \varepsilon bg$. Thus the tangent directions at every stage are exactly $C \otimes_B Bg$, naturally in C.

(4) **Compute the small-multiplier tangent.** For the second locus, translate $j_B(\mathfrak{m}g)$ by g and use the dual-number coefficient identity; its directions are exactly $C \otimes_B \mathfrak{m}g$. The construction and inverse are natural in the stage and reflect equalities of tangent coefficients.

□

Lower-bound witnesses also have a differential equation. In the incidence $g = af$, with a a bounded multiplier, the tangent equation is

$$\delta g = a\,\delta f + f\,\delta a.$$

For the strict lower-bound incidence the multiplier and its variation lie in $j_B\mathfrak{m}$. This equation is the derivative of the witnessed multiplication constraint, whose image realizes the lower-bound relation. Thus the multiplier witness is retained while the differential equation is formed.

6.10 The universal differential resource family

Theorem 6.25 (Universal differential resource family)**.** *The functor $j_{\bar{B}} : \mathrm{Mod}_{\bar{B}}(\mathscr{S}) \to \mathrm{Mod}_{\mathcal{R}_{\bar{B}}}(\mathcal{G}_{\mathfrak{C}_{\mathrm{Ev}}})$, given at a stage A by $A \otimes_{\bar{B}} -$, is exact, fully faithful, and strong symmetric monoidal. The module $\mathfrak{M} = j_{\bar{B}}\mathscr{M}$ is microlinear for all finite-projective Weil diagrams. At every generalized ideal point I, with stage codes reindexed along inverse image,*

$$I^*\mathfrak{M} \cong j_B M_I. \tag{6.15}$$

Moreover $\mathfrak{M}^D \cong \mathfrak{M} \times \mathfrak{M}$. For a polynomial resource map $F : j_{\bar{B}}\bar{H} \to j_{\bar{B}}\bar{K}$, the universal constrained locus $S_F = F^{-1}(\mathfrak{M})$ has tangent

$$T_x S_F = \ker\big(j_{\bar{B}}(q) DF_x\big), \qquad q : \bar{K} \to \bar{K}/\mathscr{M}. \tag{6.16}$$

These comparisons specialize to the corresponding budget fibres.

Proof.

(1) **Realize the universal family and prove microlinearity.** The Yoneda and finite-dual-basis proof of Theorem 6.13 is internal and applies in $\mathscr{S}$, giving the functorial properties and the Weil coefficient identities. The flatness of $\mathscr{M}$ from Theorem 3.3 lets the proof of Theorem 6.15 apply to the full projective stages.

(2) **Compute ideal fibres.** Under Theorem 6.10, inverse image at I gives

$$\varinjlim_{a\in I}(A\otimes_B M_a)\cong A\otimes_B \varinjlim_{a\in I} M_a\cong A\otimes_B M_I.$$

(3) **Verify parameter and stage coherence.** For a locally varying stage code, the same isomorphism follows by reindexing its finite matrices; inverse image preserves the idempotent equations and the balanced tensor presentation. Both comparison paths along a stage map send $c\otimes m$ to the same substituted elementary tensor. This proves the parameter and stage coherence of (6.15).

(4) **Compute tangent coefficients and the normal kernel.** The dual-number basis gives the exponential formula. Finally, exactness identifies $\mathfrak{M}$ with the kernel of $j_{\bar{B}}(q)$; expanding $qF(x+\varepsilon v)$ at a feasible point gives $\varepsilon qDF_x(v)$. Coefficient uniqueness proves (6.16). Inverse image preserves this kernel and its finite coefficient calculation.

□

6.11 Conservative realization of measured feasibility

Realization preserves the measured incidence and its feasibility image.

Theorem 6.26 (Conservative computational comparison)**.** *Positive exact, signed, and Weil-realized incidences classify the same measured implementations. At a budget I, the natural map $iK\to j_BK$ sends f to $1\otimes f$, and*

$$iP\times_{j_BK} j_BM_I\cong iZ_I,\qquad \operatorname{im}(iZ_I\to iV)\cong i\mathcal{C}_{\mathrm{cost}}(I).$$

These comparisons also hold relatively over $\mathscr{S}$ for $i_W\mathscr{Z}\cong i_W\Delta P\times_{j_{\bar{B}}\bar{K}}\mathfrak{M}$.

Proof.

(1) **Preserve and reflect exact resource membership.** Theorem 2.13 preserves and reflects membership of an exact positive cost in its marked resource. At a Weil stage A, membership of $1\otimes\Phi(e)$ in $A\otimes_B M_I$ follows from membership in M_I. Conversely augmentation $A\to B$ recovers $\Phi(e)\in M_I$ from a stage witness.

(2) **Identify the incidence subobject and its image.** Flatness of A makes the scalar-extended marking monic. Thus the incidence is the same constant subobject at each stage; images are stagewise. The identical calculation at each budget a proves the relative assertion.

□

Equivalently, the realization square

$$\begin{array}{ccc} iM_I & \longrightarrow & j_BM_I \\ \downarrow & & \downarrow \\ iK & \longrightarrow & j_BK \end{array} \tag{6.17}$$

is cartesian.

Corollary 6.27 (Preservation of implemented feasibility and optimality)**.** *Formal principal parts and de Rham coefficient constructions preserve the original implementation diagram's feasible opens in* $\mathfrak{C}_{\mathrm{Ev}}$ *and their compactness and supercompactness. Thus the optimal-witness and finite-portfolio equivalences of Section 5 apply with the same implementation fibres and computational environments.*

Proof.

(1) **Recover measured incidence and feasibility by reduction.** The coefficient constructions act on the resource-coordinate variable. Reduction recovers the measured incidence by Theorem 6.26; its image and the lattice of opens of $\mathfrak{C}_{\mathrm{Ev}}$ are unchanged.

□

The comparison uses the split Weil extension. Arbitrary coefficient changes behave differently: tensoring $B \hookrightarrow K$ with B/tB gives a nonzero source and zero target.

The finite calculus now differentiates the measured resource families and their budget constraints. We next classify variations of the embedded budgets and calculate the successive feasibility obstructions.

7 Deformations of resource-constrained computation

Ordinary budget families have already been classified by $\mathfrak{C}_{\mathrm{Ev}}$. We now ask how an embedded budget can vary over a nilpotent coefficient extension. This is a different moduli problem. The earlier point object $\mathrm{Pt}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}}) \cong \mathrm{Idl}(L)$ has constant realization, with $(i\,\mathrm{Idl}(L))^D \cong i\,\mathrm{Idl}(L)$ by Section 6.5; it carries no infinitesimal variation. We therefore define deformations directly by extending the coefficient ring and varying the marked submodule of the extended fraction algebra. Tangents and lifting obstructions will describe this functor.

7.1 The marked deformation functor

For a Weil stage $A = B \oplus \mathfrak{n}_A$ put $K_A = K \otimes_B A$ and $\mathfrak{n}_{K,A} = K \otimes_B \mathfrak{n}_A$. The split augmentation gives this kernel formula. For an idempotent e, the multiplicative group $e + e\mathfrak{n}_{K,A}$ is formed in the corner ring eK_A; inverses are finite geometric series.

Definition 7.1 (The principal embedded-resource deformation functor)**.** Fix $M = Bg \hookrightarrow K$ and its support $e = e_g$. For a finite-projective Weil stage A, let $\mathrm{Def}^{\mathrm{pr}}_M(A)$ be the object of marked images $Ax \hookrightarrow K_A$ with $x \in eK_A$ reducing to g. Generators determine the same point precisely when their embedded images agree. An augmented Weil map $A \to A'$ sends this point to $A'x' \hookrightarrow K_{A'}$, where x' is the substituted generator.

To verify functoriality, write $x = g(e + z)$ for a unique $z \in e\mathfrak{n}_{K,A}$, since $gg^\dagger = e$. The multiplier $e + z$ is invertible in the corner, and the finite-projective stage A is flat over B. Multiplication by x therefore identifies the displayed image with eA. If $Ax = Ay$, each generator is a scalar multiple of the other; substitution preserves these two equations, hence preserves equality of images. Identity and composition hold on generators and descend to the image quotient. Tight replacement of g gives the same marked functor, as proved in Corollary 7.5.

The functor thus classifies nilpotent deformations of an embedded budget, while $h_{\mathfrak{C}_{\mathrm{Ev}}}(Z)$ classifies ordinary budget families over locales. We first calculate the deformation functor at first order, then use that tangent module to describe its formal classifying disk.

7.2 First-order deformations of finite-projective cells

At first order, a deformation changes the embedding of a module while reducing to its original marking. Let $A_1 = B[\varepsilon]/(\varepsilon^2)$ and $K_1 = K \otimes_B A_1 \cong K[\varepsilon]/(\varepsilon^2)$. For a displayed finite-projective $i_M : M \hookrightarrow K$, a framed deformation is an A_1-linear monomorphism $\widetilde{i} : M \otimes_B A_1 \hookrightarrow K_1$ reducing to i_M. The frame is the displayed source $M \otimes_B A_1$. Forgetting it identifies monomorphisms with the same embedded image. The theorem calculates this image quotient.

Theorem 7.2 (The embedded resource tangent module)**.** *Every framed first-order embedded deformation has the unique form*

$$i_h(m + \varepsilon n) = m + \varepsilon(n + h(m)), \qquad h \in \mathrm{Hom}_B(M, K). \tag{7.1}$$

Two deformations have the same image if and only if $h - h'$ factors through $M \hookrightarrow K$. Consequently the object of image classes is canonically

$$T_M^{\mathrm{emb}} \cong \mathrm{Hom}_B(M, K)/\,\mathrm{Hom}_B(M, M) \cong \mathrm{Hom}_B(M, K/M). \tag{7.2}$$

For $M = Bg \cong Be_g$ this reduces to

$$T_{Bg}^{\mathrm{emb}} \cong e_g K/Bg; \qquad T_B^{\mathrm{emb}} \cong K/B. \tag{7.3}$$

These isomorphisms are natural under marked isomorphisms of finite-projective resource modules and substitution of parameters.

Proof.

(1) **Classify framed first-order deformations.** The decompositions $A_1 = B \oplus \varepsilon B$ and $K_1 = K \oplus \varepsilon K$ are finite biproducts. An A_1-linear map reducing to i_M sends m to $m + \varepsilon h(m)$ for a unique B-linear $h : M \to K$; A_1-linearity forces its value at εn to be εn. This proves (7.1).

(2) **Prove that every framed deformation is monic.** If $i_h(m + \varepsilon n) = 0$, comparison of the constant coefficients gives $m = 0$ because $M \hookrightarrow K$ is monic. Comparison of the nilpotent coefficients then gives $n = 0$. Thus every i_h is a monomorphism.

(3) **Identify changes of framing.** If i_h and $i_{h'}$ have the same image, their monomorphisms factor through each other as inverse A_1-linear automorphisms of $M \otimes_B A_1$. Reduction of such an automorphism is the identity, since i_M is monic. Its formula is necessarily $m + \varepsilon n \mapsto m + \varepsilon(n + u(m))$ for a B-linear $u : M \to M$. The equation $i_{h'} = i_h(1 + \varepsilon u)$ yields $h' - h = i_M u$.

(4) **Identify the image-equivalence quotient.** Conversely this equation constructs the required inverse automorphism $1 - \varepsilon u$. The first quotient in (7.2) is therefore the image-equivalence quotient.

(5) **Compute the quotient by finite-projective duality.** Apply $\mathrm{Hom}_B(M, -)$ to $0 \to M \to K \to K/M \to 0$. The specified finite-free retract supplies a finite dual basis, so $\mathrm{Hom}_B(M, -) \cong M^\vee \otimes_B -$ is exact: the dual module is a retract of a finite free module. Consequently the cokernel is $\mathrm{Hom}_B(M, K/M)$.

(6) **Specialize to principal resources and verify naturality.** For $M \cong Be_g$ evaluation at e_g identifies $\mathrm{Hom}_B(M, K/M)$ with $e_g(K/M) = e_g K/Bg$. All maps are finite module operations and quotient universal properties, hence commute with parameter substitution.

□

Change of frame and its effective quotient. The tangent calculation can be organized by the changes of frame that leave the image fixed. Write $T_M = T_M^{\mathrm{emb}} \cong \mathrm{Hom}_B(M, K/M)$, $\mathcal{F}_M = \mathrm{Hom}_B(M, K)$, and $\mathcal{G}_M = \mathrm{End}_B(M)$. The additive action is $u \cdot h = h + i_M u$. Its compatibility with the embedding is

$$\begin{array}{ccc} M \otimes_B A_1 & \xrightarrow[\cong]{1+\varepsilon u} & M \otimes_B A_1 \\ {\scriptstyle i_{h'}}\downarrow & & \downarrow{\scriptstyle i_h} \\ K_1 & = & K_1. \end{array} \qquad h' = h + i_M u.$$

The action is free since i_M is monic. Its effective quotient is the marked-image object, presented by the coequalizer

$$\mathcal{G}_M \times \mathcal{F}_M \underset{\pi_2}{\overset{\mathrm{act}}{\rightrightarrows}} \mathcal{F}_M \twoheadrightarrow T_M. \tag{7.4}$$

Exactness of $\mathrm{Hom}_B(M, -)$ gives the short exact sequence

$$0 \longrightarrow \mathrm{End}_B(M) \xrightarrow{i_M \circ -} \mathrm{Hom}_B(M, K) \xrightarrow{q \circ -} \mathrm{Hom}_B(M, K/M) \longrightarrow 0.$$

The cokernel therefore removes exactly the infinitesimal changes of frame. Monicity of the marking makes the action free, and the effective quotient records its image.

For a principal resource $M = Bg$, the first-order image classes in Definition 7.1 coincide with those just computed. A frame sends g to $g + \varepsilon h(g)$; B-linearity makes $h(g)$ supported on e_g. Conversely each supported generator reducing to g gives a frame on $M \otimes_B A_1 \cong e_g A_1$. Equality of images is the same equivalence in both descriptions. Thus (7.3) is the first-order value of the already defined deformation functor.

For cofinite numerical input the class of $t^{-1} = n + 1$ is nonzero in K/B; the deformation

$$B[\varepsilon](1 + \varepsilon t^{-1}) \subset K[\varepsilon]$$

therefore differs from the constant deformation $B[\varepsilon] \subset K[\varepsilon]$ while reducing to the same embedded module $B \subset K$. This exhibits a nonzero moduli tangent already on the rank-one resource cell.

7.3 Formal resource cells and their classifying disks

The first-order quotient identifies the intrinsic tangent module. For a principal resource, finite logarithm and exponential extend this description to every nilpotence order. We now construct the quotient and representing formal disk of Definition 7.1.

Theorem 7.3 (The formal classifier of a principal resource cell)**.** *For $M = Bg \hookrightarrow K$, with $e = e_g$, the image classes of principal A-submodules generated by an element reducing to g and supported in eK_A form the functor*

$$\mathrm{Def}_M^{\mathrm{pr}}(A) = \frac{e + e\mathfrak{n}_{K,A}}{e + e\mathfrak{n}_A}, \qquad [u] \longmapsto A(gu) \subseteq K_A. \tag{7.5}$$

Put $T_M = \mathrm{Hom}_B(M, K/M)$. There are canonical natural isomorphisms

$$eK/eB \xrightarrow{\cong} T_M, \qquad \mathrm{Def}_M^{\mathrm{pr}}(A) \cong \mathfrak{n}_A \otimes_B T_M. \tag{7.6}$$

The first map sends a multiplier class $[z]$ to $(m \mapsto zm \bmod M)$. With $0 : 1 \to iT_M$, the formal disk

$$\mathfrak{D}_M = 1 \times_{iT_M} jBT_M \tag{7.7}$$

represents the functor: its value at A, equivalently the internal object of maps $D_A \to \mathfrak{D}_M$, is $\mathrm{Def}_M^{\mathrm{pr}}(A)$.

Proof.

(1) **Classify supported deformations by unit cosets.** The support inverse $g^\dagger = e/g$ satisfies $gg^\dagger = e$. Thus every supported generator reducing to g is uniquely gu with $u \in e + e\mathfrak{n}_{K,A}$. Multiplication by gu embeds eA in eK_A: the inclusion $A \hookrightarrow K_A$ is monic since A is a finite-free retract and regular denominators act injectively on it. Two such images agree precisely when $u^{-1}v$ is a unit of eA reducing to e, namely an element of $e + e\mathfrak{n}_A$. This proves the quotient classification and its functoriality.

(2) **Construct finite logarithm and exponential.** Choose the displayed bound r with $\mathfrak{n}_A^{r+1} = 0$. The ideal $e\mathfrak{n}_{K,A}$ has the same nilpotence bound. For x in that ideal define in the corner

$$\log(e + x) = \sum_{k=1}^{r} \frac{(-1)^{k+1}}{k} x^k, \qquad \exp(x) = e + \sum_{k=1}^{r} \frac{x^k}{k!}.$$

(3) **Identify the multiplicative and additive quotients.** The rational polynomial identities, reduced modulo terms of degree greater than r, prove that these are inverse and that logarithm takes products to sums. Increasing the displayed bound changes neither map. They carry the subgroup over eA onto $e\mathfrak{n}_A$, so they induce

$$\mathrm{Def}_M^{\mathrm{pr}}(A) \cong (eK \otimes_B \mathfrak{n}_A)/(eB \otimes_B \mathfrak{n}_A) \cong \mathfrak{n}_A \otimes_B (eK/eB).$$

(4) **Identify the tangent module intrinsically.** Here $\mathfrak{n}_A$ is a displayed finite-free retract, hence flat. Multiplication by g identifies eK/eB with eK/Bg; the dual-basis calculation of Theorem 7.2 identifies this with T_M. Its intrinsic formula is the multiplier map in the statement, so it is independent of the chosen generator.

(5) **Identify the representing formal disk.** Finally, augmentation on $A \otimes_B T_M$ has kernel $\mathfrak{n}_A \otimes_B T_M$ by the split decomposition of A. This calculates (7.7); the internal Yoneda lemma gives its representing property. All polynomial identities and quotients commute with Weil maps and parameters.

□

At every stage the logarithm is an isomorphism of the quotient presentations. In the corner supported on e, it gives

$$\begin{array}{ccccc} e + e\mathfrak{n}_A & \hookrightarrow & e + e\mathfrak{n}_{K,A} & \twoheadrightarrow & \mathrm{Def}_M^{\mathrm{pr}}(A) \\ \cong\downarrow \log & & \cong\downarrow \log & & \downarrow\cong \\ e\mathfrak{n}_A & \hookrightarrow & e\mathfrak{n}_{K,A} & \twoheadrightarrow & \mathfrak{n}_A \otimes_B T_M. \end{array} \tag{7.8}$$

The upper row is a quotient of multiplicative groups and the lower row is its additive cokernel presentation. Naturality of the finite logarithm and exponential makes this a diagram of Weil-stage functors.

Corollary 7.4 (Universal images and finite lifts)**.** *For $z \in \mathfrak{n}_A \otimes_B T_M$, choose a representative $\widetilde{z} \in e\mathfrak{n}_{K,A}$ locally along the quotient epimorphism. The image $A(g \exp \widetilde{z})$ is independent of that representative and constructs the universal marked family over $\mathfrak{D}_M$. Surjective Weil-stage maps induce internal epimorphisms on these deformation objects. For $A_r = B[d]/d^{r+1}$ and $M = B$,*

$$\mathrm{Def}_B^{\mathrm{pr}}(A_r) \cong \frac{1 + dK[d]/d^{r+1}}{1 + dB[d]/d^{r+1}} \cong \bigoplus_{j=1}^{r} d^j(K/B). \tag{7.9}$$

Reduction to order $r - 1$ has kernel K/B.

Proof.

(1) **Descend the universal image through the logarithmic quotient.** Changing $\widetilde{z}$ by $e\mathfrak{n}_A$ multiplies its exponential by a unit of eA, hence preserves the image. These equal images descend across the quotient epi and commute with stage maps, giving a submodule of K_A over the formal classifier.

(2) **Construct lifts along surjective stage maps.** A surjective augmented algebra map is surjective on augmentation ideals: subtract the augmentation of a lift. Tensoring preserves that epimorphism. The last assertion follows by the displayed direct-sum description.

□

Corollary 7.5 (Invariance under tight replacement)**.** *If $g' \in \Theta(g)$, then $Bg' = Bg$ as marked modules and their formal classifiers, universal images, and tangent comparisons are canonically identical.*

Proof.

(1) **Identify the deformation classifier under tight replacement.** Write $g' = vg$ with $v \in B^\times$ by Theorem 2.10. The support is unchanged and $A(g'u) = A(gu)$. The multiplier description of T_M and logarithm identify the same quotient and representing disk at every stage.

□

Logarithm and exponential use finite rational arithmetic at the displayed nilpotence bound, so they act on the coefficient presentations. Subgroup membership and equality of image classes remain the associated quotient predicates, with their separately specified observation structure. A deformation of a resource need not preserve a particular resource constraint. We now vary a polynomial resource value and its budget together. At first order, the change of the value modulo the budget must match the change of the budget at that value.

7.4 Deformation of incidence with a moving budget

Let $M \hookrightarrow K$ be a displayed finite-projective resource cell, $q : K \to Q = K/M$, and let $F : j_B H \to j_B K$ be one of the polynomial resource maps constructed in Theorem 6.20 and Proposition 6.21. A reduced feasible point $x : 1 \to H$ has $F(x) \in M$. The displayed generators of M supply the principal generator $g = \sum_i |i_M r(e_i)|$, where $i_M : M \hookrightarrow K$ is the marking, $r : B^n \to M$ is its displayed retraction, and (e_i) is the standard basis. Thus $M = Bg$ by bounded division, and the preceding formal classifier applies. Its independence of the generator was proved in Corollary 7.5. Define the incidence deformation functor by

$$\mathsf{Inc}_{F,x,M}(A) = \left\{ (x_A, N_A) : \begin{array}{ll} x_A \in A \otimes_B H, & (\epsilon_A \otimes 1)x_A = x, \\ N_A \in \mathrm{Def}_M^{\mathrm{pr}}(A), & F_A(x_A) \in N_A \end{array} \right\}. \tag{7.10}$$

Here $\epsilon_A : A \to B$ is the augmentation and $N_A \hookrightarrow K_A$ is the universal marked image of Corollary 7.4. Weil substitution acts on both entries and preserves membership by naturality of F and of that image. This constructs a subfunctor of the formal disk of $j_B H$ at x times $\mathfrak{D}_M$. Source frames are forgotten by equality of embedded images. Its first-order object is denoted $T_{(x,M)}\mathsf{Inc}_F := \mathsf{Inc}_{F,x,M}(B[\varepsilon]/\varepsilon^2)$.

Theorem 7.6 (The tangent of resource incidence)**.** *The first-order incidence deformations are the kernel of the B-linear map*

$$H \oplus T_M \longrightarrow Q, \qquad (v, \theta) \longmapsto qDF_x(v) - \theta(F(x)).$$

Equivalently,

$$T_{(x,M)}\mathsf{Inc}_F = \{(v, \theta) : qDF_x(v) = \theta(F(x))\}. \tag{7.11}$$

For a fixed budget this gives

$$T_x F^{-1}(j_B M) = \ker(qDF_x). \tag{7.12}$$

Proof.

(1) **Represent the varying point and budget.** A first-order point is $x + \varepsilon v$ and its value is $F(x) + \varepsilon DF_x(v)$ by the Weil coefficient identity. By Theorem 7.2, a budget deformation is locally represented by $h : M \to K$, with image of $m + \varepsilon n \mapsto m + \varepsilon(n + h(m))$ and class $\theta = qh$.

(2) **Derive the incidence equation in the quotient.** Membership of the value is exactly $DF_x(v) - h(F(x)) \in M$. Passing to Q gives (7.11).

(3) **Recover membership and descend the calculation.** Conversely that equation supplies the unique element of M representing the displayed difference, and hence the required membership. Replacing h by a map with the same class changes neither condition. The calculations therefore descend from the quotient cover, naturally in all parameters. The fixed-budget slice is $\theta = 0$.

□

The incidence deformation complexes. The same linearized map governs first-order solutions and higher-order corrections. We record the fixed and moving versions as two-term complexes of B-modules in degrees $0, 1$:

$$\begin{aligned} \mathsf{C}^{\mathrm{fix}}_{F,x,M} &= [\, H \xrightarrow{qDF_x} Q \,], \\ \mathsf{C}^{\mathrm{mov}}_{F,x,M} &= [\, H \oplus T_M \xrightarrow{L^{\mathrm{mov}}_{(x,M)}} Q \,], \\ L^{\mathrm{mov}}_{(x,M)}(v, \theta) &= qDF_x(v) - \theta(F(x)). \end{aligned} \tag{7.13}$$

In each complex, H^0 is the kernel and H^1 the cokernel. The kernel gives the first-order incidence deformations. At higher orders, a residual can be cancelled precisely when its class in the cokernel vanishes; the lift theorems below then identify the inhabited lift object as a torsor under H^0. Allowing the budget to vary gives the comparison

$$\begin{array}{ccc} H & \xrightarrow{qDF_x} & Q \\ {\scriptstyle v \mapsto (v,0)}\big\downarrow & & \big\| \\ H \oplus T_M & \xrightarrow[L^{\mathrm{mov}}_{(x,M)}]{} & Q. \end{array} \tag{7.14}$$

This compares the tangent and obstruction modules for the two problems: the vertical inclusion allows the additional budget direction while preserving the value residual.

Proposition 7.7 (Normal directions for a fixed marked budget)**.** *For any embedded resource M and geometric valuation family $F : j_B H \to j_B K$, the tangent of $S_F = F^{-1}(j_B M)$ at a feasible point is* $\ker(j_B(q)DF_x)$.

Proof.

(1) **Realize the fixed resource as a kernel.** Exact realization identifies $j_B M$ with the kernel of $j_B(q)$. Exponentiation by the dual-number object preserves this pullback.

(2) **Compute the tangent kernel at a feasible point.** Its coefficient formula gives $(qF(x), qDF_x(v))$, whose zero fibre at a feasible point is the asserted kernel. This proves (7.12) for arbitrary embedded budgets.

□

Corollary 7.8 (Rigidity of a tight budget containing a fixed cost)**.** *For a fixed implemented cost $c \in K$, the only principal formal deformation of Bc containing the constant cost c is the constant deformation. Its first-order version is $\{\theta \in T_{Bc} : \theta(c) = 0\} = 0$.*

Proof.

(1) **Translate containment of the fixed cost into a unit equation.** Write $e = e_c$. A deformation is $A(cu)$ for $u \in e + e\mathfrak{n}_{K,A}$. If $c \in A(cu)$, multiplication by $c^\dagger$ yields $e = au$ for some $a \in eA$.

(2) **Prove triviality of the deformation.** Thus $a = u^{-1}$ reduces to e and is a nilpotent perturbation of the unit in eA. Its finite geometric inverse belongs to eA, so $u \in e + e\mathfrak{n}_A$ and its class is trivial.

(3) **Deduce first-order rigidity.** At first order, c generates Bc, so a linear map vanishing on c is zero.

□

7.5 Higher-order feasibility and lifting obstructions

A feasible deformation through order r need not extend to order $r+1$. Its lower-order coefficients determine the next failure of the constraint, and a new coefficient changes that failure through the linearized incidence map. We calculate whether this residual can be cancelled. Use the fixed budget M and polynomial F above, and put $L_x = qDF_x : H \to Q$. An order-r feasible jet is

$$x_r = x + \sum_{j=1}^{r} d^j v_j \quad \text{over } B[d]/d^{r+1}, \qquad qF(x_r) = 0.$$

View the same displayed polynomial over $B[d]/d^{r+2}$ and define

$$R_{r+1} = [d^{r+1}]qF(x_r) \in Q.$$

The coefficients v_j are uniquely determined by the jet, so this is its canonical residual.

Theorem 7.9 (The obstruction and the object of lifts)**.** *The obstruction to lifting x_r by one order is*

$$\mathrm{ob}_{r+1}(x_r) = [R_{r+1}] \in \operatorname{coker} L_x. \tag{7.15}$$

It vanishes exactly when the internal object of feasible lifts is inhabited. That object is the fibre $\{w : H : L_x(w) = -R_{r+1}\}$; when inhabited, it is a torsor under $\ker L_x$. The construction is natural in the polynomial data and reduced parameters, and applies relatively over the structured resource space.

Proof.

(1) **Compute the next-order correction equation.** A lift has the unique form $x_r+d^{r+1}w$. The polynomial Taylor identity gives, modulo d^{r+2},

$$qF(x_r + d^{r+1}w) = d^{r+1}(R_{r+1} + L_x(w)).$$

Indeed terms quadratic in the added correction have degree at least $r + 2$, and the positive-degree part of $DF_{x_r} - DF_x$ also has that property after multiplication by d^{r+1}. For $r = 0$, the same identity is the dual-number formula.

(2) **Identify lifts with the linear fibre.** Equality of polynomial coefficients proves the fibre assertion.

(3) **Characterize existence by the obstruction class.** The kernel of $Q \to \operatorname{coker} L_x$ is the regular image of L_x, so zero obstruction means exactly internal existence of w.

(4) **Identify the lift torsor and parameter stability.** Two lifts differ by a unique element of $\ker L_x$, and addition by such an element preserves a lift. This proves the torsor assertion. All maps are coefficient operations, kernels, and cokernels; their internal constructions and identities apply in $\mathscr{S}$ as well.

□

Example 7.10 (The second-order obstruction of the measured family)**.** For the family $F(\lambda) = 1 + t^{-2}\lambda^2$ of Example 6.2, take $H = B$, $M = B$, and $x = 0$. Here $L_0 = qDF_0 = 0$, and the first-order jet $\lambda_1 = d$ is feasible. Its next residual is $R_2 = [t^{-2}] \in K/B$, which is nonzero because $(n + 1)^2$ is unbounded. Every lift $d + d^2w$ has the same square modulo d^3, so no parameter correction removes the obstruction. Under the coordinate isomorphism $B \to t^{-1}B$, $\lambda \mapsto b = t^{-1}\lambda$, this is equivalently the family $1 + b^2$ with first-order direction t^{-1}.

7.6 Complete principal-resource moduli

The principal-resource classifier of Theorem 7.3 applies at every finite nilpotence order. We now assemble its image classes into a compatible formal family. The classification is taken at the level of those classes, so compatible generators need not be chosen.

Theorem 7.11 (Completed principal deformation functor)**.** *For a principal marked resource $M = Bg \hookrightarrow K$ put $T_M = \operatorname{Hom}_B(M, K/M)$ and $A_r = B[d]/d^{r+1}$. Then*

$$\widehat{\operatorname{Def}}^{\mathrm{pr}}_M := \varprojlim_r \operatorname{Def}^{\mathrm{pr}}_M(A_r) \;\cong\; d\,T_M[[d]] = \prod_{j\geq 1} d^j T_M. \tag{7.16}$$

The transition maps are projections onto initial coefficient segments. Every finite-order class admits a compatible extension to all higher orders, and the displayed comparison is natural in marked principal-resource isomorphisms and parameters.

Proof.

(1) **Compute each finite deformation stage.** The augmentation ideal of A_r is the displayed free module $\mathfrak{n}_r = dB \oplus \cdots \oplus d^r B$. Applying the finite logarithm classification of Theorem 7.3 yields

$$\operatorname{Def}^{\mathrm{pr}}_M(A_r) \cong \mathfrak{n}_r \otimes_B T_M \cong \bigoplus_{1\leq j\leq r} d^j T_M.$$

(2) **Identify the inverse limit with coefficient sequences.** Its naturality in Weil-stage maps identifies truncation with deletion of the highest coefficient. A point of the inverse limit is therefore exactly a compatible finite-prefix family of T_M-coefficients, equivalently an internal sequence $(z_j)_{j\geq 1}$. This gives (7.16).

(3) **Construct an extension of every finite prefix.** Extend a given coefficient prefix by zero coefficients to obtain an explicit compatible sequence of deformation *classes*.

□

Principal-resource image deformations are formally unobstructed by coefficientwise zero extension. The following calculation describes the quotient that supplies their coefficients.

Proposition 7.12 (B-linear nonsplitting of the normal quotient)**.** *In the cofinite numerical fibre, the quotient $q : K \twoheadrightarrow K/B$ does not admit a B-linear section.*

Proof.

(1) **Exhibit nonzero torsion in the normal quotient.** Let $t(n) = 1/(n+1) \in B$. It is regular in B and invertible in K, while $t^{-1} = n + 1 \notin B$. Therefore $\xi = q(t^{-1}) \in K/B$ is nonzero but $t\xi = q(1) = 0$.

(2) **Derive a contradiction from a linear section.** If a B-linear section $s : K/B \to K$ existed, then

$$t\, s(\xi) = s(t\xi) = 0.$$

Multiplication by t is invertible on K, so $s(\xi) = 0$, contradicting $q(s(\xi)) = \xi \neq 0$.

□

We now combine these two calculations. The budget itself admits formal extension, but its extension must still be compatible with the resource value. Allowing a budget correction enlarges the linearized incidence operator and gives the moving-budget obstruction.

7.7 Higher-order incidence with a moving principal budget

Let H be a displayed finite-projective B-module, let $F : H \to K$ be a specified polynomial resource map, and let $x \in H$ satisfy $F(x) \in M = Bg$. Write

$$Q = K/M, \quad q : K \twoheadrightarrow Q, \quad T_M = \mathrm{Hom}_B(M, Q).$$

The first-order moving-incidence operator of Section 7 is

$$L^{\mathrm{mov}}_{(x,M)} : H \oplus T_M \longrightarrow Q, \qquad (v, \theta) \longmapsto qDF_x(v) - \theta(F(x)). \tag{7.17}$$

This map is intrinsic to the marking $M \hookrightarrow K$.

Definition 7.13 (Feasible moving formal jet)**.** At order r, a moving formal jet is a valuation coefficient $x_r = x + \sum_{j=1}^{r} d^j v_j \in H \otimes_B A_r$ together with a principal deformation image $M_r \in \mathrm{Def}^{\mathrm{pr}}_M(A_r)$. It is feasible when the image of $F(x_r)$ in $(K \otimes_B A_r)/M_r$ is zero. The object of these pairs is formed as a pullback of the corresponding polynomial evaluation and marked-incidence subobjects. All its equalities are Ω-valued.

Theorem 7.14 (Moving-budget lifting obstruction)**.** *For a feasible moving jet (x_r, M_r), there is a canonical class*

$$\mathrm{ob}^{\mathrm{mov}}_{r+1}(x_r, M_r) \in \mathrm{coker}\, L^{\mathrm{mov}}_{(x,M)}. \tag{7.18}$$

Its vanishing is equivalent to internal inhabitation of the object of feasible lifts to order $r+1$. When inhabited, that object is a torsor under $\ker L^{\mathrm{mov}}_{(x,M)}$. The constructions are natural in polynomial resource maps, marked principal modules, and all supplied parameters. The fixed-budget theorem Theorem 7.9 is the specialization in which the budget tangent component is zero.

Proof.

(1) **Represent the budget by a supported logarithm.** Write $e = e_g$. A principal image class admits, locally along the quotient epimorphism of Theorem 7.3, a supported logarithmic multiplier

$$z_r(d) = \sum_{j=1}^{r} d^j z_j \in eK \otimes_B dA_r.$$

(2) **Express feasibility through an ambient unit.** In the corner algebra $eK \otimes_B A_r$, its finite exponential $\exp_e(z_r)$ has constant coefficient e. Extend it to a unit of $K \otimes_B A_r$ by

$$U_r(d) = (1-e) + \exp_e(z_r(d)), \qquad U_r^{-1} = (1-e) + \exp_e(-z_r(d)).$$

Then $M_r = U_r(d)\,(M \otimes_B A_r)$, and its membership equation is equivalent to

$$q(U_r^{-1}F(x_r)) = 0 \pmod{d^{r+1}}.$$

(3) **Define the next residual coefficient locally.** Regard the displayed polynomials as their zero-extended representatives modulo d^{r+2}, evaluate the finite exponential to that order, and put

$$R_{r+1} = [d^{r+1}]q(U_r^{-1}F(x_r)) \in Q.$$

(4) **Descend the residual through changes of representative.** This is well-defined locally. Two logarithmic representatives of the same image class differ by $b(d) \in deB[d]/d^{r+1}$, by the logarithmic quotient in Theorem 7.3. Zero-extending both logarithms and taking their exponentials modulo d^{r+2} gives

$$U_r' = U_r h, \qquad h = (1-e) + \exp_e(b) \in (B[d]/d^{r+2})^{\times}, \qquad h(0) = 1.$$

The residual series is consequently multiplied by h^{-1}. Its coefficients through degree r vanish in Q, so the coefficient of degree $r+1$ is unchanged. This equality on the kernel pair of the representative epimorphism constructs the descended map R_{r+1} by the effective-quotient property.

(5) **Represent the valuation and budget corrections.** A next-order lift adds $d^{r+1}v$ to the valuation and a new budget tangent coefficient $\theta \in T_M$. Locally represent θ by a supported multiplier $\zeta \in eK$, so its infinitesimal action on $m \in M$ is $\theta(m) = \zeta m \bmod M$.

(6) **Compute the linearized correction equation.** A finite polynomial Taylor expansion gives

$$F(x_r + d^{r+1}v) \equiv F(x_r) + d^{r+1}DF_x(v) \pmod{d^{r+2}}.$$

The inverse supported multiplier changes by $U_{r+1}^{-1} \equiv U_r^{-1}(1 - d^{r+1}\zeta)$ at that order. Since its constant term is 1 and $F(x) \in M$, the new coefficient of the feasibility equation is

$$R_{r+1} + qDF_x(v) - \theta(F(x)) = R_{r+1} + L^{\mathrm{mov}}_{(x,M)}(v,\theta).$$

(7) **Verify independence of the new representative.** Changing ζ by eB has no effect on this equation because $F(x) \in M$, so it is independent of the new local lift as well.

(8) **Identify the canonical obstruction.** It follows that a feasible lift exists internally if and only if $-R_{r+1}$ belongs to the regular image of $L^{\mathrm{mov}}_{(x,M)}$, equivalently if and only if its cokernel class vanishes. This defines the canonical obstruction (7.18).

(9) **Identify the torsor and parameter coherence.** The fibre of solutions to a linear inhomogeneous equation is a torsor under its kernel: subtraction of two solutions gives a unique kernel element, and addition of such an element preserves solutions. Every step was a finite polynomial equation or exact quotient calculation and hence commutes with parameter substitution.

□

Corollary 7.15 (Comparison of fixed and moving obstructions)**.** *For a fixed-budget feasible jet, regarded as a moving jet with constant budget, the morphism* (7.14) *carries its fixed obstruction to its moving obstruction. Explicitly the comparison square is*

$$\begin{array}{ccc} Q & \twoheadrightarrow & H^1(\mathsf{C}^{\mathrm{fix}}_{F,x,M}) \\ \| & & \downarrow \\ Q & \twoheadrightarrow & H^1(\mathsf{C}^{\mathrm{mov}}_{F,x,M}). \end{array}$$

Proof.

(1) **Identify the residual for a constant budget.** The constant budget is represented by $U_r = 1$. Both residual constructions therefore give $[d^{r+1}]qF(x_r)$ in Q.

(2) **Construct the comparison of obstruction classes.** The inclusion $\mathrm{im}(qDF_x) \subseteq \mathrm{im}(L^{\mathrm{mov}}_{(x,M)})$ induces the displayed map of cokernels and proves the assertion.

□

Example 7.16 (The measured family with a moving budget)**.** For Examples 6.2 and 7.10, choose

$$M_2 = (1 + d^2t^{-2})B[d]/d^3.$$

Then $F(d) = 1 + d^2t^{-2} \in M_2$. Here $T_B = K/B$ and $L^{\mathrm{mov}}_{(0,B)}(v, \theta) = -\theta(1)$ is surjective, so the moving obstruction vanishes. The displayed generator supplies the feasible second-order budget deformation.

The measured linear budget. Multiplication by the full-support germ $g = t^{-1}$ transports this example to the budget of the doubling program in Section 12.4. In fact

$$K/B \xrightarrow{\cong} K/Bg, \quad [a] \longmapsto [ga], \qquad F_g(b) = g(1 + b^2).$$

The map is well-defined and invertible because g is a unit of K, and it carries B onto Bg. Thus the fixed second-order residual is $[gt^{-2}]$, and the moving budget $(1 + d^2t^{-2})Bg[d]/d^3$ contains $F_g(dt^{-1})$.

7.8 Infinitesimal compatibility of budget diagrams

Proposition 7.17 (The tangent of a resource flag)**.** *For displayed finite-projective embedded cells $M_0 \subseteq M_1$, write $q_{01} : K/M_0 \to K/M_1$. The first-order deformations of this flag are the pairs*

$$(\theta_0, \theta_1) \in \mathrm{Hom}_B(M_0, K/M_0) \oplus \mathrm{Hom}_B(M_1, K/M_1) \quad \textit{with} \quad q_{01}\theta_0 = \theta_1|_{M_0}.$$

For a finite diagram, impose this equation on every inclusion.

Proof.

(1) **Derive the first-order inclusion equation.** Locally lift the classes to $h_i : M_i \to K$ using the exact finite-dual-basis presentations. The deformed image of M_0 lies in that of M_1 exactly when $h_0(m) - h_1(m) \in M_1$ for every $m \in M_0$. This is the displayed equation.

(2) **Recover the inclusion and descend the condition.** Its converse constructs the inclusion by the unique correction in M_1; changes of lifts preserve it. The condition therefore descends and is necessary and sufficient.

(3) **Extend the calculation to finite diagrams.** Imposing it for finitely many edges gives the diagram statement, with commutativity following from the common ambient marking. This is the formal incidence problem lying over the classifier of Theorem 3.8, not the tangent of its infinitesimally discrete constant point object.

□

7.9 Complete finite budget-diagram equations

A system of resource constraints may include inclusions between several budgets. Deforming it requires those inclusions to persist. For a finite diagram $M_i \subseteq M_j$ of principal marked resources, write $M_i = Bg_i$, put $e_i = e_{g_i}$ and choose local supported logarithms $z_i(d)$. Use the ambient units

$$U_i(d) = (1 - e_i) + \exp_{e_i}(z_i(d)) \in K[d]/d^{r+2}, \qquad U_i(0) = 1.$$

Their products with M_i represent the deformations. Extending by $1 - e_i$ makes every inverse below an inverse in the common ambient coefficient algebra. An edge remains an inclusion precisely when

$$q_j(U_j(d)^{-1}U_i(d)m) = 0 \quad (m \in M_i), \qquad q_j : K \to K/M_j.$$

It is enough to test a displayed finite family of generators. The coefficient at first order is the existing tangent flag equation $q_{ij}\theta_i = \theta_j|_{M_i}$. At degree $r+1$ the new coefficients occur through this same linear operator, while all lower coefficients determine an explicit residual.

Intrinsically the finite diagram has the two-term complex

$$\mathsf{C}^{\bullet}_{\mathrm{flag}} = \left[\bigoplus_i T_{M_i} \xrightarrow{\partial_{\mathrm{flag}}} \prod_{(i \to j)} \mathrm{Hom}_B(M_i, K/M_j) \right], \tag{7.19}$$

where $\partial_{\mathrm{flag}}(\theta)_{ij} = q_{ij}\theta_i - \theta_j|_{M_i}$. The products and sums here are finite. Each target is an intrinsic edge quotient module; a displayed generator identifies it with the corresponding supported quotient used in the coefficient calculation. The vanishing of one edge component is the commutative square

$$\begin{array}{ccc} M_i & \hookrightarrow & M_j \\ {\scriptstyle\theta_i}\downarrow & & \downarrow{\scriptstyle\theta_j} \\ K/M_i & \xrightarrow[q_{ij}]{} & K/M_j. \end{array}$$

Use this differential as L_{flag} below. The degree-$(r+1)$ residual on an edge is the B-linear map $m \mapsto [d^{r+1}]q_j(U_j^{-1}U_i m)$, so it has precisely the target in (7.19). Changes of next-order

coefficients act through ∂_{flag}. Thus H^0 organizes compatible tangent directions and H^1 the finite-diagram obstructions.

Proposition 7.18 (Formal obstruction for a finite budget flag)**.** *A finite diagram of principal embedded-resource inclusions has a constructed linear first-order compatibility map L_{flag} into the finite product of its edge quotient modules. Every compatible order-r moving diagram has a canonical order-$(r+1)$ obstruction in* $\operatorname{coker} L_{\mathrm{flag}}$*. Its vanishing is internal inhabitation of the compatible lift object; when inhabited that object is a torsor under* $\ker L_{\mathrm{flag}}$*. Adding polynomial valuation incidences enlarges the same finite operator by their first-order rows and gives the analogous obstruction theorem.*

Proof.

(1) **Extract the residual on each inclusion edge.** For each inclusion edge, conjugate the deformed inclusion by the target supported unit as displayed above. Its expression has zero coefficients through degree r. Extract the next coefficient.

(2) **Prove independence of representatives.** Changing logarithmic representatives gives $U_i' = U_i h_i$, where h_i is an ambient $B[d]$-unit with constant term 1 and preserves M_i. For an edge $i \to j$, commutativity of the coefficient algebra gives

$$q_j\big((U_j')^{-1}U_i' m\big) = h_j^{-1} h_i \, q_j(U_j^{-1} U_i m).$$

The quotient is a $B[d]$-module, and the original expression has zero coefficients through order r. Since $(h_j^{-1}h_i)(0) = 1$, the order-$(r+1)$ residual is unchanged. This proves descent on both the source and target representative kernel pairs.

(3) **Identify the flag correction map.** New coefficients of source and target units enter linearly and give precisely the first-order flag-tangent map.

(4) **Assemble the obstruction and lift torsor.** The finite graph of edges allows all residual coefficients and maps to be assembled into one finite product. The lift equation is therefore $L_{\mathrm{flag}}(w) = -R_{r+1}$. Its image, cokernel, and kernel supply the obstruction, inhabited fibre, and torsor statements exactly as in the preceding theorem. Adding valuation rows uses the same polynomial Taylor calculation.

□

The next example combines two measured incidences with a budget inclusion. Its obstruction will then be compared under changes of the allowed overheads.

Example 7.19 (An obstruction to jointly moving two budgets)**.** In the cofinite numerical fibre put $t(n) = 1/(n+1)$ and take

$$M_0 = B \subset M_1 = t^{-1}B, \qquad F_0(\lambda) = 1 + t^{-2}\lambda^2, \qquad F_1(\lambda) = t^{-1}.$$

The first valuation is Example 6.2; the second is measured by a program that emits $n+1$ unit-cost idle events and returns its input. At order one, $\lambda = d$ and the two constant budgets satisfy both incidences and their inclusion.

Each incidence separately extends to order two: the inner budget may be $(1 + d^2t^{-2})B[d]/d^3$, while the outer budget remains constant. For a joint lift, put $Q_i = K/M_i$, let $q_{01} : Q_0 \to Q_1$, and write $a = \theta_0(1) \in Q_0$, $b = \theta_1(t^{-1}) \in Q_1$ for the new budget coefficients. A parameter correction $d + d^2v$ has square d^2 modulo d^3. Consequently the two incidence equations and the inclusion equation require

$$a = [t^{-2}]_{Q_0}, \qquad b = 0, \qquad q_{01}a - tb = 0.$$

The last equation uses $\theta_1(1) = t\theta_1(t^{-1})$. It would imply $t^{-2} \in t^{-1}B$, contrary to the unboundedness of t^{-1}. Thus no joint lift exists.

The full obstruction calculation identifies its intrinsic quotient. The correction operator is

$$L : B \oplus Q_0 \oplus Q_1 \longrightarrow Q_0 \oplus Q_1 \oplus Q_1, \qquad L(v, a, b) = (-a, -b, q_{01}a - tb).$$

The surjection $(x, y, z) \mapsto q_{01}x - ty + z$ has kernel $\operatorname{im} L$: if its value is zero, the triple is $L(0, -x, -y)$. Hence $\operatorname{coker} L \cong Q_1$, and the residual $([t^{-2}]_{Q_0}, 0, 0)$ represents

$$o = [t^{-2}] \in K/t^{-1}B, \qquad \operatorname{Ann}_B(o) = tB. \tag{7.20}$$

Indeed $bo = 0$ means $bt^{-2} \in t^{-1}B$, equivalently $b \in tB$. The obstruction records compatibility of the budget system even though both individual moving-incidence problems are solvable.

7.10 Gauge change of infinitesimal resources

Proposition 7.20 (Localization of finite-cell deformation tangents)**.** *For a multiplicative overhead ideal P containing $[1]$ and a finite-projective resource module $M \subseteq K$, localization induces a canonical isomorphism*

$$B_P \otimes_B T_M^{\mathrm{emb}} \cong \operatorname{Hom}_{B_P}(B_P \otimes_B M,\, K/B_P M). \tag{7.21}$$

For $M = B$ it sends the tangent quotient K/B to K/B_P. The principal formal-deformation functor commutes with finite-projective Weil coefficient changes $A \mapsto B_P \otimes_B A$ at the level of its defining unit-group quotients.

Proof.

(1) **Localize the ambient module and normal quotient.** The coefficient ring B_P is a localization of B at regular denominators, and K is already the total quotient ring. Hence $B_P \otimes_B K \cong K$. Flat localization sends the quotient K/M to $K/B_P M$.

(2) **Commute tangent Hom objects with localization.** A finite-projective module has a finite dual basis, so its internal hom commutes with scalar extension:

$$B_P \otimes_B \operatorname{Hom}_B(M, K/M) \cong \operatorname{Hom}_{B_P}(B_P \otimes_B M, K/B_P M).$$

Apply Theorem 7.2 to obtain (7.21).

(3) **Construct the comparison of principal deformation functors.** For principal cells, tensor associativity identifies the extended finite Weil algebra and its nilpotent augmentation ideal. The embeddings $B_P \subset K$ and $B \subset B_P$ map the numerator and denominator unit groups of (7.5) into their localized counterparts, commuting with all stage maps and reduction. This constructs the functorial coefficient-change comparison.

□

Theorem 7.21 (Overhead localization of resource obstructions)**.** *For the fixed, moving, or finite-diagram incidence complexes constructed above, write $C^\bullet = [E^0 \xrightarrow{L} E^1]$. Let $B_P = S_P^{-1}B$ be an overhead localization from Theorem 2.23. There are natural identifications*

$$C_P^\bullet \cong B_P \otimes_B C^\bullet, \qquad H^i(C_P^\bullet) \cong B_P \otimes_B H^i(C^\bullet) \quad (i = 0, 1).$$

At every finite order, the obstruction o of a feasible jet maps to $o_P = 1 \otimes o$, and

$$o_P = 0 \iff \exists s \in S_P \; so = 0 \iff S_P \cap \operatorname{Ann}_B(o) \text{ is inhabited.} \tag{7.22}$$

The existential and the lift objects are interpreted internally.

Proof.

(1) **Compare the localized deformation complexes.** Localization identifies $B_P \otimes_B K$ with K and carries K/M to $K/B_P M$. Every Hom object in the complexes has a displayed finite-projective source. Its finite dual basis therefore gives the scalar-extension identification of Proposition 7.20. The finite sums, edge products, derivatives, and evaluation maps in L commute with these identifications, proving the complex comparison.

(2) **Preserve tangent and obstruction modules.** Flatness of localization preserves its kernel and cokernel.

(3) **Identify the localized obstruction class.** Residuals are obtained by finite coefficient extraction from polynomial valuations and finite supported exponentials. Those operations commute with scalar extension. Their equality on representative kernel pairs therefore identifies the localized obstruction with $1 \otimes o$.

(4) **Characterize vanishing by denominator annihilation.** By the fraction-equivalence relation, $o/1 = 0$ in $S_P^{-1}H^1(C)$ exactly when some $s \in S_P$ annihilates o. This gives (7.22); the localized obstruction theorem identifies its vanishing with the inhabited lift object.

□

For Example 7.19, allowing the polynomial gauge $B[t^{-1}]$ removes the obstruction, since $to = 0$. Polylogarithmic overheads do not: writing $\lambda(n) = 1 + \mathrm{bitlength}(n+1)$, no λ^{-k} belongs to tB. On $n = 2^m$ the ratio $t^{-1}\lambda^{-k}$ is unbounded, as the binomial lower bound for 2^m of degree $k+1$ shows. Thus the same finite diagram has no lift after polylogarithmic localization but has a lift after polynomial localization. The annihilator gives the precise overhead criterion.

The resource deformation problem is now expressed by tangent modules, incidence complexes, and their successive lift objects. The following sections construct the geometric function objects and transport structures associated with these families.

8 Geometric function objects and differential cohesion

The preceding calculus describes variations of measured resource values and embedded budgets. To compare functions on their infinitesimal neighbourhoods with differential forms and transport, we now represent the coordinate spaces themselves. A nilpotent arrow records both a family and its reduced boundary. Its two endpoints construct the adjunctions below, while affine representability identifies its function objects. Restriction to based probes recovers the coefficient calculations used in the resource deformation theory.

8.1 The primary infinitesimal-arrow geometry and its étale topology

A nilpotent thickening has a smaller boundary and a larger infinitesimal space. Keeping both endpoints produces the primary geometry. Its restriction to the based stages of Section 6.3 will recover their resource coefficient calculus.

Definition 8.1 (Ordinary algebraic and infinitesimal tests)**.** Let $\mathsf{Aff}_B^{\mathrm{fp}}$ be the opposite of the internal category of finitely presented commutative B-algebras. A presentation is a quotient $B[x_1, \ldots, x_n]/(f_1, \ldots, f_m)$ by a displayed finite list; its equations, homomorphisms, and changes of presentation are formed by internal finite lists, quotients, and algebraic

equality. Write $\operatorname{Spec} A$ for the corresponding object of $\mathsf{Aff}_B^{\mathrm{fp}}$. Let I_B be the internal category whose objects are surjective maps of those algebras $S \twoheadrightarrow R$ with a nilpotent kernel, written

$$j = (U = \operatorname{Spec} R \hookrightarrow T = \operatorname{Spec} S).$$

Its arrows are *all* commutative squares of the corresponding affine arrows. The nilpotence condition is an existential finite power equation, $\exists n \geq 1\ (\ker(S \to R))^n = 0$; witnesses enter the arguments by existential elimination.

Finite polynomial codes and their interpretations define these categories internally. Surjectivity and nilpotence are bounded power-object predicates. Finite polynomial and finite-projective presentations provide the computationally named tests.

Lemma 8.2 (The two affine endpoints)**.** *There are functors $c, d : \mathsf{I}_B \to \mathsf{Aff}_B^{\mathrm{fp}}$ and $\iota : \mathsf{Aff}_B^{\mathrm{fp}} \to \mathsf{I}_B$ with*

$$c(j) = T, \qquad d(j) = U, \qquad \iota(V) = (V \xrightarrow{\mathrm{id}} V), \qquad c \dashv \iota \dashv d, \quad c\iota = d\iota = \mathrm{id}\,.$$

Proof.

(1) **Identify maps into the identity arrow.** A square $j \to \iota(V)$ is uniquely specified by its larger endpoint map $T \to V$; the smaller map must be its composite with $U \to T$.

(2) **Identify maps out of the identity arrow.** A square $\iota(V) \to j$ is uniquely specified by a map $V \to U$; its larger map is forced.

(3) **Verify naturality and the endpoint identities.** These give mutually inverse internal hom-object bijections, natural in both variables; the two identities follow on identity arrows.

□

The two endpoints have different universal properties. Their adjunctions are expressed by

$$\mathrm{Hom}_{\mathsf{Aff}_B^{\mathrm{fp}}}(c(j), V) \cong \mathrm{Hom}_{\mathsf{I}_B}(j, \iota V), \qquad \mathrm{Hom}_{\mathsf{I}_B}(\iota V, j) \cong \mathrm{Hom}_{\mathsf{Aff}_B^{\mathrm{fp}}}(V, d(j)).$$

Their canonical arrows are the commutative squares

$$\begin{array}{ccc} U & \hookrightarrow & T \\ \downarrow & & \| \\ T & = & T \end{array} \qquad \begin{array}{ccc} U & = & U \\ \| & & \downarrow \\ U & \hookrightarrow & T. \end{array}$$

These are respectively the arrows $j \to \iota c(j)$ and $\iota d(j) \to j$. Their opposite directions will determine the endpoint adjunctions on sheaves.

On $\mathsf{Aff}_B^{\mathrm{fp}}$, use the *big finitely presented étale site*: a basic covering family consists of a finite presented family of finitely presented étale maps $T_\lambda \to T$ with an effective jointly covering image. Such finite families generate the topology on the finitely presented affine site; covering diagrams are indexed internal families. On I_B, declare the pullbacks

$$U_\lambda = U \times_T T_\lambda \hookrightarrow T_\lambda \quad (T_\lambda \to T \text{ an étale covering})$$

to be basic covers.

Lemma 8.3 (Étale lift of finite nilpotent covering diagrams)**.** *For a finite nilpotent thickening $U \hookrightarrow T$, base change induces an equivalence between the categories of finitely presented étale T-schemes and étale U-schemes. It preserves finite intersections and jointly covering families, and the comparison is natural in commutative squares of thickenings. Consequently the larger- and smaller-endpoint constructions carry the stated étale covers to étale covers.*

Proof.

(1) **Construct lifts across square-zero and nilpotent ideals.** For a square-zero ideal the standard-étale Jacobian presentation gives uniqueness and existence of lifts: the invertible Jacobian solves the correction equation by finite linear algebra. Iteration over the powers of a nilpotent ideal provides the finite-order statement.

(2) **Glue local lifts by uniqueness.** An arbitrary étale presentation is locally standard étale; uniqueness on overlaps and ordinary étale descent glue the local lifts and maps. This is the infinitesimal invariance of the étale site (Stacks Project Authors 2026b, Theorems 41.15.2 and 41.16.1; Section 59.45).

(3) **Preserve covers, refinements, and descent diagrams.** The equivalence identifies covering categories, refinements, and Čech diagrams. Surjectivity of a family is preserved since a nilpotent closed immersion induces the same underlying open-support lattice. The proof is finite in each nilpotence witness.

□

Theorem 8.4 (Differential cohesion over the Landau ring)**.** *The internal étale sheaf categories*

$$\mathscr{X}_{B,\mathrm{et}} = \mathrm{Sh}_{\mathrm{et}}(\mathsf{Aff}_B^{\mathrm{fp}}; \mathcal{E}), \qquad \mathscr{H}_{B,\mathrm{et}} = \mathrm{Sh}_{\mathrm{et}}(\mathsf{I}_B; \mathcal{E})$$

carry four adjoints

$$\iota_! \dashv \iota^* \dashv \iota_* \dashv \iota^!. \tag{8.1}$$

Both endpoint embeddings are fully faithful, and

$$(\iota_! F)(U \hookrightarrow T) = F(T), \qquad (\iota_* F)(U \hookrightarrow T) = F(U).$$

The first three functors preserve all internal limits and colimits; $\iota^!$ preserves all internal limits. The associated idempotent modalities

$$\mathfrak{R} = \iota_! \iota^*, \qquad \mathfrak{I}_{\mathrm{inf}} = \iota_* \iota^*, \qquad \mathfrak{C} = \iota_* \iota^!$$

obey $\mathfrak{R} \dashv \mathfrak{I}_{\mathrm{inf}} \dashv \mathfrak{C}$. The canonical $\iota_! \to \iota_*$ and $\mathfrak{R} \to \mathfrak{I}_{\mathrm{inf}}$ are not isomorphisms.

Proof.

(1) **Identify the presheaf adjoints and full faithfulness.** On presheaves, left and right Kan extension along ι^{op} are evaluation at c and d, respectively, by Lemma 8.2. Their restrictions to the identity-arrow objects recover the input functor, proving full faithfulness.

(2) **Compare the endpoint covering diagrams.** We spell out the sheafification comparison. At an arrow $j = (U \hookrightarrow T)$, basic covers and their refinements are precisely the finite étale covering diagrams of T. For the larger endpoint c this is the defining identification. For the smaller endpoint d, Lemma 8.3 identifies that category, including its intersections and refinements, with the corresponding category over U. At an identity arrow, the same diagrams are identity-arrow covers.

(3) **Commute endpoint functors with sheafification.** In each case, matching families for a precomposed presheaf are therefore the corresponding endpoint matching families. Taking their equivalence classes under refinement gives natural isomorphisms

$$(F \circ c)^+ \cong F^+ \circ c, \qquad (F \circ d)^+ \cong F^+ \circ d, \qquad (H \circ \iota)^+ \cong H^+ \circ \iota.$$

Applying the construction twice gives the same comparisons for sheafification. Thus endpoint precomposition and restriction preserve sheaves and commute with sheafification.

(4) **Verify preservation of limits and colimits.** Presheaf limits are pointwise, and sheaf colimits are sheafifications of pointwise colimits; this proves all the asserted limit and colimit preservation for $\iota_!, \iota^*, \iota_*$.

(5) **Construct the remaining adjoint.** The fourth right adjoint is constructed by the internal end

$$(\iota^! H)(T) = \mathrm{Nat}_{\mathscr{H}_{B,\mathrm{et}}}(\iota_* a y_T, H),$$

where ay_T is the sheaf represented by T: the internal Yoneda density formula represents maps out of arbitrary $\iota_* F$ as maps out of F to this end. Equalizers and power objects in the internal sheaf categories construct the end, including its naturality and adjunction laws.

(6) **Distinguish the two endpoint embeddings.** Take the affine-line sheaf $F(T) = \Gamma(T, \mathcal{O}_T)$ and $j = (\operatorname{Spec} B \hookrightarrow \operatorname{Spec}(B[\varepsilon]/\varepsilon^2))$. At j the two endpoint embeddings have components $B[\varepsilon]/\varepsilon^2$ and B, with canonical map given by reduction. This is not an isomorphism, since coefficient extraction sends ε to its nonzero coefficient 1.

(7) **Distinguish the modalities and verify their laws.** Apply the two associated modalities to $\iota_! F$ to see that their comparison is also not an isomorphism. Finally all adjunction and idempotency equations follow from the endpoint adjunctions and their full faithfulness.

□

Proposition 8.5 (Smooth automorphisms under nilpotent reduction)**.** *Over $S = B[\varepsilon]/\varepsilon^2 \twoheadrightarrow R = B$, the distinct smooth S-automorphisms of the affine line $x \mapsto x$ and $x \mapsto x + \varepsilon$ become identical after reduction. Thus the functor on categories of smooth schemes induced by a nilpotent thickening is not faithful; the proof of Theorem 8.4 cannot be obtained by substituting "smooth" for "étale".*

Proof.

(1) **Exhibit a nontrivial automorphism with identity reduction.** The inverse of $x \mapsto x + \varepsilon$ is $x \mapsto x - \varepsilon$; its reduction is the identity. The two maps differ at x by the nonzero element ε. The conclusion follows.

□

8.2 Recovery of algebraic synthetic differential geometry

The primary site gives an affine calculation of function objects. We use it to show that the coordinate line supports the algebraic SDG identities needed by the subsequent differential and formal constructions. Let $\Delta_{\mathrm{inf}} : \mathcal{E} \to \mathscr{H}_{B,\mathrm{et}}$ be structural inverse image. For a finitely presented B-algebra A, write

$$F_A(\operatorname{Spec} S) = \mathrm{Hom}_{B\text{-alg}}(A, S), \qquad X_A = \iota_! F_A, \qquad \mathcal{O}_{\mathrm{inf}} = X_{B[z]}.$$

At $j = (\operatorname{Spec} R \hookrightarrow \operatorname{Spec} S)$, these have values $\operatorname{Hom}_B(A, S)$ and S, respectively. A square from j to $\iota(\operatorname{Spec} A)$ is determined by its larger-endpoint map, so $X_A = y\iota(\operatorname{Spec} A)$. These representables are sheaves: algebra maps at both endpoints glue along the stated étale covers, and their commutative-square equation is local. In the tensor formulas below, scalar extension is along $\Delta_{\mathrm{inf}} B \to \mathcal{O}_{\mathrm{inf}}$.

Theorem 8.6 (Primary algebraic SDG recovery)**.** *For every finitely presented B-algebra A, canonical evaluation induces an isomorphism of $\mathcal{O}_{\mathrm{inf}}$-algebras*

$$\mathcal{O}_{\mathrm{inf}} \otimes_{\Delta_{\mathrm{inf}} B} \Delta_{\mathrm{inf}} A \xrightarrow{\cong} \mathcal{O}_{\mathrm{inf}}^{X_A}. \tag{8.2}$$

Its value at j is $S \otimes_B A$. Consequently, for every internal commutative $\mathcal{O}_{\mathrm{inf}}$-algebra C, coordinate evaluation gives the natural isomorphism

$$\underline{\operatorname{Hom}}_{\mathcal{O}_{\mathrm{inf}}\text{-}alg}(\mathcal{O}_{\mathrm{inf}}^{X_A}, C) \;\cong\; \operatorname{Spec}_C(A), \tag{8.3}$$

where $\operatorname{Spec}_C(A)$ is the object of solutions in C of a displayed finite presentation of A. In particular, taking $A = B \otimes_{\mathbb{Q}} A_0$ for finitely presented rational A_0 proves the comprehensive algebraic SDG axiom over $\mathbb{Q}$.

For a rational Weil algebra W, put $W_B = B \otimes_{\mathbb{Q}} W$ and $D_W^{\mathrm{all}} = X_{W_B}$. Then

$$\mathcal{O}_{\mathrm{inf}}^{D_W^{\mathrm{all}}} \cong \mathcal{O}_{\mathrm{inf}} \otimes_{\mathbb{Q}} W. \tag{8.4}$$

For $W = \mathbb{Q}[\varepsilon]/\varepsilon^2$, the probe is $D^{\mathrm{all}} = \{d \in \mathcal{O}_{\mathrm{inf}} : d^2 = 0\}$ and the comparison is the Kock–Lawvere isomorphism

$$\mathcal{O}_{\mathrm{inf}} \times \mathcal{O}_{\mathrm{inf}} \xrightarrow{\cong} \mathcal{O}_{\mathrm{inf}}^{D^{\mathrm{all}}}, \qquad (a, b) \longmapsto (d \mapsto a + db). \tag{8.5}$$

This infinitesimal object is nontrivial. The line and every displayed finite-projective affine module cell are microlinear for rational Weil diagrams: their augmented Weil-prolongation functors preserve finite limits in the slice over the cell. All comparisons are internal, natural in the displayed algebras and their maps, and uniform in parameters. The construction therefore also applies in $\mathscr{S} = \mathrm{Sh}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}})$ with coefficient ring $\bar{B}$.

Proof.

(1) **Construct the product with an affine test.** The product $j \times \iota(\operatorname{Spec} A)$ is the arrow

$$\operatorname{Spec}(R \otimes_B A) \hookrightarrow \operatorname{Spec}(S \otimes_B A).$$

Its endpoints are finitely presented. The displayed quotient is surjective, and its kernel is the ideal generated by the image of $\ker(S \to R)$, so it is nilpotent with the same bound. This is the right-exact scalar-extension formula for a quotient.

(2) **Compute the primary function algebra.** Products of represented sheaves and internal Yoneda now give

$$\begin{aligned}(\mathcal{O}_{\mathrm{inf}}^{X_A})(j) &\cong \underline{\operatorname{Hom}}_{\mathscr{H}_{B,\mathrm{et}}}(yj \times y\iota(\operatorname{Spec} A), \mathcal{O}_{\mathrm{inf}}) \\ &\cong \mathcal{O}_{\mathrm{inf}}(j \times \iota(\operatorname{Spec} A)) = S \otimes_B A.\end{aligned}$$

(3) **Verify the algebra structure and naturality.** The evaluation map sends $s \otimes a$ to the function $\alpha \mapsto s\alpha(a)$, in the corresponding parameter context. Thus the comparison preserves multiplication and units, and is natural under test substitution and maps of A.

(4) **Establish affine duality.** If $A = B[x_1, \dots, x_n]/(f_1, \dots, f_m)$, the internal scalar-extension algebra on the left of (8.2) is the sheafification of the pointwise coefficient algebras with this presentation over S. The Yoneda calculation has already identified those algebras with the sheaf $\mathcal{O}_{\mathrm{inf}}^{X_A}$. This proves (8.2).

(5) **Identify internal algebra maps independently of presentation.** An algebra map from the scalar extension to C is exactly a tuple in C^n satisfying the displayed equations. The same argument holds in every slice, so it gives the internal-hom comparison (8.3) and its presentation independence.

(6) **Recover the Weil and dual-number coefficient identities.** Specializing to W_B gives (8.4). For dual numbers, the basis $1, \varepsilon$ gives (8.5) with its displayed canonical map.

(7) **Prove nontriviality of the square-zero probe.** At the dual-number test, the generalized element ε differs from zero by coefficient extraction, since $1 \neq 0$ in B. Thus the square-zero probe is not the zero subobject.

(8) **Compute the probes of a module cell.** For a displayed finite-projective B-module E, put $X_E = X_{\mathrm{Sym}_B(E^\vee)}$. Finite duality identifies $X_E(j) = S \otimes_B E$, and the same product calculation gives

$$(X_E^{D_W^{\mathrm{all}}})(j) \cong (S \otimes_B E) \otimes_{\mathbb{Q}} W.$$

(9) **Prove rational microlinearity and parameter stability.** Finite limits of augmented rational Weil diagrams are computed by finite products over $\mathbb{Q}$ and kernels of linear differences, retaining augmentation. Their finite-dimensional rational linear algebra splits by Gaussian elimination, so tensoring with $S \otimes_B E$ preserves these augmented limits. The comparisons are therefore isomorphisms at each test and in the slice over X_E. All constructions are internal finite presentations, quotients and universal properties, and hence apply with the indicated parameters in $\mathscr{S}$ as well.

□

This establishes an algebraic SDG model over the Landau coefficient ring, in the sense of the comprehensive and Weil axioms of (Kock 2006, Sections I.12, I.16, III.1 and Appendix D). It supplies the function objects used for formal completion and jets.

Lemma 8.7 (Full and zero-boundary infinitesimal probes)**.** *For $W_B = B \otimes_{\mathbb{Q}} W$, set*

$$D_W^0 = y(\operatorname{Spec} B \hookrightarrow \operatorname{Spec} W_B).$$

There is a canonical monomorphism $D_W^0 \hookrightarrow D_W^{\mathrm{all}}$. It induces isomorphisms

$$\mathcal{O}_{\mathrm{inf}}^{D_W^{\mathrm{all}}} \xrightarrow{\cong} \mathcal{O}_{\mathrm{inf}}^{D_W^0}, \qquad X_E^{D_W^{\mathrm{all}}} \xrightarrow{\cong} X_E^{D_W^0} \tag{8.6}$$

for displayed finite-projective affine module cells X_E. The two probes themselves need not be isomorphic. For dual numbers, D^0 consists of square-zero coordinates whose reduction at the smaller endpoint is zero.

Proof.

(1) **Identify the zero-boundary condition.** At j, a W_B-point of S belongs to D_W^0 precisely when its composite to R equals the augmentation followed by $B \to R$. This is a subobject of the full set of W_B-points.

(2) **Construct the zero-boundary product test.** The product of j with the representing zero-boundary arrow has larger algebra $S \otimes_B W_B$ and smaller algebra R. Its quotient kernel is the sum of the two extended nilpotent ideals, which is nilpotent by a finite binomial bound.

(3) **Compare the two function objects.** Yoneda therefore computes the same larger-endpoint function algebra as in Theorem 8.6. The map between the products is the identity on that larger algebra, proving the first isomorphism; the identical argument with $S \otimes_B E$ proves the second.

(4) **Distinguish the full and zero-boundary probes.** At the identity test of $B[\varepsilon]/\varepsilon^2$, the element ε belongs to the full square-zero probe and has nonzero reduction, so it does not belong to D^0.

□

The zero-boundary probes describe displacements relative to the smaller endpoint of a thickening. They are the probes used by the shape relation and its boundary principal parts below.

8.3 Comparison with the based resource calculus

Proposition 8.8 (Based recovery of the primary coefficient identities)**.** *Restriction to $j_C = (\operatorname{Spec} B \hookrightarrow \operatorname{Spec} C)$, $C \in \mathsf{W}_B$, defines a functor $\operatorname{res}_W : \mathscr{H}_{B,\mathrm{et}} \to \mathcal{G}_B$ with natural identifications*

$$\operatorname{res}_W \mathcal{O}_{\mathrm{inf}} \cong \mathcal{R}_B, \qquad \operatorname{res}_W X_E \cong j_B E, \qquad \operatorname{res}_W D_W^{\mathrm{all}} \cong \operatorname{res}_W D_W^0 \cong D_{C_W}.$$

For the line and these affine module cells, restriction also identifies the exponentials in Theorem 8.6 and Lemma 8.7 with the corresponding based Weil exponentials of Theorem 6.13. These comparisons preserve augmentation, Weil substitution and polynomial coordinate maps.

Proof.

(1) **Evaluate the primary objects at based tests.** Evaluation at j_C gives C, $C \otimes_B E$ and $\operatorname{Hom}_B(C_W, C)$, respectively.

(2) **Identify the two probes on based stages.** Every B-algebra map from a Weil stage to the reduced ring B kills its nilpotent ideal. The zero-boundary condition is therefore automatic at j_C, identifying the two probes there.

(3) **Recover the coefficient identities and coherence.** Evaluating their primary function objects gives $C \otimes_{\mathbb{Q}} W$, or $(C \otimes_B E) \otimes_{\mathbb{Q}} W$ for a module cell. These are exactly the expressions obtained from Theorem 6.13 in $\mathcal{G}_B$. Every map is the same coefficient substitution or elementary-tensor map, proving the stated coherence.

□

Proposition 8.9 (Budget-indexed endpoint geometry)**.** *The constructions of Lemma 8.2 and Theorem 8.4 may be performed in the budget-diagram category $\mathscr{S} = [L, \mathcal{E}]$, yielding endpoint adjunctions on the associated indexed internal étale sheaves. Their defining formulas are pointwise in the budget parameter and compatible with $M_a \hookrightarrow M_b$. Transport of an actual name-cover and tracked operation into this indexed geometry preserves its equality of denotations, its Σ-classified domain and its original positive-observation evaluation.*

Proof.

(1) **Reindex the endpoint geometry over budgets.** A diagram $a \mapsto M_a$ reindexes every polynomial, nilpotent-arrow, and endpoint formula using the same ring B; finite algebraic operations and naturality commute with the inclusions.

(2) **Preserve classified computation and its witnesses.** The named computation equations are originally equations of maps and classified spans in $\mathcal{E}$. Applying the left-exact sheaf realization to their finite diagrams preserves equality, pullbacks and regular images; the stored classifier is transported with the span. Thus the original Σ-valued observation remains typed and its finite witnesses remain tracked.

□

The primary function objects now supply coordinate algebras for the finite neighbourhoods used above. Completing their diagonal ideals assembles the neighbourhoods of every order.

9 Formal completion of differential resource geometry

The finite calculus measures resource variation to a specified order. Completion assembles compatible orders and exposes a further resource question: whether one budget contains every coefficient of the assembled family. The measured tower in Example 9.12 has a polynomial budget at each finite order and no common polynomial growth degree. It will also supply the de Rham and crystalline example.

We construct formal neighbourhoods by powers of the diagonal ideal, first over B in $\mathcal{E}$ and then over the resource parameters in $\mathscr{S} = \mathrm{Sh}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}})$. Principal parts retain both endpoints and the full Taylor action of coordinate changes. Boundary principal parts represent pairs with equal smaller-endpoint value on the primary site; their inverse-limit function objects identify the completed infinitesimal nerve. Effective finite observations are then obtained from the same coefficient tower.

9.1 The diagonal and its finite principal parts

An order-r neighbourhood retains differences between two points through degree r. Its coordinate algebra is obtained by quotienting the corresponding power of the diagonal ideal. Let E be a displayed finite-projective B-module, let $V = E^\vee$, and write $A_E = \mathrm{Sym}_B(V)$ for its coordinate algebra, with the variance of Proposition 6.19. Put

$$D_E = A_E \otimes_B A_E, \qquad \mu_E : D_E \longrightarrow A_E, \qquad I_E = \ker \mu_E.$$

Both tensor-factor inclusions will be retained; they give the source and target actions on every principal-parts object.

Definition 9.1 (Formal diagonal filtration)**.** For $r \in \mathbb{N}$ construct the A_E-algebra

$$P^r_{E/B} = D_E / I_E^{r+1}.$$

The transition maps $P^{r+1}_{E/B} \twoheadrightarrow P^r_{E/B}$ define the pro-algebra $\boldsymbol{P}_E = (P^r_{E/B})_r$ and its internal inverse-limit realization

$$\widehat{P}_{E/B} = \varprojlim_r P^r_{E/B}.$$

The pro-algebra is retained along with its realization. A B-algebra map $P^r_{E/B} \to C$ is an *order-r formal neighbourhood point* of the diagonal.

Theorem 9.2 (Polynomial calculation of formal principal parts)**.** *There are natural filtered A_E-algebra isomorphisms*

$$\begin{aligned} P^r_{E/B} &\cong \bigoplus_{0\le j\le r} A_E \otimes_B \mathrm{Sym}^j_B(V), \\ \widehat{P}_{E/B} &\cong \prod_{j\ge 0} A_E \otimes_B \mathrm{Sym}^j_B(V), \\ \mathrm{gr}_{I_E}(D_E) &\cong \mathrm{Sym}_{A_E}(\Omega^1_{A_E/B}), \qquad I_E/I_E^2 \cong \Omega^1_{A_E/B}. \end{aligned} \tag{9.1}$$

Each finite degree is a finite-projective A_E-module; multiplication on the first line is truncated by total degree and on the second is coefficientwise finite convolution. The filtered isomorphisms are natural under polynomial coordinate substitution with its full Taylor transport: for $y = p(x)$ the displacement is sent to $p(x+u) - p(x)$. On the associated graded this transport is the symmetric algebra of the cotangent map. All comparisons also commute with changes of displayed finite-projective presentation.

Proof.

(1) **Introduce displacement coordinates.** The symmetric-algebra universal property gives $D_E \cong \mathrm{Sym}_B(V \oplus V)$. Apply the invertible linear transformation $(v_0, v_1) \mapsto (v_0, v_1 - v_0)$, with inverse $(v_0, d) \mapsto (v_0, v_0 + d)$. In the resulting coordinates, μ_E is evaluation at $d = 0$. Hence

$$D_E \cong A_E \otimes_B \mathrm{Sym}_B(V), \qquad I_E^k = \bigoplus_{j\ge k} A_E \otimes_B \mathrm{Sym}^j_B(V).$$

(2) **Compute principal parts and finite-projective coefficients.** The quotient and product formulas follow; each multiplication coefficient uses finitely many summands. By the finite-free retract for V, every tensor power $V^{\otimes j}$ is finite-projective. Since $j!$ is invertible in the rational ring B, the averaging idempotent $\frac{1}{j!}\sum_{\sigma\in S_j}\sigma$ exhibits $\mathrm{Sym}^j_B(V)$ as a finite-projective retract of $V^{\otimes j}$.

(3) **Identify the cotangent module and associated graded algebra.** The free-algebra derivation formula $\Omega^1_{A_E/B} \cong A_E \otimes_B V$ identifies the first quotient I_E/I_E^2 with the cotangent module, and identifies the whole graded multiplication with its symmetric algebra.

(4) **Transport linear substitutions and projective retractions.** The linear changes and symmetric-power maps commute with displayed linear substitutions and retractions.

(5) **Compute polynomial transport in displacement coordinates.** For a polynomial map p, the two-vertex algebra map sends (y_0, y_1) to $(p(x_0), p(x_1))$. In displacement coordinates this is exactly

$$y = p(x), \qquad v = p(x+u) - p(x). \tag{9.2}$$

(6) **Preserve the displacement filtration.** The second expression has zero constant term in u, so substitution preserves every power of the displacement ideal and defines compatible maps on finite quotients and their inverse limits.

(7) **Verify composition and identity coherence.** For composable polynomial maps p, q, the displacement in the composite is

$$q(p(x) + [p(x+u) - p(x)]) - q(p(x)) = q(p(x+u)) - q(p(x)),$$

which proves composition coherence; the identity map gives the unit equation.

(8) **Identify the associated-graded transport.** Its degree-one part is $dp_x(u)$, so the associated-graded map is the symmetric algebra of the induced cotangent map. These formulas are expressions of the two-vertex universal property and therefore descend along every displayed projective retraction.

(9) **Distinguish filtered transport from its linear part.** For example $p(x) = x^2$ sends v to $2xu + u^2$; retaining only $2xu$ gives the associated-graded transport, not the filtered one. This proves polynomial naturality with its specified Taylor action.

□

The coordinate diagram below fixes the two actions and reduction.

$$\begin{array}{ccccc} A_E & \underset{t^\sharp}{\overset{s^\sharp}{\rightrightarrows}} & A_E \otimes_B A_E & \twoheadrightarrow & P^r_{E/B} \\ & & & \searrow^{\mu_E} & \downarrow{\scriptstyle \mathrm{aug}_r} \\ & & & & A_E. \end{array} \tag{9.3}$$

The augmentation ideal of $P^r_{E/B}$ has $(r+1)$-st power zero. The quotient $P^0_{E/B} = A_E$, so this filtration begins at the ordinary resource-coordinate space and adds no independently specified infinitesimals.

9.2 Finite formal neighbourhoods on based Weil probes

Principal parts describe a neighbourhood over the whole polynomial cell: their reduction is A_E. They need not be finite-projective over B and hence need not be B-Weil stages themselves. We realize them by evaluating their functor of points on the already constructed category $\mathsf{W}_B^{\mathrm{proj}}$:

$$\mathcal{D}_E^{\leq r}(C) = \mathrm{Hom}_{B\text{-alg}}(P^r_{E/B}, C), \qquad C \in \mathsf{W}_B^{\mathrm{proj}}.$$

The quotient map makes this a subobject of $j_B E \times j_B E$.

Theorem 9.3 (Completion–Weil comparison)**.** *There is a canonical isomorphism in $\mathcal{G}_B$,*

$$\varinjlim_r \mathcal{D}_E^{\leq r} \;\cong\; j_B E \times_{\mathfrak{I}_W(j_B E)} j_B E. \tag{9.4}$$

It is natural in finite-projective polynomial coordinate maps and is compatible with the source and target projections. For every finite ordinal $[q]$, completion of the higher diagonal $A_E^{\otimes_B(q+1)} \to A_E$ gives a functor of points naturally identified with the $(q+1)$-fold fibre product of $j_B E$ over $\mathfrak{I}_W(j_B E)$. These isomorphisms respect every simplicial operator.

Proof.

(1) **Factor equal-reduction pairs through finite principal parts.** Fix a stage $C = B \oplus N$, with the supplied finite nilpotence bound $N^{s+1} = 0$. By finite duality, a pair of points of $j_B E(C)$ is a pair of B-algebra maps $A_E \to C$. It belongs to the right-hand side of (9.4) precisely when its two reductions to B agree. Equivalently the difference of the two evaluations of every $v \in V$ belongs to N. The induced map $A_E \otimes_B A_E \to C$ therefore sends I_E to N and kills I_E^{s+1}; it factors through $P^s_{E/B}$.

(2) **Recover equal reductions from finite factorization.** Conversely, if the map factors through $P^r_{E/B}$, the images of the coordinate differences are nilpotent. Their reductions in B are zero because B is reduced, by Proposition 2.7. Duality then implies equality of the reduced E-points. The implications hold internally, uniformly in C and its nilpotence datum. They identify the subobjects at every stage, and hence the corresponding functors.

(3) **Extend the comparison to higher diagonals.** For $q+1$ points use differences with the first point, or successive differences; both generate the kernel of the higher multiplication map. The same finite nilpotence argument identifies its completed formal points with points having equal reductions.

(4) **Verify simplicial coherence.** Deleting, repeating, or permuting vertices acts by substitution of these differences and preserves all powers of the higher diagonal ideal. The stagewise identifications therefore commute with faces, degeneracies, and permutations.

□

9.3 Formal completion in the primary geometry

The based calculation records displacements at fixed-base probes. We now identify the full relation on the primary site, where both endpoints of a thickening remain visible. For $A = \mathrm{Sym}_B(E^\vee)$, retain the affine sheaf $F_A \in \mathscr{X}_{B,\mathrm{et}}$ of Section 8.2, with $F_A(\operatorname{Spec} S) = \mathrm{Hom}_{B\text{-alg}}(A, S)$, and put

$$X_A = \iota_! F_A \in \mathscr{H}_{B,\mathrm{et}}, \qquad \Im_{\inf} X_A = \iota_* F_A.$$

The shape unit $\eta_{X_A} : X_A \to \Im_{\inf} X_A$ evaluated at $j = (\operatorname{Spec} R \hookrightarrow \operatorname{Spec} S)$ is the restriction map $\mathrm{Hom}_B(A, S) \to \mathrm{Hom}_B(A, R)$.

Theorem 9.4 (Étale completion of the full infinitesimal relation)**.** *For each finite order r, define a subobject of $X_A \times X_A$ by*

$$\mathcal{D}^{\le r}_{A,\mathrm{et}}(j) = \mathrm{Hom}_{B\text{-}alg}(P^r_{A/B}, S) \underset{\mathrm{Hom}_{B\text{-}alg}(A\otimes_B A, R)}{\times} \mathrm{Hom}_{B\text{-}alg}(A, R), \tag{9.5}$$

where the last arrow is the diagonal and the first arrow reduces both vertices modulo $S \to R$. These are étale sheaves, they form an increasing union, and the comparison

$$\varinjlim_r \mathcal{D}^{\le r}_{A,\mathrm{et}} \xrightarrow{\cong} \mathcal{D}_{X_A} := X_A \times_{\Im_{\inf} X_A} X_A \tag{9.6}$$

is an isomorphism of internal étale sheaves. It is natural under maps of displayed polynomial coordinate algebras, retains both vertex maps, and identifies completed higher diagonals with the complete infinitesimal Čech nerve of the shape unit, including all face and degeneracy maps.

Restriction to the based probe $\operatorname{Spec} B \hookrightarrow \operatorname{Spec} C$, $C \in \mathsf{W}_B^{\mathrm{proj}}$, identifies (9.5) with the previously constructed $\mathrm{Hom}_B(P^r_{A/B}, C)$, and (9.6) with (9.4).

Proof.

(1) **Factor equal-boundary pairs through finite principal parts.** At j, a pair of A-points of S lies in the right side exactly when its reductions in R agree. Its algebra map $A \otimes_B A \to S$ therefore takes the diagonal ideal I into $J = \ker(S \to R)$. For a nilpotence witness $J^N = 0$, it annihilates I^N, so the pair comes from P^{N-1}.

(2) **Recover boundary equality and identify the sheaves.** Conversely every element of (9.5) has equal reduced vertices by its displayed fibre-product equation. There is no chosen global N: these implications prove equality of subobjects by existential elimination in $\mathcal{E}$. Pullback of nilpotent quotients along étale maps preserves their power filtration, so the formulas define sheaves and the stagewise isomorphism is natural in coverings and refinement.

(3) **Extend the comparison to the completed nerve.** For $q+1$ vertices use the kernel of $A^{\otimes_B(q+1)} \to A$. Equality of all reductions places every difference into J, and the same nilpotence argument applies. Face, degeneracy, and permutation maps are algebra substitutions preserving powers of this ideal; therefore the comparison is simplicial.

(4) **Recover the based-Weil formula.** On the based arrow $B \hookrightarrow C = B \oplus N$, the image of every difference under a P^r-point is nilpotent. The reduced ring B has no nonzero nilpotents, so the boundary equality is automatic. These identities give the based-Weil formula directly at every stage.

□

The completed relation is the kernel pair of the primary shape unit:

$$\begin{array}{ccc} \mathcal{D}_{X_A} & \xrightarrow{\ t\ } & X_A \\ {\scriptstyle s}\downarrow & & \downarrow{\scriptstyle \eta_{X_A}} \\ X_A & \xrightarrow[\ \eta_{X_A}\ ]{} & \Im_{\inf} X_A. \end{array}$$

The theorem identifies its finite boundary filtration with $\varinjlim_r \mathcal{D}^{\leq r}_{A,\mathrm{et}}$ inside this cartesian square. Higher fibre products give its full nerve.

9.4 Represented principal parts and synthetic formal functions

Identifying a formal relation is not yet a calculation of its functions. Representability supplies that calculation: finite neighbourhoods are primary tests, so Yoneda evaluates their function objects; compatible truncations then give the functions on the formal union. Retain the coordinate-line ring object

$$\mathcal{O}_{\inf} = \iota_! F_{B[z]}, \qquad \mathcal{O}_{\inf}(\operatorname{Spec} R \hookrightarrow \operatorname{Spec} S) = S.$$

For $q \geq 0$ put

$$P^r_q = A^{\otimes_B(q+1)}/I_q^{r+1}, \qquad I_q = \ker(A^{\otimes_B(q+1)} \to A), \qquad k_{q,r} = (\operatorname{Spec} A \hookrightarrow \operatorname{Spec} P^r_q).$$

The finite-projective polynomial presentation supplies finitely many generators for I_q and for each of its finite powers. Thus P^r_q is finitely presented over B, and $k_{q,r}$ is an actual object of I_B. Write $\mathcal{D}^{\leq r}_{A,q}$ for the higher-vertex version of (9.5), and $\mathcal{D}^{(q)}_{X_A} = X_A \times_{\Im_{\inf} X_A} \cdots \times_{\Im_{\inf} X_A} X_A$ with $q+1$ factors.

Lemma 9.5 (Primary representability of finite boundary principal parts)**.** *There are natural isomorphisms of primary sheaves*

$$yk_{q,r} \cong \mathcal{D}^{\leq r}_{A,q}, \qquad \varinjlim_r yk_{q,r} \cong \mathcal{D}^{(q)}_{X_A}.$$

They preserve all vertex maps, truncations, and polynomial coordinate substitutions. The representing sheaves are already étale sheaves.

Proof.

(1) **Identify the representing boundary square.** A square from $j = (\operatorname{Spec} R \hookrightarrow \operatorname{Spec} S)$ to $k_{q,r}$ is precisely a pair of algebra maps $P^r_q \to S$, $A \to R$ whose composites from $A^{\otimes_B(q+1)}$ agree. This is the displayed boundary fibre product, including equality of all reduced vertices.

(2) **Verify the sheaf condition.** Algebra maps into an étale covering diagram glue uniquely at both endpoints; the square equation glues because equality is local. Hence the representable is a sheaf.

(3) **Identify the completed union and coherence.** The union assertion is the higher-diagonal case of Theorem 9.4. Every comparison is induced by the same algebra substitution on the two endpoints, which proves the stated coherence.

□

Theorem 9.6 (Synthetic functions on the completed primary nerve)**.** *Let* $\underline{\mathrm{Hom}}_{\mathscr{H}_{B,\mathrm{et}}}$ *denote the internal* $\mathcal{E}$*-object of sheaf maps, and define the ring objects*

$$\mathcal{C}_r^q = \mathcal{O}_{\mathrm{inf}}^{\mathcal{D}_{A,q}^{\leq r}}, \qquad \widehat{\mathcal{C}}^q = \mathcal{O}_{\mathrm{inf}}^{\mathcal{D}_{X_A}^{(q)}}.$$

Then

$$\underline{\mathrm{Hom}}_{\mathscr{H}_{B,\mathrm{et}}}(\mathcal{D}_{A,q}^{\leq r}, \mathcal{O}_{\mathrm{inf}}) \cong P_q^r, \tag{9.7}$$

$$\underline{\mathrm{Hom}}_{\mathscr{H}_{B,\mathrm{et}}}(\mathcal{D}_{X_A}^{(q)}, \mathcal{O}_{\mathrm{inf}}) \cong \varprojlim_r P_q^r, \tag{9.8}$$

$$\widehat{\mathcal{C}}^q \cong \varprojlim_r \mathcal{C}_r^q. \tag{9.9}$$

At $j = (\operatorname{Spec} R \hookrightarrow \operatorname{Spec} S)$*, these coefficient sheaves have the explicit values*

$$\mathcal{C}_r^q(j) \cong S \otimes_B P_q^r, \qquad \widehat{\mathcal{C}}^q(j) \cong \varprojlim_r (S \otimes_B P_q^r).$$

All comparisons respect the cosimplicial maps, algebra operations, and polynomial coordinate substitutions with their full Taylor law.

Proof.

(1) **Compute finite and completed internal function objects.** Internal Yoneda and Lemma 9.5 give $\underline{\mathrm{Hom}}(yk_{q,r}, \mathcal{O}_{\mathrm{inf}}) = \mathcal{O}_{\mathrm{inf}}(k_{q,r}) = P_q^r$. Mapping out of the internally indexed union gives the inverse-limit formula, uniformly in any additional $\mathcal{E}$-parameter.

(2) **Compute the completed coefficient sheaf.** In the sheaf topos, product with a fixed object preserves colimits. Applying this fact to the exponential adjunction proves (9.9).

(3) **Construct the product test.** The product $j \times k_{q,r}$ has larger-endpoint algebra $S \otimes_B P_q^r$ and smaller-endpoint algebra $R \otimes_B A$. The kernel of its quotient is generated by the two extended nilpotent kernels, whose sum is nilpotent by a finite binomial bound. Both endpoints are finitely presented, so this is a product in I_B.

(4) **Compute stagewise coefficients and their limit.** Yoneda therefore calculates

$$(\mathcal{O}_{\mathrm{inf}}^{yk_{q,r}})(j) = \underline{\mathrm{Hom}}(yj \times yk_{q,r}, \mathcal{O}_{\mathrm{inf}}) = S \otimes_B P_q^r.$$

Sheaf limits are computed pointwise, giving the completed value.

(5) **Recover internal maps and verify coherence.** At the terminal test $\iota(\operatorname{Spec} B)$ this recovers the two internal map objects above. Algebra substitutions represent the vertex and coordinate maps, so Yoneda identifies every associated coherence diagram, including multiplication of functions.

□

Restriction of functions along a finite-neighbourhood inclusion is the truncation map under Yoneda:

$$\begin{array}{ccc} \underline{\mathrm{Hom}}(yk_{q,r+1}, \mathcal{O}_{\mathrm{inf}}) & \longrightarrow & \underline{\mathrm{Hom}}(yk_{q,r}, \mathcal{O}_{\mathrm{inf}}) \\ \cong \downarrow & & \downarrow \cong \\ P_q^{r+1} & \twoheadrightarrow & P_q^r. \end{array}$$

Here the hom objects are those of $\mathscr{H}_{B,\mathrm{et}}$ internal to $\mathcal{E}$. Taking the inverse limit of this natural diagram gives the completed function comparison, including its cosimplicial structure.

The function calculation must also respect the endpoint structure. We next compare the endpoint functors with inverse limits of the coefficient sheaves, including the units, counits, and maps of the infinitesimal nerve.

9.5 The endpoint modalities through formal completion

Theorem 9.7 (Endpoint adjunctions on formal towers and their realizations)**.** *The endpoint quadruple of Theorem 8.4 acts levelwise on internally $\mathbb{N}^{\mathrm{op}}$-indexed towers, with the same four adjunctions, full-faithfulness equations, and idempotent modalities. Every endpoint functor commutes with realization of these towers by internal inverse limits. In particular*

$$\mathfrak{M}(\varprojlim_r H_r) \cong \varprojlim_r \mathfrak{M}(H_r), \qquad \mathfrak{M} \in \{\mathfrak{R}, \mathfrak{I}_{\mathrm{inf}}, \mathfrak{C}\}. \tag{9.10}$$

These comparisons commute with all adjunction units and counits, the modality multiplication and comultiplication, and maps of towers. They apply to the completed coefficient sheaves $\widehat{\mathcal{C}}^q$ of Theorem 9.6, including their cosimplicial operations and normalized additive cochains. Rings and modules are transported over the corresponding transported coefficient rings.

Proof.

(1) **Prove limit preservation for the endpoint functors.** Limits of internal sheaves are computed pointwise. The endpoint formulas identify $\iota_!$, ι^*, and ι_* with precomposition on sheaves, so each commutes with internally indexed limits.

(2) **Prove limit preservation for the end-defined adjoint.** For the fourth functor, its constructed end gives directly

$$(\iota^! \varprojlim_r H_r)(T) = \underline{\mathrm{Hom}}(\iota_* a y_T, \varprojlim_r H_r) \cong \varprojlim_r \underline{\mathrm{Hom}}(\iota_* a y_T, H_r).$$

This is the limit-preservation equation for $\iota^!$, natural in T.

(3) **Lift the adjunctions to towers.** The componentwise adjunction bijections carry natural transformations of towers to natural transformations of towers, by naturality of the unit and counit. They therefore construct the four tower adjunctions; their triangle and full-faithfulness equations are componentwise. Composing the four limit comparisons proves (9.10).

(4) **Verify compatibility with modal structure maps.** For each unit, counit, or modality structure map, both paths through the claimed comparison have the same composites with every limit projection. The limit universal property proves their equality. The same argument applies to any map of towers.

(5) **Preserve algebraic and normalized cochain structures.** Ring operations are finite-product diagrams, and additive normalization is a finite intersection of degeneracy

kernels in each cochain degree. All four functors preserve those limits, so the comparisons preserve these structures as well. Apply the result to (9.9); the identities for cofaces, degeneracies, and coordinate changes are identities of maps of the finite towers and hence persist in their realizations.

□

For the shape modality, compatibility with the unit is the square

$$\begin{array}{ccc} \varprojlim_r H_r & \xrightarrow{\eta} & \Im_{\mathrm{inf}}(\varprojlim_r H_r) \\ \| & & \downarrow \cong \\ \varprojlim_r H_r & \xrightarrow[\varprojlim_r \eta_{H_r}]{} & \varprojlim_r \Im_{\mathrm{inf}} H_r. \end{array}$$

The analogous counit squares and their mates express the compatibility for the other endpoint modalities.

There are two related limiting operations here. Formal spaces are unions of finite represented neighbourhoods, identified in Theorem 9.4; their function objects are inverse limits. The theorem applies to the inverse limits of coefficient sheaves.

Proposition 9.8 (Completion of geometric coefficients)**.** *For an internal coefficient tower P_r, scalar extension has a canonical comparison*

$$S \otimes_B \varprojlim_r P_r \longrightarrow \varprojlim_r (S \otimes_B P_r).$$

It is not always an isomorphism, even for a finitely presented polynomial algebra S. The completed coefficient sheaf in Theorem 9.6 is the right-hand construction.

Proof.

(1) **Construct the coefficient-completion comparison.** Tensor the limit projections and use the target limit universal property to construct the comparison.

(2) **Exhibit the polynomial-series comparison.** Take $S = B[z]$ and $P_r = B[u]/u^{r+1}$. It becomes

$$B[[u]][z] \longrightarrow B[z][[u]].$$

(3) **Prove injectivity and failure of surjectivity.** It is injective by coefficient extraction. Every element of the source has a finite bound on its z-degree, whereas $\sum_{m\geq 0} z^m u^m$ has none: its coefficient at any larger u-degree has that larger z-degree, and $1 \neq 0$ in B. Thus the comparison is not surjective.

(4) **Recover the stagewise sheaf calculation.** The sheaf formula is the pointwise calculation already proved.

□

Proposition 9.9 (Polynomial lifting, effective descent, and a singular counterexample)**.** *For a displayed finite-projective E, the unit $\eta_{X_A} : X_A \to \Im_{\mathrm{inf}} X_A$ is an effective (epimorphic) descent morphism of $\mathscr{H}_{B,\mathrm{et}}$. This assertion fails for general finitely presented affine algebras: the unit of $X = \iota_! \operatorname{Spec}(B[x]/x^2)$ is not an epimorphism.*

Proof.

(1) **Prove effective lifting for finite-projective affine cells.** The functor of points of $A = \mathrm{Sym}_B(E^\vee)$ is $S \mapsto E \otimes_B S$. A displayed finite-free retract for E exhibits $E \otimes_B S \to E \otimes_B R$ as a retract of a finite power of the surjection $S \to R$; hence it is a regular epimorphism in $\mathcal{E}$, uniformly in j. The sheaf unit is thus locally surjective, hence effective epic. In an elementary topos every epimorphism is an effective descent morphism for objects, by the regular kernel-pair and descent construction.

(2) **Construct the singular lifting problem.** For the counterexample use $S = B[t]/(t^3) \twoheadrightarrow R = B[t]/(t^2)$. The element $t \in R$ is a point of $\mathrm{Spec}(B[x]/x^2)$.

(3) **Show that the lift fails locally.** Any lift to S would have the form $t + at^2$, but $(t + at^2)^2 = t^2 \neq 0$ in S; the equation cannot hold even after a faithfully flat étale extension of S. Thus the associated component of the unit is not locally surjective. The obstruction is algebraic, not a failure of an implementation name.

□

Proposition 9.10 (A natural nonpolynomial Weil transformation)**.** *The reduction–constant-extension transformation $R = \epsilon_{\mathcal{R}_B}\eta_{\mathcal{R}_B} : \mathcal{R}_B \to \mathcal{R}_B$ is natural in projective Weil stages, satisfies $R_B = \mathrm{id}_B$, and annihilates every nilpotent coefficient. It is not represented by any polynomial in $B[x]$. Hence the algebraic higher-diagonal cochains form a proper subcalculus of all maps of Weil functors.*

Proof.

(1) **Verify naturality of reduction.** At a stage $C = B \oplus N$ the map is $R_C(b + n) = b$ under constant extension. Naturality is the equation $u\eta_C = \eta_{C'}u$ for augmented stage morphisms $u : C \to C'$.

(2) **Force a polynomial representative to be the identity.** Suppose a polynomial $p(x) = \sum_{j=0}^{m} b_j x^j$ induced R. Evaluating at $C = B$ yields $p(b) = b$ for all $b \in B$, in particular at $m + 1$ distinct rational constants. Subtracting x gives a polynomial of degree at most $\max(m, 1)$; evaluating at one more rational constant if necessary supplies that many more than its degree. The Vandermonde determinant is a product of nonzero rational differences and thus a unit of B, so all coefficients of $p - x$ vanish. Hence $p = x$.

(3) **Contradict polynomiality on a nilpotent coefficient.** At the existing dual-number stage $B[\varepsilon]/\varepsilon^2$, however, $R(\varepsilon) = 0$ while $p(\varepsilon) = \varepsilon \neq 0$. The inequality holds because $1, \varepsilon$ are independent components of the underlying free B-module and $1 \neq 0$ in the rational-germ ring. This is a contradiction.

(4) **Place the example in the two geometric settings.** The map R is therefore a nonpolynomial morphism of the based-Weil geometry $\mathcal{G}_B$. On the primary site, Theorem 9.6 identifies the completed coordinate cochains with all functions on the represented formal nerve with values in $\mathcal{O}_{\mathrm{inf}}$.

□

At each finite order, the coefficient constructions commute with the stated gauge and input changes. Passing to an infinite coefficient object raises a further question: can one denominator serve all orders? The next comparison records precisely where finite-order transport and completed transport differ.

9.6 Formal-order coherence, gauges, and exact coefficients

Proposition 9.11 (Finite-order coefficient change)**.** *For a localization $B \to B_P$ constructed by an overhead gauge, put $A_P = B_P \otimes_B A_E$. For every r, the canonical map*

$$B_P \otimes_B P^r_{E/B} \xrightarrow{\cong} P^r_{A_P/B_P}$$

is an isomorphism of augmented algebras. It commutes with the source and target actions, the differential comparison, truncation, and all higher-diagonal substitution maps.

Proof.

(1) **Localize the diagonal ideal and its powers.** Localization is flat by Theorem 2.23. It preserves the kernel of multiplication μ_E, and hence carries its ideal to the kernel of the base-changed multiplication. Products of a finite number of ideal elements commute with tensor extension, so it carries I_E^{r+1} onto the $(r+1)$-st power of the new ideal.

(2) **Compare finite principal parts and their structure maps.** Flatness preserves the resulting quotient sequence. Every map in the comparison is induced by the universal algebra maps and finite multiplication, so the stated diagrams commute.

□

For the completed algebra the finite comparisons induce

$$B_P \otimes_B \widehat{P}_{E/B} \longrightarrow \varprojlim_r (B_P \otimes_B P^r_{E/B}). \tag{9.11}$$

The following measured family identifies the resource distinction between its two sides.

Example 9.12 (Measured finite programs and a completed resource family)**.** Work in the cofinite numerical instance and put $t(n) = 1/(n+1)$, $q = t^{-1}$, and $R = B[t^{-1}]$. Use an idle invocation carrying a natural-number tag; it leaves the returned input unchanged, and its tag is retained in the history. NNO recursion constructs a finite description e_r which, for each $0 \le j \le r$, emits $(n+1)^j$ copies of the event tagged j and then returns its input. Exponentiation and the finite concatenation are implemented by the retained iteration constructors. The map $r \mapsto e_r$ is itself given by finite code construction.

Give event j the weight z^j and other invocation events weight zero. For each nonnegative rational z, the exact measured cost and its germ polynomial are

$$T_{r,z}(n) = \sum_{j=0}^{r} (n+1)^j z^j, \qquad F_r(z) = \sum_{j=0}^{r} q^j z^j. \tag{9.12}$$

The polynomial coefficient extension is calculated in the exact signed algebra. The histories have the same returned value, and Theorem 6.1 identifies their measured coefficients with geometric Taylor coefficients. Every coefficient of F_r belongs to $t^{-r}B$, since $q^j = t^{-r}t^{r-j}$ for $j \le r$. Truncation gives $F_{r+1} \bmod z^{r+1} = F_r$, so these polynomials determine

$$F(z) = \sum_{j \ge 0} q^j z^j \in R[[z]], \qquad (1 - qz)F(z) = 1.$$

The measured objects are the finite programs e_r and their costs; F is the compatible formal generating function of their coefficients.

Proposition 9.13 (Finite polynomial budgets without a common degree)**.** *The canonical map*

$$B[[z]][t^{-1}] \longrightarrow B[t^{-1}][[z]] \tag{9.13}$$

is injective. Its image consists exactly of coefficient families contained in one budget $t^{-N}B$ for some $N \in \mathbb{N}$. The measured formal family F of Example 9.12 lies outside this image, although every finite truncation lies in it.

Proof.

(1) **Characterize series with a common denominator.** A fraction $t^{-N}b(z)$ has all its coefficients in $t^{-N}B$. Conversely, a coefficient family with that common budget becomes a unique B-coefficient series after multiplication by t^N, giving the required fraction.

(2) **Prove injectivity and exclude the displayed series.** Regularity of t proves injectivity coefficientwise. If F had such a denominator, its coefficient at $j = N + 1$ would give $t^N q^{N+1} = t^{-1} \in B$, a contradiction.

(3) **Recover the finite-order budget bounds.** The finite budget calculation in Example 9.12 proves the last assertion.

□

Thus localization and completion distinguish an order-dependent polynomial growth degree from a single degree valid for every coefficient. The constants in each coefficient's Landau estimate may depend on that coefficient. Formal order is also distinct from merely quotienting by asymptotically vanishing errors, as the next calculation shows.

Proposition 9.14 (The Landau vanishing ideal is idempotent)**.** *For every proper-filter rational germ context,*

$$\mathfrak{m}^2 = \mathfrak{m}, \qquad \varprojlim_r B/\mathfrak{m}^{r+1} \cong B/\mathfrak{m}.$$

Consequently completion at the original small-error ideal cannot replace the diagonal-adic formal-order tower.

Proof.

(1) **Construct a rational square-root bound.** Let $f \in \mathfrak{m}$. There is a uniform rational operation $q \mapsto s(q)$ on nonnegative rationals satisfying

$$s(0) = 0, \qquad q \le s(q)^2 \le 4q \quad (q > 0),$$

obtained by locating a positive rational between successive powers of two using finite rational comparison and doubling.

(2) **Factor a vanishing germ.** Apply it pointwise to $|f|$ and write $g = s(|f|)$. Define $h = f/g$ on the nonzero support, zero elsewhere. These are rational profiles and respect eventual equality. One has $f = gh$ and $|h| \le g$ pointwise.

(3) **Prove that both factors vanish.** For every $k \ge 1$, the smallness bound $4k^2|f| \le 1$ on an eventual extent yields $kg \le 1$ there, and hence also $k|h| \le 1$. Thus $g, h \in o(1) = \mathfrak{m}$. So $f \in \mathfrak{m}^2$.

(4) **Deduce idempotence and the constant quotient tower.** The opposite inclusion follows from ideality. All positive powers of the idempotent ideal equal $\mathfrak{m}$, and the inverse system of quotients is constant at $B/\mathfrak{m}$.

□

9.7 Formal coordinates on indexed resource cells

We now apply formal coordinates to the universal resource family. The budget classifier remains $\mathfrak{C}_{\mathrm{Ev}}$; the new objects are neighbourhoods of its resource cells. At a principal budget $a \in L$, the module is $\mathscr{M}(a) = M_a$. Over the regular-epimorphic cover supplying a representative g, it is the displayed finite-projective module $Bg \cong Be_g$. Construct $A_a = \mathrm{Sym}_B(M_a^\vee)$ and its principal parts locally, then descend along the representative kernel pair. An inclusion $M_a \hookrightarrow M_b$ induces the contravariant coordinate map

$$A_b \longrightarrow A_a.$$

Free algebras, diagonal ideals, finite quotients, and completed higher diagonals are functorial in these coordinate maps. The coordinate algebras therefore vary contravariantly with the budget order, while their represented formal spaces vary covariantly. This gives the geometric budget diagram with the required variance.

The descent depends only on the marked module. Equal growth classes give the same embedded image $Bg \subseteq K$, and finite-projective duality and principal parts are intrinsic to that image. Their local presentations consequently agree on kernel pairs and descend by effective quotients in $\mathcal{E}$.

Proposition 9.15 (Preservation of the global resource classifiers)**.** *The functors and formal systems of this section leave unchanged the classifying equivalence* $\mathrm{Loc}_{\mathcal{E}}(Z, \mathfrak{C}_{\mathrm{Ev}}) \cong \mathrm{Sub}_{\Delta_Z B}(\Delta_Z K)$*, the universal marked module, all finite resource-diagram classifiers, and the feasibility classifier* $\mathfrak{F} = \mathbb{S}^{\mathfrak{C}_{\mathrm{Ev}}}$*. The formal differential objects are constructed* over *their already classified parameters.*

Proof.

(1) **Preserve the indexed classifying data under completion.** The classifiers are defined from B, K, L and their marked subobjects. Formal completion acts on the polynomial coordinate algebras over these parameters. The finite algebraic constructions commute with parameter pullback, so the original classifying maps and universal properties continue to classify the indexed data.

□

9.8 Observed formal towers and their effective points

A compatible family of formal coefficients also needs an observation structure. We generate it from the finite-order observations and apply the effective-point construction of Section 1.3. The generator test of Lemma 1.7 then reduces the limit comparison to the observations at each finite stage.

Theorem 9.16 (Effective points of an observed inverse tower)**.** *For an internal tower* $(D_r, \mathcal{O}_r)$ *in* $\mathbf{EDcpo}_\Sigma$*, form the underlying compatible-family poset* $D_\infty = \varprojlim_r D_r$*. Its pointwise directed suprema and the observation structure*

$$\mathcal{O}_\infty = \mathrm{cl}_\Sigma\{\pi_r^* U : r \in \mathbb{N},\ U \in \mathcal{O}_r\} \tag{9.14}$$

construct its inverse limit in $\mathbf{EDcpo}_\Sigma$*. There is a canonical isomorphism over* D_∞*,*

$$\mathrm{Pt}_\Sigma D_\infty \cong \varprojlim_r \mathrm{Pt}_\Sigma D_r. \tag{9.15}$$

It is uniform in parameters and natural in maps of observed towers. On the generating observations its positive evaluation is

$$\mathrm{Obs}_{D_\infty}(x, \pi_r^* U) = \mathrm{Obs}_{D_r}(\pi_r x, U).$$

Proof.

(1) **Construct directed suprema coordinatewise.** For an inhabited directed ideal of compatible families, take its lower image under each projection. This is an inhabited directed ideal in D_r; take its supremum there. Scott continuity of the transition maps makes these suprema a compatible family. Their coordinatewise universal property proves that this family is the supremum in D_∞.

(2) **Construct the observation closure and limit property.** Every projection is Scott-continuous. Its pulled-back observations are therefore Scott open; their Σ-observation closure remains Scott open by the closure construction of Section 1.3. For a cone of observation-preserving maps, inverse image preserves each generating observation and hence its closure. This proves the limit universal property in $\mathbf{EDcpo}_\Sigma$.

(3) **Define the forward effective-point comparison.** The effective projections give the forward comparison in (9.15).

(4) **Construct the profile on generating observations.** Conversely work with a parameterized element $(x_r)_r$ of the limit on the right. Evaluation on the internally indexed family $\sum_r \mathcal{O}_r$ is the single map $(r, U) \mapsto \mathrm{Obs}_{D_r}(x_r, U)$ into Σ.

(5) **Descend and extend the observation profile.** If two generators have the same pullback to D_∞, these evaluations agree in Ω and hence in Σ, since $\Sigma \hookrightarrow \Omega$ is monic. They therefore descend to the image family of generating observations in (9.14). The testing-generators lemma Lemma 1.7 extends this profile to the whole observation closure. The effective-point pullback then gives a unique point of $\mathrm{Pt}_\Sigma D_\infty$ over $(x_r)_r$.

(6) **Verify the isomorphism and parameter coherence.** The two maps are inverse over the compatible-family object, proving the isomorphism. All maps are constructed in the parameter context; uniqueness in the effective-point pullback proves naturality and coherence.

□

Construction 9.17 (Canonical observed exact coefficient towers)**.** For an exact coefficient object C of $\mathcal{E}$, write C_{disc} for its discrete poset and set $\mathsf{D}(C) = F_\Sigma(C_{\mathrm{disc}})$, using the canonical observed free completion of Section 1.3. An inhabited directed lower ideal in a discrete poset is a singleton. Thus the underlying dcpo of $\mathsf{D}(C)$ is canonically C, and its observations are precisely the predicates represented by maps $C \to \Sigma$, viewed in Ω. In particular

$$\mathrm{Pt}_\Sigma\, \mathsf{D}(C) \cong C.$$

Every exact arrow $f : C \to C'$ preserves these observations by precomposition and defines $\mathsf{D}(f)$. This fixes the finite observation structure by the canonical free construction.

Apply this construction to $C = P_q^r$ and its truncation maps, and then use Theorem 9.16. It constructs an observed dcpo $\widehat{\mathsf{D}}^{\,q}$ with

$$\mathrm{Pt}_\Sigma\, \widehat{\mathsf{D}}^{\,q} \cong \varprojlim_r \mathrm{Pt}_\Sigma\, \mathsf{D}(P_q^r) \cong \varprojlim_r P_q^r \cong \underline{\mathrm{Hom}}_{\mathscr{H}_{B,\mathrm{et}}}(\mathcal{D}_{X_A}^{(q)}, \mathcal{O}_{\mathrm{inf}}). \tag{9.16}$$

The same construction applies to finite normalized coefficient objects and their compatible-family limit. The final comparison is Theorem 9.6.

Proposition 9.18 (Finite-output observation and name transport)**.** *An internal map between constructed coefficient towers, or between their limits whose order-r output is given uniformly by a finite coefficient map at a constructed order $b(r)$, preserves the generated*

observations. It therefore induces the corresponding map of effective points. Finite arithmetic, polynomial Taylor substitution, truncation, and the coface maps have this property; the de Rham differentials and completed homotopies constructed below have the same property by their stated finite coefficient formulas.

In the effective and relative-realizability instances, the retained exact arithmetic name maps implement these operations on their projective name presentations, uniformly in the order and supplied parameters. A uniformly implemented compatible input family consequently gives uniformly implemented finite output observations of the resulting formal datum.

Proof.

(1) **Transport finite-output observations.** For a generating output observation $\pi_r^* U$, its pullback is $\pi_{b(r)}^*(U \circ f_r)$, where f_r is the displayed finite coefficient map. Precomposition is an allowed observation in the canonical exact coefficient structure. For a map of general observed towers the same assertion is its finite-stage observation-preservation equation. Lemma 1.7 extends the result to all generated observations; its effective-point functor then supplies the claimed map.

(2) **Compose the retained arithmetic name maps.** Each finite formula is a composition of the retained exact arithmetic, finite-list, rational-coefficient, and matrix operations. Their name maps compose by Lemma E.3.

(3) **Construct one uniform operation on finite outputs.** The order bound $b(r)$ and the finite lists of coefficient operations are constructed by NNO recursion, so the index is an input to one uniform operation.

(4) **Verify products, descent, and dependent name transport.** A map defined on several coefficient objects uses their finite product presentation. Passing from presentations to denotations preserves its equation; dependent inputs use the projective refinement of Proposition E.1 before taking a tracker. This proves the finite-output comparison on names.

□

The limit classifies compatible effective points with the generated observations. A uniform compatible family of names carries its tracker; projective presentations of the limit are supplied by Proposition E.1.

We can now use the formal neighbourhoods and their identified function objects to build cochains on the infinitesimal nerve. The next section compares those cochains with algebraic differential forms.

10 Synthetic and algebraic de Rham resource geometry

Resource variations have differential coefficients, and compatible changes of those coefficients require transport. De Rham forms express the infinitesimal data; curvature tests the compatibility of a proposed connection, while a stratification records coherent transport at every formal order. The event instrumentations of Proposition 10.7 exhibit this compatibility test. The completed measured family of Example 9.12 provides a nonzero logarithmic class and an explicit flat transport (Proposition 10.10).

Functions on the completed infinitesimal nerve and algebraic Kähler forms give two descriptions of this differential information. We construct a chain-homotopy equivalence between them over the exact resource coefficients. The geometric function calculation is

Theorems 8.6 and 9.6; its effective finite observations are Construction 9.17 and Proposition 9.18. The algebraic precedents are (Grothendieck 1966; Berthelot and Ogus 1978), and the finite rational contraction uses a bar–Koszul calculation related to (Hochschild et al. 1962).

Relative forms differentiate valuation parameters while fixing B. Absolute forms over $\mathbb{Q}$ also differentiate resource coefficients; the logarithmic connections on $t^{-k}B$ in Proposition 12.14 use that second calculus. The primary site supplies the function objects for the comparison, and based-Weil restriction recovers their finite polynomial coefficient formulas.

10.1 Completed higher-diagonal cochains

A cochain evaluates a function on a finite sequence of infinitesimally related points. The completed higher diagonals provide these configurations and their coordinate algebras. Fix E finite-projective, $V = E^\vee$, and $A = \mathrm{Sym}_B(V)$. For $q \geq 0$ let

$$D_q = A^{\otimes_B(q+1)}, \quad I_q = \ker(D_q \xrightarrow{\mu_q} A), \quad \widehat{C}^q(A/B) = \varprojlim_r D_q/I_q^{r+1}.$$

By Theorem 9.6 there is a natural cosimplicial algebra identification

$$\widehat{C}^q(A/B) \cong \underline{\mathrm{Hom}}_{\mathscr{H}_{B,\mathrm{et}}}(\mathcal{D}^{(q)}_{X_A}, \mathcal{O}_{\mathrm{inf}}).$$

These limits are taken in internal B-modules or algebras; (9.9) gives their coefficient-sheaf realization. To calculate their maps, write the successive coordinate differences as $u_i = x_i - x_{i-1}$, $1 \leq i \leq q$. The finite-projective change of variables gives

$$\widehat{C}^q(A/B) \cong A[[u_1, \ldots, u_q]], \tag{10.1}$$

where each u_i denotes a block of variables dual to E and the completion is in the total difference degree. The degree $q = 0$ algebra is A.

A simplicial map of the higher diagonal sends a vertex to a selected or repeated vertex, hence sends each difference to a finite sum of successive differences. This preserves the powers of I_q and constructs the cosimplicial algebra

$$\widehat{C}^\bullet(A/B) : \Delta \longrightarrow \mathrm{CAlg}_B(\mathcal{E}).$$

Its normalization and differential are

$$N^q_{\mathrm{inf}}(A/B) = \bigcap_{i=0}^{q-1} \ker(s_i : \widehat{C}^q \to \widehat{C}^{q-1}), \qquad \delta = \sum_{i=0}^{q+1} (-1)^i d_i. \tag{10.2}$$

By the cosimplicial identities $\delta^2 = 0$ and $\delta N^q \subseteq N^{q+1}$. These are equations of maps, proved by pairing the two terms corresponding to any two deleted vertices.

Proposition 10.1 (Explicit completed coface formula)**.** *For a normalized q-cochain $f(x; u_1, \ldots, u_q)$,*

$$\begin{aligned} \delta f(x; u_1, \ldots, u_{q+1}) = {} & f(x + u_1; u_2, \ldots, u_{q+1}) \\ & + \sum_{i=1}^{q} (-1)^i f(x; u_1, \ldots, u_i + u_{i+1}, \ldots, u_{q+1}) \\ & + (-1)^{q+1} f(x; u_1, \ldots, u_q). \end{aligned} \tag{10.3}$$

Normalization consists exactly of series having positive degree in each successive difference block u_i. All displayed operations are defined by finite polynomial substitution at each fixed homogeneous degree and commute with truncation.

Proof.

(1) **Compute the coface substitutions.** Deleting the first vertex changes the base point from x to $x+u_1$; deleting the i-th interior vertex replaces the two adjacent differences by their sum; deleting the last vertex removes the last difference. This proves the formula.

(2) **Identify normalized cochains.** Repeating one vertex sets one of its adjacent differences to zero. Being in all codegeneracy kernels therefore means that every monomial contains a positive contribution from each difference block.

(3) **Extend the formulas to completed cochains.** The substitutions respect the total-degree filtration, and each resulting coefficient is a finite sum, proving the completion assertion.

□

The normalized-completed comparison is summarized by the commuting diagram

$$\begin{array}{ccc} N^{\bullet}_{\mathrm{inf}}(A/B) & \xrightarrow{\ \rho\ } & \Omega^{\bullet}_{A/B} \\ \Big\downarrow{\scriptstyle \delta} & & \Big\downarrow{\scriptstyle d} \\ N^{\bullet+1}_{\mathrm{inf}}(A/B) & \xrightarrow[\ \rho\ ]{} & \Omega^{\bullet+1}_{A/B}. \end{array} \tag{10.4}$$

The maps will be constructed and compared in Theorem 10.3.

The normalized complex has an Alexander–Whitney cup product,

$$(f \smile g)(x; u_1, \dots, u_{p+q}) = f(x; u_1, \dots, u_p)\, g(x + u_1 + \cdots + u_p; u_{p+1}, \dots, u_{p+q}). \tag{10.5}$$

Associativity is reassociation of the same ordered vertices. The coface calculation gives $\delta(f \smile g) = \delta f \smile g + (-1)^p f \smile \delta g$.

10.2 Relative forms and the infinitesimal comparison

We now compare the normalized cochains with the algebraic de Rham complex. The Kähler module of Section 6.6 already supplies its degree-one forms. Define

$$\Omega^p_{A/B} = \bigwedge_A^p \Omega^1_{A/B}, \qquad d : A \to \Omega^1_{A/B}$$

and extend d as the unique graded derivation of degree one with $d(da) = 0$. The alternating Leibniz identities give $d^2 = 0$. For a polynomial resource cell,

$$\Omega^{\bullet}_{A/B} \cong A \otimes_B \bigwedge_B^{\bullet} V. \tag{10.6}$$

This construction is internal to the signed rational algebraic sector and imposes no completion on the quantitative scalar semiring S.

The comparison extracts the first-order contribution in each successive displacement and alternates it. For a normalized q-cochain f, a temporary finite free presentation gives the coefficient formula

$$\rho_q(f) = \sum_{a_1, \dots, a_q} [u_1^{a_1} \cdots u_q^{a_q}] f \; dx^{a_1} \wedge \cdots \wedge dx^{a_q}. \tag{10.7}$$

The exterior quotient makes this alternating tensor formula independent of the temporary basis. At degree zero it is $\rho_0 = \mathrm{id}_A$. The theorem proves that this map retains the cohomological information of the completed cochains.

Lemma 10.2 (Homogeneous additive-cochain calculation and natural contraction)**.** *Let V be a displayed finite-projective module over the rational commutative ring B. In homogeneous difference degree m, put*

$$C_0^{q,m}(V) = \bigoplus_{\substack{m_1+\cdots+m_q=m \\ m_i\geq 1}} \mathrm{Sym}_B^{m_1}(V) \otimes_B \cdots \otimes_B \mathrm{Sym}_B^{m_q}(V)$$

with $C_0^{0,0} = B$ and all other degree-zero summands absent. The degree-preserving component δ_0 of (10.3) has cohomology

$$H^q(C_0^{\bullet,m}(V)) \cong \begin{cases} \bigwedge_B^m V, & q = m, \\ 0, & q \neq m. \end{cases}$$

There are constructed*, B-linear homogeneous operators i_m, p_m, h_m of cochain degrees $0, 0, -1$, respectively, with target complex $\bigwedge_B^m V[-m]$ (and $B[0]$ at $m = 0$), such that*

$$\begin{aligned} p_m i_m = 1, \quad \delta_0 i_m = 0, \quad p_m \delta_0 = 0, \\ \delta_0 h_m + h_m \delta_0 = 1 - i_m p_m, \qquad h_m^2 = h_m i_m = p_m h_m = 0. \end{aligned} \tag{10.8}$$

They are natural under linear maps of displayed finite-projective modules, and their coefficients are finite rational expressions constructed uniformly in m.

Proof.

(1) **Identify the homogeneous bar–cobar complex.** The differential is the reduced cobar differential of the graded symmetric coalgebra on V, with coproduct induced by $u \mapsto u \otimes 1 + 1 \otimes u$. In homogeneous degree m this is a finite complex. Its graded dual is the reduced bar complex of $\mathrm{Sym}_B(V^\vee)$: the divided-power dual is identified with the symmetric algebra using invertible factorials.

(2) **Compute its cohomology by the bar–Koszul comparison.** The Koszul resolution on a displayed free basis of $V^\vee$ resolves its augmentation B; exactness follows by tensoring the one-variable free resolutions. The normalized bar resolution resolves the same augmentation by its extra-degeneracy contraction. Their antisymmetrization comparison is a quasi-isomorphism of free resolutions (the usual bar–Koszul calculation (Hochschild et al. 1962)); tensoring with B and taking finite graded duals gives the stated cohomology, concentrated at $q = m$.

(3) **Present the finite permutation complexes.** We construct the homotopy on finite permutation modules. Give the set of ordered block decompositions of $\{1, \ldots, m\}$ its decidable finite permutation presentation. In degree q the induced permutation modules

$$P_m^q = \bigoplus_{\substack{m_1+\cdots+m_q=m \\ m_i\geq 1}} \mathbb{Q}[S_m] \otimes_{\mathbb{Q}[S_{m_1}\times\cdots\times S_{m_q}]} \mathbb{Q}$$

carry the bar–cobar differential by splitting adjacent blocks into their shuffles with the signs of δ_0. With right/left permutation conventions matched, symmetrization gives a natural isomorphism

$$C_0^{q,m}(V) \cong V^{\otimes_B m} \otimes_{\mathbb{Q}[S_m]} P_m^q.$$

The alternating representation sgn_m gives the corresponding presentation of $\bigwedge_B^m V$. All these representations are finite-dimensional rational vector spaces with specified permutation bases.

(4) **Identify the alternating cohomology representation.** To identify its cohomology equivariantly, evaluate the free-module bar–Koszul calculation at $V = \mathbb{Q}^m$ and take the multidegree $(1, \dots, 1)$ in its m labelled generators. The resulting summand of $C_0^{\bullet,m}(V)$ is precisely $P_m^\bullet$, and the corresponding summand of $\bigwedge^m V$ is the sign representation sgn_m. Thus the calculation identifies the cohomology of $P_m^\bullet$ as sgn_m in degree m and zero elsewhere, equivariantly under permutations.

(5) **Construct the projection and antisymmetrization.** Take $p : P_m^\bullet \to \mathrm{sgn}_m[-m]$ to be the exterior quotient and take i to be normalized antisymmetrization:

$$i(v_1 \wedge \cdots \wedge v_m) = \frac{1}{m!} \sum_{\sigma \in S_m} \mathrm{sgn}(\sigma) v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(m)}.$$

These equivariant chain maps satisfy $pi = 1$. The cohomology calculation above shows that $K_m^\bullet = \ker p$ is acyclic.

(6) **Construct equivariant splittings of the acyclic kernel.** Write $B_m^q = \mathrm{im}(\delta_0 : K_m^{q-1} \to K_m^q)$. In every degree there is an exact sequence of finite rational representations

$$0 \longrightarrow B_m^q \longrightarrow K_m^q \xrightarrow{\delta_0} B_m^{q+1} \longrightarrow 0.$$

Lexicographically ordered rational Gaussian elimination constructs a linear section $s_q : B_m^{q+1} \to K_m^q$ of this last surjection. Replace it by the finite average

$$\bar{s}_q = \frac{1}{m!} \sum_{\sigma \in S_m} \sigma s_q \sigma^{-1}.$$

Since δ_0 is equivariant, $\delta_0 \bar{s}_q = 1$ still holds. Consequently $K_m^q = B_m^q \oplus \bar{s}_q(B_m^{q+1})$ is an equivariant direct-sum decomposition.

(7) **Define the contracting homotopy.** For $z \in P_m^q$ set

$$h_q(z) = \bar{s}_{q-1}(z - ipz - \bar{s}_q \delta_0 z).$$

The argument of $\bar{s}_{q-1}$ belongs to B_m^q: it lies in $\ker p$ and its differential is zero.

(8) **Verify the homotopy equation and side conditions.** Now $\delta_0 h_q z = z - ipz - \bar{s}_q \delta_0 z$, while $h_{q+1} \delta_0 z = \bar{s}_q \delta_0 z$. Thus $\delta_0 h + h \delta_0 = 1 - ip$. The image of h_q is the complementary summand $\bar{s}_{q-1}(B_m^q)$, on which the next h vanishes; also $hi = 0$ and $ph = 0$. This proves all the side conditions in (10.8). The $m = 0$ case is $i = p = 1$, $h = 0$.

(9) **Transport the contraction naturally to coefficient modules.** Finally apply $V^{\otimes m} \otimes_{\mathbb{Q}[S_m]} (-)$. The homotopy equations remain equations of finite matrices, valid over every rational B; all operations are natural in V because S_m acts by permuting tensor factors. If V is a retract of a finite free module, the same functorial maps restrict along its displayed idempotent.

(10) **Construct the family uniformly in the degree.** The lexicographic elimination and finite average are algorithms on m and its finite permutation lists, so the entire family is constructed by NNO recursion, not by selecting one splitting for each m.

□

Theorem 10.3 (Completed infinitesimal de Rham comparison)**.** *For every displayed finite-projective B-module E, the maps* (10.7) *assemble into a natural cochain map*

$$\rho : N_{\mathrm{inf}}^\bullet(\mathrm{Sym}_B(E^\vee)/B) \longrightarrow \Omega^\bullet_{\mathrm{Sym}_B(E^\vee)/B}.$$

It is a chain-homotopy equivalence*, with a parameter-uniform homotopy constructed on compatible truncations. In particular it is a quasi-isomorphism, natural under polynomial substitution and in internal resource parameters. Through Theorem 9.6, this is the comparison from normalized synthetic functions on the full primary infinitesimal nerve to algebraic relative forms. It carries the Alexander–Whitney product to the exterior product and induces an isomorphism of graded cohomology rings.*

Proof.

(1) **Decompose normalized completed cochains by degree.** Put $V = E^\vee$, $A = \mathrm{Sym}_B(V)$ and use the total difference-degree filtration. Normalization and the finite symmetric-power decomposition of the previous lemma identify

$$N^q_{\mathrm{inf}}(A/B) \cong \prod_{m \geq q} A \otimes_B C_0^{q,m}(V).$$

(2) **Separate the homogeneous differential and perturbation.** Split the coface differential as $\delta = \delta_0 + \Delta$. The first part keeps the base point x fixed and has homogeneous difference degree zero. The perturbation Δ comes only from expanding the coefficients of $f(x + u_1; \ldots)$ around x: it raises total difference degree by at least one.

(3) **Assemble the homogeneous contraction.** The contraction operators of Lemma 10.2, extended over A and assembled as product maps i, p, h, obey

$$pi = 1, \quad \delta_0 h + h\delta_0 = 1 - ip, \quad h^2 = hi = ph = 0, \quad p = \rho.$$

The target is $\Omega^\bullet_{A/B}$ with zero homogeneous differential before perturbation.

(4) **Construct the completed perturbation maps.** In every finite truncation, $h\Delta$ and Δh are nilpotent because Δ raises filtration. Therefore the following sums are finite modulo any specified difference degree, and define internal maps of the completed product:

$$\begin{aligned} J &= (1 + h\Delta)^{-1} i = \sum_{k \geq 0} (-h\Delta)^k i, \\ H &= h(1 + \Delta h)^{-1} = \sum_{k \geq 0} (-1)^k h (\Delta h)^k, \\ p' &= p(1 + \Delta h)^{-1}, \qquad d' = p\Delta J. \end{aligned} \tag{10.9}$$

(5) **Verify the perturbation identities coefficientwise.** These are the finite-geometric-series formulas of the basic perturbation lemma, applied on each finite truncation (Miller and Rahmati 2024). Substitution into $(\delta_0 + \Delta)^2 = 0$ and the homogeneous contraction equations gives $\delta J = Jd'$, $p'\delta = d'p'$, $p'J = 1$, and $\delta H + H\delta = 1 - Jp'$. The equations agree under truncation and hence hold in the internal inverse limit. This proves the perturbation identities coefficientwise.

(6) **Identify the transferred projection.** Filtration identifies their transferred terms exactly. The map p extracts difference degree equal to cochain degree; h preserves difference degree but lowers cochain degree, so $p\Delta h = 0$. Thus $p' = p = \rho$.

(7) **Identify the transferred de Rham differential.** Terms in $p\Delta(h\Delta)^k i$ with $k \geq 1$ have difference degree strictly above target cochain degree, and hence vanish. Only $p\Delta i$ survives in $d' = p\Delta J$. For a polynomial coefficient a, the component of degree one of $a(x + u) - a(x)$ is da; applying alternating projection to the remaining linear difference coefficients gives precisely

$$d' = p\Delta i = d_{\mathrm{dR}}.$$

(8) **Establish the completed homotopy equivalence.** Consequently the actual completed maps satisfy

$$\boxed{\rho J = 1, \qquad \delta J = J d_{\mathrm{dR}}, \qquad \rho\delta = d_{\mathrm{dR}}\rho, \qquad \delta H + H\delta = 1 - J\rho.} \tag{10.10}$$

This proves the asserted homotopy equivalence, including injectivity and surjectivity on all cohomology objects.

(9) **Verify finite-output construction and naturality.** All operators in a fixed output difference degree depend on only finitely many lower coefficients. Their defining algorithms use finite rational matrices and sums and commute with parameter substitution. In particular the compatible infinite corrections are produced by the NNO and the product universal property. The comparison ρ commutes with polynomial coordinate substitutions by the chain rule and the universal alternating quotient. The auxiliary maps H, J have the linear naturality established by their finite coefficient construction.

(10) **Verify compatibility with cup and wedge products.** Finally the degree-$(p+q)$ component of (10.5) linear in every difference block is the concatenation of those of f and g. Alternation gives $\rho(f \smile g) = \rho(f) \wedge \rho(g)$, as claimed.

□

For a polynomial coordinate map $p : E \to F$, naturality of the comparison itself is the commutative square of complexes

$$\begin{array}{ccc} N^{\bullet}_{\mathrm{inf}}(A_F/B) & \xrightarrow{\rho_F} & \Omega^{\bullet}_{A_F/B} \\ \downarrow{\scriptstyle p^*} & & \downarrow{\scriptstyle p^*} \\ N^{\bullet}_{\mathrm{inf}}(A_E/B) & \xrightarrow[\rho_E]{} & \Omega^{\bullet}_{A_E/B}. \end{array}$$

The finite homogeneous contraction constructs its homotopy equivalence; the full Taylor substitution supplies the vertical cochain map.

10.3 Relative formal Poincaré contraction

The preceding comparison relates two descriptions of forms. A second calculation checks the local formal neighbourhood itself by contracting its relative displacement directions. For n finite free formal variables put $R = A[[u_1, \dots, u_n]]$ and define the continuous relative forms

$$\widehat{\Omega}^p_{R/A} = R \otimes_A \bigwedge_A^p \Big(\bigoplus_{i=1}^n A\, du_i \Big).$$

The differential is termwise formal differentiation, which is well-defined because each output coefficient is a finite sum.

Theorem 10.4 (Formal Poincaré lemma)**.** *The inclusion $A \to \widehat{\Omega}^{\bullet}_{R/A}$ is a chain-homotopy equivalence. Its relative de Rham cohomology is A in degree zero and zero in positive degrees. The result is natural in finite-projective formal displacement modules.*

Proof.

(1) **Compute the Euler contraction identity.** Let $\mathsf{E} = \sum_i u_i \partial/\partial u_i$ and let ι_{E} be contraction. The graded Leibniz rule gives

$$d\iota_{\mathsf{E}} + \iota_{\mathsf{E}} d = \mathcal{L}_{\mathsf{E}}.$$

(2) **Define the contraction on positive weights.** On homogeneous coefficient degree m and form degree p the Lie derivative is multiplication by $m+p$. For $m+p>0$ define $h=(m+p)^{-1}\iota_{\mathsf{E}}$, and set $h=0$ on degree-zero constants. Rational invertibility and coefficientwise finite arithmetic make this an internal operator on the completed product.

(3) **Verify the deformation retraction and projective descent.** Its identity is

$$dh+hd=\mathrm{id}-\mathrm{const}\,\mathrm{ev}_0\,.$$

This exhibits the required deformation retraction. The Euler operator and the homogeneous decomposition are intrinsic to symmetric powers and commute with finite-projective retractions.

□

10.4 Crystalline stratifications and integrable resource connections

Differential coefficients can also be transported between infinitesimally related points. A stratification records that transport and its composition law; an integrable connection records its first-order differential data. We construct their comparison for a displayed finite-projective module N over $A=\mathrm{Sym}_B(E^\vee)$. Pull N back along the two coefficient actions on principal parts:

$$s_r^*N=P^r_{A/B}\otimes_{A,s_r}N,\qquad t_r^*N=P^r_{A/B}\otimes_{A,t_r}N.$$

Definition 10.5 (Formal resource stratification)**.** With $s_r(a)=a(x)$ and $t_r(a)=a(x+u)$, a formal stratification in the source-to-target convention is a family of $P^r_{A/B}$-linear isomorphisms

$$\varepsilon_r : s_r^*N\xrightarrow{\cong}t_r^*N$$

compatible with truncation and equal to the identity on the reduced diagonal. At every finite completed triple diagonal, writing ε_{xy} for transport from the fibre over x to the fibre over y, its cocycle equation is

$$\varepsilon_{yz}\,\varepsilon_{xy}=\varepsilon_{xz}.$$

The morphisms are A-linear maps commuting with this family.

An integrable relative connection is a B-linear map

$$\nabla : N\to N\otimes_A\Omega^1_{A/B}$$

satisfying the Leibniz identity $\nabla(an)=a\nabla(n)+n\otimes da$ and whose graded extension satisfies $\nabla^2=0$. Curvature is an A-module map.

Theorem 10.6 (Connections are formal stratifications)**.** *For displayed finite-projective E and finite-projective $A=\mathrm{Sym}_B(E^\vee)$-modules N, integrable B-connections on N and formal stratifications of N form naturally equivalent categories. The equivalence commutes with polynomial substitution, parameter substitution, and finite-order coefficient base change.*

Proof.

(1) **Derive the iterated covariant Leibniz formula.** Write $P^r=P^r_{A/B}$ and $u=y-x$. We first construct the transport in the direction forced by its semilinearity. In a finite-free

coordinate presentation write $\nabla n = \sum_i \nabla_i(n) \otimes dx^i$. The Leibniz identity and flatness say

$$\nabla_i(an) = \partial_i(a)n + a\nabla_i(n), \qquad [\nabla_i, \nabla_j] = 0.$$

The finite multi-index Leibniz formula follows by induction:

$$\nabla^\alpha(an) = \sum_{\beta \le \alpha} \binom{\alpha}{\beta} \partial^\beta(a) \nabla^{\alpha-\beta}(n).$$

(2) **Construct the finite Taylor transport.** Define a P^r-linear map by its values on t_r-generators,

$$\begin{aligned} T_r : t_r^* N &\longrightarrow s_r^* N, \\ T_r(1 \otimes_{t_r} n) &= \sum_{|\alpha| \le r} \frac{u^\alpha}{\alpha!} (1 \otimes_{s_r} \nabla^\alpha n). \end{aligned} \tag{10.11}$$

(3) **Verify balancing and reduction.** Indeed finite reindexing of the Leibniz sum gives

$$T_r(1 \otimes_{t_r} an) = a(x+u) T_r(1 \otimes_{t_r} n) = t_r(a) T_r(1 \otimes_{t_r} n),$$

so the defining formula respects the t_r-balanced relation. Its reduction modulo I is the identity.

(4) **Prove the transport cocycle.** On the finite completed triple diagonal use vertices x, y, z, with differences $u = y - x$ and $v = z - y$. Expanding the iterated covariant derivatives and using their commutation, the ordinary multinomial identity gives

$$T_{x \leftarrow y} T_{y \leftarrow z} = T_{x \leftarrow z}, \qquad T_{x \leftarrow x} = 1.$$

Each coefficient is a finite sum, since total difference degree is truncated.

(5) **Construct the inverse stratification.** Swapping the vertices gives the reverse transport; the cocycle with $z = x$ proves that T_r is invertible. Define

$$\varepsilon_r = T_r^{-1} : s_r^* N \xrightarrow{\cong} t_r^* N.$$

The displayed identity for T now yields the required source-to-target cocycle, while reduction and truncation are immediate. In particular no arbitrary lifting of an inverse is needed.

(6) **Extract the connection from first-order transport.** Conversely, given ε, set $T = \varepsilon^{-1}$. Working in $s_1^* N$, define ∇n by

$$T_1(1 \otimes_{t_1} n) - (1 \otimes_{s_1} n) \mod I^2 \ \in \ (I/I^2) \otimes_A N \cong \Omega^1_{A/B} \otimes_A N.$$

(7) **Verify Leibniz and zero curvature.** The equality $T_1(1 \otimes_{t_1} an) = t_1(a) T_1(1 \otimes_{t_1} n)$ gives $\nabla(an) = a\nabla(n) + n \otimes da$ after taking the first difference coefficient. The alternating coefficient of $u_i v_j$ in the triple cocycle is $[\nabla_i, \nabla_j]$, so the cocycle gives zero curvature.

(8) **Recover every higher Taylor coefficient.** Differentiating the same triple cocycle once in the final nilpotent displacement recursively gives the higher coefficient rule $T_{\alpha + e_i} = (\alpha_i + 1)^{-1} \nabla_i T_\alpha$. It recovers uniquely the finite Taylor polynomial (10.11) because all positive integers are invertible in B.

(9) **Descend the construction and verify naturality.** For a displayed projective retract, replace the temporary free basis by a finite dual basis of E; constant directional derivations commute, and their symmetric tensor expressions factor through the displayed idempotent. Every identity above is an identity of finite symmetric-power morphisms. The construction therefore descends to the finite-projective cell, commutes with coordinate substitutions and coefficient base change, and is functorial on connection-preserving module maps.

(10) **Verify that the constructions are inverse.** The two procedures are mutually inverse by the first-order coefficient and the unique higher-coefficient recurrence.

□

10.5 Measured instrumentation, curvature, and coherent transport

Two changes of event measurement can preserve returned values while their order changes the resource data. Use the idle events a, b and let $N = (N_a, N_b)^{\mathsf{T}}$ be the corresponding count profiles of a retained program. For natural numbers u, v, the finite substitutions

$$\sigma_u(a) = a, \quad \sigma_u(b) = ba^u, \qquad \tau_v(a) = ab^v, \quad \tau_v(b) = b$$

are implemented by inserting the indicated idle blocks. Other primitives are unchanged. Recursion on finite descriptions performs the replacement under sequencing, branching, and iteration. Since the inserted blocks are finite and leave the state unchanged, the finite return comparison identifies the return domains and values. The returning histories are the displayed free-monoid substitutions.

Writing count vectors as columns, their exact actions are

$$U_u = I + uP, \qquad V_v = I + vQ, \qquad P = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

For instance σ_u changes (N_a, N_b) to $(N_a + uN_b, N_b)$. These polynomial laws extend to resource coefficients. The count profiles lie in a common principal cell, so the two-component calculation is realized on a finite-projective resource bundle.

Proposition 10.7 (Instrumentation curvature and ordered transport)**.** *The two instrumentations have mixed-jet discrepancy*

$$U_\varepsilon V_\delta - V_\delta U_\varepsilon = \varepsilon\delta[P, Q], \qquad [P, Q] = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{10.12}$$

over $B[\varepsilon, \delta]/(\varepsilon^2, \delta^2)$. On the coefficient bundle $B[u, v]^2$, the connection prescribing these constant infinitesimal updates,

$$\nabla_0 = d - P\,du - Q\,dv,$$

has curvature $[P, Q]\,du \wedge dv$ and admits no formal stratification inducing it. The ordered family $G(u, v) = V_v U_u$ instead constructs the integrable connection

$$\nabla_G = d - (dG)G^{-1}$$

and the formal transport $\varepsilon_{x,y} = G(y)G(x)^{-1}$. Its associated crystalline coaction is given by Proposition 11.6.

Proof.

(1) **Compute the measured mixed coefficient.** Expand the two products and compare the coefficient of $\varepsilon\delta$. The signed difference is (10.12). These are measured coefficients by the count formulas and Theorem 6.1.

(2) **Compute curvature and exclude a constant-rule stratification.** The differential of $-P\,du - Q\,dv$ is zero, and its exterior square is $[P, Q]\,du \wedge dv$, which is nonzero in the free coefficient bundle. The connection–stratification equivalence therefore rules out a stratification with these constant update rules.

(3) **Construct the polynomial frame and flat connection.** Since $P^2 = Q^2 = 0$, the inverse of G is $(I - uP)(I - vQ)$, an actual polynomial matrix. For a column h, $\nabla_G(Gh) = G\,dh$; applying the graded connection again gives zero curvature.

(4) **Verify the stratification cocycle.** The transport $G(y)G(x)^{-1}$ is the identity on the diagonal, is compatible with every truncation, and satisfies

$$G(z)G(y)^{-1}\,G(y)G(x)^{-1} = G(z)G(x)^{-1}.$$

It therefore gives the required stratification.

(5) **Identify the ordered update rules.** Its v-update is Q, while its u-update is $V_v P V_v^{-1}$; at the origin these are the original two updates. The conjugation records their specified order and supplies the compatible transport.

□

This example distinguishes the mixed response of two implemented instrumentations from their common returned computation. It also exhibits, on the same resource coefficients, a curvature obstruction and a constructed flat transport. Formal stratifications act on the primary infinitesimal relation by Proposition 11.6; based restriction gives the displayed Taylor matrices.

10.6 Global resource coefficients, absolute forms, and a nonzero class

The construction is uniform in E, its displayed idempotent, and its coefficient maps. It therefore applies at each budget to $A_a = \mathrm{Sym}_B(M_a^\vee)$. Budget transitions act by polynomial coordinate restriction, and Theorem 10.3 identifies the induced maps on cochains and forms. This assembles the relative complexes and crystalline transport over the resource cells. The parameter space is still $\mathfrak{C}_{\mathrm{Ev}}$, with its previously specified points and positive observations.

Theorem 10.8 (Formal principal parts and de Rham comparison after localization)**.** *Let $A = \mathrm{Sym}_B(V)$ for a displayed finite-projective V, let $f \in A$, and put $A_f = A[f^{-1}]$. Write $D_q = A^{\otimes_B(q+1)}$ and $I_q = \ker(D_q \to A)$, with analogous notation $I_{q,f}$ over A_f. At every q, r there are natural isomorphisms*

$$\frac{A_f^{\otimes_B(q+1)}}{I_{q,f}^{r+1}} \;\cong\; A_f \otimes_A \frac{D_q}{I_q^{r+1}}. \tag{10.13}$$

They respect every vertex and truncation map. Consequently normalized completed coordinate cochains of A_f are chain-homotopy equivalent to $\Omega^\bullet_{A_f/B}$. The connection–stratification equivalence and finite Taylor transport extend to finite-projective A_f-modules.

Proof.

(1) **Construct inverses at every vertex.** Write f_i for the image of f at vertex i in the higher-diagonal algebra. Modulo I_q^{r+1}, the element $f_i - f_0$ is nilpotent of exponent $r+1$. After inverting f_0, the inverse of every other f_i is constructed by

$$f_i^{-1} = f_0^{-1} \sum_{k=0}^{r} \left(-\frac{f_i - f_0}{f_0} \right)^k \pmod{I_q^{r+1}}.$$

(2) **Identify the localized principal-parts algebra.** Thus the universal algebra obtained by inverting all the f_i at finite order is precisely the one obtained by inverting f_0. Since localization commutes with quotients and with finite powers of their defining ideals, the two universal properties give the inverse maps of (10.13).

(3) **Verify simplicial and truncation coherence.** The finite inverse is unique, so the comparison commutes with all simplicial vertex maps and truncation.

(4) **Extend the homogeneous contraction over the localization.** The normalized completed complex is consequently the inverse limit of its finite localizations $A_f \otimes_A (D_q/I_q^{r+1})$. In difference coordinates its homogeneous additive complex is $A_f \otimes_B C_0^{\bullet,m}(V)$. The maps of Equation (10.8) extend over A_f.

(5) **Construct the localized de Rham retraction.** Expanding the inverse $f(x+u)^{-1}$ by the displayed finite geometric formula produces a perturbation strictly raising formal difference degree. The same finite-order perturbation construction (10.9) therefore gives a deformation retraction onto $\Omega^{\bullet}_{A_f/B}$, whose transferred differential is the universal localized derivation $d(f^{-1}) = -f^{-2}df$. Localization is performed at each finite diagonal order before taking the limit.

(6) **Extend connection transport over the localization.** The Taylor transport of Theorem 10.6 uses the same finite inverse identities; it therefore extends and satisfies its unit and cocycle equations over A_f.

□

Proposition 10.9 (A logarithmic relative de Rham class)**.** *On $A = B[x, x^{-1}]$, relative to B,*

$$H^0_{\mathrm{dR}}(A/B) \cong B, \qquad H^1_{\mathrm{dR}}(A/B) \cong B \cdot [d\log x],$$

and the higher cohomology objects vanish. The formal diagonal has $P^r_{A/B} \cong A[u]/(u^{r+1})$, and $d\log x = dx/x$ is represented by the completed normalized cocycle

$$\ell(x,y) = \log\left(1 + \frac{y-x}{x}\right) = \sum_{j\geq 1} \frac{(-1)^{j+1}}{j} \left(\frac{y-x}{x}\right)^j.$$

Proof.

(1) **Compute the Laurent de Rham cohomology.** Termwise relative differentiation gives $d(b_j x^j) = j b_j x^{j-1} dx$. Every $j \neq 0$ is invertible in B and the derivative has no $x^{-1}dx$ coefficient. This computes the two nonzero cohomology objects; the cotangent module is free of rank one, so higher forms vanish.

(2) **Compute the finite and completed principal parts.** Put $y = x + u$. At finite order y has the explicitly constructed inverse

$$y^{-1} = x^{-1} \sum_{k=0}^{r} (-u/x)^k \pmod{u^{r+1}}.$$

It follows from Theorem 10.8 that $P^r_{A/B} \cong A[u]/(u^{r+1})$ and $\widehat{P}_{A/B} \cong A[[u]]$.

(3) **Construct the logarithmic cocycle and identify its class.** The formal logarithm $\ell(x, y)$ is therefore a legitimate completed coordinate cochain. Its finite logarithm identity $\ell(x, y) + \ell(y, z) = \ell(x, z)$ holds modulo each higher diagonal power, so $\delta\ell = 0$. Its first difference coefficient is x^{-1}; therefore $\rho_1(\ell) = dx/x$, as claimed.

□

Proposition 10.10 (A measured logarithmic class and its transport)**.** *For the measured family of Example 9.12, retain $R = B[t^{-1}]$, $q = t^{-1}$, and put*

$$A = R[z, (1 - qz)^{-1}], \qquad f(z) = (1 - qz)^{-1}, \qquad \omega = \frac{q}{1 - qz}\, dz.$$

Relative to R, the form ω is closed and has nonzero de Rham class. Its completed diagonal representative is

$$\ell(z, z + u) = \log \frac{1 - qz}{1 - q(z + u)} = \sum_{j \geq 1} \frac{1}{j} \left(\frac{qu}{1 - qz} \right)^j. \tag{10.14}$$

The rank-one connection $\nabla = d - \omega$ is integrable, f is horizontal, and its source-to-target stratification is multiplication by

$$\tau(z, z') = \frac{f(z')}{f(z)} = \frac{1 - qz}{1 - qz'}. \tag{10.15}$$

The Taylor coefficients of f at zero are exactly the measured coefficients q^j of the family F_r.

Proof.

(1) **Identify the nonzero logarithmic de Rham class.** Since q is a unit of R, $x = 1 - qz$ identifies A with $R[x, x^{-1}]$ and ω with $-dx/x$. Proposition 10.9, applied over R, identifies its nonzero class. The same relative calculation is available over B: localization gives $\Omega^1_{R/B} = 0$. Alternatively $A = B[z, (t(t - z))^{-1}]$, a principal open of a polynomial B-cell.

(2) **Construct the measured logarithmic cocycle.** At every finite diagonal order, $qu/(1 - qz)$ is nilpotent, so (10.14) is a finite logarithm there. Its truncations are compatible. Multiplication of the ratios in (10.15) gives the additive cocycle identity for their formal logarithms, and the first difference coefficient is $q/(1 - qz)$. Thus the de Rham comparison sends ℓ to ω.

(3) **Verify integrability and the horizontal frame.** The relative exterior square vanishes in this one-coordinate algebra, and $d\omega = 0$, giving integrability. Direct differentiation gives $df = \omega f$. The unit f is a horizontal frame, so transport of its scalar coefficients is multiplication by $f(z')/f(z)$.

(4) **Verify transport coherence and the measured expansion.** These ratios satisfy the identity and triple cocycle equations at every finite order. Expanding $(1 - qz)^{-1}$ at zero gives q^j, which agrees with (9.12).

□

The de Rham class concerns the absence of an algebraic additive primitive for ω. The displayed multiplicative transport exists and will give the crystalline jet coaction in Example 11.9. Its horizontal generating function retains the complete coefficient family whose resource bounds were calculated in Proposition 9.13.

The absolute algebraic de Rham complex on these cells has an Euler contraction.

Theorem 10.11 (Absolute Euler contraction and the Laurent class)**.** *For every displayed finite-projective V over the rational ring B, inclusion and zero-section augmentation construct a natural chain-homotopy equivalence*

$$\Omega^{\bullet}_{B/\mathbb{Q}} \;\simeq\; \Omega^{\bullet}_{\mathrm{Sym}_B(V)/\mathbb{Q}}.$$

For $L = B[x, x^{-1}]$, there is a natural chain-homotopy equivalence

$$\Omega^{\bullet}_{L/\mathbb{Q}} \;\simeq\; \Omega^{\bullet}_{B/\mathbb{Q}} \oplus \Omega^{\bullet-1}_{B/\mathbb{Q}} \cdot d\log x,$$

where the shifted summand has the corresponding shifted differential. Consequently

$$H^k_{\mathrm{dR}}(L/\mathbb{Q}) \cong H^k_{\mathrm{dR}}(B/\mathbb{Q}) \oplus H^{k-1}_{\mathrm{dR}}(B/\mathbb{Q})\,[d\log x].$$

Proof.

(1) **Decompose the absolute de Rham complex by weight.** The symmetric algebra has its canonical weight grading $A = \bigoplus_{m\geq 0} \mathrm{Sym}^m_B(V)$. Give the universal differential da the same weight as homogeneous a. The relations defining $\Omega^1_{A/\mathbb{Q}}$ and its exterior powers are homogeneous, so the absolute de Rham complex is a direct sum by nonnegative weight. Its weight-zero subcomplex is exactly $\Omega^{\bullet}_{B/\mathbb{Q}}$, with inclusion and projection induced by $B \to A \to B$.

(2) **Construct the Euler contraction.** The Euler derivation $\mathcal{E}$ is zero on B and acts as multiplication by m on $\mathrm{Sym}^m_B(V)$. Its interior contraction on forms obeys the Cartan identity

$$d\iota_{\mathcal{E}} + \iota_{\mathcal{E}} d = \mathcal{L}_{\mathcal{E}}.$$

(3) **Contract the positive-weight summands.** On weight $m > 0$ the right side equals m. Define $h = m^{-1}\iota_{\mathcal{E}}$ on this weight and zero on weight zero. Since each algebraic form has finitely many weights, h is an actual map of internal B-modules. It gives $dh + hd = 1 - \iota_B \epsilon_B$, proving the first homotopy equivalence. The grading and Euler map are intrinsic and commute with finite-projective retractions.

(4) **Contract the nonzero Laurent weights.** For $L = B[x, x^{-1}]$, give x weight 1, x^{-1} weight -1, and dx weight 1. The same Euler contraction kills every nonzero integer weight, since every nonzero integer is a unit of B. The weight-zero complex is

$$\Omega^{\bullet}_{B/\mathbb{Q}} \oplus \Omega^{\bullet-1}_{B/\mathbb{Q}} \wedge \frac{dx}{x}.$$

(5) **Identify the remaining complex and its cohomology.** The absolute differential preserves this direct sum and $d(dx/x) = 0$. More explicitly, a chain isomorphism from the standard cochain shift (whose differential is $-d_B$) sends α of degree $k-1$ to $(-1)^{k-1}\alpha \wedge d\log x$ in degree k. Hence the same Cartan identity supplies the asserted deformation retraction and its cohomology calculation.

□

Relative formal cochains use B as base. The absolute retractions above use $\mathbb{Q}$ and apply to the numerical scale differentials and logarithmic connections of Section 12.10.

10.7 The smooth-groupoid Artin sector

A finite-projective resource-coordinate module may be described in local frames. Changes of frame act on its coordinate expression and on every polynomial resource map defined on it. To organize these isomorphisms and their stabilizers, we use algebraic stacks in groupoids on the same big étale site. An internal groupoid-valued presheaf is constructed as a finite-limit category object in $\mathcal{E}$. Descent requires its values on an étale Čech nerve to give the groupoid of compatible objects and arrows, and stackification glues these finite covering data. This is a further sector alongside the $\mathcal{E}$-valued sheaf categories of Theorem 8.4.

Construction 10.12 (Smooth Artin quotients)**.** For a displayed smooth affine groupoid $R \rightrightarrows U$ of finitely presented B-schemes, with smooth source and target, let $[U/R]$ be the stack of local U-objects and their R-valued change-of-trivialization arrows, glued along étale covering diagrams. Equivalently, on a test T its objects are groupoid-principal R-bundles $P \to T$ with equivariant map $P \to U$, and its arrows are their equivariant isomorphisms. The atlas $U \to [U/R]$ has kernel-pair groupoid R and is smooth and surjective in the ordinary algebraic-stack sense. The construction includes the full simplicial nerve $U_\bullet$ with $U_0 = U$, $U_1 = R$ and $U_q = R \times_U \cdots \times_U R$.

Proof.

(1) **Verify descent for the torsor quotient.** The torsor cocycle on a covering family is finite diagram data; refining two coverings compares them by their pullbacks. The sheaf condition for U and R supplies effectiveness of object and arrow descent, and torsor descent makes the displayed assignment a stack.

(2) **Recover the groupoid nerve and smooth atlas.** Pullback of the universal torsor along $U \to [U/R]$ reconstructs the groupoid arrows and its full nerve. Smoothness and surjectivity of the atlas follow by pulling back to a torsor trivializing cover; this is the standard ordinary smooth-groupoid presentation theorem (Stacks Project Authors 2026c, Sections 94.12 and 94.17).

□

Theorem 10.13 (Endpoint cohesion on groupoid-valued étale stacks)**.** *Let* $\mathrm{St}^{(1)}_{\mathrm{et}}(\mathsf{Aff}^{\mathrm{fp}}_B; \mathcal{E})$ *and* $\mathrm{St}^{(1)}_{\mathrm{et}}(\mathsf{I}_B; \mathcal{E})$ *denote the 2-categories of internal groupoid-valued étale stacks, with pseudonatural maps and modifications. The endpoint construction lifts to a string of 2-adjoints*

$$\iota_! \dashv \iota^* \dashv \iota_* \dashv \iota^!,$$

with the same larger- and smaller-endpoint formulas as (8.1)*. On a representable affine* T*, the rightmost functor is determined by*

$$(\iota^! \mathcal{H})(T) = \mathrm{PsNat}(\iota_* y_T, \mathcal{H}),$$

the groupoid of pseudonatural transformations and their invertible modifications. The two endpoint embeddings remain nonisomorphic, witnessed by the discrete affine-line stack. The full Artin quotient stacks from Construction 10.12 occur as objects of the base stack 2-category.

Proof.

(1) **Lift the endpoint adjunctions to groupoid-valued presheaves.** The endpoint universal properties of Lemma 8.2 apply to pseudonatural transformations of groupoid-valued presheaves, yielding the first three 2-adjunctions and full faithfulness by the two identities $c\iota = d\iota = 1$.

(2) **Compare endpoint descent and stackification.** The étale lift of Lemma 8.3 identifies not only covering families but their complete descent groupoids; therefore endpoint precomposition commutes with stackification by gluing effective descent data. It preserves the 2-colimits of stack diagrams and consequently has a right 2-adjoint.

(3) **Construct the remaining right 2-adjoint.** To construct it, express a groupoid-valued presheaf by the Grothendieck construction on its representable objects and arrows, with their specified invertible coherence. Its 2-colimit is the original presheaf by the bicategorical Yoneda calculation. Applying ι_* to that presentation and mapping to $\mathcal{H}$ identifies the resulting groupoid of maps with the pseudonatural transformations into the displayed rightmost formula.

(4) **Descend the adjunction and distinguish the embeddings.** Stackification gives the same adjunction on étale stacks. The affine-line example in Theorem 8.4 consists of discrete groupoids, so the endpoint embeddings are also distinct here.

□

Proposition 10.14 (Infinitesimal symmetries of resource frames)**.** *The finitely presented group scheme*

$$\mathrm{GL}_{n,B} = \operatorname{Spec} B[x_{ij}, \det(x_{ij})^{-1}]$$

is smooth, and its quotient $B\mathrm{GL}_{n,B} = [\operatorname{Spec} B/\mathrm{GL}_{n,B}]$ is a displayed Artin stack. Its atlas has self-intersection $\mathrm{GL}_{n,B}$. Over any square-zero algebra $R[\varepsilon]/\varepsilon^2$, its infinitesimal stabilizer at the identity is

$$\ker\big(\mathrm{GL}_n(R[\varepsilon]/\varepsilon^2) \to \mathrm{GL}_n(R)\big) = \{1 + \varepsilon M : \ M \in M_n(R)\},$$

with multiplication $(1+\varepsilon M)(1+\varepsilon N) = 1 + \varepsilon(M+N)$. In particular the displayed atlas is not étale for $n > 0$.

Proof.

(1) **Represent invertible matrices by a smooth principal open.** Localization at the determinant represents invertible matrices. The Jacobian criterion makes the principal open in a finite affine space smooth over B.

(2) **Compute the infinitesimal group law.** A matrix congruent to the identity modulo a square-zero ideal is uniquely $1 + \varepsilon M$, with inverse $1 - \varepsilon M$. Multiplying and using $\varepsilon^2 = 0$ gives the stated group law.

(3) **Identify the nonzero tangent and quotient atlas.** The group scheme has a nonzero relative tangent module, so its atlas cannot be étale. Its torsor quotient and atlas are Construction 10.12.

□

Theorem 10.15 (The infinitesimal relation on presented Artin groupoids)**.** *For the displayed quotient stack $\mathcal{X} = [U/R]$ of Construction 10.12, define on every infinitesimal arrow $j = (\operatorname{Spec} R_0 \hookrightarrow \operatorname{Spec} S_0)$ the groupoid*

$$\mathcal{D}_{\mathcal{X}}(j) = \mathcal{X}(S_0) \overset{\mathrm{h}}{\underset{\mathcal{X}(R_0)}{\times}} \mathcal{X}(S_0).$$

An object is a pair of S_0-families with a specified isomorphism between their reductions to R_0; arrows are pairs of isomorphisms compatible with that boundary isomorphism. The assignment is an étale stack on I_B. Its iterated homotopy fibre products give a simplicial

infinitesimal groupoid, with unit and composition induced by identity and composition of the specified boundary isomorphisms. On the ordinary affine sheaf F_A this groupoid reduces to the fibre-product relation of Theorem 9.4, and consequently has its explicit finite principal-parts filtration.

Proof.

(1) **Construct the functorial infinitesimal groupoid relation.** The restriction $\mathcal{X}(S_0) \to \mathcal{X}(R_0)$ is functorial in commutative squares of thickenings. Homotopy pullbacks of groupoids may be constructed as the groupoids of triples (x, y, α) with $\alpha : x|_{R_0} \cong y|_{R_0}$; their morphisms are compatible pairs. This construction proves functoriality.

(2) **Verify descent and the simplicial operators.** On an étale covering diagram of j, each endpoint gives a covering diagram by Lemma 8.3. Torsors and their isomorphisms descend along those covers, so the groupoids at both endpoints satisfy descent; finite homotopy pullbacks preserve the required limit diagrams. Composition and inversion of the boundary isomorphisms obey the groupoid laws, giving all the simplicial operators.

(3) **Recover the affine principal-parts relation.** For a represented ordinary affine scheme F_A, its point groupoids are discrete. An isomorphism between their reductions is precisely equality, so the displayed homotopy pullback is the ordinary kernel pair $F_A(S_0) \times_{F_A(R_0)} F_A(S_0)$. Apply the finite nilpotence proof of Theorem 9.4 for its principal-parts filtration. For a general Artin quotient the boundary isomorphism remains part of the groupoid data.

□

10.8 Étale descent and the smooth Artin groupoid de Rham interface

The smooth-groupoid sector extends the coefficient calculation to presentations with stabilizers. To retain that information we use the full simplicial atlas of the Artin presentations from Construction 10.12. The following construction provides its internal coefficient diagram over B. The broader de Rham formalism for ordinary Artin stacks is the cited theory of (Behrend 2005; Stacks Project Authors 2026c).

Construction 10.16 (De Rham cochains of a displayed smooth affine groupoid)**.** Let $R \rightrightarrows U$ be a smooth affine groupoid over B whose nerve has finitely presented smooth affine terms U_q; put $A_q = \mathcal{O}(U_q)$. The face maps of this full simplicial scheme induce a cosimplicial diagram of relative differential graded algebras

$$[q] \longmapsto \Omega^{\bullet}_{A_q/B}.$$

Define its cochains by the total complex of the normalized cosimplicial complex, with the de Rham differential d and the alternating coface differential ∂; explicitly

$$\mathrm{DR}_{\mathrm{grp}}([U/R]/B) = \mathrm{Tot}_{q\geq 0}\, N^q \Omega^{\bullet}_{A_\bullet/B}, \qquad D|_{(q,p)} = \partial + (-1)^q d.$$

Normalization is the intersection of degeneracy kernels. In each fixed total degree only finitely many terms occur, so D and all its signs are maps of internal B-modules. For polynomial and principal-localization terms, the natural comparison of Theorems 10.3 and 10.8 replaces each relative-form complex with its completed formal coordinate-cochain complex. Its simplicial naturality gives a morphism of the corresponding total complexes.

Proof.

(1) **Verify the total differential.** All simplicial maps are maps of B-algebras and commute strictly with universal derivations. The simplicial identities imply $\partial^2 = 0$; naturality gives $d\partial = \partial d$, so the displayed signed total differential squares to zero.

(2) **Normalize the simplicial complex.** Degenerate summands split off by the standard finite normalization identities.

(3) **Compare completed infinitesimal cochains with the total de Rham complex.** The comparison ρ from completed normalized infinitesimal cochains to forms is natural under polynomial substitution and algebraic localization, hence is compatible with all cofaces wherever those local charts apply. Levelwise chain homotopy equivalence then gives a total-complex quasi-isomorphism by the first-quadrant finite-total-degree filtration.

□

Proposition 10.17 (Étale refinement and infinitesimal stabilizer retention)**.** *The above cosimplicial coefficient construction is functorial under maps of displayed smooth groupoids and under étale refinement of their affine nerve terms. The augmentation of an étale Čech refinement is a quasi-isomorphism of the corresponding total complexes of forms. For the classifying stack $B\mathrm{GL}_{n,B}$, the nerve is*

$$U_q = (\mathrm{GL}_{n,B})^q,$$

and its first nonconstant term has differential module $\Omega^1_{\mathrm{GL}_{n,B}/B}$; its cotangent at the unit contains $\mathfrak{gl}_n(B)^\vee$. Hence this construction retains the nonzero infinitesimal stabilizers which would be lost by using only the atlas $U_0 = \operatorname{Spec} B$.

Proof.

(1) **Construct the map induced by a groupoid morphism.** A morphism of groupoids induces a map of whole nerves, so it induces every coface and degeneracy map and hence a map of the total complexes.

(2) **Apply differential base change to an étale refinement.** Under an étale affine refinement $A \to A'$, the vanishing of $\Omega^1_{A'/A}$ and the flat base-change formula give $\Omega^p_{A'/B} \cong A' \otimes_A \Omega^p_{A/B}$.

(3) **Prove the refinement quasi-isomorphism.** Faithfully flat Čech descent is exact on each displayed coefficient module; the spectral sequence of the resulting first-quadrant double complex proves the claimed quasi-isomorphism.

(4) **Identify the stabilizer tangent retained by the nerve.** For the last assertion the groupoid nerve has $U_0 = \operatorname{Spec} B$, $U_1 = \mathrm{GL}_{n,B}$ and the indicated higher fibre products. Differentiating the matrices $1 + \varepsilon M$ of Proposition 10.14 identifies the identity tangent with $M_n(B)$ and its dual with the identity cotangent module. The full simplicial diagram therefore records this information although $\Omega^1_{B/B} = 0$ at the zero-simplicial atlas.

□

For ordinary smooth Artin stacks, presentation independence follows from Behrend (2005). The internal nerve construction above gives functoriality under displayed groupoid maps, invariance under the stated étale refinements, and the completed-polynomial comparison.

10.9 Effective observations of differential coefficients

The observation structures of Theorem 9.16 and Construction 9.17 apply to the differential coefficient objects. Their finite formulas give the following tracking result.

Proposition 10.18 (Positive tests and completed differential data)**.** *The finite coefficient operations defining P^r, normalized cochains, d, truncation, and formal Taylor transport commute with substitution of all computational parameters. Their realizations on supplied effective coefficient names are uniformly tracked when the underlying exact arithmetic names are supplied. Their semantic equalities, closed-form predicates, and cohomology classes remain predicates and quotients in $\mathcal{E}$; their classifiers take values in Ω.*

Proof.

(1) **Implement the finite differential operations.** The finite operations are compositions of the exact scalar constructors, finite lists, sums, products, and finite matrix identities implemented in Section E. The tracked structural maps are closed under these finite compositions and substitute uniformly in the ambient parameters. The cochain differentials and truncated Taylor maps are given by the same finite expressions at each order, proving the first two assertions.

(2) **Identify the logical status of coefficient equations.** Equality and closedness are kernel and equalizer conditions on coefficient maps; their classifiers are generally Ω-valued.

(3) **Verify finite coefficient dependence of the completed maps.** For the normalized coefficient tower, use the canonical discrete observed presentations of Construction 9.17. Truncation, δ, and ρ have finite coefficient dependence; the maps J, H of (10.9) have the same property because the perturbation raises difference degree. Each fixed output coefficient therefore uses finitely many of its geometric-series terms. Formal displacement differentiation uses one additional input degree. The order bounds and rational finite matrices are uniform in the indices.

(4) **Obtain effective-point and name comparisons.** Applying Proposition 9.18 proves preservation of the generated observations and the corresponding effective-point and finite-output name comparisons for the completed maps.

□

We now have formal functions, their comparison with differential forms, and transport of algebraic coefficients along the completed relation. The next section uses that same relation to describe jets: sections over infinitesimal neighbourhoods and the operators determined by their finite coefficients.

11 Crystalline jets and formal differential operators

A resource jet collects the finite response coefficients of a valuation or a section of a resource bundle. Theorem 6.1 identifies these coefficients with Taylor-valued execution measurements. Principal-parts correspondences now specify their geometric re-centring and composition. Their full compatible system constructs the jet comonad, and flat resource transport gives its crystalline coalgebras. The family of Example 9.12 will realize all these comparisons explicitly in Example 11.9.

The shape-based jet construction follows the synthetic perspective of Khavkine and Schreiber (2027). Here the primary endpoint model supplies represented algebraic function

objects and exact resource coefficients. Finite principal parts identify fibrewise-linear jet operators with algebraic differential operators. Effective descent identifies the full coalgebra category on the stated cells, while the connection–stratification comparison supplies its finite-projective algebraic sector.

11.1 Finite principal-parts correspondences

For a finite-projective coordinate cell E, write $A_E = \mathrm{Sym}_B(E^\vee)$ and $X_E = X_{A_E} = \iota_! F_{A_E}$ in the primary infinitesimal-arrow category. Write $\mathcal{D}_E^{\le r}$ for the étale sheaf (9.5); its restriction to based Weil probes is the previously constructed finite diagonal of Section 9.2. The two projections

$$s_r, t_r : \mathcal{D}_E^{\le r} \rightrightarrows X_E$$

form the order-r formal correspondence. At $r = 0$ both projections are the identity; for $r > 0$ they retain nilpotent differences through order r.

The correspondence acts on sections by an adjoint pair. Dependent sum carries data along it, and dependent product collects the corresponding sections:

$$\mathbb{T}^r_{X_E} = (t_r)_! s_r^* \quad \dashv \quad \mathbb{J}^r_{X_E} = (s_r)_* t_r^*. \tag{11.1}$$

Here $(t_r)_!$ is dependent sum and $(s_r)_* = \Pi_{s_r}$ is dependent product. The adjunction follows by composing $(t_r)_! \dashv t_r^*$ and $s_r^* \dashv (s_r)_*$. The common underlying correspondence is

$$X_E \xleftarrow{\ s_r\ } \mathcal{D}_E^{\le r} \xrightarrow{\ t_r\ } X_E.$$

Composing two correspondences pulls back over their common endpoint. The next calculation maps that composite to the order-$(r+s)$ neighbourhood; taking mates will give the jet comparison in the reverse direction.

Theorem 11.1 (Addition of formal order)**.** *Composition of adjacent formal displacements defines natural maps*

$$\mathcal{D}_E^{\le r} \times_{X_E} \mathcal{D}_E^{\le s} \longrightarrow \mathcal{D}_E^{\le r+s},$$

compatible with order truncation, unit, associativity, and interchange of the two vertices. Applying dependent sums gives $\mathbb{T}^s_{X_E}\mathbb{T}^r_{X_E} \to \mathbb{T}^{r+s}_{X_E}$. Their right-adjoint mates give the filtered jet comparison maps

$$\mathbb{J}^{r+s}_{X_E} Y \longrightarrow \mathbb{J}^r_{X_E}\mathbb{J}^s_{X_E} Y, \qquad \mathbb{J}^r_{X_E} Y = \Pi_{s_r}(t_r^* Y). \tag{11.2}$$

At order zero $\mathbb{J}^0_{X_E} \cong \mathrm{Id}$. The displayed comparisons obey the filtered comonad associativity and counit identities.

Proof.

(1) **Bound the order of a composite displacement.** At any nilpotent arrow, two successive differences u, v have nilpotence bounds $u^{r+1} = 0$ and $v^{s+1} = 0$ in the sense that all their monomials of the indicated total degrees vanish. Every monomial of total degree $r + s + 1$ in $u + v$ contains at least $r + 1$ factors from u or at least $s + 1$ factors from v. Thus $u + v$ has order at most $r + s$.

(2) **Verify units, associativity, and vertex interchange.** This defines the displayed map, naturally in the stage. The unit is displacement zero, associativity is $(u+v)+w = u+(v+w)$, and vertex interchange sends u to $-u$. These polynomial equations commute with all truncation maps.

(3) **Compose the finite correspondences.** Put $Q_{r,s} = \mathcal{D}_E^{\leq r} \times_{t_r, X_E, s_s} \mathcal{D}_E^{\leq s}$. The cartesian square

$$\begin{array}{ccc} Q_{r,s} & \xrightarrow{\pi_2} & \mathcal{D}_E^{\leq s} \\ {\scriptstyle \pi_1}\downarrow & & \downarrow{\scriptstyle s_s} \\ \mathcal{D}_E^{\leq r} & \xrightarrow[t_r]{} & X_E \end{array}$$

gives dependent-sum Beck–Chevalley. Consequently the composite $(t_s)_! s_s^* (t_r)_! s_r^*$ is the action of the composite correspondence. The map $Q_{r,s} \to \mathcal{D}_E^{\leq r+s}$ therefore induces $\mathbb{T}^s_{X_E} \mathbb{T}^r_{X_E} \to \mathbb{T}^{r+s}_{X_E}$.

(4) **Construct the jet comparison and its coherence.** Taking right-adjoint mates under (11.1) gives (11.2). Mates preserve the equations between these composites, proving the filtered associativity and unit identities.

□

Write $\delta_{r,s} : \mathbb{J}^{r+s} \to \mathbb{J}^r \mathbb{J}^s$. The mate of associativity is the commutative square

$$\begin{array}{ccc} \mathbb{J}^{r+s+t} & \xrightarrow{\delta_{r+s,t}} & \mathbb{J}^{r+s}\mathbb{J}^t \\ {\scriptstyle \delta_{r,s+t}}\downarrow & & \downarrow{\scriptstyle \delta_{r,s}\mathbb{J}^t} \\ \mathbb{J}^r \mathbb{J}^{s+t} & \xrightarrow[\mathbb{J}^r \delta_{s,t}]{} & \mathbb{J}^r \mathbb{J}^s \mathbb{J}^t. \end{array} \tag{11.3}$$

Order-zero factors give the counit identities. Thus the filtered comonad equations are the mates of the associativity and unit equations for formal correspondences. The addition of orders records the bound on a composite displacement.

The corresponding algebra maps of principal parts are the duals of the finite correspondence compositions. They retain the two A_E-actions; replacing their target by an unaugmented symmetric power would discard the source–target information needed for substitution.

11.2 Formal disks and jets in the primary geometry

For $X \in \mathscr{H}_{B,\mathrm{et}}$ let $\eta_X : X \to \mathfrak{I}_{\mathrm{inf}} X$ be the unit of the middle endpoint adjunction, and put

$$\mathcal{D}_X = X \times_{\mathfrak{I}_{\mathrm{inf}} X} X, \qquad s, t : \mathcal{D}_X \rightrightarrows X.$$

Because $\mathscr{H}_{B,\mathrm{et}}$ is an internal sheaf topos, its slices admit dependent sums and products $(\eta_X)_! \dashv \eta_X^* \dashv (\eta_X)_*$. Define the formal-disk monad and jet comonad by

$$\mathbb{T}^\infty_X = \eta_X^* (\eta_X)_!, \qquad \mathbb{J}^\infty_X = \eta_X^* (\eta_X)_*.$$

Theorem 11.2 (Full jet and precise coalgebra descent)**.** *The displayed functors are a monad and a comonad on $\mathscr{H}_{B,\mathrm{et}}/X$, with*

$$\mathbb{T}^\infty_X \dashv \mathbb{J}^\infty_X, \qquad \mathbb{J}^\infty_X Y \cong \Pi_s (t^* Y).$$

For every X there is a canonical comparison functor

$$\mathscr{H}_{B,\mathrm{et}}/\mathfrak{I}_{\mathrm{inf}} X \longrightarrow (\mathscr{H}_{B,\mathrm{et}}/X)^{\mathbb{J}^\infty_X}.$$

It is an equivalence if η_X is an effective epimorphism, in particular for the polynomial finite-projective cells $X = X_A$ of Proposition 9.9.

Proof.

(1) **Construct the monad, comonad, and their adjunction.** The adjunctions construct the monad and comonad, with multiplication and comultiplication given by the unit and counit mates. The triangle identities prove both unit and counit equations and the associativity/coassociativity equations. Successive slice adjunctions give

$$\mathrm{Hom}_X(\eta_X^*(\eta_X)_!E, F) \cong \mathrm{Hom}_{\Im_{\mathrm{inf}}X}((\eta_X)_!E, (\eta_X)_*F) \cong \mathrm{Hom}_X(E, \eta_X^*(\eta_X)_*F).$$

(2) **Identify the formal-pair correspondence and comultiplication.** Dependent-product Beck–Chevalley for the defining pullback of $\mathcal{D}_X$ gives the correspondence formula $\Pi_s(t^*Y)$. Comparing the two groupoid action diagrams determines the jet comultiplication by re-centering along composable formal pairs.

(3) **Construct the descent comparison.** The comparison functor sends $H \to \Im_{\mathrm{inf}}X$ to its pullback η_X^*H with the canonical unit-induced coaction.

(4) **Prove comonadicity under effective descent.** If η_X is an effective epimorphism, it is an effective descent morphism in the sheaf topos: its entire Čech kernel-pair diagram realizes the target, and objects with compatible descent action are precisely its pullbacks. The same Čech action is the coalgebra structure of $\eta_X^*(\eta_X)_*$, proving comonadicity in this case. The effective-epimorphism calculation on polynomial cells is Proposition 9.9.

□

The jet counit evaluates a formal-disk section at its centre. Comultiplication re-centres it along composable nilpotent pairs. A section $\sigma : X \to Y$ prolongs by $X \cong \mathbb{J}^\infty_X X \to \mathbb{J}^\infty_X Y$, and coKleisli composition uses that comultiplication. This is the crystalline differential-operator calculus of the primary endpoint geometry. Restricting its relations to $\operatorname{Spec} B \hookrightarrow \operatorname{Spec} C$ recovers the based-Weil calculation of Theorem 6.11.

11.3 Primary jet coefficients and algebraic principal parts

To calculate the jets of an algebraic bundle, we realize its coefficient module as an affine bundle sheaf. For a displayed finite-projective A-module N, the fibre at $x : A \to S$ is $S \otimes_{A,x} N$. Equivalently, for a displayed idempotent $e_N \in M_k(A)$, put

$$Y_N^0(\operatorname{Spec} S) = \{(x, z) : x : A \to S,\ z \in S^k,\ e_N(x)z = z\}, \qquad Y_N = \iota_! Y_N^0 \longrightarrow X_A.$$

This bundle is represented by $\operatorname{Spec} \mathrm{Sym}_A(N^\vee)$ over $\operatorname{Spec} A$; in particular it satisfies étale descent. The following calculation takes place in the primary slice throughout.

Theorem 11.3 (Primary finite jets are algebraic principal parts)**.** *Let $j = (\operatorname{Spec} R \hookrightarrow \operatorname{Spec} S)$ and let $x : A \to S$ be a generalized point of X_A. Define*

$$Q_{j,x,r} = S \otimes_{A,x,s} P^r_{A/B}, \qquad k_{j,x,r} = (\operatorname{Spec} R \hookrightarrow \operatorname{Spec} Q_{j,x,r}),$$

where $Q_{j,x,r} \to R$ is reduction after the augmentation to S. Then $k_{j,x,r}$ is a primary test and represents the pullback $yj \times_{X_A, s_r} \mathcal{D}_A^{\leq r}$. For the bundle Y_N,

$$(\mathbb{J}^r_{X_A} Y_N)(j, x) \cong Q_{j,x,r} \otimes_{A,t} N. \tag{11.4}$$

Here the left side denotes the fibre over the generalized point x. Consequently, with source A-action,

$$P^r_{A/B}(N) := P^r_{A/B} \otimes_{A,t} N, \qquad \mathbb{J}^r_{X_A} Y_N \cong Y_{P^r_{A/B}(N)}. \tag{11.5}$$

For $A = B[x_1, \dots, x_n]$ and $N = A \otimes_B M$ with M displayed finite-projective over B, the fibre formula is

$$(\mathbb{J}^r_{X_A} Y_N)(j, x) \cong \prod_{|\alpha| \le r} (S \otimes_B M).$$

All comparisons preserve truncation and the full Taylor coordinate-substitution law of Theorem 9.2.

Proof.

(1) **Construct the representing nilpotent test.** The algebra $Q_{j,x,r}$ is finitely presented over B by its finite tensor-product presentation. Let L be its displacement ideal and $J = \ker(S \to R)$. Then $L^{r+1} = 0$ and $Q/L = S$. If $J^a = 0$, the kernel $K = \ker(Q \to R)$ satisfies $K^a \subseteq L$, so $K^{a(r+1)} = 0$. The quotient to R is surjective. Thus $k_{j,x,r}$ is an object of I_B.

(2) **Identify the pullback of finite principal parts.** The endpoint pullback of j and $k_{1,r} = (\operatorname{Spec} A \hookrightarrow \operatorname{Spec} P^r)$ over $\iota(\operatorname{Spec} A)$ has larger algebra Q and smaller algebra $R \otimes_A A = R$. It is exactly $k_{j,x,r}$. The representing property of Lemma 9.5 therefore proves the asserted sheaf pullback.

(3) **Compute the primary jet fibre.** Dependent-product adjunction and Yoneda give

$$\underline{\mathrm{Hom}}_{X_A}(yj, \Pi_{s_r} t_r^* Y_N) \cong \underline{\mathrm{Hom}}_{X_A}(yk_{j,x,r}, Y_N) \cong Q \otimes_{A,t} N,$$

where the map from $yk_{j,x,r}$ to X_A on the right is its target vertex. This proves (11.4).

(4) **Identify the represented principal-parts bundle.** Tensor associativity identifies this fibre with $S \otimes_{A,x} P^r_{A/B}(N)$. The module $P^r_{A/B}(N)$ is a retract of a finite power of P^r, which is finite-projective as a source A-module; hence its bundle is defined above and the fibrewise comparison proves (11.5).

(5) **Recover monomial coefficients and coherence.** For free coordinates, the truncated monomials in $u = y - x$ are a finite basis of Q over S, giving the product formula. Every map is induced by a representing pullback or algebra substitution; these universal properties prove all coherence assertions.

□

The coefficient algebra of a jet at $x : A \to S$ is the pushout

$$\begin{array}{ccc} A & \xrightarrow{s} & P^r_{A/B} \\ {\scriptstyle x}\downarrow & & \downarrow \\ S & \longrightarrow & Q_{j,x,r}. \end{array}$$

The source action fixes the base point; the target action $A \xrightarrow{t} P^r_{A/B} \to Q_{j,x,r}$ pulls back the bundle. Its fibre is therefore $Q_{j,x,r} \otimes_{A,t} N$. Keeping the two actions visible explains the tensor convention and the Taylor evaluation at the target.

At the universal target point of $k_{j,x,r}$, a polynomial section f is evaluated by $f(x + u)$. The finite binomial identity therefore gives its primary Taylor coefficients directly:

$$f(x + u) = \sum_{|\alpha| \le r} \frac{u^\alpha}{\alpha!} \partial^\alpha f(x) \pmod{(u)^{r+1}}. \tag{11.6}$$

The symmetric-power expression descends along displayed projective retracts. Thus primary polynomial Taylor transport is proved on the representing test, independently of any preservation of dependent products by restriction to based-Weil probes.

Theorem 11.4 (Algebraic differential operators and primary jets)**.** *For A-modules N, M, let $\mathrm{Diff}^{\leq r}_{A/B}(N, M)$ be the B-linear maps whose $(r+1)$-fold iterated commutators with multiplication by elements of A vanish. There is a natural isomorphism*

$$\mathrm{Diff}^{\leq r}_{A/B}(N, M) \cong \mathrm{Hom}_A(P^r_{A/B} \otimes_{A,t} N, M), \tag{11.7}$$

where the source A-action is used on the tensor product. For displayed finite-projective N, M, these are exactly the fibrewise-linear primary geometric maps $\mathbb{J}^r_{X_A} Y_N \to Y_M$. Their action on sections is composition with jet prolongation, and their composition agrees with the order-additive principal-parts comparisons.

Proof.

(1) **Translate a linear operator into a tensor map.** For a B-linear $D : N \to M$, define the source-A-linear map $\widetilde{D} : A \otimes_B N \to M$ by $\widetilde{D}(a \otimes n) = aD(n)$.

(2) **Identify iterated commutators with diagonal powers.** The diagonal generator $\Delta b = 1 \otimes b - b \otimes 1$ acts on $A \otimes_B N$ through its natural $A \otimes_B A$-action, and

$$\widetilde{D}\big(\Delta b\,(a \otimes n)\big) = a\big(D(bn) - bD(n)\big).$$

Induction identifies a product of $r+1$ such generators with the corresponding iterated commutator. They generate I^{r+1}, so the commutator condition is precisely that $\widetilde{D}$ annihilate $I^{r+1}(A \otimes_B N)$.

(3) **Prove the principal-parts universal property.** The quotient is $P^r_{A/B} \otimes_{A,t} N$ by the right-exact balanced tensor presentation. Conversely any map from that quotient gives $D(n) = \widetilde{D}(1 \otimes n)$ and satisfies the same condition. These inverse constructions prove (11.7), including its universal order-r map $n \mapsto 1 \otimes n$. This is the internal principal-parts universal property (Stacks Project Authors 2026a, Tag 09CH).

(4) **Identify fibrewise-linear primary bundle maps.** For finite-projective coefficients, affine-bundle Yoneda identifies fibrewise-linear bundle maps with module maps: on the symmetric coordinate algebras, fibrewise linearity selects the degree-one maps, and finite duality reverses them to the asserted module maps. Full faithfulness of $\iota_!$ and Theorem 11.3 now give the primary comparison.

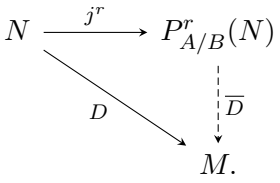
(5) **Identify the universal prolongation.** On a section n, its prolongation evaluates n at the universal target vertex, hence is $1 \otimes n$.

(6) **Compare operator composition and its order bound.** For two operators, pulling back to successive universal vertices and then composing evaluates the same two maps on those vertices. The algebra substitution representing composition of the finite correspondences is the order-additive map of Theorem 11.1. Its dependent-product mate thus gives exactly the same composite operator and the bound $r + s$, with all associativity equations following from the three-vertex substitution.

□

The operator classification is the following factorization: every order-r operator is

obtained uniquely by applying an A-linear map after the universal B-linear jet map.



Here $j^r(n) = 1 \otimes n$, and $\overline{D}$ is the unique A-linear arrow with this composite, using the source A-action on principal parts. The triangle is interpreted in B-modules, with the stronger linearity of its dashed leg specified.

11.4 Finite formal coefficients and exact Taylor formulas

Put

$$W_{n,r} = \mathbb{Q}[d_1, \dots, d_n]/(d_1, \dots, d_n)^{r+1}, \qquad D_{n,r} = D_{W_{n,r}}, \qquad D_{n,\infty} = \bigcup_r D_{n,r}.$$

This is the based-Weil specialization of the primary construction. Its polynomial coefficient formulas agree with Theorem 11.3; the proposition below also treats arbitrary B-modules and arbitrary natural maps in the based-Weil category, on that category's own domain. At a based stage C, the last union consists of the nilpotent n-tuples. A tuple with zero augmentation lies in the nilpotent augmentation ideal and hence in some $D_{n,r}(C)$. Conversely a nilpotent tuple has zero augmentation because B is reduced. Therefore

$$\mathcal{D}_{\mathcal{R}^n_B} \cong \mathcal{R}^n_B \times D_{n,\infty}, \qquad (x, y) \longmapsto (x, y - x). \tag{11.8}$$

For the finite relation let $s_r(x, d) = x$ and $t_r(x, d) = x + d$, and define $\mathbb{J}^r_{\mathcal{R}^n_B} E = \Pi_{s_r}(t_r^* E)$.

Proposition 11.5 (Finite jet coefficients)**.** *For every B-module M, the original based-Weil coefficient functor, considered in its own slice category, has the explicit finite coefficients*

$$\mathbb{J}^r_{\mathcal{R}^n_B}(\mathcal{R}^n_B \times j_B M) \cong \mathcal{R}^n_B \times \prod_{\substack{\alpha \in \mathbb{N}^n \\ |\alpha| \le r}} j_B M.$$

For a geometric map $F : j_B M \to j_B N$ and directions $v_1, \dots, v_n$, its finite Taylor formula is

$$F\left(x + \sum_i d_i v_i\right) = \sum_{|\alpha| \le r} \frac{d^\alpha}{\alpha!} D^{|\alpha|} F(x)(\underbrace{v_1, \dots, v_1}_{\alpha_1}, \dots, \underbrace{v_n, \dots, v_n}_{\alpha_n}) \quad (d \in D_{n,r}).$$

The term of degree zero is $F(x)$.

Proof.

(1) **Identify finite jets with monomial coefficients.** The formal-affine trivialization reduces the first assertion to $(j_B M)^{D_{n,r}}$ and its monomial coefficient basis in (6.12).

(2) **Compute coefficients using square-zero substitutions.** For one direction, write its uniquely represented polynomial as $\sum_{j=0}^r a_j d^j$. Substitute $d = \varepsilon_1 + \cdots + \varepsilon_r$ in the Weil algebra with $\varepsilon_i^2 = 0$. Iterating the first-order expansion identifies the coefficient of $\varepsilon_1 \cdots \varepsilon_j$ with $D^j F(x)(v, \dots, v)$. In the polynomial expression this coefficient is $j! a_j$.

(3) **Compare truncations and divide by factorials.** The substitution into j square-zero variables is defined on the order-j truncation, and truncation identifies its coefficients with the first j coefficients of the order-r expression. Rational factorials are invertible.

(4) **Recover symmetric mixed derivatives.** For several directions use disjoint groups of square-zero variables. The same coefficient comparison gives $\alpha!$ and the displayed mixed derivative. Symmetry proved above makes the formula independent of the order within each group.

□

The directed union in (11.8) gives

$$\mathbb{J}^{\infty}_{\mathcal{R}^n_B} E \cong \varprojlim_r \mathbb{J}^r_{\mathcal{R}^n_B} E.$$

Indeed maps out of its union are compatible families of maps out of the finite disks; the dependent-product adjunction gives the assertion in the slice. All countable coefficient objects here are internal dependent products indexed by the NNO.

11.5 Exact, signed and completed coefficient comparison

For a positive exact algebra A, the truncated polynomial algebra $A[d]/d^{r+1}$ has underlying module A^{r+1} and finite convolution multiplication. Hence

$$\mathrm{Gr}(A[d]/d^{r+1}) \cong (\mathrm{Gr}\, A)[d]/d^{r+1}.$$

Finite biproducts, multiplication, and the derivative comparisons give the same finite coefficient compatibility with E_S. Several variables use their finite monomial lists. The comparisons commute with truncation and every substitution represented by these polynomial presentations. Finite Weil coefficient modules also commute with gauge and filtered coefficient change by tensor associativity.

At every finite order these comparisons preserve the measured coefficients of Theorem 6.1. At infinite order the distinction is (9.13): the measured family of Example 9.12 has a polynomial budget at each finite order but no single polynomial budget for all its coefficients.

11.6 Formal-order completion and its de Rham interface

Proposition 11.6 (Algebraic crystalline bundles give primary jet coalgebras)**.** *Let N be a displayed finite-projective module over $A = \mathrm{Sym}_B(E^\vee)$, represented by $e_N \in M_k(A)$. Use the affine bundle $Y^0_N \to F_A$ and its primary realization $Y_N = \iota_! Y^0_N \to X_A$ constructed in Section 11.3. Every formal stratification on N gives naturally a $\mathbb{J}^\infty_{X_A}$-coalgebra structure*

$$\gamma_N : Y_N \longrightarrow \mathbb{J}^\infty_{X_A} Y_N$$

in the primary étale infinitesimal-arrow geometry. Morphisms of stratifications induce coalgebra morphisms. Its restriction to based B-Weil arrows is exactly the earlier coefficient-functor bundle and Taylor transport.

Proof.

(1) **Construct the coefficient bundle and its sheaf.** The displayed idempotent equation makes Y^0_N functorial in S; its fibre is $S \otimes_{A,\xi} N$. Since this is represented by an affine bundle over $\mathrm{Spec}\, A$, it satisfies étale descent and defines the claimed sheaf.

(2) **Pull back finite stratification transport.** For a nilpotent arrow $j = (R \leftarrow S)$ and a pair of S-points x, y agreeing in R, Theorem 9.4 gives a finite r and an algebra map $P^r_{A/B} \to S$. Pull back the P^r-linear stratification to obtain the fibre isomorphism

$$\varepsilon_{x,y} : (Y_N)_x \xrightarrow{\cong} (Y_N)_y.$$

(3) **Extend transport to the full relation.** Compatibility with principal-parts truncation shows that this transport is independent of the exhibited bound; it defines a map $s^*Y_N \to t^*Y_N$ on the entire infinitesimal relation.

(4) **Construct the jet coaction and verify its laws.** Transpose through $s^* \dashv \Pi_s$ to obtain γ_N. The identity pair produces the coalgebra counit equation; the triple-diagonal stratification cocycle produces $\delta\gamma_N = \mathbb{J}(\gamma_N)\gamma_N$.

(5) **Verify functoriality and based restriction.** For a morphism of stratifications f, the equation $f_y\varepsilon_{x,y} = \varepsilon'_{x,y}f_x$ proves the induced jet-coalgebra square. At a based arrow the matrix fibres and finite transport evaluate on $S = C$ exactly as in the based-Weil construction.

□

Lemma 11.7 (Crystalline restriction to a principal open)**.** *Let $A = \mathrm{Sym}_B(E^\vee)$ be a displayed finite-projective polynomial coordinate algebra and $h \in A$. For its principal open A_h, the finite principal-parts coefficients, formal stratifications, and crystalline jet coactions have the same comparisons after restriction to X_{A_h}. In particular the shape unit of X_{A_h} is an effective epimorphism, and a finite-projective A_h-module with integrable connection gives a primary jet coalgebra. Finite and full jet functors restrict along $X_{A_h} \hookrightarrow X_A$ by the corresponding slice Beck–Chevalley maps.*

Proof.

(1) **Lift invertibility through a nilpotent quotient.** At a nilpotent arrow $S \twoheadrightarrow R_0$, an element of S is a unit exactly when its image in R_0 is a unit. To construct its inverse, lift an inverse modulo the nilpotent kernel; the product differs from 1 by a nilpotent element, whose finite geometric series corrects the lift.

(2) **Identify the restricted infinitesimal relation.** Thus, for two infinitesimally related A-points, h is invertible at one endpoint exactly when it is invertible at the other. The full infinitesimal relation over X_{A_h} is therefore both the source and the target pullback of the relation over X_A. The same statement holds at each finite diagonal order, with the finite inverse formula of Theorem 10.8.

(3) **Construct the jet restriction comparisons.** These cartesian squares give the slice dependent-product Beck–Chevalley maps for the finite and full jet correspondences.

(4) **Identify principal-parts bundles and their coactions.** The finite coordinate algebras are $A_h \otimes_A P^r_{A/B}$ by Theorem 10.8; their represented bundle calculation is the finite-dual-basis calculation of Theorem 11.3. For a finite-projective A_h-module its displayed idempotent has entries in A_h and defines the same affine-bundle presentation on this open. A stratification therefore gives its full-relation transport and its jet coaction by the transposition of Proposition 11.6.

(5) **Prove effective lifting and restriction coherence.** Finally, an A_h-point of R_0 first gives an A-point. The polynomial lifting construction of Proposition 9.9 lifts it locally to S; the lifted value of h is a unit by the preceding inverse construction. Hence the lift factors through A_h. This proves that its shape unit is an effective epimorphism and establishes the stated descent comparison by Theorem 11.2. All inverse and transport formulas are compatible with truncation and parameter substitution, proving the claimed restriction laws.

□

Theorem 11.8 (Formal-order completion of the primary crystalline jets)**.** *For $X_E = X_{A_E}$ a finite-projective polynomial resource-coordinate sheaf in the primary étale infinitesimal-arrow geometry,*

$$\mathbb{J}^{\infty}_{X_E} Y \cong \varprojlim_r \mathbb{J}^r_{X_E} Y \tag{11.9}$$

naturally in every $Y \to X_E$. The order-additive comparison maps of Theorem 11.1 construct the comultiplication of the jet comonad of Theorem 11.2. Its jet-coalgebra descent equivalence holds here by Proposition 9.9. For finite-projective algebraic coefficients it is the completed primary principal-parts bundle

$$\mathbb{J}^{\infty}_{X_A} Y_N \cong \varprojlim_r Y_{P^r_{A/B}(N)},$$

with finite coefficients calculated by Theorem 11.3. Algebraic stratifications construct its crystalline jet coactions by Proposition 11.6.

Proof.

(1) **Express full jets as a limit of finite jets.** By Theorem 9.4, the full relation is the internally increasing union $\mathcal{D}_{X_E} = \varinjlim_r \mathcal{D}_E^{\leq r}$. For every object $Z \to X_E$, the universal property of the union gives compatible maps on its finite stages:

$$\begin{aligned} \mathrm{Hom}_{X_E}(Z, \mathbb{J}^{\infty}_{X_E} Y) &\cong \mathrm{Hom}_{\mathcal{D}_{X_E}}(s^* Z, t^* Y) \\ &\cong \varprojlim_r \mathrm{Hom}_{\mathcal{D}_E^{\leq r}}(s_r^* Z, t_r^* Y) \\ &\cong \mathrm{Hom}_{X_E}\left(Z, \varprojlim_r \mathbb{J}^r_{X_E} Y\right). \end{aligned}$$

(2) **Identify the natural isomorphism and comultiplication.** Internal Yoneda proves the natural isomorphism. Addition of displacements and ordinary composition of formal pairs are the same maps in the union, so their dependent-product mates identify the limit comultiplication with the comultiplication of Theorem 11.2.

(3) **Recover crystalline descent and based coefficients.** The effective descent and crystalline coaction clauses are the explicit constructions cited in the statement. Restriction to based B-Weil probes recovers the original finite principal-parts maps by the displayed stagewise coefficient formulas.

□

Example 11.9 (The measured family as a crystalline resource line)**.** Continue with $R = B[t^{-1}]$, $q = t^{-1}$, and $A = R[z, (1-qz)^{-1}] = B[z, (t(t-z))^{-1}]$ from Proposition 10.10. The principal-open comparison of Lemma 11.7 realizes the line $N = A$ and its integrable connection $d - \omega$ in the primary resource geometry. Its stratification $\tau(z, z') = (1-qz)/(1-qz')$ induces the coaction $Y_N \to \mathbb{J}^{\infty}_{X_A} Y_N$.

The horizontal section $f = (1-qz)^{-1}$ has finite jet

$$f(z+u) = \sum_{j=0}^{r} \frac{q^j}{(1-qz)^{j+1}} u^j \pmod{u^{r+1}}. \tag{11.10}$$

This follows by multiplying the finite geometric-series inverse of $1 - qu/(1-qz)$ by $f(z)$. At $z = 0$ it is precisely $F_r(u)$, so Theorem 6.1 identifies this geometric jet with the measured coefficients of the program e_r. The complete jet is the compatible family of these

finite expressions. Its fibre at zero has coefficients in $R[[u]]$ and is outside $B[[u]][t^{-1}]$ by Proposition 9.13.

Re-centring from z to z' and then to z'' multiplies the transport factors to $\tau(z, z'')$. Transposition is exactly the coalgebra coassociativity equation. The logarithm of the same factor is the completed cocycle (10.14), whose de Rham image is ω. Thus measured coefficients, completed functions, forms, and crystalline transport are calculated on one resource family.

Corollary 11.10 (Finite principal-parts differential operators)**.** *For a finite-projective polynomial resource cell, an order-r operator on a geometric bundle $Y \to X_E$ with target bundle $Z \to X_E$ is a map $\mathbb{J}^r_{X_E} Y \to Z$. The maps (11.2) compose an order-s operator after an order-r operator into an operator of order at most $r + s$. The corresponding complete operators are maps $\mathbb{J}^\infty_{X_E} Y \to Z$ and compose by the jet comonad. Polynomial resource operators have their previously constructed finite Taylor coefficients under this correspondence.*

Proof.

(1) **Construct composition using order addition.** The first assertion is the definition via the finite formal disk adjunction. For $\alpha : \mathbb{J}^r Y \to Z$ and $\beta : \mathbb{J}^s Z \to W$, compose

$$\mathbb{J}^{r+s} Y \longrightarrow \mathbb{J}^s \mathbb{J}^r Y \xrightarrow{\mathbb{J}^s \alpha} \mathbb{J}^s Z \xrightarrow{\beta} W.$$

The comparison is (11.2) with its indices interchanged if necessary.

(2) **Verify finite and completed composition laws.** The finite associativity theorem proves the operator composition laws; the completed comonad gives the same argument without an order bound.

(3) **Compare polynomial and fibrewise-linear algebraic operators.** Primary polynomial agreement is the representing-test Taylor calculation (11.6) and Theorems 6.20 and 11.3. For fibrewise-linear algebraic operators, the universal-property identification and its composition coherence are Theorem 11.4. Their based-Weil specializations agree with Proposition 11.5.

□

The completed relation now has both a modal and an algebraic description. Where the shape unit is an effective epimorphism, including polynomial finite-projective cells, its jet coalgebras are identified with $\mathscr{H}_{B,\mathrm{et}}/\mathfrak{I}_{\mathrm{inf}} X$. Within the algebraic coefficient sector, Theorem 10.6 identifies finite-projective stratifications with integrable B-connections. These comparisons use the same infinitesimal relation but have different domains; an arbitrary semantic jet coalgebra need not be an algebraic connection.

The measured family now has an operational coefficient calculation, a completed function, a de Rham cocycle, and crystalline transport. The remaining instances relate this geometry to numerical clocks, input sizes, and the logarithmic connections of asymptotic scales.

12 Effective instances and geometric calculations

This section connects the constructions to two kinds of examples. The numerical execution instance supplies measured costs, input-size comparisons, and finite resource certificates. The later geometric calculations exhibit normal resource directions, asymptotic scales, and logarithmic connections. Each example keeps its specified implementation family and coefficient context, so its conclusion can be read directly against the general classification.

We begin with an operational realization of the measured span. The machine configuration and update map determine its histories; the first-hit clock then identifies the numerical cost of a returning execution.

12.1 The numerical execution instance

Construction 12.1 (Numerical configurations and measured return)**.** In the effective numerical fibre, use the finite register-program syntax of Proposition E.5: increment, zero test with decrement, jumps, and halt. Finite instruction lists have primitive-recursive codes; invalid codes denote a permanently looping program. A configuration consists of its program code, instruction pointer, and finite register data, with the Boolean halted flag. Reading an instruction and updating these finite data constructs the total primitive-recursive maps

$$h : \mathsf{Conf} \to 2, \qquad s : \mathsf{Conf} \to \mathsf{Conf}, \qquad \mathrm{init} : \mathbb{N} \times \mathbb{N} \to \mathsf{Conf},$$

and an output map $o : h^{-1}(1) \to \mathbb{N}$. Extend s by the identity on halted configurations. These are the numerical code and configuration objects specified by these finite instruction clauses.

Define an equation morphism with additive natural cost by

$$b : \mathsf{Conf} \to L_{\Sigma,\mathbb{N}}(\mathbb{N} + \mathsf{Conf}), \qquad b(c) = \begin{cases} \eta(0, \mathsf{inl}\, o(c)), & h(c) = 1, \\ \eta(1, \mathsf{inr}\, s(c)), & h(c) = 0. \end{cases}$$

Finite case analysis constructs this map; the discreteness lemma makes the indicated total data maps computational algebra morphisms. Then

$$\mathrm{ev}_{\mathrm{time}} = b^{\dagger_{\mathbb{N}}} \mathrm{init} : \mathbb{N} \times \mathbb{N} \to L_\Sigma(\mathbb{N} \times \mathbb{N})$$

returns the number of updates before the halted state and the output. The cost is zero for an initially halted configuration. Tagging each branch by its current configuration instead constructs

$$\mathrm{ev}_H : \mathbb{N} \times \mathbb{N} \to L_\Sigma(\mathsf{Conf}^* \times \mathbb{N})$$

by Theorems 4.4 and 4.6. Its word includes the initial, intermediate, and final configurations. The valuation that assigns 1 to a nonhalted configuration and 0 to a halted one recovers $\mathrm{ev}_{\mathrm{time}}$.

A fixed total primitive-recursive occupancy function $\mathrm{occ} : \mathsf{Conf} \to \mathbb{N}$, defined from the chosen finite register representation, folds these same recorded configurations by maximum. The product monoid records time and that peak occupancy on one execution. Other finite measurement expressions built from the recorded fields, constants, addition, maximum, and finite folds have total interpretations $\nu : \mathsf{Conf}^* \to \mathbb{N}$ by finite recursion. Here the expression is selected before forming its implementation image.

The first halting stage defines the partial evaluator of Proposition E.5. Its finite run is exactly the corresponding return component of Lemma 1.6, so erasure has the same value and domain. The regular epimorphism $\mathbb{N} \twoheadrightarrow \mathsf{PC} = J_\Sigma \mathsf{FG}$ of that proposition supplies the numerical code presentation. The added time and occupancy values are the folds just defined. The oracle-relative instance uses the same relative code construction and records the declared oracle-invocation events.

Proposition 12.2 (The bounded numerical clock)**.** *The NNO recursion* $\mathrm{run}(e, x, k) = s^k(\mathrm{init}(e, x))$ *constructs the monotone Boolean clock*

$$\alpha(e, x, k) = h(\mathrm{run}(e, x, k)).$$

Its first-hit domain is the return domain of the preceding execution, and its unique first-hit index is the time value of $\mathrm{ev}_{\mathrm{time}}$*. Write its domain and index as*

$$D_\alpha = \{(e,x) : \exists k : \mathbb{N},\ \alpha(e,x,k) = 1\} \hookrightarrow \mathbb{N} \times \mathbb{N}, \qquad \tau_\alpha : D_\alpha \to \mathbb{N}.$$

Proof.

(1) **Establish monotonicity and define the first-hit test.** The halted state is absorbing, so the clock is monotone. Define the Boolean predicate

$$\mathrm{First}_\alpha(e,x,k) = \alpha(e,x,k) \wedge \bigwedge_{j<k} \neg\alpha(e,x,j).$$

(2) **Construct the unique first hit.** Finite induction through a successful stage produces a least one, and linearity of the NNO order proves uniqueness.

(3) **Identify the classified return domain and its cost.** The corresponding subobject therefore projects isomorphically onto $\{(e,x) : \exists k\, \alpha(e,x,k) = 1\}$, whose classifier lies in Σ. On its kth summand there are exactly k update branches followed by the zero-cost return branch. This is R_k of Theorem 4.4, with the same output and cost k.

□

12.2 Numerical cost adequacy

Theorem 12.3 (Numerical cost adequacy)**.** *In the numerical instance, the measured evaluator erases to the partial evaluator* $\varphi(e,x)$ *of Proposition E.5, with exactly the same domain. Its partial time map* $T(e,x)$ *has that domain and satisfies*

$$T(e,x) = k \quad \Longleftrightarrow \quad \mathrm{First}_\alpha(e,x,k) = 1. \tag{12.1}$$

The graph predicate on the right is primitive recursive, uniformly in e, x, k*. The bounded-return predicate*

$$\mathsf{Within}(e,x,b) = \alpha(e,x,b)$$

is primitive recursive and is true exactly when the execution returns with time at most b*. In the oracle-effective instance these same claims hold relative to its fixed oracle.*

Proof.

(1) **Identify erasure and the return domain.** The first-hit definition of φ and Lemma 1.6 identify erasure and its domain; the weighted-return comparison retains the same components.

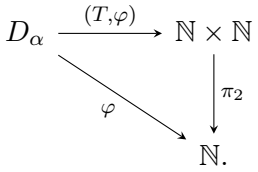
(2) **Construct the bounded first-hit test.** The primitive-recursive configuration step gives primitive-recursive bounded iteration run by NNO recursion. Thus α and its finite first-hit test are primitive recursive.

(3) **Identify the numerical output and measured cost.** On the exact-return component R_k, the retained history contains k update events of weight 1 and a final event of weight 0; its fold is k and its returned output is $\varphi(e,x)$. Conversely a first hit at k belongs to that component, by Proposition 12.2. This proves (12.1).

(4) **Prove the bounded-time equivalence and oracle version.** A true $\alpha(e,x,b)$ has a unique first hit at some $k \le b$ by bounded finite search; a first hit at $k \le b$ makes $\alpha(e,x,b)$ true by monotonicity. The oracle proof uses the same bounded simulation relative to the fixed oracle-query instruction of Proposition E.5.

□

On the returning domain, the measured and unmeasured evaluators are related by the commuting triangle



Both partial maps use the same domain mono $D_\alpha \hookrightarrow \mathbb{N}\times\mathbb{N}$ of program–input pairs. Thus the measured span is obtained from an exact operational domain-and-value comparison.

Corollary 12.4 (The numerical time is a Blum measure)**.** *For the register-program enumeration of Proposition E.5, the numerical time $T_e(x)$ is a Blum complexity measure: its domain is the domain of φ_e, and the relation $T_e(x) = k$ is decidable uniformly in e, x, k. The same statement holds relative to a fixed oracle.*

Proof.

(1) **Verify the two defining measure properties and their relative form.** Domain equality and the primitive-recursive first-hit test are Theorem 12.3. They are the two defining properties of an abstract complexity measure in (Blum 1967). Oracle-relative bounded simulation gives the relative test.

□

This corollary concerns the constructed numerical time; an arbitrary valuation of histories need not have decidable exact-cost fibres. For example, assigning zero to every event leaves exact cost zero on precisely the halting domain.

The classical external interpretation of the numerical resource space also gives a geometric reading of Blum speedup. Blum's theorem provides computable functions for which every implementation admits an asymptotic improvement, with a prescribed recursive speedup. Choosing an unbounded multiplicative improvement rules out an O-optimal implementation. In the ordinary external growth locale, Theorem 5.2 then identifies its feasibility open as non-supercompact: an optimal implementation would generate that open and contradict speedup. The numerical measurement throughout this instance counts the updates of its displayed register machine. Other valuations use their own history folds, as in Theorem 4.6.

Measured execution gives a profile on actual inputs. Complexity bounds expressed in an input size must be pulled back along the chosen size map. Keeping that map explicit distinguishes, for example, polynomial bounds in numerical value from polynomial bounds in binary length.

12.3 Input sizes and effective bound profiles

Proposition 12.5 (Size-indexed bounds)**.** *An implemented total size map $\ell : X \to \mathbb{N}$ and a displayed total numerical bound expression $b_\beta : \mathbb{N} \to \mathbb{N}$ construct the raw profile $g_\beta = b_\beta \ell$ and its exact germ on the supplied proper-filter context of actual inputs. They therefore construct the principal admitted budget $\downarrow [g_\beta]$. For numerical inputs,*

$$\forall x : \mathbb{N}\ \alpha(e, x, g_\beta(x)) = 1 \quad \Longleftrightarrow \quad e \in P \ \text{ and } \ \forall x : \mathbb{N}\ T_e(x) \le g_\beta(x). \tag{12.2}$$

Every such pointwise bound gives $\Phi(e) \in Bg_\beta$ and hence the corresponding upper-feasibility judgment for $p(e)$.

Proof.

(1) **Construct the size-indexed profile and admitted budget.** Composition of the retained size and bound descriptions computes the raw profile; rational inclusion and the germ quotient give its exact class. The principal admitted-ideal constructor and its name implementation from Proposition E.5 give the budget. Its pairing is $u \mapsto u([g_\beta])$, so admission is explicit.

(2) **Identify the bounded clock with the time inequality.** For each x, Theorem 12.3 identifies the bounded clock with return and the stated time inequality.

(3) **Deduce incidence and the represented-input version.** Quantifying over x proves (12.2); a global bound is eventual with constant 1, giving the incidence. For represented inputs the same profile construction is on $X = C_X$, with no quotienting of their costs.

□

Finite-word length and binary bit length are constructed by finite recursion on the chosen representations. Thus a polynomial in numerical value uses $\ell(n) = n$, whereas a polynomial in binary input length uses $\ell(n) = \mathrm{bitlength}(n)$, with the encoding's declared convention at zero. These are different profiles. On higher-type or redundantly named inputs the pointwise formulation does not require finite size fibres or an enumeration of all names of a given size. An extensional size map is pulled back along q_X when available; an encoding size may instead depend on the name. Computing a bound for purposes of checking it also has its own recorded execution cost.

12.4 A numerical program and two size conventions

The following program makes the distinction concrete. Work in the register-machine environment of Construction 12.1 and Theorem 12.3. Place the natural input n in an input register and 0 in an output register. The instruction scheme uses the existing zero-test with decrement, increment, jump, and halt instructions:

loop :	if the input register is zero, go to halt;
	otherwise decrement the input register;
	increment the output register twice; jump to loop;
halt :	return the output register.

Its finite code $e_{2\cdot}$ has denotation $v(n) = 2n$. For each input n there are exactly n nonzero loop passes followed by the final zero branch. Counting the fixed finite instruction block gives fixed integers $a > 0$ and $b \geq 0$, independent of n, with

$$T_{e_{2\cdot}}(0) = b, \qquad T_{e_{2\cdot}}(n+1) = T_{e_{2\cdot}}(n) + a, \qquad T_{e_{2\cdot}}(n) = an + b. \tag{12.3}$$

For the literal four-update loop consisting of a test/decrement, two increments, and a jump, with one final zero test and zero cost for the halted state, $a = 4$ and $b = 1$. If the fixed encoding expands one of these commands into several primitive updates, its finite expansion merely changes a, b and not the growth class. The clock adequacy theorem identifies each first-hit index with this update count, so (12.3) is a statement about *measured execution*. As $a > 0$,

$$\kappa(e_{2\cdot}) = [n+1].$$

Write

$$I_{\mathrm{lin}} =\!\downarrow [n+1], \qquad I_{\mathrm{poly,val}} =\!\downarrow \{[(n+1)^k] : k \in \mathbb{N}\}.$$

Both are actual ideal points of $\mathfrak{C}_{\mathrm{Ev}}$, and the latter is admitted in the overt-NNO instance: its positive pairing with a monotone $u : L \to \Sigma$ is $\exists k\, u([(n+1)^k])$, which factors through Σ by Proposition 1.2. The displayed code gives concrete witnesses

$$I_{\mathrm{lin}} \models \mathcal{U}_v, \qquad I_{\mathrm{poly,val}} \models \mathcal{U}_v. \tag{12.4}$$

Indeed $\Phi(e_{2\cdot}) \in B(n+1)$, and this same cost belongs to the larger polynomial module. The first assertion tests the principal linear budget; the second tests the nonprincipal ideal of all polynomial bounds in *numerical input value*.

Now put $\lambda(n) = 1+\mathrm{bitlength}(n+1)$, as in Section 2.7, and define the different polynomial budget

$$I_{\mathrm{poly,bit}} =\downarrow \{[\lambda^k] : k \in \mathbb{N}\}.$$

For every fixed k, the inequality $n+1 \le c\lambda(n)^k$ fails on arbitrarily large inputs, as can be seen along $n = 2^m$ from exponential growth against fixed polynomials in m. Hence

$$\kappa(e_{2\cdot}) \notin I_{\mathrm{poly,bit}}.$$

The displayed code therefore meets the polynomial budget for numerical value and fails it for binary length. The problem's feasible open is the union over its implementation fibre, as in Theorem 4.14.

The numerical implementation over a budget family. We can also evaluate this measured code on the nonconstant budget family of (3.7). There $b : \mathbb{S} \to \mathfrak{C}_{\mathrm{Ev}}$ uses $a_0 = [1]$, $a_1 = [n+1]$, and distinguished open u. The one-description presentation containing $e_{2\cdot}$ has denotation image $\{v\}$ and feasible open U_{a_1}. By (3.8), its pullback is u: the code is feasible exactly where the budget becomes linear. In the full implementation fibre this same witness proves $u \le b^*\mathcal{U}_v$; equality additionally depends on the other implementations of v.

12.5 Closed resource certificates and semantic incidence

Finite proof certificates provide another way to witness a resource assertion. We use the checker and ordinary-context interpretation of Section F, in a fixed observational environment. This gives a concrete comparison with the cost-aware logics of (Guéneau et al. 2018; Niu et al. 2022): the assertions and their proofs are descriptions in the stated language.

Fix a total bound expression $g_\beta(x)$ built from implemented numerical constructors, composition, and primitive recursion. The parameter β records its constructor data and may include a fixed implemented size expression. Constructor quotation represents e and β in the language; arbitrary semantic cost predicates are not supplied with such quotation.

The bounded simulation and first-hit tests are fixed implemented data programs. Ordinary identity types over 2 therefore define

$$\mathsf{Bound}(e,\beta) := \Pi_{x:\mathbb{N}}\, \mathrm{Id}_2(\alpha(e,x,g_\beta(x)),1), \tag{12.5}$$

$$\mathsf{Total}(e) := \Pi_{x:\mathbb{N}}\Sigma_{k:\mathbb{N}}\, \mathrm{Id}_2(\mathrm{First}_\alpha(e,x,k),1). \tag{12.6}$$

For cofinite numerical input define the Boolean program

$$\mathsf{Tail}(e,\beta,a,N,x) := (x < N) \vee \alpha(e,x,(a+1)g_\beta(x))$$

and the ordinary type

$$\mathsf{Land}(e,\beta) := \mathsf{Total}(e) \times \Sigma_{a:\mathbb{N}}\Sigma_{N:\mathbb{N}}\Pi_{x:\mathbb{N}}\, \mathrm{Id}_2(\mathsf{Tail}(e,\beta,a,N,x),1). \tag{12.7}$$

All of these descriptions use the existing ordinary sums, products, and Boolean identity types; the notation abbreviates these finite descriptions.

Proposition 12.6 (The interpreted resource assertions)**.** *In the concrete model, the support of* $\mathsf{Total}(e)$ *is precisely* $P(e)$*. The supports of* Bound *and* Land *are respectively*

$$\{(e,\beta) : e \in P,\ \forall x\ T_e(x) \le g_\beta(x)\}, \qquad \{(e,\beta) : e \in P,\ \Phi(e) \in Bg_\beta\}.$$

Here support means the image of the interpreted type's projection to its parameter object, and g_β *in the second expression denotes its exact germ.*

Proof.

(1) **Interpret totality using the unique first hit.** At a returning input there is exactly one first-hit index. The first-hit subobject therefore projects isomorphically onto the return domain. Unique choice supplies the dependent function in (12.6) exactly on P;

(2) **Interpret the global bound.** Boolean identity is interpreted by its equality subobject, so Theorem 12.3 proves the first bound formula.

(3) **Extract eventual domination from a tail witness.** In (12.7), Total retains termination also at the finitely many inputs below N. Above N, the Boolean tail test is exactly $T_e(x) \le (a+1)g_\beta(x)$. Such a witness is an eventual domination witness.

(4) **Construct a tail witness from eventual domination.** Conversely a rational domination coefficient is bounded above by a positive integer $a+1$, and a cofinite extent supplies its cutoff N. These data give the tail factor. The totality factor was already constructed uniquely on P. Taking supports proves the second formula, uniformly in the data parameters.

□

For either ordinary type description $A(e,\beta)$, use the ordinary inhabitance tag of the mixed checker. Constructor substitution defines the closed-instance boundary $j_A^{\mathrm{cl}}(e,\beta)$: its described ordinary and linear telescopes are both empty, and its type is $A(\bar{e},\bar{\beta})$. Define, with the pointwise closed-instance convention of Section F,

$$\begin{aligned} C_{\mathcal{T}}^{\mathrm{cl}}[A;e,\beta] &:= \Sigma_{d:\mathsf{Tree}}\,\mathrm{Id}_2(\mathsf{Acc}_{\mathcal{T},c}(d, j_A^{\mathrm{cl}}(e,\beta)),1),\\ \mathbb{B}^{\mathrm{cl}}[A;e,\beta] &:= F_d C_{\mathcal{T}}^{\mathrm{cl}}[A;e,\beta]. \end{aligned} \tag{12.8}$$

The ordinary boundary tag uses its ordinary checker; the empty linear boundary is retained as in the mixed presentation. The environment c, included among the fixed supplied parameters, is not quoted as constructor data. The ambient variables e,β index closed instances; they are not hypothetical premises in the described proof.

Theorem 12.7 (Resource-certificate soundness and finite replay)**.** *The interpreted supports of the ordinary certificate families* $C_{\mathcal{T}}^{\mathrm{cl}}[\mathsf{Bound};e,\beta]$ *and* $C_{\mathcal{T}}^{\mathrm{cl}}[\mathsf{Land};e,\beta]$ *lie in the respective resource incidences of Proposition 12.6. Their denotation images therefore lie in* Upper_{g_β}*. The comparison commutes with constructor-data substitution and with the verified proof transformations of Theorem F.2.*

The numerical first-hit and bounded-return facts have uniform finite replay: a true test at supplied numerical data constructs a closed certificate of its Boolean identity at the exact stated root. The corresponding finite-source resource certificates, and finite positive-probability witnesses for closed bound certificates, also map into these semantic incidences.

Proof.

(1) **Apply soundness to an accepted closed certificate.** An accepted ordinary root is an actual derivation of the instantiated type with both described telescopes empty. Apply the concrete soundness theorem F.2. The interpretation inhabits that instantiated type; its support is the incidence calculated in Proposition 12.6. This is a semantic soundness comparison, not a reflection rule in the described theory.

(2) **Identify the feasibility image and compiler stability.** For bound certificates it factors through P and then through the upper-bound subobject of P; composition with p gives the asserted denotation image. Soundness is uniform in the retained data, and the closed-instance instantiation program commutes with their substitution. Verified proof compilers preserve their specified roots, so the same soundness calculation applies to their outputs.

(3) **Construct a checked derivation from the numerical trace.** For finite replay, run the actual bounded numerical test, retaining its finite configuration trace. On its true branch the verified fixed-program calculation construction of Lemma F.3 constructs the derivation of the corresponding Boolean identity. Its exact-root invariant gives the required closed boundary. It branches on the computed Boolean result; no runtime acceptance proof or arbitrary dependent value is quoted.

(4) **Extract the exact return history.** For a bounded-return test, bounded search also extracts the unique first hit and its finite return history.

(5) **Forget publication data while preserving acceptance.** A finite-source resource certificate of Section F retains the accepted tree along with its publication data; forgetting that data gives the ordinary certificate above.

(6) **Extract an accepted tree from positive publication weight.** A finite witness $\Theta^{\mathrm{cl}}(q; A; e, \beta)$, with rational $q > 0$, contains a finite list of accepted trees with total weight exceeding q. The list cannot be empty, so finite inspection extracts one accepted tree. These explicit maps followed by the preceding support comparison give the remaining assertions. They retain the same closed boundary and supplied environment.

□

Finite replay is complete for the stated finite numerical observations. The universally quantified Bound and Land assertions retain a different interpretation: sound finite proofs witness their truth, but truth is not identified with finite provability. The free quantitative certificate module likewise has no rule evaluating an arbitrary certificate into its described type. Contextual and pointwise closed certificate families keep their respective meanings. Search has a positive return domain, whereas the universal assertions have the quantified supports calculated above. Thus the relation between proof-source probability and execution time is carried by finite witnesses and semantic incidence.

12.6 Exact overhead constructors

An overhead bound must account for the events introduced by its implementation. The basic operation is concatenation of recorded words. In the fixed finite-description signature, include the idle primitive idle $= \eta_1 : 1 \to L_\Sigma 1$ with event $a = \iota_{\mathrm{idle}}(*, *)$. For a supplied total nonnegative integer $b(x)$, finite recursion constructs $\mathrm{rep}(a, b(x))$ by that many invocations. Prefixing it gives the history transformation

$$D_b(H) = \mathrm{rep}(a, b(x)) {+\!+} H, \qquad D_b D_c = D_{b+c}$$

on a fixed input. For the selected valuation its exact formula is

$$T_{\nu,D_b e}(x) = \nu(\mathrm{rep}(a, b(x)) +\!+ H_e(x)). \tag{12.9}$$

Here $D_b e$ denotes the retained execution description that prefixes these events. For a monoid fold this is $\overline{w}(\mathrm{rep}(a, b(x)))\, T_{\overline{w},e}(x)$. For additive time with $w(a) = 1$ it is $b(x) + T_e(x)$; for a maximum fold it is the maximum of the recorded prefix and original occupancies. The equality follows from the fold clauses, so no additive formula is assigned to an arbitrary measurement ν.

If $b(x)$ must itself be computed, the execution of that computation is prefixed too. In a numerical code translation, the actual dispatch and loop instructions of its finite wrapper also contribute their recorded events. Thus (12.9) describes the specified execution word; a concrete wrapper's bound is obtained by evaluating the valuation on all of its events. The cost of executing a code transformation and the execution cost of its output program have different inputs and are kept separate.

For the bounded clock in Proposition 12.2, a numerical reindexing is also available. Define

$$\beta(e, x, k) = \begin{cases} 0, & k < b(x), \\ \alpha(e, x, k - b(x)), & k \geq b(x). \end{cases}$$

Its return extent is unchanged, and its first-hit index satisfies

$$\tau_\beta(e, x) = b(x) + \tau_\alpha(e, x). \tag{12.10}$$

This follows by the two Boolean cases and uniqueness of the first hit. For a nonnegative integral polynomial $P(x, c)$, the inflationary map $r_x(c) = c + P(x, c)$ gives the monotone Boolean clock

$$\beta_P(e, x, k) = \bigvee_{j \leq k} (\mathrm{First}_\alpha(e, x, j) \wedge r_x(j) \leq k).$$

A first hit at j becomes a first hit at $r_x(j)$, since $j \leq r_x(j)$ and the original first hit is unique. Conversely a hit contains the original returning witness. This proves equality of extents and the exact reindexed clock value. The procedure computing this Boolean predicate has its own execution history; its runtime is not inferred from the reindexed clock value.

12.7 Polynomial valuations and compiler comparisons

For $F(x) = \sum_{j=0}^{d} b_j x^j$ and a marked resource M,

$$F(M) \subseteq Bb_0 + \sum_{j=1}^{d} (Bb_j) M^j, \qquad DF(x, v) = \sum_{j=1}^{d} j b_j x^{j-1} v.$$

The first follows from resource multiplication, and the second from the differential comparison theorem. Substituting polynomials commutes with their exact, quantitative, signed, and geometric interpretations.

Let $\gamma : P \to P'$ be a finite implemented translation with the measured comparison obtained from its compiled histories, $T_{\gamma(e)} \leq a_e T_e + b_e$ eventually, where $a_e \in \mathbb{Q}_{\geq 0}$ is a supplied numerical constant (more generally a supplied nonnegative bounded coefficient) and b_e is an exact nonnegative remainder profile. An exact bound $b_e \in O(T_e)$ gives $T_{\gamma(e)} \in O(T_e)$. A reverse measured comparison gives Θ. An exact remainder $T_{\gamma(e)} - T_e \in o(T_e)$ gives $\sim$. These implications follow from Proposition 2.5 with the implementation parameter retained.

Description measurements use the same calculus with their own exact profiles. The prefix-free word family $1^n 0$ has length $n + 1$. Prefixing a fixed word of length c gives length $n + 1 + c \sim n + 1$, with error constant c. The execution cost of prefixing is calculated in Section F.

12.8 Numerical and oscillating examples

Example 12.8 (Leading coefficients and errors)**.** Let $g = n + 1$. Then $n + 2 \sim g$, whereas $2g \in \Theta(g)$ and $2g \not\sim g$. The first difference is $1 \in o(g)$; the second is g, which fails its smallness inequality at tolerance $1/2$. In the orbit description the first multiplier is $1 + t$, and the second is the unit 2 outside $1 + \mathfrak{m}$.

Example 12.9 (Polynomial, logarithmic, and exponential orders)**.** For integers $0 \leq r < s$, $(n+1)^r \in o((n+1)^s)$, by the direct cutoff for $(n+1)^{s-r}$. The profile $(n+1)\lambda$ is in $\widetilde{\Theta}(n+1)$ and in $\omega(n+1)$. Every fixed polynomial belongs to $o(2^n)$ by the finite binomial estimate of Section 12.10. The parity examples of Section 2.4 give comparisons which have no single limiting ratio.

Example 12.10 (Tight versus asymptotically exact variation)**.** At a reference resource g, first-order variation within the tight-bound orbit has direction module Bg. Variation within its asymptotic-equivalence class has direction module $\mathfrak{m}g$ by (6.14). For $g = t^{-1}$ these are $t^{-1}B$ and $t^{-1}\mathfrak{m}$. The direction g is allowed by the first and excluded by the second, whereas the constant direction $1 = tg$ belongs to both. This is the differential form of the leading-coefficient distinction.

The next calculation returns to the geometric interpretation of resource constraints. It exhibits a direction excluded by a bounded budget and shows how allowing a chosen overhead removes that normal obstruction.

12.9 An explicit normal resource direction

For cofinite input put $H = t^{-1}B$ and take a fixed implementation of bounded cost $c = \Phi(e) \in B$. The valuation family $F_e(b) = c + b$ has budget-B tangent $B \subset H$ at $b = 0$. The direction t^{-1} gives a nonzero class in H/B, since $n + 1$ is not bounded. After localization at t, this quotient becomes zero by Theorem 2.23. This computes an explicit normal resource obstruction and its change under overhead inversion.

A numerical asymptotic scale organizes resource coefficients by order of decay. Absolute differentials describe logarithmic connections along this scale.

12.10 Asymptotic expansions, scale filtrations, and connections

12.10.1 The numerical scale and its remainders

Work in the cofinite numerical fibre and put $t(n) = 1/(n + 1)$. It is a regular element of B, belongs to $\mathfrak{m}$, and is not a unit: its reciprocal $n + 1$ is unbounded. Thus its powers form a strict asymptotic scale, $t^{r+1} \in o(t^r)$.

Definition 12.11 (Asymptotic expansion)**.** For a rational sequence $(a_j)_{j\in\mathbb{N}}$, the formal expansion relation (distinct from relative-error equivalence $\sim$) is written

$$f \sim_t \sum_{j\geq 0} a_j t^j$$

and means the family of finite remainder statements

$$f - \sum_{j=0}^{r} a_j t^j \in o(t^r) \qquad (r \in \mathbb{N}).$$

No value for an infinite sum is part of this definition. More generally a displayed scale (φ_j) with $\varphi_{j+1} \in o(\varphi_j)$ defines its expansion by the corresponding remainders. This is the finite-remainder use of asymptotic expansion (NIST 2026).

Proposition 12.12 (Coefficient uniqueness and arithmetic)**.** *The rational coefficients of a t-expansion are unique. Expansions add and multiply with the usual finite coefficient formulas. On the subobject of expansions with $a_0 \in \mathbb{Q}^\times$, their germs are units of B and their inverses have the recursively determined reciprocal expansion.*

Proof.

(1) **Exclude nonzero rational constants from the vanishing ideal.** A nonzero rational constant cannot belong to $o(1)$: choose k with $k|a| > 1$; its required eventual extent would be empty, contrary to properness.

(2) **Prove coefficient uniqueness by induction.** Suppose coefficients agree below degree r in two expansions of f. Subtract their order-r remainder equations and divide by the everywhere nonzero t^r. Then $a_r - b_r \in o(1)$, hence $a_r = b_r$. Induction proves uniqueness.

(3) **Verify addition of expansions.** Addition follows from the small-error sum law.

(4) **Compute the product expansion.** For multiplication fix order r. The order-zero estimates make both germs bounded. Multiplying the two order-r remainders by bounded germs gives errors in $o(t^r)$. Terms of the finite product with degree greater than r belong to $t^{r+1}B \subseteq o(t^r)$. The remaining coefficients are the finite convolution.

(5) **Construct the reciprocal expansion.** For $a_0 \neq 0$, the order-zero expansion gives $f = a_0(1 + a)$ with $a \in \mathfrak{m}$. Hence f is a unit by Theorem 2.9. Set $b_0 = a_0^{-1}$ and $b_r = -a_0^{-1} \sum_{j=1}^{r} a_j b_{r-j}$. At each finite order, convolution gives $f \sum_{j=0}^{r} b_j t^j = 1 + o(t^r)$. Multiply the remainder by the bounded inverse of f. This is the reciprocal expansion.

□

An expansion need not determine the exact germ. The rational profile 2^{-n} is nonzero but belongs to $o(t^r)$ for every r. Indeed for $k = r + 1$ and $n \geq 2k$,

$$2^n \geq \binom{n}{k} \geq (n/2)^k/k!,$$

so $(n + 1)^r 2^{-n}$ is bounded by a rational constant times $1/n$ eventually. Each tolerance has an explicit integer cutoff. Thus its expansion has all coefficients zero, while its exact resource remains visible.

12.10.2 Filtered coefficients and differentiation

The filtration $F^r B = t^r B$ records a specified scale of error. It is finer than declaring an error merely vanishing. For example $b(n) = 1/(1 + \lfloor\sqrt{n+1}\rfloor)$ belongs to $\mathfrak{m}$ but not to tB: square inputs make b/t arbitrarily large. The floor square root is a bounded finite search, so this is an exact rational profile.

Proposition 12.13 (Scale boundary)**.** *There are natural isomorphisms and exact sequences*

$$t^r B/t^{r+1}B \cong B/tB, \qquad \mathrm{gr}_{tB} B \cong (B/tB)[z],$$

$$0 \longrightarrow M/tM \xrightarrow{t} M/t^2M \longrightarrow M/tM \longrightarrow 0$$

for every embedded resource $M \subseteq K$.

Proof.

(1) **Identify the associated graded algebra by cancellation.** Multiplication by t^r is injective on K and hence on B and M. It identifies the quotient in degree r with B/tB. Products of these identifications add degrees, giving the polynomial graded algebra.

(2) **Verify exactness of the scale sequence.** In the displayed sequence, the first map sends $[m]$ to $[tm]$. If $tm \in t^2M$, cancellation gives $m \in tM$. Its image is exactly the kernel of reduction modulo tM. Surjectivity of the last map is quotient surjectivity.

□

For a finite polynomial parameter algebra $A = \mathrm{Sym}_B(E)$ and an ideal $J \subseteq B$, relative differentiation gives

$$d_{/B}(JA) \subseteq J(A \otimes_B E).$$

Indeed $d_{/B}(bp) = b\, d_{/B}p$ for $b \in J$. This proves coefficientwise stability of remainders in t^rA and $\mathfrak{m}A$ under polynomial parameter differentiation. It supplies an explicit differential calculus for resource-valued asymptotic expansions.

The nilpotent Taylor parameter d of Section 11 and the regular asymptotic scale t have different algebraic roles. Relative Weil differentiation fixes t as a coefficient. Absolute coefficient differentiation is constructed next.

12.10.3 Logarithmic differentials and pole resources

Let

$$L_t = (\Omega^1_{B/\mathbb{Q}} \oplus B\ell)/(dt - t\ell),$$

and write $d : B \to L_t$ for the universal differential followed by this quotient. A B-linear map $L_t \to N$ is exactly a pair (δ, ω) with $\delta : B \to N$ a $\mathbb{Q}$-derivation and $\delta(t) = t\omega$. This follows directly from the generators and the one displayed relation.

For $k \geq 0$ let $M_k = t^{-k}B \subseteq K$.

Proposition 12.14 (Logarithmic connections)**.** *The formula*

$$\nabla_k(bt^{-k}) = t^{-k} \otimes (db - kb\ell) \tag{12.11}$$

defines an L_t-connection on M_k. It is compatible with $M_k \hookrightarrow M_{k+1}$ and with $M_k \otimes_B M_l \cong M_{k+l}$. Its residue in the generator t^{-k} is $-k$.

Proof.

(1) **Verify the connection and Leibniz identity.** The coefficient b is unique because t is regular. The derivation identity gives $\nabla_k(am) = a\nabla_k(m) + m \otimes da$.

(2) **Verify compatibility with inclusions.** Under the inclusion, $bt^{-k} = tb\, t^{-(k+1)}$. Substitute $d(tb) = t\, db + b\, dt$ and $dt = t\ell$ into the right-hand formula at $k+1$; the result is exactly the formula at k.

(3) **Verify tensor compatibility.** For a product $bt^{-k} \cdot ct^{-l}$, the Leibniz rule for $d(bc)$ adds the two logarithmic terms, giving the tensor compatibility.

(4) **Construct the residue map.** Define $\mathrm{res} : L_t \to B/tB$ by $db \mapsto 0$ and $\ell \mapsto 1$. The relation maps to zero, so this is well-defined and factors through L_t/tL_t.

(5) **Compute the residue of the resource line.** Applying it to the coefficient in (12.11) gives $-k\bar{b}$. Under the generator identification of M_k/tM_k with B/tB, this is multiplication by $-k$.

□

Corollary 12.15 (Residues distinguish logarithmic resource lines)**.** *In the cofinite numerical fibre, the residue of ∇_k is $-k$ in every bounded-unit frame of $M_k = t^{-k}B$. The logarithmic resource lines (M_k, ∇_k) and (M_l, ∇_l) are isomorphic as L_t-connections exactly when $k = l$. Tensor product adds their residues.*

Proof.

(1) **Show that the residue is independent of the generator.** Every generator of the free rank-one module M_k has the form ut^{-k} with $u \in B^\times$. Its connection form is $u^{-1}du - k\ell$. Since $\operatorname{res}(du) = 0$, its residue is $-k$.

(2) **Use residues to distinguish the indices.** An isomorphism $M_k \to M_l$ has the form $t^{-k} \mapsto ut^{-l}$ with $u \in B^\times$. Compatibility with the connections gives $du + (k-l)u\ell = 0$; applying residue yields $(k-l)\bar{u} = 0$ in B/tB. This quotient is nonzero because t is not a unit, and every nonzero rational remains a unit in it. As $\bar{u}$ is also a unit, $k = l$.

(3) **Verify the converse and tensor rule.** Conversely the identity is an isomorphism when the indices agree. Tensor compatibility follows from Proposition 12.14, and its residues are $-(k+l)$.

□

12.10.4 Principal parts and geometric realization

Put $A_t = B \ltimes L_t$ and use the two algebra maps

$$\lambda(b) = (b, 0), \qquad \rho(b) = (b, db).$$

For a B-module M define $P^1_t(M) = A_t \otimes_{B,\rho} M$, with its left B-module structure induced by λ. The right-B sequence $0 \to L_t \to A_t \to B \to 0$ splits through ρ. Tensoring therefore gives a left-B exact sequence

$$0 \to L_t \otimes_B M \to P^1_t(M) \to M \to 0. \tag{12.12}$$

Its canonical $\mathbb{Q}$-linear map $j(m) = 1 \otimes m$ obeys $j(bm) = b\,j(m) + db \otimes m$. Consequently a connection $\nabla : M \to L_t \otimes_B M$ gives the B-linear splitting $j - \nabla$, and a splitting s gives the connection $j - s$. These formulas are mutually inverse. Exact full faithfulness of j_B realizes the sequence and this correspondence inside $\mathcal{G}_B$. This constructs the differential resource bundle with its actual extension class and splittings.

13 Conclusion

The construction gives a differential geometry of computational resources with a common exact coefficient interpretation. Measured execution determines cost profiles and implementations; embedded resource modules and their locale classify budgets and feasibility. Taylor-valued history instrumentation identifies finite geometric jets with measured response coefficients. Resource deformation complexes then describe constrained variations, including compatibility between moving budgets and the effect of permitted overheads on obstructions.

The primary infinitesimal geometry represents the function objects needed for completion, de Rham cochains, and crystalline jets. The comparison with algebraic forms, the equivalence between connections and stratifications, and the principal-parts description of differential operators retain the same resource parameters and effective finite observations.

The examples exercise these structures together. A measured family has polynomial bounds at every finite order without a common polynomial degree for its complete coefficient

family. Its generating function carries a logarithmic de Rham class and a crystalline resource line. Event instrumentations exhibit a mixed-jet curvature defect and an explicit coherent ordered transport. Coupled budgets can obstruct a joint lift despite separate feasibility, and logarithmic residues distinguish asymptotic pole orders. These calculations connect the geometric foundations to the measured resources from which they were constructed.

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A Admitted presentations and parameter coherence

This appendix supplies the admitted presentation theorem Theorem 1.8 used to construct the continuous linear and algebraic objects in the main text. The external dcpo method is due to Jung–Moshier–Vickers (Jung et al. 2008). We give its internal selected-ideal construction, including the parameter comparisons needed by tensors and differentials.

For a preorder take its effective antisymmetrization; lower subsets and monotone maps into posets descend uniquely. Thus it suffices to write P for a poset. For $C \subseteq P \times J_\Sigma P$, let $L_C(P)$ be the object of lower subsets $Z \subseteq P$ satisfying

$$(p, I) \in C, \quad I \subseteq Z \quad \Longrightarrow \quad p \in Z.$$

Intersections define a closure c_C on lower subsets. The object $L_C(P)$ is a complete lattice, with joins given by closure of unions, and $\eta_C(p) = c_C(\downarrow p)$ is monotone. A complete lattice is a J_Σ-algebra by restriction of its joins.

Construction A.1 (Generated computational subalgebra)**.** Intersect all subobjects of $L_C(P)$ containing $\eta_C(P)$ and closed under images of their intrinsic admitted ideals. Denote the intersection by G. For $I \in J_\Sigma G$, its image in each such subobject is admitted, and its ambient supremum lies in that subobject. Hence it lies in G. The restricted supremum defines a J_Σ-algebra on G, because the least-upper-bound characterization proves its unit and associativity laws. Set $\mathsf{Pres}_{J_\Sigma}(P, C) = G$ and $j = \eta_C$ with this restricted codomain.

The use of intrinsic weights matters: an ideal admitted on the ambient complete lattice and supported in a subobject has not thereby been shown admitted on that subobject. For a cover $(p, I) \in C$, the ambient join of $\eta_C(x)$, $x \in I$, contains I, so its cover closure contains p and $\eta_C(p)$. The lower image of I under j is admitted. It follows that j respects every cover.

Lemma A.2 (Extension and uniqueness)**.** *Restriction along j gives the order isomorphism* (1.4)*.*

Proof.

(1) **Embed the target in its lattice of admitted-closed subsets.** For a target algebra A, let $\mathrm{Cl}_{J_\Sigma}(A)$ be the complete lattice of lower subsets closed under every admitted ideal of A that they contain. The order embedding $a \mapsto{\downarrow} a$ preserves admitted suprema. Indeed, the least such closed lower subset containing an admitted ideal I is $\downarrow \sup I$: closure forces $\sup I$ and then its lower set, while this principal lower set is itself closed.

(2) **Define the extension on closed lower subsets.** For a monotone cover-respecting $f : P \to A$, define

$$f^\sharp : L_C(P) \longrightarrow \mathrm{Cl}_{J_\Sigma}(A), \qquad f^\sharp(Z) = \bigvee_{p \in Z} {\downarrow} f(p).$$

(3) **Verify cover compatibility and join preservation.** If $W \in \mathrm{Cl}_{J_\Sigma}(A)$, then $f^{-1}W$ is lower and cover-closed. In fact, $f(I) \subseteq W$ implies $\sup J_\Sigma f(I) \in W$, and the cover inequality puts $f(p)$ in W. Consequently closing a union in $L_C(P)$ introduces no additional image outside any upper bound W. This proves that $f^\sharp$ preserves all joins and that $f^\sharp \eta_C(p) = {\downarrow} f(p)$.

(4) **Restrict the extension to the generated algebra.** The embedded copy of A in $\mathrm{Cl}_{J_\Sigma}(A)$ is closed under intrinsic admitted suprema. Its inverse image under $f^\sharp$ is therefore a computational subalgebra containing every generator, so contains G. Restricting and inverting the order embedding gives $\bar{f} : G \to A$, with $\bar{f} j = f$.

(5) **Prove uniqueness and the order isomorphism.** For two such maps the equality locus is admitted-closed and contains all generators, hence is all of G. The same argument with the inequality locus proves the order isomorphism, including monotonicity of extension.

□

Lemma A.3 (Canonical presentations)**.** *Every algebra (A, a) has the canonical presentation on its underlying poset with covers $(a(I), I)$ for $I \in J_\Sigma A$. Its presented algebra is canonically A.*

Proof.

(1) **Construct the comparison and prove continuity.** The identity respects the covers, giving $h : \mathsf{Pres}_{J_\Sigma}(A, C_A) \to A$. The cover gives $j(aI) \leq \sup J_\Sigma j(I)$; monotonicity gives the reverse inequality. Thus j is an algebra morphism.

(2) **Verify the inverse equations and naturality.** We have $hj = \mathrm{id}_A$ on generators, and $jh = \mathrm{id}$ by uniqueness of extension. The same uniqueness proves naturality.

□

Lemma A.4 (Products and operations)**.** *Finite products of generated algebras are generated by the products of their generators. Cover-compatible finitary operations extend, and their equations and inequalities extend.*

Proof.

(1) **Generate the product from pairs of generators.** Let A, B be generated by G_A, G_B, and let D be the least admitted-closed subobject of $A \times B$ containing $G_A \times G_B$. For $a \in G_A$, the section $\{b : (a, b) \in D\}$ is admitted-closed, because pairing with a fixed element preserves inhabited admitted suprema. It contains G_B, hence equals B.

(2) **Complete the finite-product induction.** For arbitrary $b \in B$, the other section now contains G_A and is admitted-closed, hence equals A. Thus $D = A \times B$. Induction proves the finite-product assertion.

(3) **Define the extended operation.** For a monotone operation $\omega : P^n \to P$ respecting covers separately, put

$$\widetilde{\omega}(Z_1, \dots, Z_n) = c_C \downarrow \{\omega(p_1, \dots, p_n) : p_i \in Z_i\}.$$

(4) **Prove separate continuity.** Cover compatibility says that the inverse image of a closed lower subset under a section at generators is cover-closed. Applying the closure universal property in each variable proves that $\widetilde{\omega}$ preserves joins separately.

(5) **Deduce joint continuity and restrict to the generated algebra.** For a directed family in a finite product, common upper bounds identify the iterated coordinate suprema with the joint supremum. Thus separate continuity is joint continuity. Product generation now shows that $\widetilde{\omega}$ restricts to $G^n \to G$.

(6) **Extend the algebraic laws.** Equalities and inequalities between the extended term maps have admitted-closed loci. They contain the tuples of generators when the original laws hold, so contain all tuples. This also handles laws with repeated variables.

□

For an algebraic signature with several sorts, use one such lattice per sort. The same section proof extends each typed finite-arity operation. Finite term objects exist by encoding well-founded finite trees using finite lists and natural-number recursion. Closing covers under one-hole contexts is an internal relation on these terms; its weights remain admitted by functoriality of J_Σ. Equations are imposed by inequalities in both directions before antisymmetrization.

For the tensor presentation of Proposition 1.9, the relations are precisely additivity, zero, and $(sm) \otimes n = m \otimes (sn)$, together with the two domain-continuity and scalar-continuity covers. A map out of it is therefore a continuous balanced bilinear map. Iterated tensors represent the same multilinear maps, so associativity and symmetry are canonical. The pentagon, triangles, and symmetry equations hold on elementary tensors and then everywhere by Lemma A.4. A relative tensor adds the balance relations for its coefficient algebra. A quotient module adds its displayed equations. For differentials these are additivity, scalar linearity, and the Leibniz relations; the presentation consequently represents continuous derivations. This constructs the signed quotients used in the differential sector.

Proposition A.5 (Parameter coherence)**.** *The presentation, free-module, tensor, and quotient universal properties above hold in each parameter slice. Their induced comparison maps commute with substitution, and their coherence diagrams commute.*

Proof.

(1) **Construct the slice substitution comparisons.** Every carrier is a specified subobject of a power object, every cover retains its parameters, and every extension is determined by a unique factorization. For a substitution $\sigma : \Delta \to \Gamma$, pullback between the slices of $\mathcal{E}$ preserves finite limits and regular images. The slice power-object and dependent-product Beck–Chevalley comparisons give

$$\sigma^* \mathcal{P}_\Gamma(P) \cong \mathcal{P}_\Delta(\sigma^* P), \qquad \sigma^*(Y^X) \cong (\sigma^* Y)^{\sigma^* X}.$$

Here all objects and exponentials on the left are in $\mathcal{E}/\Gamma$, and those on the right are in $\mathcal{E}/\Delta$.

(2) **Transport admission and its monad structure.** The same comparisons transport the predicates for inhabitedness, directedness, lower closure and the complete upper-test profile. They therefore identify $\sigma^*(J_\Sigma^\Gamma P) \cong J_\Sigma^\Delta(\sigma^* P)$, including its unit, multiplication and admitted cover relation.

(3) **Transport cover closure and generated presentations.** Universal quantification over the slice power object transports the intersections defining cover closure and the generated subalgebra. Hence the resulting presentation comparison

$$\sigma^* \mathsf{Pres}^\Gamma_{J_\Sigma}(P, C) \cong \mathsf{Pres}^\Delta_{J_\Sigma}(\sigma^* P, \sigma^* C)$$

sends each pulled-back generator to the corresponding generator.

(4) **Verify extension and substitution coherence.** The extension universal property supplies its inverse and shows that extension commutes with this comparison. Identity and composite substitutions, tensor coherence, and substitution through extension agree on those generators and hence everywhere. These comparisons are formed in parameter slices of the fixed ambient topos.

□

B Support, representatives, and finite denominator witnesses

The resource calculations use formulas on representatives and then descend them to germs. This appendix records the support operations, common principal stages, and finite denominators that justify those steps. The pointwise operations are rational formulas with decidable cases:

$$e_g(x) = \begin{cases} 1, & g(x) \neq 0, \\ 0, & g(x) = 0, \end{cases} \qquad q(f,g)(x) = \begin{cases} f(x)/g(x), & g(x) \neq 0, \\ 0, & g(x) = 0. \end{cases}$$

Changing either representative on a noneventual complement changes these formulas only there. Intersecting the representative-equality extents proves that they descend. Under a domination witness $f \preceq g$, the equation $q(f,g)g = f$ holds on the same eventual comparison extent.

For a finite list $f_1, \dots, f_r$ in a signed resource, the nonnegative germ $g = \sum_i |f_i|$ belongs to that resource and dominates the list. Each f_i is a bounded multiple of g. This is the common principal stage used to verify finite tensor relations and exactness. The stage is constructed from the finite list; no simultaneous selection of principal generators for arbitrary submodules is involved.

A quotient argument with local representatives is an internal argument over the effective epimorphism from profiles to germs. To define a function on the quotient we prove its formula invariant under the equivalence relation. To prove existence of a multiplier, we construct it over a representative and then use the image predicate of its defining equation. This proves internal existence.

Similarly, a finite list of localization equations has a common denominator given by the product of its denominators. For equations involving a fixed scale t, take a common maximum exponent. These finite operations justify the flatness and finite-cover arguments. The common-denominator condition for an infinite coefficient family is retained as an existential statement about one exponent, as in (9.13); it is not obtained by taking a maximum of infinitely many witnesses.

C Finite Weil duality and coefficient coherence

The linear adjoints and coefficient-change comparisons in the main text use finite-projective duality. We record the transpose formulas and check their independence of the chosen dual basis, then verify compatibility with stage and coefficient maps. Let C be a finite projective B-algebra, F a C-module, and M a B-module. For a dual basis (c_i, ϕ_i), the transpose of a balanced map $h : C^\vee \otimes_C F \to M$ is

$$\widetilde{h}(x) = \sum_i c_i \otimes h(\phi_i \otimes x).$$

For $\psi \in C^\vee$,

$$(\psi \otimes 1)\widetilde{h}(x) = h\left(\sum_i \psi(c_i)\phi_i \otimes x\right) = h(\psi \otimes x).$$

The same identity with the right C-action on $C^\vee$ shows $\widetilde{h}(cx) = c\widetilde{h}(x)$. It also proves independence of the basis: every contraction with a functional has the specified value, and finite duality detects equality in $C \otimes_B M$. This gives the inverse equations used in (6.13).

For a stage map $u : C \to C'$, contracting the naturality equation of a family $v_C : F(C) \to C \otimes_B M$ with $\phi \in (C')^\vee$ gives

$$h_C((\phi u) \otimes x) = h_{C'}(\phi \otimes F(u)x).$$

These are exactly the coequalizer relations defining Λ_{lin}. Thus its adjunction is natural in both the coefficient module and the geometric module.

For a coefficient map $\beta : B \to B'$ and a rational Weil presentation W, tensor associativity gives

$$(B' \otimes_{\mathbb{Q}} W) \otimes_{B'} (B' \otimes_B M) \cong B' \otimes_B ((B \otimes_{\mathbb{Q}} W) \otimes_B M).$$

On a pure tensor both sides have the same coefficient, Weil basis element, and resource element. The diagrams for two coefficient maps, two Weil substitutions, truncation, and interchange therefore commute on these generators. This proves the finite coefficient coherence used for gauge and input transport.

D History substitution and numerical clock regrading

Two distinct comparisons are used in the execution examples: reassociating a recorded history preserves its fold, while regrading a numerical clock changes its index. The following calculations establish their composition laws and keep their respective measurements explicit.

D.1 Coherence of the retained resource valuation

For a monoid (C, e, μ), an event map $w : A \to C$, and returning words $s, t, u \in A^*$, the recursive fold satisfies

$$\overline{w}(s{+}{+}t) = \overline{w}(s)\overline{w}(t), \qquad \overline{w}((s{+}{+}t){+}{+}u) = \overline{w}(s{+}{+}(t{+}{+}u)).$$

The first equation is induction on s; the second follows from word associativity and monoid associativity. Empty words give the two unit equations. Each interface-cut transformation in Theorem 4.5 preserves the flattened word, so these formulas prove its resource coherence. Input substitution precomposes the state, event, and valuation maps in these same formulas; induction on the finite descriptions proves its compatibility with composition and recursive execution.

For a finite-history measurement ν that is not a monoid fold, the assembled cost remains $\nu(H_{\text{assembled}})$. No monoidal law is deduced for ν. A richer finite accumulator may instead construct its stated law. For the sequential time convention, an overhead execution of cost $H(e,x)$ returning a waiting amount $b(e,x)$ contributes $H(e,x)+b(e,x)+T_e(x)$ together with every recorded dispatch transition. This is the sum of the actual component words' time folds.

For two verified measured comparisons $T_2 \le aT_1+b$ and $T_3 \le cT_2+d$, on the intersection of their eventual extents,

$$T_3 \le caT_1 + (cb+d).$$

The comparison retains the same implementation parameter and the nonnegative coefficients. For general monotone growth transformations, their composition is the one in Construction 4.22.

D.2 The bounded numerical calculation

For the numerical clock $\alpha(e,x,k)$ and a supplied inflationary regrading $r_x : \mathbb{N} \to \mathbb{N}$ with its finite Boolean tests, set

$$R_r\alpha(e,x,k) = \bigvee_{j \le k} (\mathrm{First}_\alpha(e,x,j) \wedge r_x(j) \le k).$$

This is monotone and Boolean by its finite formula. A first hit of α at j becomes the first hit of $R_r\alpha$ at $r_x(j)$; inflationarity puts j inside the bounded search and uniqueness excludes another contributor. A hit of the new clock contains a first-hit witness of the original, so their classified extents agree.

For inflationary r_x, s_x, the two clocks

$$R_s(R_r\alpha) = R_{s\circ r}\alpha$$

are true at exactly the indices at least $s_x(r_x(j))$ when the original first hit is j. Existence and uniqueness of that j prove the equality in every parameter context. The identity regrading gives the original monotone clock. Input substitution acts on the same bounded formula, so commutes with regrading. This is a clock-index calculation for the constructed numerical instance; measured compiler costs are supplied by the retained histories and their valuations above.

E Name presentations, finite descriptions, and numerical codes

The implementation construction requires names with exact domain comparisons, finite typed descriptions, and a numerical program presentation. This appendix supplies those three interfaces from the realizability and exact-completion constructions of (Hyland 1982; Birkedal and van Oosten 2002; van Oosten 2008). Costs are subsequently assigned by the execution histories and valuations of the main text.

Proposition E.1 (Names and tracked lifts)**.** *Every object in the displayed realizability models has a regular-epimorphic partitioned-assembly presentation. On a retained projective source, an arrow has a tracked lift through a target name cover. Partial arrows have implementations with the same classified domain. Finite products and substitutions admit partitioned refinements; equations hold after realization and on common refinements.*

Proof.

(1) **Lift total arrows through projective presentations.** Use the projective cover supplied by the exact-completion presentation (Birkedal 2000, Corollaries 10.6.4–10.6.5). For $q : C \twoheadrightarrow X$, $q' : C' \twoheadrightarrow Y$, and $f : X \to Y$, pull back q' along fq. The resulting regular epimorphism onto the projective C splits. Composing its section with the other projection gives $\widehat{f} : C \to C'$ and $q'\widehat{f} = fq$. Fullness of the partitioned-assembly embedding supplies its single permitted tracker.

(2) **Construct a projective presentation of the hit domain.** For a partial arrow, pull its domain back to C and lift its Σ-classifier through $2^{\mathbb{N}} \twoheadrightarrow \Sigma$. Let α_c be the resulting total stream. Its hit assembly $H = \{(c, n) : \alpha_c(n) = 1\}$ is partitioned and projective. The semantic value on H therefore lifts to C'.

(3) **Implement the partial lift by first-hit search.** On input c, search for the least hit and apply that lift to (c, n). Search terminates exactly on the original domain; the returned name realizes the original value. It defines a classified partial arrow by the functional first-hit relation.

(4) **Verify the domain-sensitive implementation.** The first-hit search is implemented by represented natural-number recursion with delayed recursive calls. Every value used after a hit has its stated domain witness, so no test for arbitrary pca application definedness is used.

(5) **Construct dependent lifts and common refinements.** Finite products of partitioned presentations are partitioned. For a dependent pullback, cover the actual pullback by a partitioned assembly and perform the same construction there. Two presentations have the common refinement obtained by covering their fibre product over the denotation object.

(6) **Verify coherence after realization.** Identity and composition of retained lifts satisfy their squares by composition; different selected lifts need only agree after realization. This is precisely the coherence asserted, and requires no functorial selection of one name per denotation.

□

For a modest object one can use its usual partial-equivalence-relation presentation on ambient names: the cover consists of the names in the domain of the PER, and its kernel pair is the PER. An extensional tracker descends through this pair; a morphism lifts by the preceding proposition. The same statement for a relation retains a name for its membership proof. An inclusion of represented relations lifts after pulling back their proof-name covers, so internal witness transformations used by finite algorithms can be implemented on those actual presentations.

Construction E.2 (Typed finite descriptions)**.** For a fixed finite typed computational signature, let $\mathcal{S}$ contain its constructor tags, arities, source and target sorts, and supplied parameter slots. A relative raw tree is a finite prefix list of constructor nodes, with its finite binding and child-position data, together with the tuple of supplied literal labels required at its leaves. The valid-tree condition is the finite stack-parsing condition: each node consumes one pending place and opens the number specified by its tag; the stack is empty exactly at the end. Thus the carrier is a subobject of a sum of finite products of label objects and numerical data. It exists using the NNO and finite lists and has a partitioned presentation in the realizability models.

Type annotations record the types at every node, including the types of its operation slots. The well-typed subobject is defined by the constructor typing equations. Semantic equalities of arbitrary literal labels are not used as Boolean grammar tests. An environment object Θ retains those labels and operation parameters. Its well-typed descriptions over $\theta : \Theta$ form the objects $E_{X,Y,\theta}$.

Lemma E.3 (Interpretation and substitution)**.** *Finite recursion constructs the interpretation of the displayed descriptions in the computational function objects, and tracked interpretations on their name presentations. Interpretation commutes with finite typed substitution. Pure structural maps, composition, branching, sequential pairing, and recursive execution have actual constructors.*

Proof.

(1) **Interpret the finite description tree.** A leaf is its retained parameter or primitive; a node applies the designated constructor morphism to the interpretations of its children. Products and exponentials of Theorem 1.4, lifted strength, and Proposition 1.5 give those morphisms. A recursion node denotes the least solution of its body, rather than a recursive definition of the tree interpreter itself. Evaluate the finite tree from its leaves to its root. This constructs its semantic interpretation with the stated type.

(2) **Implement the constructors on names.** On names, retain the primitive implementations and their classified-domain squares from Proposition E.1. Composition runs the first implementation and then the second on the returned name; branching inspects its finite tag; a recursive execution runs its body until its exit tag is returned. The exact finite-return construction supplies the latter partial map.

(3) **Prove domain and value adequacy.** Induction on the number of transitions identifies its domain and returned denotation with the semantic iteration. Structural induction then proves the same for the whole finite description.

(4) **Verify identity and composite substitution.** Substitution replaces slot leaves by their designated trees and shifts bound indices by the depth of the enclosing context. At a leaf the interpretation equation is its definition. At a compound node it follows from the equation for the children and the same constructor morphism. This proves identity and composite substitution, including the iteration node.

(5) **Retain parameters and construct code-output descriptions.** The algorithms traverse only finite trees; arbitrary supplied names remain labels. A code-output extension of the signature uses these same trees as data sorts, and its identity and composition constructors provide the compiler descriptions used in Construction 4.22.

□

For pure syntax, all literal numerical data are constructor strings over a finite alphabet. Prefix serialization and stack parsing therefore give total primitive-recursive maps $\mathsf{ser} : \mathsf{Tree} \to \mathbb{N}$ and $\mathsf{parse} : \mathbb{N} \to \mathsf{Tree}$, with $\mathsf{parse}\,\mathsf{ser} = \mathrm{id}$ (invalid words parse to a fixed dummy tree). Enumerating finite words up to a given length gives finite lists $E(s)$ containing every tree at some stage. Consequently pure trees are overt. This assertion concerns pure descriptions, not a numerical enumeration of arbitrary relative literal carriers.

E.1 The numerical program presentation

Let FG be the decidably ordered poset of finite functional graphs in $\mathbb{N} \times \mathbb{N}$, ordered by inclusion. Its empty graph is least. An ideal of FG determines a partial function by taking

its union, and every partial function determines its ideal of finite subgraphs. Directedness proves functionality and reconstructs every finite subgraph; these operations are inverse.

Lemma E.4 (Recognizing admitted finite-information ideals)**.** *For an overt poset P with Σ-classified order, an ideal belongs to $J_\Sigma P$ exactly when its membership predicate is uniformly Σ-valued.*

Proof.

(1) **Recover uniform membership from admitted pairings.** The positive upper test $u_p(q) = (p \leq q)$ has pairing $\chi_I(u_p) \leftrightarrow I(p)$, by lower closure and reflexivity. Admission therefore gives membership uniformly in p.

(2) **Construct admitted pairings from membership.** Conversely, uniform membership and an upper test have Σ-valued conjunction; overtness of P gives their existential pairing uniformly in the test.

□

Proposition E.5 (Numerical codes and extensional denotation)**.** *In* **Eff***, finite register programs construct a partial evaluator $\varphi(e, x)$ and a regular epimorphism*

$$q : \mathbb{N} \twoheadrightarrow \mathsf{PC} = J_\Sigma \mathsf{FG}, \qquad q(e) = \{g : g \subseteq \mathrm{graph}(\varphi_e)\}.$$

Its kernel pair is extensional equality of partial functions. Specialization and sequential composition have total recursive code constructors. The construction relativizes to a fixed oracle.

Proof.

(1) **Construct the register-program enumeration.** Encode finite instruction lists with instructions for increment, zero-test with decrement, jumps, and halt. A configuration stores the program, instruction pointer, and finite register array. The total primitive-recursive transition reads one instruction, updates that finite data, and leaves a halted configuration fixed. Invalid codes are assigned a fixed looping configuration. Primitive recursion constructs bounded runs. Minimizing the first halting stage and reading the output defines φ.

(2) **Implement the partial-recursive constructors.** Conversely, zero, successor, projections, composition, primitive recursion, and minimization have register-program implementations: use finite register blocks for intermediate values, counted loops for recursion, and an increasing search loop for minimization. Thus this is the ordinary partial-recursive enumeration underlying K_1 (van Oosten 2008).

(3) **Recognize finite subgraphs and prove admission.** For fixed e, g, the total stream whose stage-s value tests whether every pair in g is verified by an s-step run of e has hit support $g \subseteq \mathrm{graph}(\varphi_e)$. The empty graph is accepted immediately. The preceding lemma proves that $q(e)$ is admitted, uniformly in e.

(4) **Construct a uniform numerical code from a name.** To prove epicity, cover $J_\Sigma \mathsf{FG}$ by a partitioned assembly. Its uniform membership classifier on that cover times FG lifts to a total stream $\theta(a, g, k)$. On input x, search all graph codes and hit stages by finite diagonals until a hit is found for a graph containing (x, y), and return y. Directedness makes different possible discoveries agree; every pair in the represented partial function is eventually found. The search is one program with the supplied point name a as parameter, hence compiles uniformly into a numerical code. Its composite with q is the original cover, proving that q is regular epic.

(5) **Identify extensional equality and its name comparison.** Equality of the finite-subgraph ideals is exactly equality of their unions. Uniformly, a finite convergence witness for one code can be transported to an extensionally equal code by running the latter on the finitely many named inputs until the same outputs appear. This gives the equality comparison on names as well as the semantic kernel-pair equation.

(6) **Implement parameter specialization.** Specialization prefixes instructions loading the fixed numerical parameter and then runs the original code on its paired input.

(7) **Implement composition and oracle relativization.** Composition renames the scratch registers of two finite programs, runs the first, copies its output to the second input block, and runs the second. These are total recursive operations on instruction lists; their return domains and values follow from the two executions. An oracle-query instruction uses the same finite syntax and finite witnessed transitions relative to that oracle, so all comparisons relativize.

□

Lemma E.6 (Diagonal nontermination is not positive)**.** *In the numerical effective model, the complement of* $H = \{e : \varphi_e(e) \downarrow\}$ *is not* Σ*-classified.*

Proof.

(1) **Recognize diagonal termination positively.** H is classified by its bounded-run hit stream.

(2) **Derive a total decision from positive nontermination.** A Σ-classifier of its complement lifts on the projective numerical presentation to a total recursive hit stream. Interleaving the two searches would decide H.

(3) **Obtain the diagonal contradiction.** The register program which loops on input e when that decision says $e \in H$, and halts otherwise, has an index d by Proposition E.5. Evaluating at d contradicts either Boolean answer. This proves the claim.

□

The numerical and relative name comparisons above concern denotations and exact partial domains. Costs are supplied by the retained instruction events and valuations in Section 4.5 and Construction 12.1. Neither an extensional quotient nor projective lifting identifies the costs of its different representatives.

F The finite resource-certificate calculus

The certificate comparison in Section 12.5 uses a finite language whose proofs can be checked and interpreted. We construct it from the indexed adjunction of Proposition 1.10, specifying ordinary and linear families before the checking and replay procedures. As in (Niu et al. 2022; Grodin et al. 2024), costs and behavioral assertions have separate semantic roles. The annotation on a linear judgment is its augmentation grade; an execution-time interpretation requires the measured comparison.

F.1 Ordinary and linear families

Ordinary types over Γ are the split cartesian families of Proposition 1.10. The universe-free ordinary syntax has $0, 1, 2, \mathbb{N}$, finite lists and trees, dependent Π, Σ, Id, and Boolean type case. Its terms have variables, the corresponding constructors and eliminators, and

function symbols for finite descriptions of total data programs. Such programs are built from constructors, composition, Boolean case, and parameterized structural recursion, together with a finite supplied family of total observation maps $a_i(c, -)$; c remains an environment parameter. The defining equations for a program are read from its finite description, so their recognition does not require executing an arbitrary program. The scalar object S is also an ordinary sort, with rational constants and its specified finite-operation symbols. Scalar conversions require displayed equality derivations. Arbitrary semantic scalars are neither constructor data nor assumed to have decidable equality.

The ordinary rules are formation, introduction, elimination, their computation equations, congruence, symmetry and transitivity of equality, and conversion by a displayed equality derivation. For Π these are dependent abstraction, application, and their beta and eta equations; for Σ, dependent pairing and elimination with the corresponding equations; for Id, reflexivity and identity elimination with its reflexivity equation. The data sorts have their constructor and structural-recursion rules. Boolean type case has $\mathrm{If}(0; A, B) = A$ and $\mathrm{If}(1; A, B) = B$, including substitution. There is no equality-reflection or semantic-truth rule.

These rules are interpreted by composition and dependent adjoints in $\mathcal{E}/\Gamma$, and diagonals for identity. To specify split Boolean case, decompose a parameter object by the two fibres of its Boolean term and glue the two displays. Choose the constant-zero case to be its first display and the constant-one case to be its second; over an initial base both are the identity display. The unique cartesian comparisons give the substitution equations. This external normalization introduces no Boolean test for equality of semantic types.

For the linear fragment put $G_b M = \{m : \mathrm{aug}_M(m) = b\}$, and interpret

$$\Gamma; \Delta \vdash_b t : M, \qquad \Delta = x_1 : M_1, \ldots, x_k : M_k,$$

by a continuous linear map $\bigotimes_i M_i \to M$ satisfying $\mathrm{aug}_M\, t = b\ \mathrm{aug}_{\otimes_i M_i}$. All displayed linear types and b depend only on the ordinary context. The empty tensor is $I = (S, \mathrm{id})$. The linear type formers are

$$I, \quad 0_\ell, \quad M \otimes N, \quad M \multimap N, \quad F_d A, \quad \Sigma^\ell_{a:A} M(a), \quad \Pi^\ell_{a:A} M(a),$$

and $G_b M$ is an ordinary type.

The dependent linear sum is presented by generators $[a, m]$, linearity in m, and the continuity covers in each fibre; its augmentation is $\mathrm{aug}[a, m] = \mathrm{aug}_{M(a)}\, m$. The dependent product is the continuous submodule

$$\left\{ (r, m) : r : S,\ m : \prod_{a:A} UM(a),\ \forall a\ \mathrm{aug}_{M(a)}\, m(a) = r \right\}, \qquad \mathrm{aug}(r, m) = r. \tag{F.1}$$

The scalar coordinate is retained even if A is empty. Consequently the empty sum is 0_ℓ and the empty product is I. Maps out of the sum are the corresponding families of maps; maps into the product are the corresponding families with one common augmentation. These facts prove the dependent adjunctions. Tensor Frobenius sends $[a, m] \otimes n$ to $[a, m \otimes n]$, with inverse given by the opposite formula. The presentation theorem and pointwise product formula prove Beck–Chevalley and coherence. Defining these operations componentwise on split cartesian families gives their substitution comparisons.

The following rule schemes specify the linear fragment; every conversion mentioned in them is a supplied finite equality tree. A variable and the unit introduction have grade 1. Tensor introduction uses disjoint linear contexts and multiplies grades. Linear abstraction preserves the grade; application uses disjoint contexts and multiplies grades. Tensor and

unit elimination use the same context splitting and grade multiplication. There is the unique unit-grade discard $\epsilon_M : M \to I$ and the initial-type eliminator from 0_ℓ. Weakening is its explicit elaboration $\mathrm{weak}_x(t) = \mathrm{let} * = \epsilon_N x \text{ in } t$. There is no unrestricted contraction of a linear variable. Addition uses the same input context in both premises and adds their grades; scalar multiplication multiplies the grade; the zero map has grade 0.

For the ordinary–linear interface, an ordinary $a : A$ gives $[a] : F_d A$ at empty linear context and grade 1. A term $t : M$ at empty linear context and grade b gives $\mathrm{up}_b(t) : G_b M$, with inverse down_b. Eliminating $F_d A$ uses its generator and linear universal property: from a major premise of grade c and a body of grade b, the result has grade cb; the other context, result type, and body grade are pulled back from the context before introducing the bound ordinary variable. Dependent linear sum introduction uses a supplied ordinary index and the grade of its component; its eliminator has the same independence and multiplication rules. Dependent product abstraction preserves a grade pulled back from the outer context, and evaluation at an ordinary index preserves it. Include their beta and eta equations, typed exchange, congruence, the semimodule equations with their displayed grades, and identity transport into linear families with its reflexivity equation. These rules are exactly the adjunction, tensor, and dependent universal-property maps just constructed.

Lemma F.1 (Graded substitution)**.** *Ordinary substitution and graded linear substitution are finite operations on derivations. Substituting a grade-c term for a linear variable in a grade-b judgment yields grade bc. They have finite derivations of their typed identity and composition equations, and preserve concrete interpretation.*

Proof.

(1) **Define substitution at binders and variables.** Use de Bruijn indices and recurse simultaneously on formation, typing, and equality trees. At a binder, shift indices and substitute in the type annotations and conversion derivations. At a variable, insert the supplied derivation.

(2) **Preserve grades at split-context nodes.** At a split-context node, transform the unique premise containing the substituted variable and insert the associativity and commutativity equations for the grades.

(3) **Preserve additive and zero grades.** At addition, transform both premise trees and insert $(b_1 + b_2)c = b_1 c + b_2 c$; at zero use $0c = 0$.

(4) **Execute the discarded argument and retain its grade.** At weakening of the substituted variable the output is

$$(\mathrm{weak}_x\, t)\langle u/x\rangle = \mathrm{let} * = \epsilon_M u \text{ in } t.$$

The discarded argument is executed as the stated linear morphism; its grade c remains, and unit elimination has grade cb, converted to bc.

(5) **Handle the remaining rule templates.** At weakening of another variable rebuild that node after the recursive substitution. The scalar and dependent eliminator cases use their displayed scalar and substitution equations. These are finite rule templates.

(6) **Prove the substitution equations.** The identity and composition equations follow by the same induction. At weakening, unit beta, eta, and commuting conversions identify the iterated discards; at addition use distributivity and congruence. These are equality derivations, not literal equalities of different proof trees.

(7) **Verify preservation of interpretation.** The interpretation of each step is composition, evaluation, or the relevant dependent universal property, which proves preservation of interpretation.

□

F.2 Annotated checking and calculation replay

Pure sort, type, term, context, judgment, and derivation descriptions are finite constructor trees. A derivation node stores its exact root, rule tag, premise trees, context splits, and all conversion proofs. Distinct tags distinguish ordinary and linear judgments, formation, typing, and equality. An observation leaf stores $(i, \vec{x}, b)$ with constructor data $\vec{x}$ and Boolean b; its acceptance locus is the result of actually testing $a_i(c, \vec{x}) = b$. The environment c is supplied to the checker and is not quoted.

Theorem F.2 (Finite checker and concrete soundness)**.** *There is a total finite data program* $\mathsf{Check}_{\mathcal{T},c}(d, j) : \mathsf{Tree} \times \mathsf{Tree} \to 2$ *whose accepted trees are exactly the annotated derivations at the stated root* j*, on their observation loci. Accepted derivations have the stated interpretations in the indexed ordinary and graded linear model. Renaming, ordinary substitution, graded linear substitution, application, and their finite compositions are implemented proof transformations with verified exact roots.*

Proof.

(1) **Check the root and inference annotations.** Parse both trees and check that the stored root of d is literally j. At a node compare its annotations with the finite rule schema, check the context splits and binder indices, and recursively check its proper premise subtrees.

(2) **Check supplied equality and program equations.** A conversion node must supply an equality derivation; a scalar-equation leaf must be an instance of a semiring law or a proved equation from its defining finite program. It never tests equality of arbitrary scalar denotations. The program-symbol scheme checks its finite source description and the appropriate defining equation.

(3) **Check observations and establish totality.** An observation leaf makes its designated total observation call and compares with the recorded Boolean. Malformed trees return zero. This is structural recursion with total observation calls, so it is a total data program.

(4) **Prove checker exactness.** Induction on a derivation proves acceptance. Conversely induction on an accepted tree reconstructs each inference from its checked premises and exact annotations. At an observation leaf the actual Boolean test supplies its claimed validity locus. This proves checker exactness.

(5) **Interpret the rules and their grades.** For soundness, interpret contexts by comprehension, ordinary substitution by pullback, ordinary rules by their dependent universal properties, and linear rules by the maps above. At an application or eliminator, augmentation grades multiply; at addition they add. The weakening case is composition with discard and retains its grade.

(6) **Verify equality, observations, and semantic soundness.** The semimodule and adjunction equations interpret the equality rules; observation leaves are valid precisely on their tested loci. Rule induction proves soundness, simultaneously with equality and substitution.

(7) **Verify the proof-transformer invariants.** The explicit transformations of Lemma F.1 rebuild these same finite rule templates with their exact annotations; node induction proves their checker-correctness invariants.

□

Lemma F.3 (Fixed-program calculation)**.** *For a fixed total data program $f : X \to D$ on the displayed constructor data sorts there is a total finite program $\mathsf{calc}_f(x)$ constructing an accepted closed derivation of*

$$\mathrm{Id}_D(\mathsf{f}(\bar{x}), \overline{f(x)})$$

at that exact root, with the supplied environment retained. The root and acceptance identities have finite proofs in the same calculus.

Proof.

(1) **Quote constructor data and initialize structural induction.** Constructor quotation is recursion on the data value; it does not quote an arbitrary semantic element or the environment. Induct on the finite description of f. For a constructor, use its defining equation and reflexivity.

(2) **Construct the certificate for composition.** For composition, recursively calculate the intermediate data, then use congruence and transitivity.

(3) **Construct the certificate for branching.** For a Boolean case, calculate its selector and use the equation for the selected branch.

(4) **Construct the certificate for recursion.** For structural recursion, recurse on the immediate smaller data, apply the defining successor equation, and combine the induction certificates.

(5) **Construct the certificate for an observed answer.** The observation case calls a_i and emits the leaf with its actual answer; its check is reflexive.

(6) **Verify exact roots and checker acceptance.** Each node template has its exact annotated root, and its checker equation reduces to the comparisons of those annotations and the premise checks. Induction gives both invariants as finite derivations using the data eliminators.

(7) **Apply the construction to fixed proof-processing programs.** This also applies to a fixed checker or proof-transformer program: it proves the Boolean calculation of that program, not the conclusion of an arbitrary accepted derivation by a reflection rule.

□

An exact judgment and an inhabitance boundary have distinct tags. At an ordinary boundary $(\gamma; a)$, extract the term from an ordinary typing root and check that exact root in context γ. At a linear boundary $(\gamma; \delta; m; b)$, do the same while retaining the linear context and grade. A mismatching tag returns zero. Denote these total tests, together with exact-root checking, by $\mathsf{Acc}_{\mathcal{T},c}(d, j)$. For the resource types, constructor substitution of e, β constructs $j_A^{\mathrm{cl}}(e, \beta)$ with both described contexts empty. The ambient parameters still index the family of these closed instances. The ordinary certificate family and its free quantitative object are therefore exactly

$$C_{\mathcal{T}}^{\mathrm{cl}}[A; e, \beta] = \sum_{d:\mathsf{Tree}} \mathrm{Id}_2(\mathsf{Acc}_{\mathcal{T},c}(d, j_A^{\mathrm{cl}}(e, \beta)), 1), \qquad \mathbb{B}^{\mathrm{cl}}[A; e, \beta] = F_d C_{\mathcal{T}}^{\mathrm{cl}}[A; e, \beta].$$

Their supports and denotation images are ordinary regular images.

A finite-source certificate retains such an accepted tree, a finite source-program description, its finite generation trace, and its declared publication fields. Its forgetful map returns the tree. A finite positive-probability witness $\Theta^{\mathrm{cl}}(q; A; e, \beta)$ retains a finite list of such accepted trees, their nonnegative rational weights, and the checked rational inequality that their sum exceeds $q > 0$ (and the source trace establishing those weights when specified). Its list is nonempty; returning its first entry gives the certificate map used in the soundness theorem. Neither construction requires a probability-to-time identification.

Finally, finite search through the lists $E(s)$ of pure trees has support $\exists d\, \mathsf{Acc}_{\mathcal{T},c}(d, j) = 1$, by totality of the checker and exhaustiveness of that enumeration. The support is Σ-classified, uniformly in j, c. This identifies certificate search with certificate existence.

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